2026-11-06 15:44:11 You

I am a scientist of geoscience section. Recently, AI is being developed rapidly and applied to almost of sections. So, I want to ask some questions.

1. AI systems which deal with the geological exploration such as ground survey, geochemical prospecting and geophysical prospecting.

2. From what and how to start to develop such AI systems for geological exploration?

3. Titles and contents of the thesis of Doctor's degree which are made through the development of AI technology in the mineral exploration section.

4. How to apply AI to the subjects of our faculty which is named "Faculty of geoscience and technology". Our faculty gives lectures on fundamental subjects such as geology, mineralogy and petrology, geological structure and major subjects such as geology of ore deposits, mineral exploration and synthetic assessment, mineral engineering, applied geochemistry, geophysical prospecting, remote sensing, hydro-geochemistry, civil engineering.

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Application of Artificial Intelligence in Geological Sciences and Mineral Exploration

Your questions are very timely and important. I believe that Artificial Intelligence (AI) can become a major research and educational direction for a Faculty of Geoscience and Technology.

In particular, AI has considerable potential in mineral exploration because geological exploration involves the integration of many different types of information:

- Geological maps and field observations.
- Mineralogical and petrological data.
- Geochemical analyses of rocks, soils, stream sediments and water.
- Geophysical data, including magnetic, gravity, electrical, electromagnetic and radiometric surveys.

- Satellite images and remote sensing data.
- Borehole logs, drilling results and mineral resource information.
- Structural geology and tectonic information.

Traditionally, geologists and exploration specialists interpret these datasets separately and then integrate them using geological knowledge and experience. AI can help integrate large, complicated datasets, identify hidden relationships, predict prospective areas and support exploration decisions.

I will discuss your four questions from both a **scientific research perspective and a university education perspective**.

1. AI systems for geological exploration

1.1. Main categories of AI applications

AI applications in geological exploration can be divided into seven major categories.



A. AI for geological field surveys

- Automatic identification and classification of rock types from photographs.
- Recognition of geological boundaries, faults, folds and fractures.
- Identification of alteration zones and mineralization indicators.
- Digital geological mapping using mobile devices, GPS and AI.
- Interpretation of geological field photographs and outcrop images.



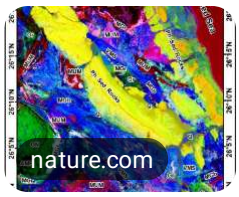
B. AI for geochemical prospecting

- Identification of geochemical anomalies.
- Classification of mineralized and background samples.
- Analysis of multi-element geochemical associations.
- Prediction of mineralization from soil, stream sediment and rock geochemistry.
- Identification of geochemical halos around ore deposits.
- Detection of relationships between pathfinder elements and economic metals.



C. AI for geophysical prospecting

- Interpretation of magnetic and gravity anomalies.
- Automatic identification of geological structures.
- Classification of geophysical signatures associated with mineral deposits.
- Interpretation of electrical resistivity and induced polarization (IP) data.
- Integration of seismic, electromagnetic and radiometric data.
- Prediction of subsurface geological structures and mineralized zones.



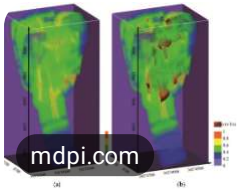
D. AI for remote sensing and satellite imagery

- Mapping hydrothermal alteration zones.
- Identification of iron oxides, clay minerals and other alteration minerals.
- Lithological classification.
- Structural lineament extraction.
- Mapping of potential mineralized zones using multispectral and hyperspectral images.
- Integration of satellite images with geological and geochemical data.



E. AI for drilling and subsurface exploration

- Automatic drill-core logging and rock classification.
- Prediction of ore intervals from borehole data.
- Estimation of mineral grades.
- Three-dimensional geological modelling.
- Prediction of mineralized bodies between drill holes.
- Optimization of drilling locations.



F. AI for mineral prospectivity mapping

- Integration of geological, geochemical, geophysical and remote sensing datasets.
- Prediction of areas favorable for specific types of mineral deposits.
- Ranking of prospective areas.
- Production of mineral potential maps.
- Identification of unexplored targets.



G. AI for exploration decision support

- Selection of the next exploration locations.
- Optimization of survey programs and sampling density.
- Estimation of exploration uncertainty.
- Cost-effective exploration planning.
- Integration of geological knowledge with AI predictions.

1.2. Important AI techniques

You do not need to develop a completely new AI algorithm for every geological application. Many established AI methods can be adapted to geological problems.

AI technique	Geological application
Random Forest (RF)	Geochemical anomaly detection and mineral prospectivity

Support Vector Machine (SVM)	Rock classification and mineral potential prediction
XGBoost / LightGBM	Mineral prospectivity and geochemical prediction
Artificial Neural Networks (ANN)	Complex relationships in geological datasets
Convolutional Neural Networks (CNN)	Geological photographs, satellite images and drill-core images
Deep Learning	Multisource geological data interpretation
Autoencoders	Anomaly detection and dimensionality reduction
Clustering (K-means, DBSCAN)	Geochemical and mineralogical classification
Graph Neural Networks (GNN)	Relationships among geological structures and spatial data
Transformer models	Multimodal geological data and scientific document interpretation
Bayesian Machine Learning	Probability estimation and exploration uncertainty
Reinforcement Learning	Exploration planning and sequential decision-making

My recommendation: Start with Random Forest, XGBoost, SVM and basic CNN models. These methods are relatively accessible, have substantial research literature and can provide useful baseline results. More complicated deep-learning approaches can follow when sufficient data become available.

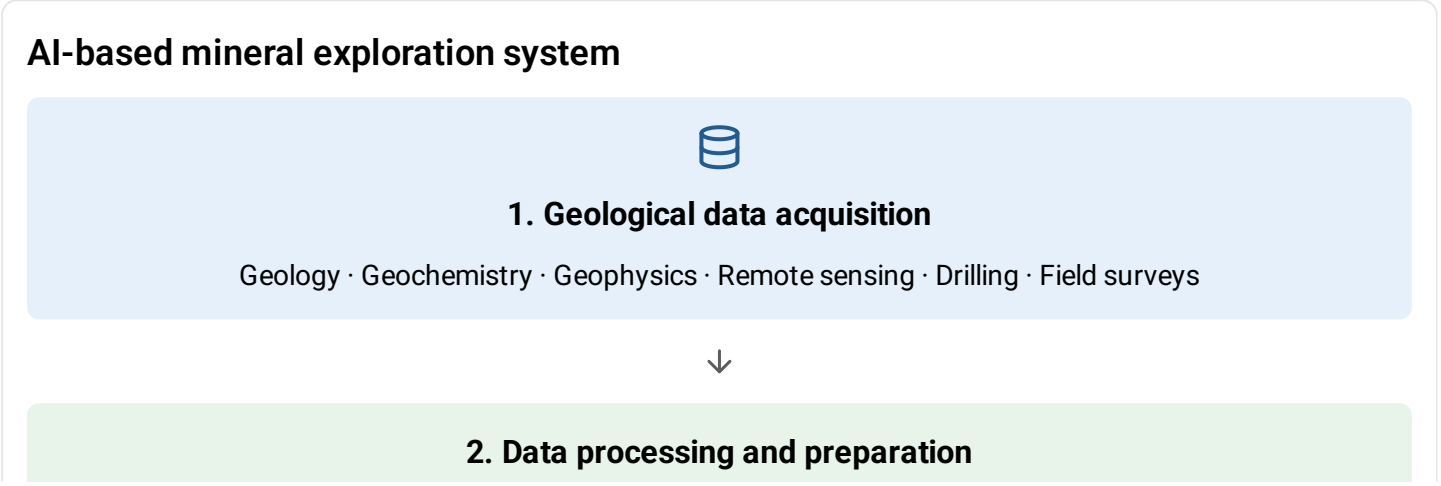
2. How to start developing AI systems for geological exploration

This is probably the most important question from a research and institutional development perspective.

My advice is **not to start by developing a complicated deep-learning system**. Start by developing a relatively simple AI application that solves a real geological exploration problem. Gradually expand it into a comprehensive exploration system.

2.1. Basic architecture of an AI-based geological exploration system

A typical system should contain six major components.



Data cleaning · GIS processing · Coordinate matching · Feature extraction · Quality control

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3. AI and machine-learning models

Random Forest · XGBoost · SVM · CNN · Deep Learning · Bayesian Models

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4. Data integration and geological interpretation

Multisource integration · Geological knowledge · Model interpretation · Uncertainty assessment

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5. Exploration prediction and visualization

Mineral potential maps · Geological models · Anomaly maps · 3D visualization

↓

6. Geological verification and decision-making

Field investigation · Sampling · Geophysical surveys · Drilling · Model updating

Conceptual architecture. The workflow is iterative: new field and drilling results should be fed back into the database and models.

2.2. A practical development roadmap

I suggest the following seven steps.

Step 1. Select a specific geological problem

Choose one problem that has scientific importance and sufficient available data.

Possible starting projects:

Project	Main objective	Difficulty
Geochemical anomaly detection	Identify anomalous elemental concentrations	Low–medium
Rock classification	Classify rocks using chemical and mineralogical data	Low–medium
Alteration mapping	Identify hydrothermal alteration from satellite images	Medium
Geophysical anomaly interpretation	Identify structures associated with mineralization	Medium
Mineral prospectivity mapping	Predict areas favorable for ore deposits	Medium–high

3D mineralization prediction	Predict ore bodies at depth	High

I would recommend beginning with **AI-based geochemical anomaly detection**, followed by integrated mineral prospectivity mapping.

The reasons are:

- Geochemical data are relatively straightforward to organize.
- You can begin with conventional statistical methods.
- You can compare traditional geochemical interpretation with machine learning.
- The results can be evaluated using known mineral occurrences.
- It provides a foundation for integrating geophysics and remote sensing later.

Step 2. Collect geological datasets

You need to establish a geological database.

For example, a copper exploration project might require:

Dataset	Important parameters
Geological	Lithology, faults, folds, geological boundaries
Geochemical	Cu, Au, Mo, Pb, Zn, Ag, As, Sb, etc.
Geophysical	Magnetic intensity, gravity, resistivity, IP chargeability
Remote sensing	ASTER, Landsat, Sentinel-2, hyperspectral data
Mineral occurrences	Known deposits, prospects and mineral showings
Drilling	Lithology, alteration, assay results, mineralized intervals

For publicly accessible datasets, I recommend examining these sources:

- United States Geological Survey (USGS): Mineral Resources Data. It provides geological, geochemical, geophysical and mineral occurrence information. U.S. Geolog... +1
- Geoscience Australia: Geoscience Australia. Its precompetitive geoscience datasets include geology, geophysics, geochemistry and drilling information. Geoscience A...
- Geological Survey of Queensland: GSQ Open Data Portal. It provides geochemistry, geological mapping, mineral occurrences, boreholes and geophysical data. Business Que...

You should also investigate the geological survey and mineral resource databases of your own country. Local data are often more useful than global datasets for developing an operational exploration system.

Step 3. Learn programming and data science

You do not necessarily need to be an expert computer scientist. However, someone on your research team should have practical programming and machine-learning skills.

I recommend the following software environment:

Software / language	Purpose
Python	Main AI programming language
Jupyter Notebook	Data analysis and experiments
Pandas / NumPy	Data management and numerical calculations
Scikit-learn	Conventional machine learning
XGBoost	Advanced tree-based prediction
PyTorch	Deep learning
QGIS	Geological and spatial data processing
GDAL / Rasterio	Raster data processing
GeoPandas	Geological spatial data
Google Earth Engine	Satellite and remote-sensing analysis

Useful learning resources:

- Python Getting Started
- Scikit-learn tutorials
- PyTorch tutorials
- QGIS
- Google Earth Engine

Step 4. Develop the first machine-learning model

Consider a simple example.

Suppose you have 1,000 geochemical sampling points containing Cu, Au, Mo, Pb, Zn, As and other elements.

You want to identify areas that might contain copper mineralization.

Your initial procedure could be:

1. Organize the geochemical data in a spreadsheet or database.
2. Remove duplicate records and examine missing values.
3. Check analytical quality, detection limits and sampling locations.
4. Examine elemental distributions and correlations.
5. Identify known copper deposits and prospects.

6. Develop training datasets using geological knowledge.
7. Train Random Forest and XGBoost models.
8. Compare their predictive performance.
9. Produce geochemical anomaly and prospectivity maps.
10. Compare predictions with independent geological evidence.

An illustrative machine-learning dataset might look like this:

Cu (ppm)	Au (ppb)	Mo (ppm)	Pb (ppm)	Zn (ppm)	Known mineralization
450	85	25	120	180	1
25	4	2	18	35	0
680	110	40	90	210	1
35	6	3	22	40	0

Here, 1 represents a sample associated with known mineralization and 0 represents a background or selected non-mineralized sample. In real exploration, an unrecorded deposit is not necessarily a true negative, so the way negative samples are selected is an important research issue.

Step 5. Evaluate the model scientifically

This is an especially important point for doctoral research.

You should not simply demonstrate that an AI model produces an attractive mineral potential map.

You must demonstrate that the prediction is scientifically reliable.

Important evaluation methods include:

- Accuracy, precision, recall and F1-score.
- ROC-AUC and precision-recall AUC.
- Confusion matrix.
- Spatial cross-validation.
- Independent validation using previously withheld deposits or exploration areas.
- Geological interpretation of the important variables.
- Uncertainty estimation.

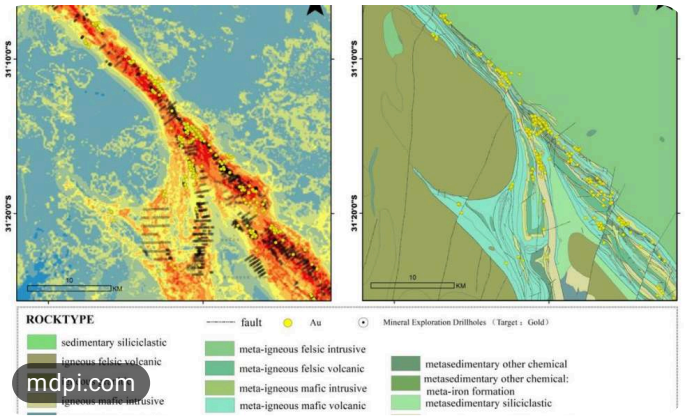
One major problem in geological AI is spatial autocorrelation. Nearby samples may have very similar characteristics. Randomly dividing neighboring samples between training and testing can therefore produce overly optimistic results. Spatially separated validation is essential when the intended application is prediction in unexplored areas. ScienceDir... +1

Step 6. Integrate multiple geological datasets

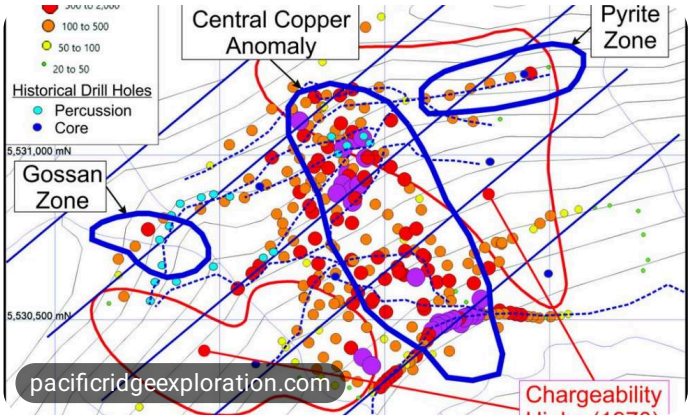
After developing individual models, combine the different sources of evidence.

For example:

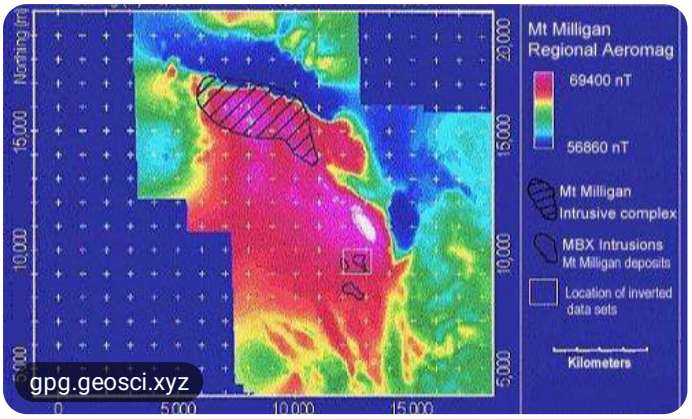
Integrated copper prospectivity prediction



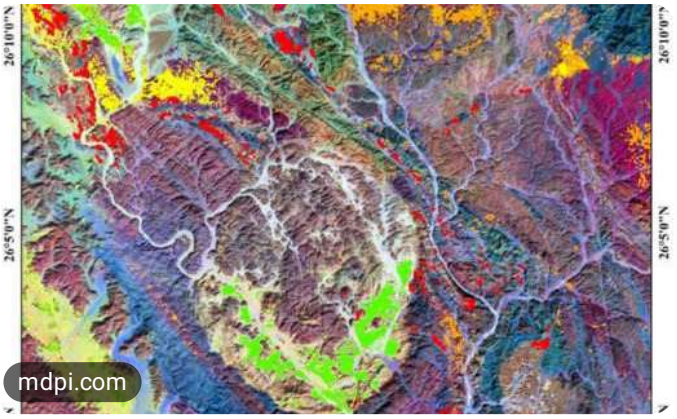
Geological data



Geochemical data



Geophysical data



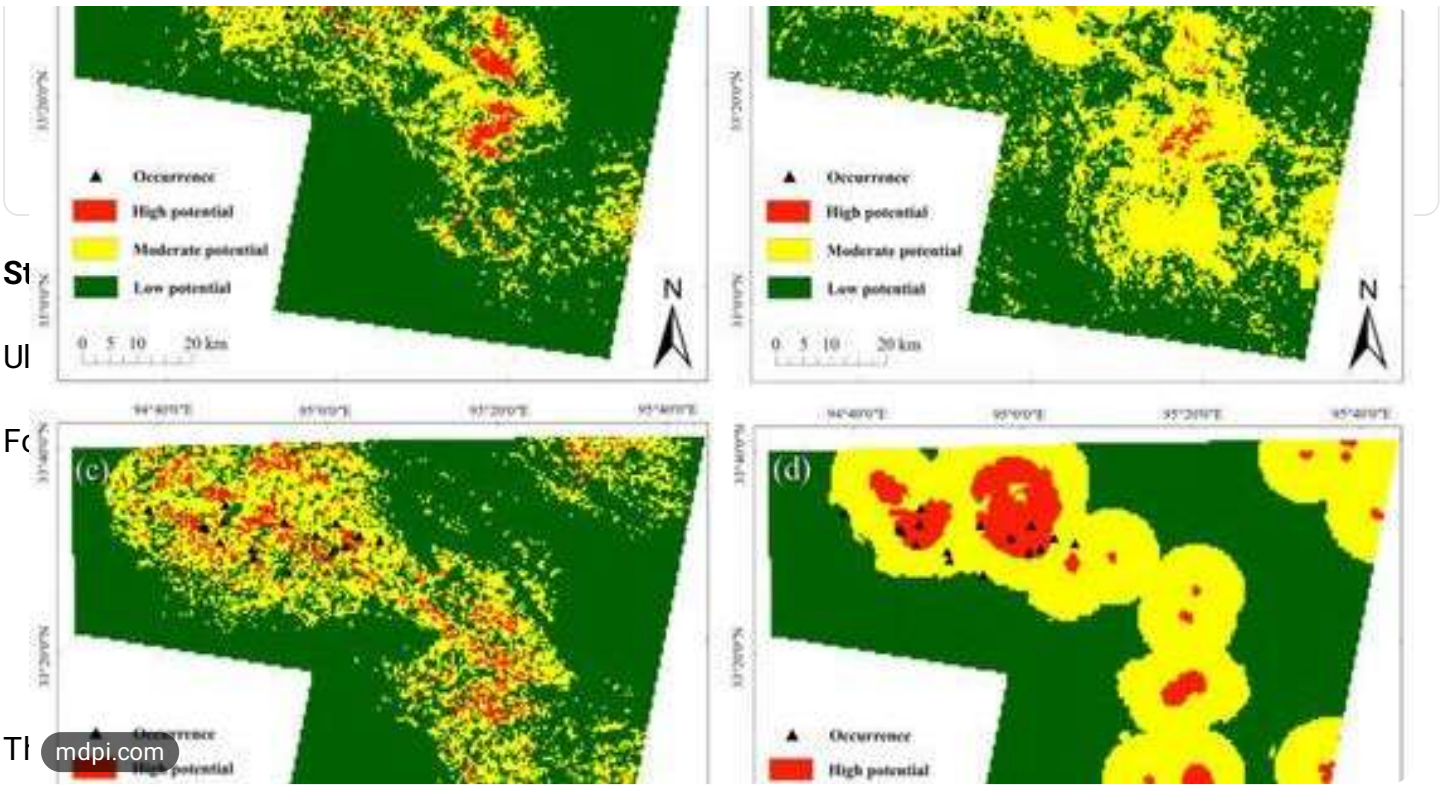
Remote sensing

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Multisource AI data-fusion model

Random Forest / XGBoost / CNN / Hybrid models

↓



Final mineral prospectivity map

Hybrid map potential, uncertainty and recommended exploration targets

- Geological feature maps.
- Geochemical anomaly maps.
- Geophysical interpretation maps.
- Mineral potential maps.
- Ranked exploration targets.
- Explanations of the main factors influencing predictions.

This could become a dedicated software product for your faculty or a research platform for cooperation with geological surveys and mineral exploration companies.

2.3. How should your research team be organized?

I recommend an interdisciplinary team rather than placing the entire responsibility on geologists.

Researcher	Main responsibility
Economic geologist	Deposit models and mineralization interpretation
Exploration geologist	Field surveys and validation
Geochemist	Geochemical database and anomaly interpretation
Geophysicist	Geophysical data and interpretation
Remote-sensing specialist	Satellite and hyperspectral data
Computer scientist / AI specialist	Algorithms and software development
GIS specialist	Spatial database and visualization
Mineral engineer	Mineral processing and resource applications

Exported with AI Exporter

One of the most important principles is that AI specialists must work closely with geologists. A model can achieve good statistical performance while producing geologically unreasonable predictions.

3. Doctoral thesis topics and examples of previous research

There are already several doctoral studies demonstrating how AI and data science can be integrated into mineral exploration.

I have identified some actual doctoral dissertations that may be useful for your literature review and for designing future doctoral research in your faculty.

3.1. Examples of completed doctoral theses

2026 · Doctoral dissertation

1. LLM-Assisted Neuro-Symbolic AI Models for Advanced Mineral Exploration

Weilin Chen, University of Idaho, USA.

Main research contents:

- Extracting geological knowledge from mineral deposit literature using Large Language Models (LLMs).
- Constructing geological knowledge graphs.
- Integrating geological rules with machine-learning models.
- Applying Random Forest to geochemical data from the USGS National Geochemical Database.
- Using SHAP to interpret predictions and examine their consistency with geological knowledge.

This is an interesting example of combining AI with conventional geological expertise. University of I...

University repository:

2024 · PhD thesis

2. A Data Science Framework for Mineral Resource Exploration and Estimation Using Remote Sensing and Machine Learning

Muhammad Ahsan Mahboob, University of the Witwatersrand, South Africa.

Main research contents:

- Satellite image processing for hydrothermal alteration mapping.
- Geochemical and drilling data analysis.
- Application of CNN, Random Forest, SVM and regression models.
- Comparison of machine-learning and geostatistical approaches.
- Prediction of copper and other mineralization indicators.

Particularly relevant to your subjects of remote sensing, applied geochemistry and mineral exploration. WIReDSpace

Full thesis:

2023 · PhD thesis

3. Application of Neural Networks for Geophysical Interpretation – A Case Study of Basement Delineation around a Detrital Iron Ore Deposit, Hamersley Province, Western Australia

Tasman Gillfeather-Clark, University of Western Australia.

Main research contents:

- Development of three neural-network approaches.
- Interpretation of seismic data.
- Integration of airborne electromagnetic inversion.
- Modelling of the basement surface.
- Application of AI to subsurface geological interpretation.

This is a good model for doctoral research in geophysical prospecting. the UWA Profil...

Full thesis:

2023 · PhD thesis

4. Factors Contributing to Metal Endowment in the Western Wabigoon and Southern Abitibi Subprovinces: A Machine Learning Approach to Precambrian Greenstone Belts

Becki Montsion, University of Western Australia and Laurentian University.

Main research contents:

- Machine-learning analysis of geological controls on mineralization.
- Geochemical classification.
- Mapping structural complexity.
- Analysis of hydrothermal alteration.
- Random Forest-based ranking of geological factors controlling mineralization.
- Application to magmatic Ni-Cu-PGE, VMS and orogenic gold deposits.

This thesis is especially relevant to ore deposit geology and integrated mineral exploration.
the UWA Profil...

Full thesis:

2021 · PhD thesis

5. Geological Knowledge Graph Construction from Mineral Exploration Text

Majigsuren Enkhsaikhan, University of Western Australia.

Main research contents:

- Extraction of geological information from exploration reports.
- Automatic recognition of geological entities and relationships.
- Construction of geological knowledge graphs.
- Development of machine-learning and natural-language-processing workflows.
- Integration of geological knowledge into AI systems.

This research is particularly interesting for developing an AI system that can learn from geological reports and exploration records. the UWA Profil...

Full thesis:

These examples demonstrate that doctoral research can focus on different components of AI-assisted mineral exploration. A PhD does not necessarily need to develop an entire exploration software platform. It can make an original scientific contribution to one important component.

3.2. Suggested doctoral thesis titles for your faculty

I would propose the following topics. I have grouped them according to their research focus.

Group A. Geochemical exploration

Proposed title 1

Development of an Artificial Intelligence-Based Integrated Geochemical Anomaly Detection System for Mineral Exploration

Possible thesis contents:

1. Review of conventional geochemical anomaly detection methods.
2. Development of a geochemical database.
3. Statistical and spatial analysis of multi-element geochemical data.
4. Application of Random Forest, XGBoost and deep learning.
5. Development of algorithms for detecting weak and concealed geochemical anomalies.
6. Integration of geological and geochemical information.
7. Development of a geochemical anomaly prediction system.
8. Validation using known mineral deposits and independent field samples.

Proposed title 2

Application of Explainable Artificial Intelligence for Identification of Geochemical Signatures Associated with Hydrothermal Ore Deposits

Possible research emphasis:

- Element associations and pathfinder elements.
- Geochemical zoning.
- Hydrothermal alteration signatures.
- Explainable AI and geological interpretation.
- Identification of geochemical indicators for different deposit types.

Group B. Geophysical exploration

Proposed title 3

Development of Deep Learning Models for Integrated Interpretation of Magnetic, Gravity and Electrical Geophysical Data in Mineral Exploration

Possible thesis contents:

1. Compilation of geophysical datasets.
2. Processing and enhancement of geophysical anomalies.
3. Identification of geological structures.
4. Development of CNN and other deep-learning models.
5. Integration of multiple geophysical methods.
6. Prediction of subsurface geological structures.
7. Comparison with conventional geophysical interpretation.
8. Application to a selected mineralized district.

Proposed title 4

AI-Based Three-Dimensional Modelling and Prediction of Concealed Mineral Deposits Using Integrated Geophysical and Geological Data

Research emphasis:

- 3D geological modelling.
- Borehole and geophysical data integration.
- Deep-learning-based subsurface prediction.
- Uncertainty estimation.
- Selection of optimal drilling targets.

Group C. Remote sensing and mineral exploration

Proposed title 5

Development of an AI-Based Remote Sensing System for Identification and Mapping of Hydrothermal Alteration Zones Associated with Mineral Deposits

Possible thesis contents:

1. Processing of ASTER, Landsat and Sentinel-2 images.
2. Spectral analysis of alteration minerals.
3. Identification of iron oxides, clay minerals and other alteration indicators.
4. CNN-based image classification.
5. Hyperspectral image analysis.
6. Integration of remote sensing and geological information.
7. Development of an alteration mapping system.
8. Field verification of predicted alteration zones.

Group D. Integrated mineral exploration

Proposed title 6

Development of an Explainable Artificial Intelligence-Based Multisource Mineral Prospectivity Mapping System Integrating Geological, Geochemical, Geophysical and Remote Sensing Data

Strong candidate for a flagship doctoral research project

This topic is particularly attractive because it integrates several subjects taught in your faculty.

Possible thesis contents:

1. Theoretical background of mineral prospectivity mapping and AI.
2. Geological and mineral-system modelling.
3. Development of geological, geochemical, geophysical and remote sensing databases.
4. Development of individual AI models for each dataset.
5. Development of multisource data-fusion algorithms.
6. Development of explainable AI methods.
7. Production of mineral prospectivity maps.
8. Spatial validation and uncertainty analysis.
9. Field investigation and verification.
10. Development of a prototype exploration decision-support system.

Group E. Advanced AI and geological knowledge

Proposed title 7

Development of a Geological Knowledge Graph and Large Language Model-Based Intelligent System for Mineral Exploration

Possible thesis contents:

1. Collection of geological maps, reports, publications and exploration records.
2. Extraction of geological entities and relationships using natural language processing.
3. Development of a mineral deposit knowledge graph.
4. Integration of deposit models and geological rules.
5. Development of an LLM-based geological information retrieval system.
6. Integration with machine-learning prediction models.
7. Evaluation by expert geologists.

Proposed title 8

Development of a Physics-Informed Artificial Intelligence Framework for Predicting Mineralization and Subsurface Geological Structures

Possible thesis contents:

- Physical constraints from geophysics.
- Geological structural constraints.
- Physics-informed neural networks.
- Inversion and forward modelling.
- Geological uncertainty and model validation.

3.3. A recommended structure for a doctoral thesis

For the integrated mineral prospectivity mapping topic, I suggest the following chapter structure:

Chapter	Proposed contents
1. Introduction	Research background, problems, objectives and significance
2. Literature review	Mineral exploration, AI, machine learning and mineral systems
3. Geological setting	Study area, tectonic setting and ore deposit geology
4. Data preparation	Geological, geochemical, geophysical and remote sensing databases
5. Individual AI models	Development of models for individual datasets
6. Multisource integration	Development of integrated AI prediction algorithms
7. Model evaluation	Accuracy, spatial validation, uncertainty and explainability
8. Mineral prospectivity mapping	Production and interpretation of mineral potential maps
9. Field verification	Geological investigation and evaluation of predicted targets
10. Conclusions	Scientific contributions, limitations and future applications

The most important doctoral contribution should be a new method, model, scientific understanding or validated framework—not simply the application of an existing algorithm to a new dataset.

4. How to apply AI to the subjects of the Faculty of Geoscience and Technology

This is where I see the greatest long-term opportunity.

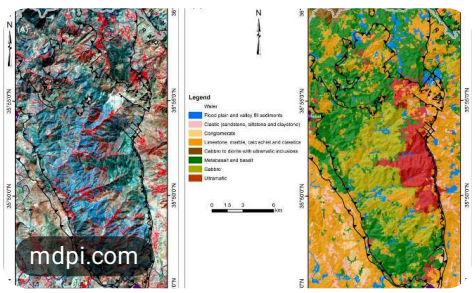
Your faculty has a particularly suitable range of subjects for AI applications because the subjects are closely interconnected. For example, mineral exploration requires knowledge of geology, mineralogy, petrology, geochemistry, geophysics and remote sensing.

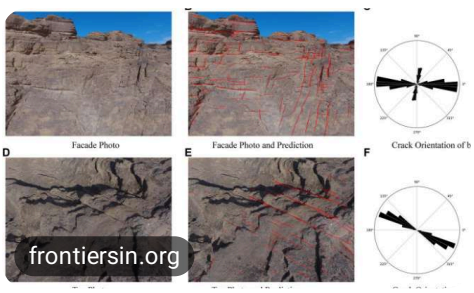
Rather than introducing AI only as a separate subject, I recommend introducing AI into **almost every existing subject as a new research and teaching methodology**.

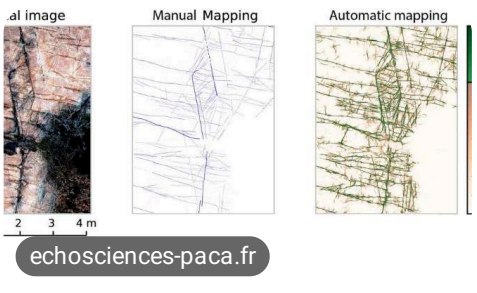
I will discuss the applications subject by subject.

4.1. Fundamental subjects

A. Geology







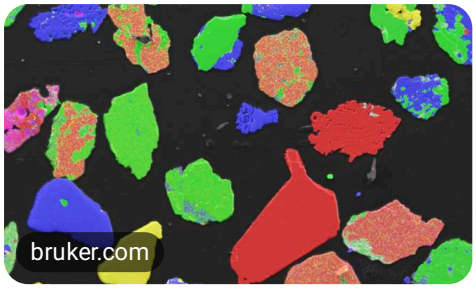
Possible AI applications:

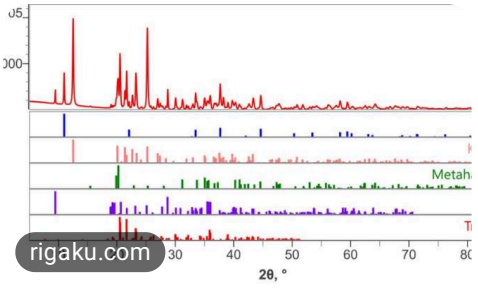
1. Automatic classification of geological formations.
2. Identification of lithological boundaries from geological maps.
3. Recognition of faults, folds, fractures and lineaments.
4. Automatic interpretation of geological field photographs.
5. Prediction of geological formations from spatial datasets.
6. Development of 3D geological models.
7. Integration of geological maps with satellite imagery.

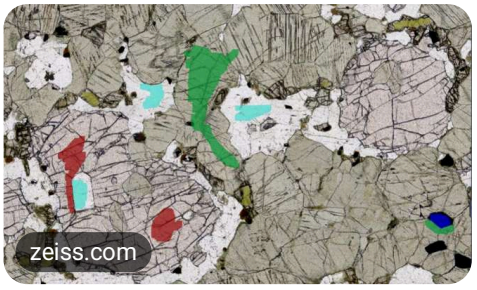
Suggested student research topics:

- AI-Based Classification of Geological Formations Using Digital Geological Maps.
- Automatic Recognition of Geological Structures Using Deep Learning.
- Development of an Intelligent Geological Mapping System.

B. Mineralogy







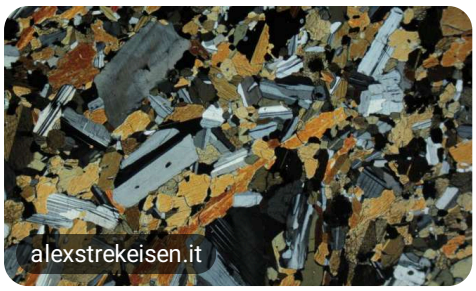
Possible AI applications:

1. Automatic mineral identification from microscopic photographs.
2. Mineral classification using SEM-EDS data.
3. Identification of minerals using X-ray diffraction (XRD) patterns.
4. Classification of mineral assemblages.
5. Prediction of mineral composition.
6. Automatic identification of ore minerals and gangue minerals.
7. Analysis of mineral textures and grain-size distributions.

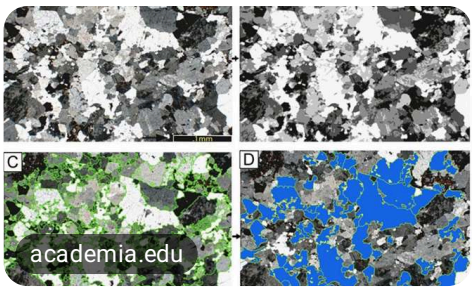
Suggested student research topics:

- Deep Learning-Based Identification of Rock-Forming and Ore Minerals from Microscopic Images.
- AI-Based Classification of Minerals Using XRD and SEM-EDS Data.
- Development of an Intelligent Mineral Identification System.

C. Petrology







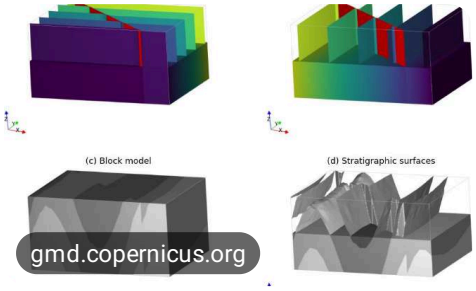
Possible AI applications:

1. Automatic classification of igneous, sedimentary and metamorphic rocks.
2. Identification of minerals in thin sections.
3. Quantitative analysis of mineral proportions.
4. Recognition of rock textures and microstructures.
5. Classification of magma series using whole-rock geochemistry.
6. Prediction of metamorphic facies and mineral assemblages.
7. Analysis of relationships between rock chemistry and mineral composition.

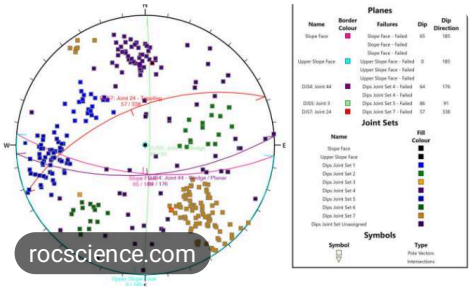
Suggested student research topic:

Development of an AI-Based Petrographic Image Analysis System for Automatic Rock Classification and Mineral Identification.

D. Geological Structure







Possible AI applications:

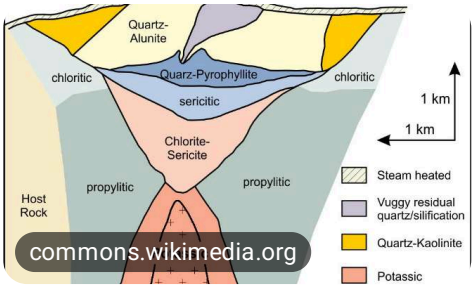
1. Automatic extraction of faults and lineaments.
2. Recognition of folds and fault systems.
3. Classification of structural patterns.
4. Analysis of fracture density and orientation.
5. Three-dimensional structural modelling.
6. Prediction of favorable structural environments for mineralization.
7. Integration of structural geology with geophysical data.

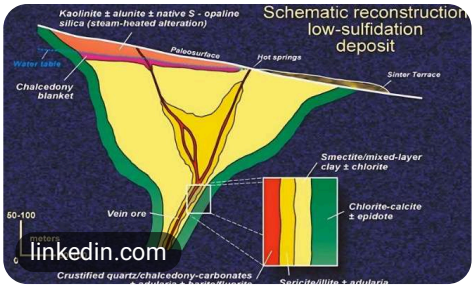
Suggested student research topic:

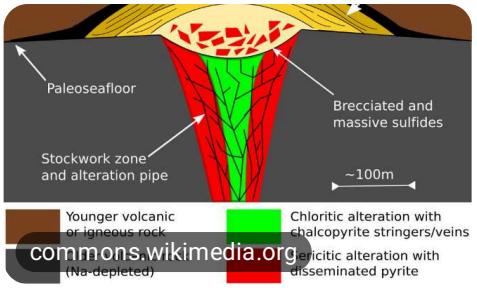
Application of Deep Learning and Spatial AI for Automatic Identification and Three-Dimensional Modelling of Geological Structures.

4.2. Major subjects

E. Geology of Ore Deposits







AI applications:

- Classification of mineral deposits according to geological characteristics.
- Identification of mineralization-related alteration zones.
- Analysis of relationships between host rocks and mineralization.
- Recognition of ore-forming processes and mineral assemblages.
- Prediction of deposit types from geological and geochemical characteristics.
- Development of mineral deposit knowledge graphs.
- AI-assisted comparison of geological models from different deposits.

An especially interesting direction is to develop AI systems that can learn from known mineral deposit models.

For example, a system could be trained using information about porphyry copper deposits, including:

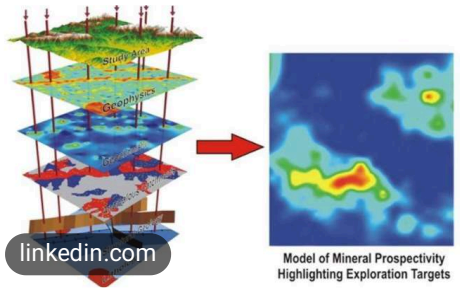
- Intrusive rock types.
- Alteration zones.
- Structural controls.
- Ore minerals.
- Geochemical signatures.
- Geophysical characteristics.

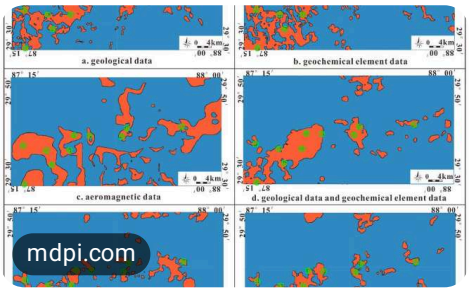
The system could then evaluate whether an unexplored region exhibits similar geological characteristics.

Suggested research topic:

Development of an AI-Based Mineral Deposit Classification and Geological Knowledge System for Identification of Potential Ore Deposits.

F. Mineral Exploration and Synthetic Assessment







This subject has perhaps the most direct connection with AI.

Possible applications:

1. Integration of geological, geochemical and geophysical information.
2. Development of mineral prospectivity maps.
3. Identification of favorable mineralization environments.
4. Ranking of exploration targets.
5. Prediction of concealed mineral deposits.
6. Optimization of exploration survey locations.
7. Integration of field observations with AI predictions.

Suggested research topic:

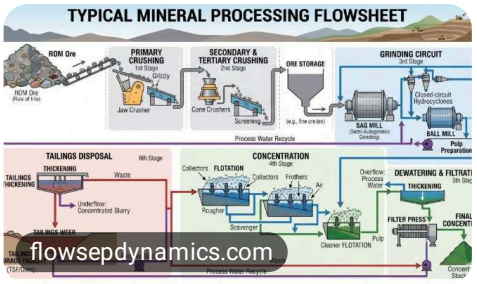
Development of an Integrated Artificial Intelligence System for Mineral Exploration and Comprehensive Evaluation of Mineral Potential.

I would make this subject the main practical application of AI in your faculty.

G. Mineral Engineering







AI is also very useful after the discovery of mineral deposits.

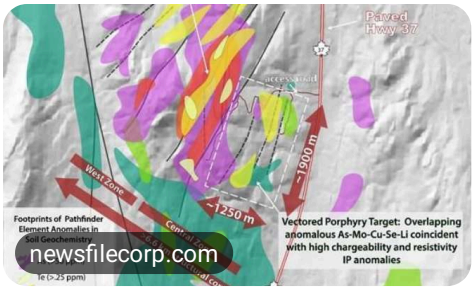
Possible applications:

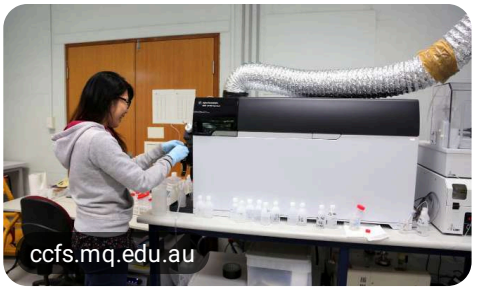
1. Prediction of ore grades.
2. Mineral processing optimization.
3. Prediction of flotation recovery.
4. Optimization of grinding and crushing processes.
5. Automatic identification of ore and gangue minerals.
6. AI-based ore sorting.
7. Prediction of processing plant performance.
8. Monitoring and optimization of tailings management.

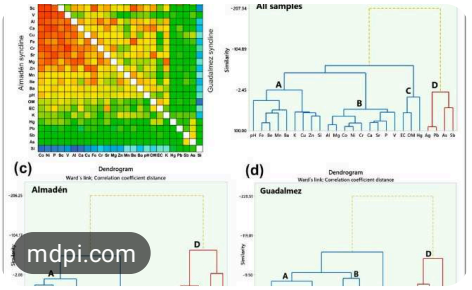
Suggested research topics:

- Machine Learning-Based Prediction of Mineral Processing Performance.
- AI-Based Optimization of Flotation Processes.
- Deep Learning-Based Ore Classification and Sensor-Based Sorting.

H. Applied Geochemistry







Possible applications:

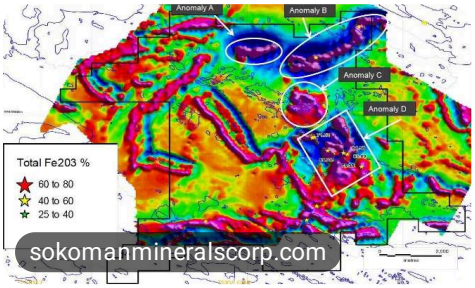
1. Detection of weak geochemical anomalies.
2. Classification of geochemical samples.
3. Identification of pathfinder elements.
4. Analysis of multi-element associations.
5. Identification of geochemical halos.
6. Prediction of elemental concentrations.
7. Analysis of geochemical dispersion patterns.
8. Identification of relationships between geochemistry and mineralization.

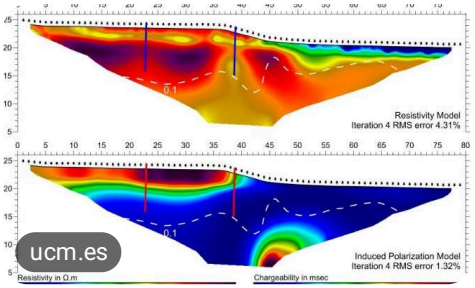
Suggested research topic:

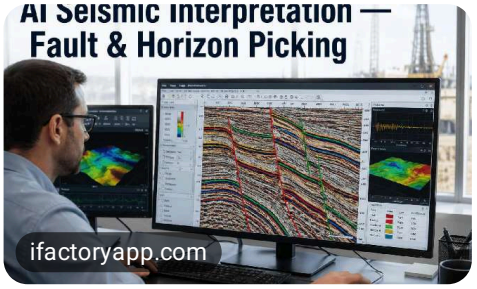
Application of Machine Learning for Identification and Interpretation of Multielement Geochemical Anomalies Associated with Mineral Deposits.

An important scientific consideration is that geochemical data are often compositional. Students should learn appropriate statistical transformations, such as log-ratio methods, before applying AI models. They should also consider detection limits, missing data and sampling bias.

I. Geophysical Prospecting







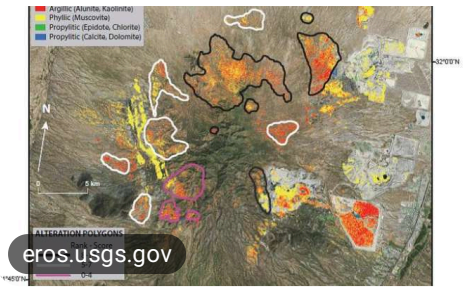
Possible applications:

1. Automatic interpretation of magnetic anomalies.
2. Classification of gravity anomalies.
3. Prediction of geological structures from geophysical measurements.
4. Interpretation of induced polarization and resistivity data.
5. Automatic seismic horizon and fault identification.
6. AI-assisted geophysical inversion.
7. Integration of multiple geophysical datasets.
8. Three-dimensional subsurface prediction.

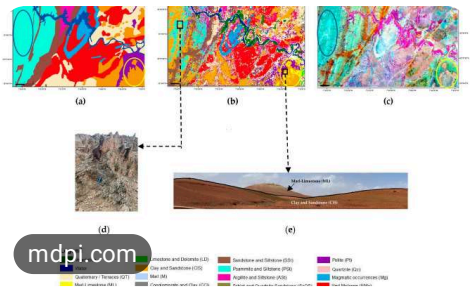
Suggested research topic:

Development of an AI-Based Integrated Interpretation System for Geophysical Data in Mineral Exploration.

J. Remote Sensing







Possible applications:

1. Automatic lithological mapping.
2. Identification of hydrothermal alteration minerals.
3. Extraction of structural lineaments.
4. Identification of iron oxide zones.
5. Mapping of clay minerals.
6. Hyperspectral mineral classification.
7. Integration of satellite images with geochemical data.
8. Prediction of mineralized zones.

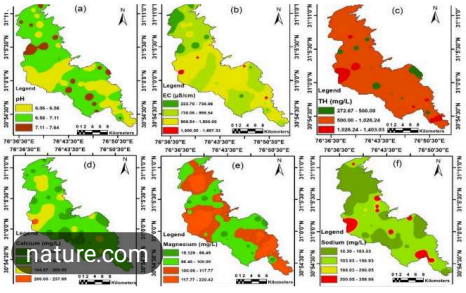
Suggested research topic:

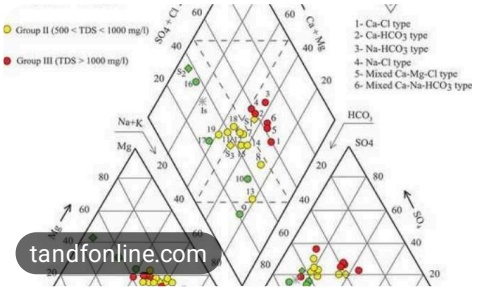
Development of a Deep Learning-Based Remote Sensing System for Lithological and Hydrothermal Alteration Mapping in Mineral Exploration.

This subject could become an important part of your faculty's AI research program because satellite data are increasingly accessible.

K. Hydro-geochemistry







Possible applications:

1. Classification of groundwater chemical characteristics.
2. Identification of groundwater quality patterns.
3. Prediction of hydrochemical parameters.
4. Identification of relationships between geological formations and groundwater chemistry.

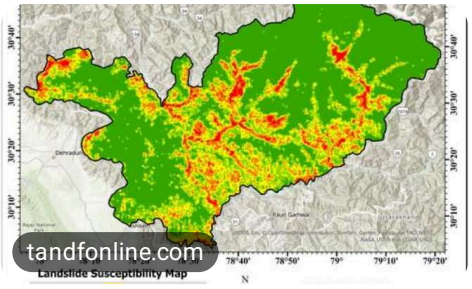
5. Prediction of groundwater contamination.
6. Mapping of groundwater recharge and discharge patterns.
7. Analysis of temporal changes in groundwater chemistry.
8. Identification of possible geochemical anomalies associated with mineralization.

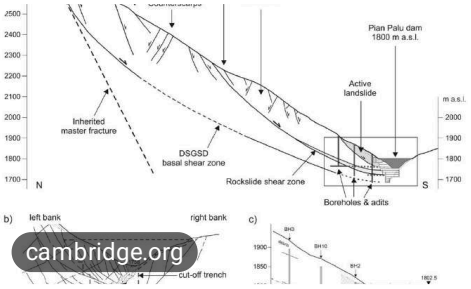
Suggested research topic:

Application of Artificial Intelligence for Integrated Analysis and Prediction of Hydro-geochemical Characteristics of Groundwater Systems.

L. Civil Engineering







AI applications:

1. Classification of soil and rock materials.
2. Prediction of geotechnical parameters.
3. Landslide susceptibility mapping.
4. Slope stability assessment.
5. Prediction of soil settlement.
6. Identification of geological hazards.
7. Analysis of foundation conditions.
8. Prediction of rock mass behavior.
9. Monitoring of dams, tunnels and underground structures.

Suggested research topic:

Development of AI-Based Geological and Geotechnical Assessment Systems for Civil Engineering Applications.

All safety-critical predictions should be checked by qualified engineering professionals and validated against appropriate field and laboratory measurements.

4.3. Summary: AI application matrix for the faculty

The following table could be useful when preparing a faculty-level AI research and curriculum plan.

Faculty subject	Main AI technology	Expected outcome
Geology	CNN, computer vision, GIS-AI	Automatic geological mapping

Mineralogy	CNN, image classification, spectral ML	Intelligent mineral identification
Petrology	CNN, clustering, regression	Automatic rock classification
Geological structure	Computer vision, GNN, 3D AI	Structural interpretation
Ore deposits	Knowledge graphs, ML, LLM	Deposit classification and knowledge system
Mineral exploration	RF, XGBoost, deep learning	Mineral prospectivity maps
Mineral engineering	Regression, reinforcement learning	Processing optimization
Applied geochemistry	Clustering, RF, anomaly detection	Geochemical anomaly identification
Geophysical prospecting	CNN, ANN, physics-informed ML	Geophysical interpretation
Remote sensing	CNN, Transformers, spectral ML	Lithological and alteration mapping
Hydro-geochemistry	Time-series ML, regression, clustering	Groundwater prediction
Civil engineering	ML, computer vision, predictive models	Geotechnical and hazard assessment

5. My recommendations for developing an AI research program in your faculty

I would suggest that your faculty approach AI development at three levels:

1. **AI as a new educational subject:** Students learn programming, statistics, machine learning and AI fundamentals.
2. **AI as a research methodology:** Each academic department applies AI to its own geological and engineering research problems.
3. **AI as a practical technological system:** Your faculty develops integrated software and decision-support systems for geological exploration and engineering applications.

5.1. Establish an AI for Geoscience and Technology Research Laboratory

I would recommend establishing a laboratory with a name such as:

AI for Geoscience and Mineral Resources Laboratory

Proposed interdisciplinary research facility

Five research divisions

1. **Intelligent Geological Mapping**Geological mapping, lithological classification, structural interpretation and 3D geological modelling.
2. **AI-Based Geochemical Exploration**Geochemical anomaly detection, elemental associations and mineralization prediction.
3. **AI-Based Geophysical Exploration**Geophysical data interpretation, inversion and subsurface modelling.

4. **Remote Sensing and Earth Observation AI**Satellite image processing, hyperspectral mineral mapping and alteration detection.
5. **Integrated Mineral Exploration and Engineering AI**Mineral prospectivity mapping, mineral processing, hydro-geochemistry and geotechnical applications.

The laboratory does not initially need expensive equipment. A shared computer laboratory, appropriate data storage, several GPU-equipped workstations or access to cloud computing, and open-source software would be sufficient to start many projects.

5.2. Introduce AI into the university curriculum

I suggest developing a new subject called:

Artificial Intelligence and Data Science for Geoscience and Technology

The course could be organized into the following modules:

Module	Contents
1. Introduction to AI	AI, machine learning and deep learning
2. Programming	Python and scientific computing
3. Statistics and data science	Statistical analysis and data preparation
4. GIS and spatial AI	Spatial databases and geographical analysis
5. Machine learning	RF, SVM, XGBoost and clustering
6. Deep learning	Neural networks and CNN
7. Remote sensing AI	Satellite and hyperspectral image analysis
8. Geoscience applications	Geological, geochemical and geophysical AI
9. Integrated AI systems	Multisource data integration
10. Research project	Development of a practical geological AI application

I would also recommend introducing practical AI exercises into existing subjects rather than relying exclusively on this new course.

For example:

- Geology students develop a geological map classification model.
- Mineralogy students develop an automatic mineral identification model.
- Petrology students analyze thin-section photographs.
- Geochemistry students develop anomaly detection models.
- Geophysics students interpret magnetic and electrical data using machine learning.
- Remote sensing students classify alteration zones using satellite images.

- Mineral exploration students develop mineral potential maps.
- Civil engineering students develop landslide susceptibility models.

This approach would allow students to understand both the conventional scientific methods and the advantages and limitations of AI.

5.3. Suggested implementation schedule

I recommend a gradual three-year implementation program.

Three-year development plan

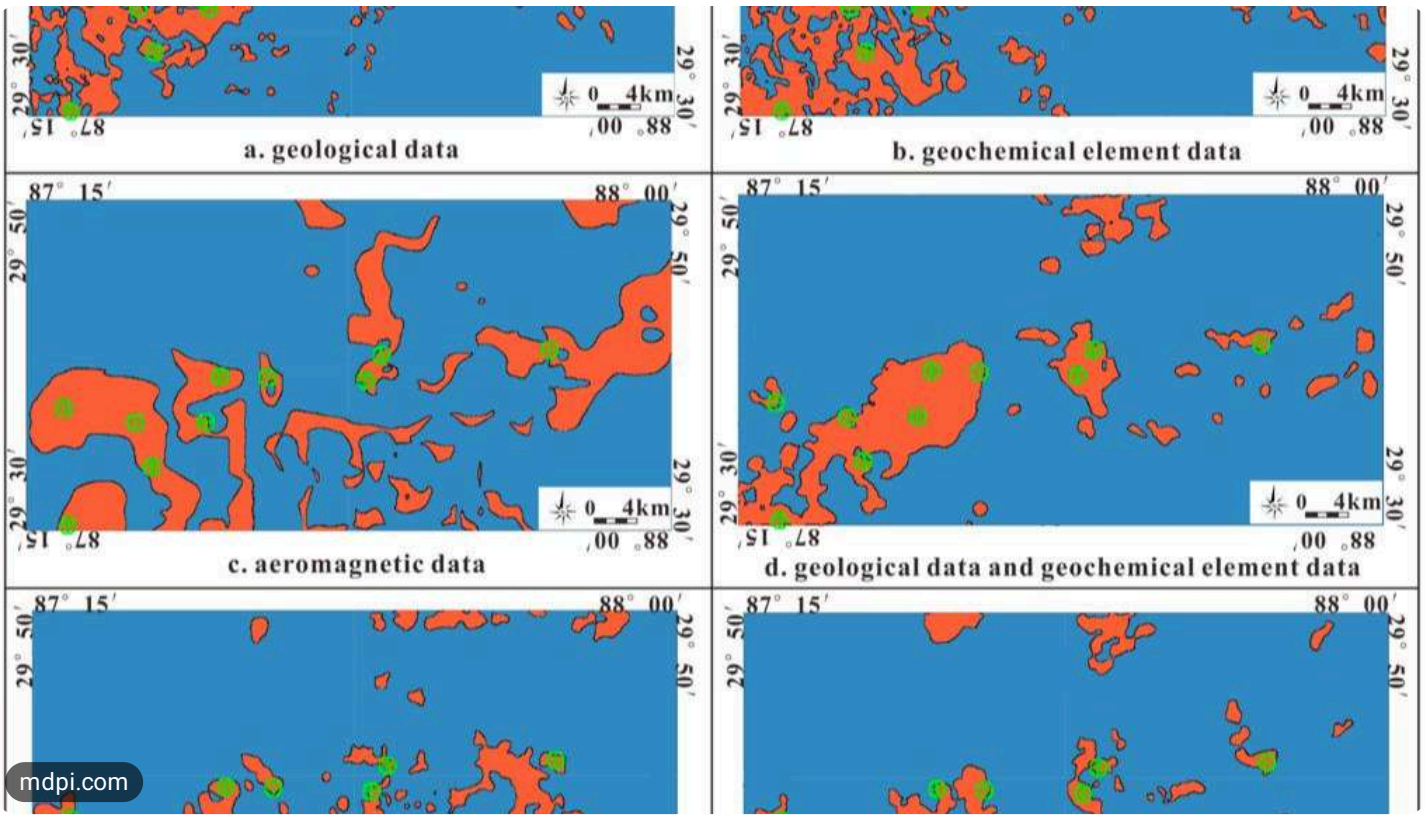
1. **Year 1 – Establish the foundation** Priority: Basic capacity
    - Establish an interdisciplinary AI working group.
    - Identify available geological datasets.
    - Train interested faculty members in Python and machine learning.
    - Establish the initial AI laboratory.
    - Introduce AI into two or three existing subjects.
    - Start two pilot research projects.
2. **Year 2 – Develop research applications** Priority: Research development
    - Develop geochemical anomaly detection models.
    - Develop AI-based remote sensing applications.
    - Develop geophysical interpretation models.
    - Start master's and doctoral research projects.
    - Establish collaboration with geological surveys and mineral exploration companies.
    - Organize an international AI and geoscience research workshop.
3. **Year 3 – Develop integrated systems** Priority: Practical systems
    - Develop integrated mineral prospectivity mapping.
    - Combine geological, geochemical, geophysical and remote sensing information.
    - Develop 3D geological prediction systems.
    - Establish a geological AI database.
    - Develop prototype software for exploration decision support.
    - Publish research papers and pursue joint research projects.

6. Three particularly promising flagship research projects

If I were responsible for coordinating AI research in your faculty, I would select three projects to connect the different departments.

Flagship project 1

Development of an Integrated AI-Based Mineral Exploration System



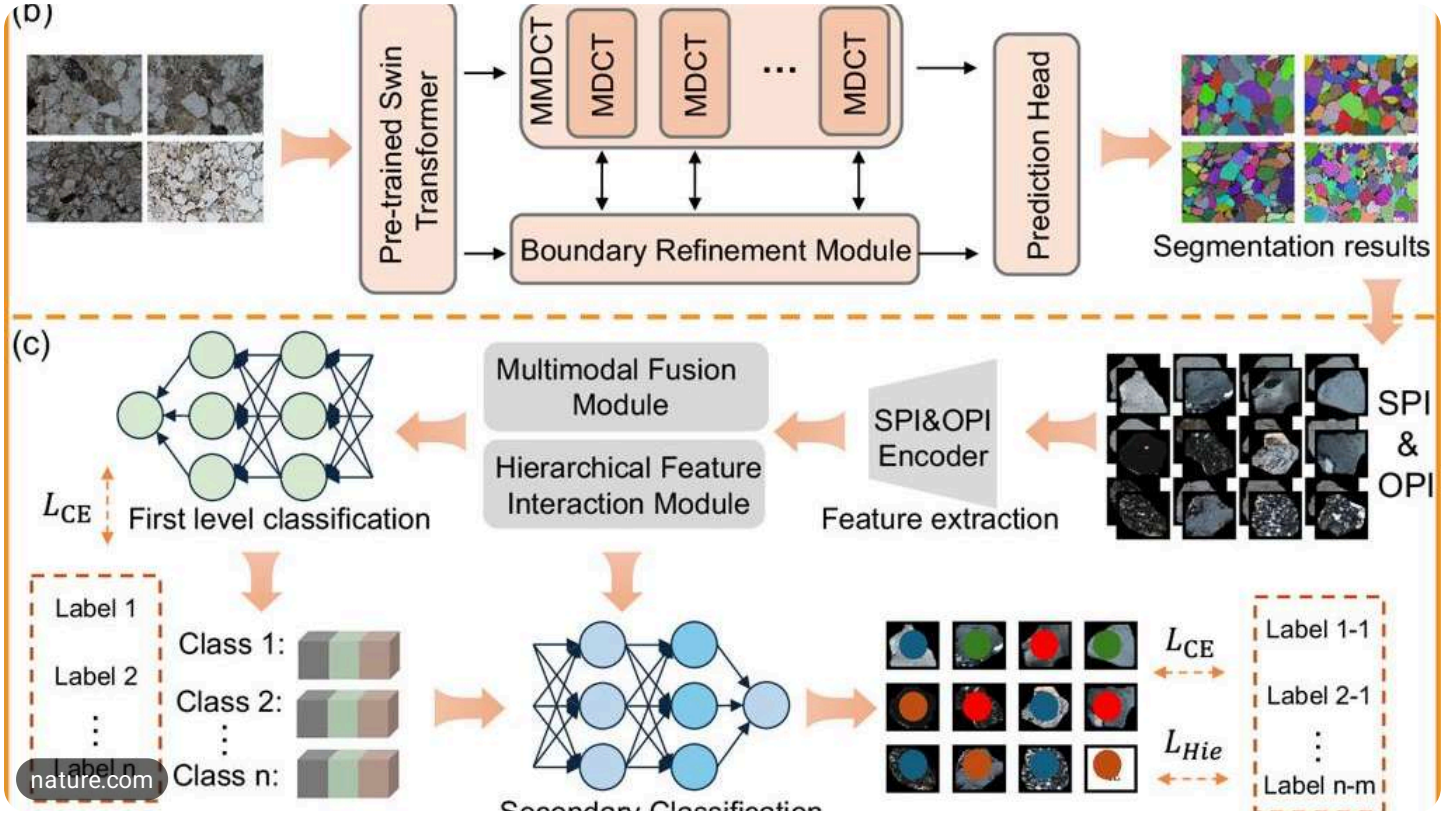
Participating subjects:

- Geology of ore deposits.
- Applied geochemistry.
- Geophysical prospecting.
- Remote sensing.
- Mineral exploration and synthetic assessment.

Expected output: An AI system that integrates different geological datasets and predicts prospective areas for mineral exploration.

Flagship project 2

Development of an AI-Based Intelligent Geological and Mineralogical Identification System



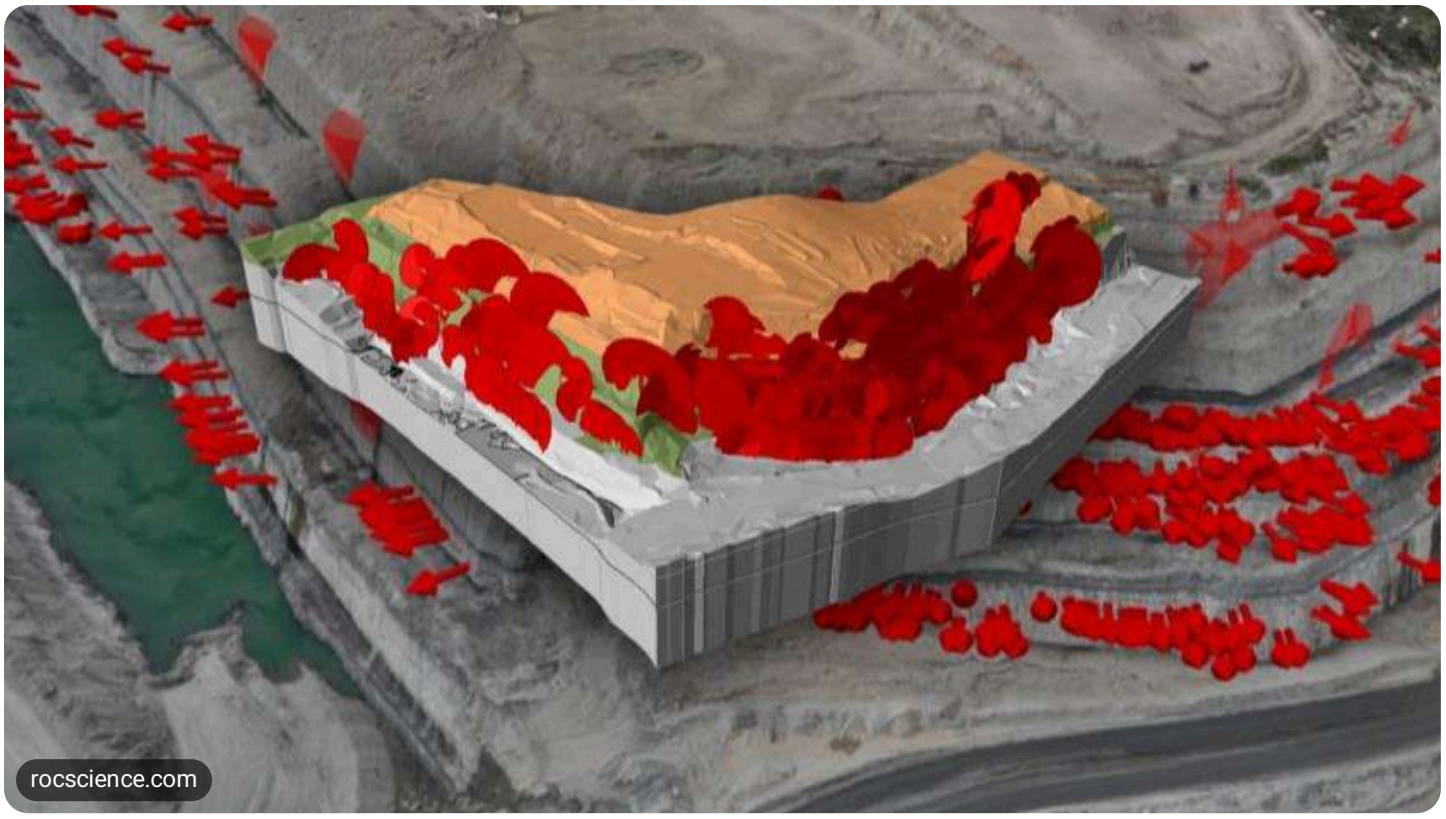
Participating subjects:

- Geology.
- Mineralogy.
- Petrology.
- Ore deposit geology.

Expected output: Software that identifies rock types, minerals and petrographic characteristics from digital images and laboratory analytical data.

Flagship project 3

Development of an AI-Based Integrated Geological and Geotechnical Assessment System



Participating subjects:

- Geological structure.
- Civil engineering.
- Hydro-geochemistry.
- Geophysics.

Expected output: A system for geological hazard assessment, geotechnical prediction and subsurface characterization.

7. Important scientific challenges that your faculty should consider

AI is promising, but several issues should be considered carefully when developing doctoral research and practical systems.

A systematic review published in 2026 examined 349 studies on deep learning for mineral exploration. It highlighted the importance of integrating multiple geological datasets and identified continuing

challenges involving limited training data, class imbalance, interpretability and prediction across different geological regions. ScienceDirect

I would particularly emphasize the following six issues:

Scientific challenge	Recommended approach
Insufficient geological data	Combine public datasets, historical reports and new field surveys
Unbalanced mineralization data	Use appropriate sampling, class weighting and positive-unlabelled learning
Different data resolutions	Standardize spatial support and coordinate systems
Unreliable predictions	Use spatial cross-validation and independent testing
Black-box AI models	Use explainable AI, including SHAP and feature importance
Geological uncertainty	Combine AI predictions with geological interpretation and field verification

An additional important direction is the integration of geological knowledge with AI. Rather than allowing AI to make predictions using statistical relationships alone, geological principles and mineral deposit models can be incorporated into the modelling process. Recent research on explainable mineral prospectivity mapping emphasizes this need. MDPI +1

In my opinion, this is a particularly promising research direction for geoscience faculties because it makes AI research scientifically meaningful to geologists.

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8. Final recommendations: Where should you personally start?

Considering the subjects in your faculty, I would prioritize the following sequence.

Priority	Research direction	Expected contribution
1	AI-based geochemical anomaly detection	First practical AI research project
2	AI-based remote sensing and alteration mapping	New exploration mapping technology
3	AI-based geophysical interpretation	Improved subsurface understanding
4	Integrated mineral prospectivity mapping	Main faculty research system
5	AI-based geological and mineralogical identification	Intelligent geological laboratory
6	Geological knowledge graphs and LLMs	Advanced intelligent geological information system
7	AI applications in mineral engineering and civil engineering	Expansion into other faculty disciplines

I would also recommend reading these three recent research reviews as starting points for your research group:

1. (2026). This is particularly useful for understanding the different AI methods and geological datasets.
2. (2026). This is useful for understanding how geological knowledge can be combined with explainable AI.
3. (2024). This is useful for understanding one of the major technical problems in mineral exploration AI.

9. A few questions that would help me prepare a more specific research and development plan

I would be very interested in understanding your faculty's current situation because the best AI development strategy depends considerably on the available geological datasets, staff expertise and research priorities.

Could you tell me the following?

1. In which country is your university located?

Not specified

2. What geological data are currently available to your faculty?

- (●) Geological and geochemical data
- ( ) Geological, geochemical and geophysical data
- ( ) All four types of exploration data
- ( ) Limited geological data

3. What is the current expertise of your academic staff?

- (●) Mainly geologists and geoscientists
- ( ) Geoscientists and computer scientists
- ( ) Geoscientists with some programming experience
- ( ) An established AI research group

4. What is your main objective?

- (●) Both research and education

- ○ Development of doctoral research
- ○ Practical exploration software
- ○ Modernization of the faculty curriculum

5. Which research direction interests you most?

- ◉ Integrated AI-based mineral exploration
- ○ AI-based geochemical prospecting
- ○ AI-based geophysical prospecting
- ○ AI-based remote sensing
- ○ AI applications across all faculty subjects

Submit answers

With these details, I can help you develop a more concrete **five-year AI development master plan for your Faculty of Geoscience and Technology**, including:

- Research themes and projects for each academic department.
- Specific PhD and master's thesis titles and research contents.
- Required AI software, equipment and databases.
- Staff training and student curriculum.
- International research collaboration opportunities.
- A flagship AI-based mineral exploration system to be developed by your faculty.

I think the most valuable long-term objective would be to make your faculty a research center where traditional geological science and modern AI technology are developed together, rather than treating AI as an isolated computer-science subject.

2026-11-06 15:50:57 **You**

Here is information about my Faculty of Geoscience and Technology:
Country: Not specified
Available data: Geological, geochemical and geophysical data
Academic staff: Geoscientists with programming experience
Main objective: Both research and education
Research priority: Integrated mineral exploration AI

Please develop a detailed institutional AI research, education and implementation plan based on these answers.

ChatGPT gpt-6-thinking

Five-Year Institutional Master Plan for Artificial Intelligence in the Faculty of Geoscience and Technology

Proposed institutional strategy · 2027–2031

Proposed strategic name: AI-Driven Geoscience, Mineral Exploration and Earth Technology Development Program

Based on the information you provided, I recommend that your faculty establish an integrated AI research and education program, with mineral exploration as its central research direction.

Your faculty already has three important advantages:

1. You have geological, geochemical and geophysical datasets. This means that research can begin with actual geological problems rather than artificial examples.
2. Your academic staff have some programming experience. Therefore, you can build on existing capabilities instead of starting computer science training from zero.
3. Your objective includes both research and education. This creates an opportunity to develop AI research projects while simultaneously training undergraduate, master's and doctoral students.

There is one important strategic consideration: because your faculty has not yet identified its country or specific mineral resource priorities, I will design this as a flexible institutional plan. It can subsequently be adapted to your country's geology, mineral resources, geological survey infrastructure and university regulations.

Part I. Institutional vision and strategic objectives

1. Proposed vision

AI for Geoscience and Mineral Exploration

Proposed institutional vision

To establish an interdisciplinary center of excellence in Artificial Intelligence for Geological Sciences, Mineral Exploration and Earth Technology.

The center will integrate conventional geological knowledge, scientific data analysis, machine learning and advanced AI technologies to improve mineral exploration, geological understanding, engineering applications and geoscience education.

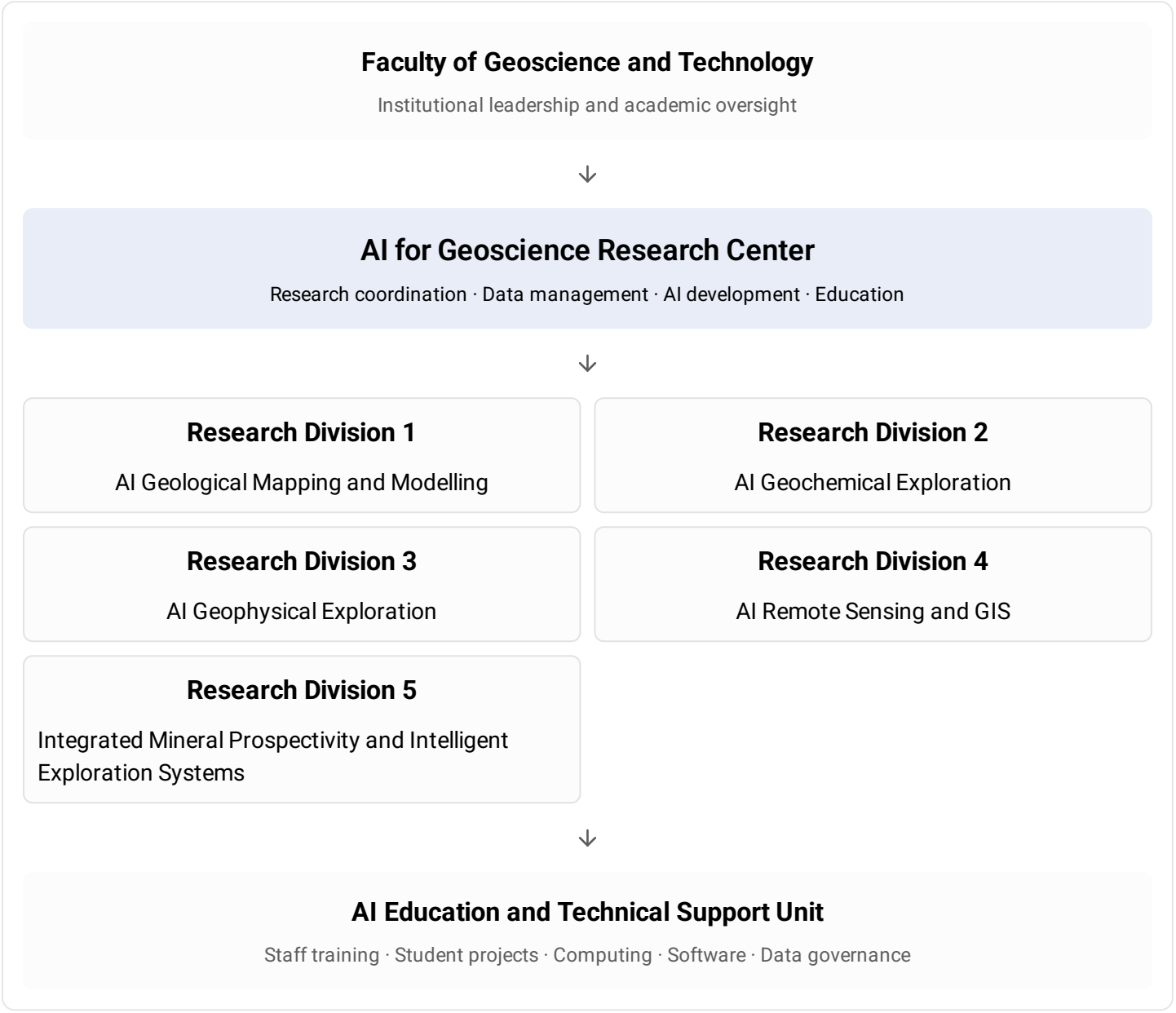
2. Five strategic objectives

Objective	Expected institutional achievement
1. Research development	Develop new AI methods for mineral exploration
2. Data infrastructure	Establish an integrated geological database
3. Education modernization	Introduce AI into all relevant faculty subjects
4. Human resource development	Train academic staff and students in geoscience AI
5. Technology development	Develop practical AI software for exploration and geological assessment

3. Proposed institutional structure

I recommend establishing an **AI for Geoscience Research Center (AIGRC)** within the faculty, initially as a coordinating research unit rather than a completely independent institute.

It should connect the existing academic departments.



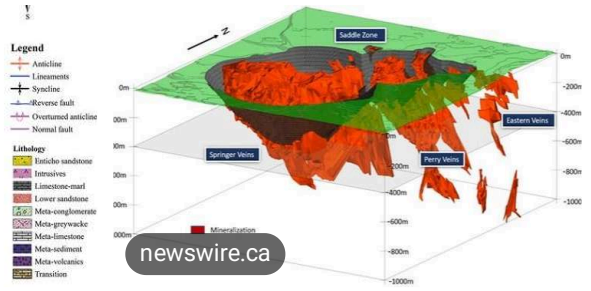
The divisions do not need to be separate departments. Academic staff can participate in several divisions according to their expertise.

Part II. Five major research programs

I recommend that the faculty concentrate its research resources on five connected programs.

Research Program 1. AI-based geological and mineral-system modelling







Main objective: Develop intelligent systems for interpreting geological information and identifying geological environments favorable for mineralization.

Proposed research projects:

Project	Main research contents
1.1	AI-based lithological classification
1.2	Automatic geological boundary and fault identification
1.3	AI-based structural geology interpretation
1.4	3D geological modelling using borehole and surface information
1.5	Geological knowledge graphs for mineral deposit modelling

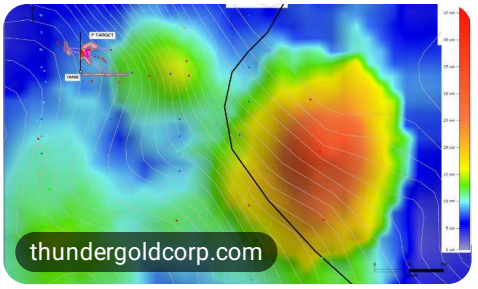
Suggested doctoral research title:

Development of an AI-Based Geological Modelling Framework for Identification of Mineralization-Favorable Geological Environments.

Participating subjects: Geology, geological structure, petrology and geology of ore deposits.

Research Program 2. AI-based geochemical exploration







Main objective: Develop AI methods to identify geochemical anomalies and understand the geochemical characteristics of mineral deposits.

Proposed projects:

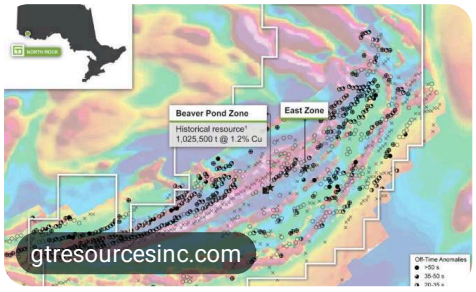
- Multielement geochemical anomaly detection.
- AI-based identification of pathfinder elements.
- Geochemical classification of mineralized and background environments.
- Prediction of concealed mineralization from surface geochemical information.
- Geochemical dispersion and hydrothermal alteration modelling.
- Integration of geochemical and geological datasets.

Suggested doctoral research title:

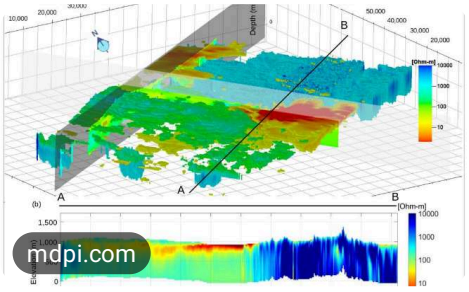
Development of an Explainable Machine Learning System for Multielement Geochemical Anomaly Detection and Mineral Exploration.

This should be one of your first research programs because your faculty already has geochemical data and the required expertise can be developed using conventional machine-learning methods.

Research Program 3. AI-based geophysical exploration







Main objective: Improve the interpretation of geophysical information and develop models for identifying concealed geological structures and mineralized zones.

Proposed projects:

- AI interpretation of magnetic data.
- AI interpretation of gravity data.
- Machine-learning analysis of electrical resistivity and IP data.
- Automatic recognition of geophysical anomalies.
- AI-assisted geophysical inversion.

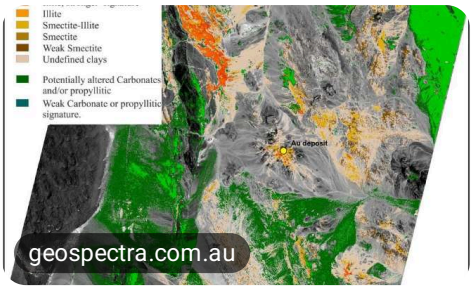
- Integration of geophysical and geological information.
- 3D subsurface geological prediction.

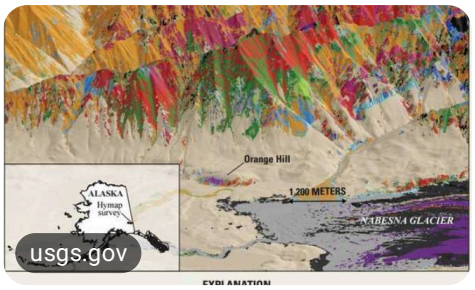
Suggested doctoral research title:

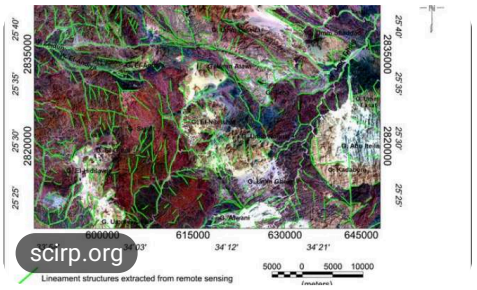
Development of an AI-Based Integrated Geophysical Interpretation System for Prediction of Concealed Mineralization.

An important condition is to obtain the original geophysical measurements and processing metadata, rather than relying only on images of interpreted geophysical maps.

Research Program 4. AI-based remote sensing and GIS







Main objective: Develop AI systems for geological mapping, mineral alteration identification and regional exploration targeting.

Proposed projects:

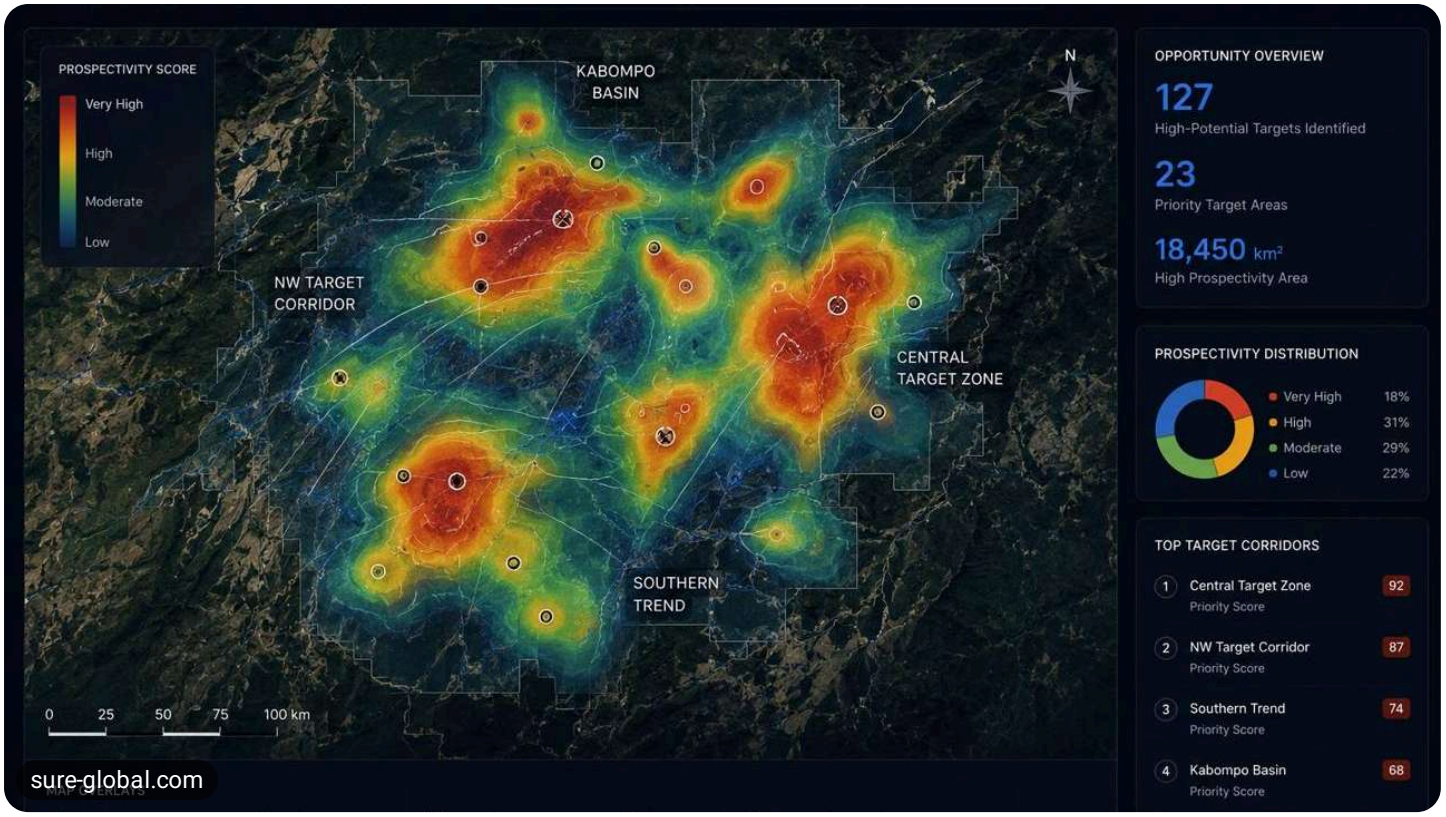
- Satellite-based lithological classification.
- Identification of hydrothermal alteration zones.
- Automatic extraction of geological lineaments.
- Hyperspectral mineral identification.
- Integration of remote sensing and geochemical information.
- Development of GIS-based mineral exploration tools.

Suggested doctoral research title:

Development of a Deep Learning-Based Remote Sensing Framework for Identification of Hydrothermal Alteration and Mineralization-Favorable Zones.

Research Program 5. Integrated AI-based mineral prospectivity mapping

Central flagship research program



Main objective: Integrate the research outcomes from Programs 1–4 into one comprehensive mineral exploration system.

This should be the central research project of your faculty because it provides a common scientific objective for the different academic departments.

The system will integrate:

- Geological lithology and structures.
- Geochemical elemental concentrations and anomalies.
- Magnetic, gravity, electrical and other geophysical information.
- Remote sensing and alteration information when these datasets become available.
- Known mineral occurrences and deposit characteristics.
- Drilling information, when available.

The system should produce:

1. Mineral prospectivity maps.
2. Predicted exploration targets.
3. Probability and uncertainty maps.
4. Explanations of the geological factors controlling predictions.
5. Recommended locations for additional surveys and field investigation.

Proposed flagship doctoral thesis:

Development of an Explainable Multisource Artificial Intelligence System for Integrated Mineral Prospectivity Mapping and Exploration Targeting.

This could also become the basis of a long-term institutional software development project.

Part III. Research implementation and institutional development

4. The first practical project: develop a prototype mineral exploration AI system

I would not begin by trying to integrate every available dataset into one large AI model.

Instead, I recommend selecting one mineralized district or geological exploration area and developing a working prototype there.

This will be the first demonstration that your faculty can conduct meaningful AI-based mineral exploration research.

4.1. Proposed pilot project

Recommended first institutional project

AI-Based Integrated Mineral Prospectivity Mapping Using Geological, Geochemical and Geophysical Data

Research objective

Develop a machine-learning system that integrates existing geological, geochemical and geophysical datasets to identify areas favorable for mineralization.

Initial study area

One well-documented mineralized district, preferably with:

- Geological maps.
- Geochemical sampling results.
- Geophysical survey results.
- Known mineral occurrences or deposits.
- Some independent field or drilling evidence for validation.

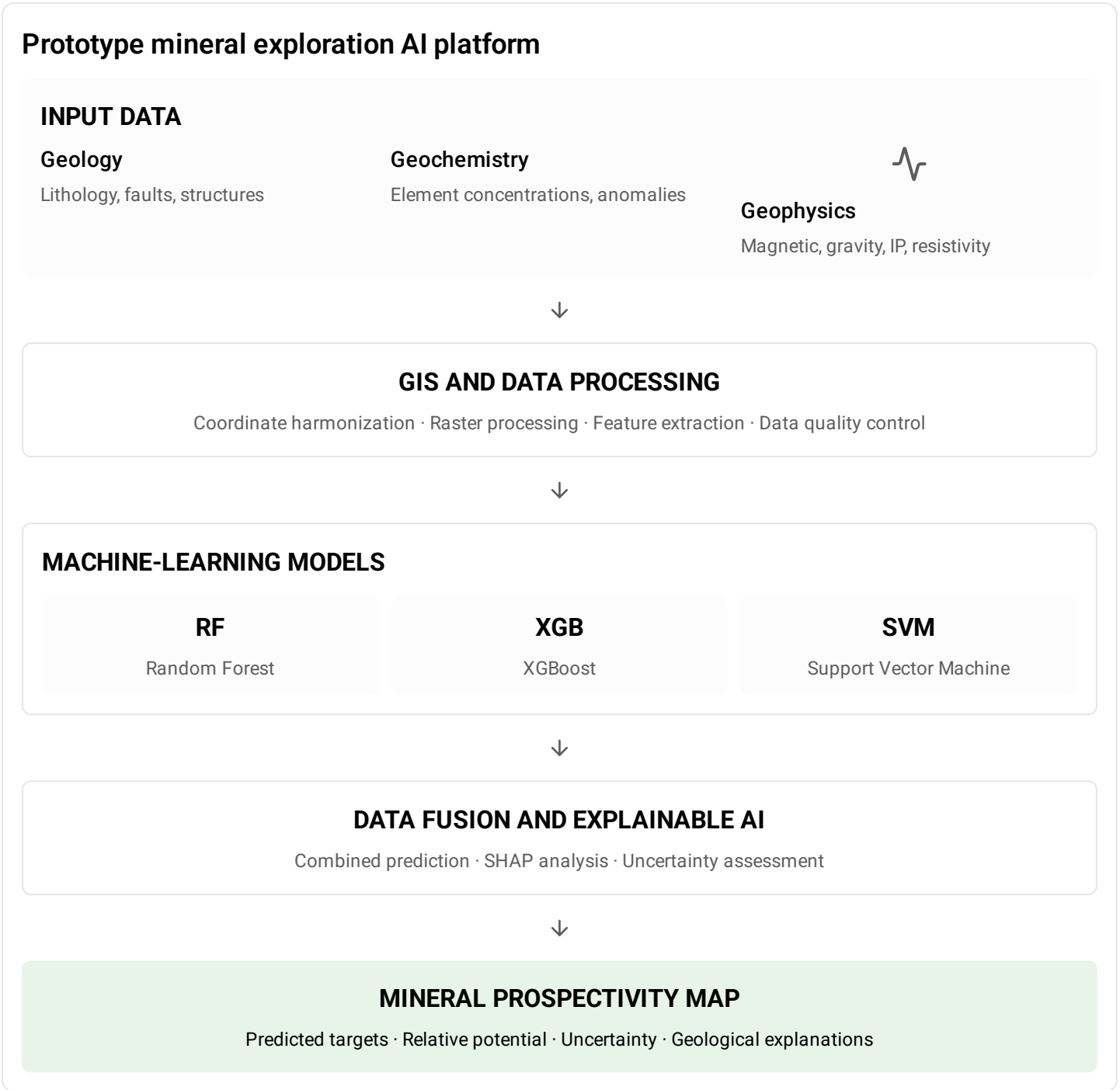
4.2. Technical development stages

Stage	Technical activities	Expected output
1	Select study area and deposit type	Defined research area
2	Compile and audit existing data	Geological database
3	Standardize spatial information	GIS-ready datasets
4	Analyze individual datasets	Geological evidence maps
5	Develop statistical baseline models	Initial anomaly maps

6	Train RF, XGBoost and SVM	Individual AI models
7	Integrate the three data sources	Integrated prospectivity model
8	Apply explainable AI	Geological interpretation
9	Perform independent spatial validation	Evaluated prediction maps
10	Develop software prototype	Demonstration AI system

4.3. How should the first AI system work?

The following is a proposed technical design.



One important recommendation is to maintain two levels of prediction:

- Individual models: geology-only, geochemistry-only and geophysics-only.

- Integrated model: combined evidence from all available sources.

This allows your researchers to investigate whether combining the datasets actually improves exploration performance, and which sources contribute useful additional information.

4.4. Essential scientific safeguards

I would make these requirements compulsory for all doctoral and master's research involving mineral prospectivity mapping.

1. **Data quality:** Check coordinate systems, sampling locations, analytical methods, detection limits, missing values and measurement units.
2. **Spatial resolution:** Ensure that geological, geochemical and geophysical information is integrated at scientifically appropriate spatial scales.
3. **Label quality:** Distinguish known mineralization, verified barren areas and locations where mineralization is simply unknown.
4. **Spatial validation:** Use geographically separated training and testing areas to evaluate predictions for unexplored regions.
5. **Geological interpretation:** Investigate whether the model's important features make geological sense.
6. **Uncertainty:** Report uncertainty and limitations, not just a colored prediction map.

These safeguards are particularly important because recent research reviews identify spatial data leakage, class imbalance, limited labeled data and poor transferability between geological regions as major challenges in mineral prospectivity mapping. ScienceDir… +2

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5. Establish an integrated geological database

I consider this one of the most important institutional investments.

Your faculty may already have geological reports, geochemical spreadsheets, geophysical survey files, geological maps and research datasets distributed among different departments and individual researchers.

Before developing sophisticated AI models, you should organize these resources into a common research database.

5.1. Proposed database structure

Database	Information
Geological database	Lithology, stratigraphy, geological boundaries, faults and folds
Geochemical database	Sample ID, coordinates, elements, concentrations, analytical methods

Geophysical database	Survey information, magnetic, gravity, electrical and other measurements
Mineral occurrence database	Deposit locations, deposit types, ore minerals and host rocks
Drilling database	Borehole coordinates, depths, lithology and assay intervals, if available
Research database	Reports, publications, field observations and research results
AI model database	Training data versions, model versions, parameters and validation results

I recommend adopting the FAIR data principles:

- Findable.
- Accessible.
- Interoperable.
- Reusable.

These principles are relevant to university research because datasets should remain understandable and reusable by different researchers and departments. The USGS also recommends documented data standards, metadata, open formats, quality management, preservation and security in scientific data management. NSF - U.S. … +1

5.2. Recommended technical infrastructure

Component	Recommended technology
Spatial database	PostgreSQL + PostGIS
GIS	QGIS
Geological raster data	GeoTIFF / Cloud Optimized GeoTIFF
Vector data	GeoPackage
Scientific programming	Python
Machine learning	Scikit-learn, XGBoost
Deep learning	PyTorch
Data visualization	Python, QGIS
Model tracking	MLflow
Code management	Git
Documentation	Standardized metadata and data dictionaries

These are suggested open-source technologies, not mandatory products. The faculty should select formats and systems compatible with its existing data and IT environment.

Part IV. AI education and academic curriculum development

6. Proposed AI education program

Your faculty should not simply introduce a single AI course and expect students to become capable of applying AI in geological research.

I recommend a three-level educational system.

Level 1 · Undergraduate

AI Fundamentals for Geoscientists

Students learn Python, statistics, GIS, data analysis and introductory machine learning.

Level 2 · Master's

Applied AI in Geological Sciences

Students develop practical AI applications using geological, geochemical and geophysical datasets.

Level 3 · Doctoral

Advanced AI Research in Geoscience

Students develop new AI algorithms, integrated exploration systems, geological knowledge models and advanced predictive methods.

6.1. Recommended new university courses

I suggest initially introducing three new courses.

Course	Level	Suggested credits
Introduction to Artificial Intelligence and Data Science for Geoscientists	Undergraduate	3
Machine Learning Applications in Mineral Exploration	Master's	3
Advanced AI and Multisource Data Integration in Geoscience	Doctoral	3

Credit requirements should be adapted to your university's curriculum regulations.

6.2. Detailed contents of the main undergraduate course

Course title: Introduction to Artificial Intelligence and Data Science for Geoscientists

Suggested 14-week structure:

Week	Lecture and practical contents
1	Introduction to AI and its applications in geoscience

2	Python programming for geological research
3	Geological databases and data management
4	Statistics and exploratory data analysis
5	GIS and spatial data analysis
6	Introduction to machine learning
7	Classification and regression
8	Random Forest and XGBoost
9	Clustering and anomaly detection
10	Introduction to neural networks
11	AI applications in geochemical exploration
12	AI applications in geophysical exploration
13	Integrated mineral prospectivity mapping
14	Student research presentations

Practical student assignment:

Each student should develop a small AI project using real geological data.

For example:

- Predict geological lithology from geological and geochemical attributes.
- Identify anomalous copper concentrations.
- Classify geophysical anomalies.
- Produce a mineral prospectivity map.

6.3. Integration of AI into existing subjects

I would also recommend revising the practical exercises in your existing courses.

Existing subject	Proposed AI practical exercise
Geology	Geological map classification using Python
Mineralogy	Mineral identification using image classification
Petrology	Automated thin-section image analysis
Geological structure	Automatic fault and lineament extraction
Ore deposit geology	AI classification of mineral deposit types
Mineral exploration	Mineral prospectivity modelling

Mineral engineering	Prediction of flotation recovery
Applied geochemistry	Multielement anomaly detection
Geophysics	Magnetic and electrical anomaly classification
Remote sensing	Satellite-based alteration mapping
Hydro-geochemistry	Groundwater chemistry prediction
Civil engineering	Landslide susceptibility modelling

The most important educational principle is that students should first understand the geological or engineering problem and then use AI as a scientific tool to solve it.

AI should strengthen geological education, not replace fundamental geology, mineralogy, petrology or conventional exploration methods.

Part V. Doctoral and master's research development

7. Proposed graduate research program

I recommend developing a coordinated graduate research program in which master's and doctoral students work on related problems.

For example, five students could work on different components of the same integrated mineral exploration project.

Graduate research theme

Integrated AI-Based Mineral Exploration

Common research area and geological database

PhD 1

AI-based geochemical anomaly detection

PhD 2

AI-based geophysical interpretation

PhD 3

Geological and structural modelling

PhD 4

Multisource AI data integration

Master's research students

Individual algorithms, data preparation, GIS, validation and geological case studies

↓

Integrated faculty research output

AI-based mineral exploration platform

7.1. Proposed doctoral thesis titles

I recommend initially offering the following six doctoral topics:

No.	Proposed doctoral research title
1	Development of Machine Learning Algorithms for Multielement Geochemical Anomaly Detection
2	Deep Learning-Based Interpretation of Geophysical Data for Concealed Mineral Exploration
3	AI-Based Geological and Structural Modelling of Mineralization-Favorable Environments
4	Explainable AI for Geological Interpretation and Mineral Prospectivity Assessment
5	Development of Multisource AI Data-Fusion Models for Mineral Prospectivity Mapping
6	Development of an Intelligent Geological Exploration Decision-Support System

7.2. Proposed master's thesis topics

Master's research should be more focused and manageable.

Examples:

1. Application of Random Forest for Copper Geochemical Anomaly Detection.
2. Application of XGBoost for Mineral Prospectivity Mapping.
3. Machine Learning-Based Classification of Magnetic Anomalies.
4. Automatic Extraction of Geological Lineaments Using Satellite Images.
5. Application of SVM for Lithological Classification.
6. GIS-Based Integration of Geological and Geochemical Information.
7. Application of Explainable AI for Interpretation of Geochemical Data.
8. Prediction of Mineralized Zones Using Geophysical and Geological Data.

The advantage of this arrangement is that master's students can produce useful datasets, algorithms and case studies that contribute to the larger doctoral research program.

Part VI. Human resources and technical capacity development

8. Staff training program

Since your academic staff already have some programming experience, I would avoid spending too much time on general computer literacy.

Instead, establish a structured AI training program for geoscientists.

8.1. Four levels of staff development

Training level	Contents	Expected capability
Level 1	Python, data science and GIS	Analyze geological datasets
Level 2	Machine learning and statistics	Develop basic AI models
Level 3	Deep learning and data fusion	Develop advanced geological AI applications
Level 4	Explainable AI, physics-informed AI and research software engineering	Lead advanced research projects

8.2. Proposed first-year training schedule

Period	Training activity
Months 1–2	Python for geoscience and data preparation
Months 3–4	Machine learning and statistical modelling
Months 5–6	GIS and spatial machine learning
Months 7–8	Geochemical and geophysical AI applications
Months 9–10	Deep learning and neural networks
Months 11–12	Integrated mineral prospectivity research

I would recommend selecting approximately 6–10 interested academic staff members as the first AI research group. This is an initial planning target, not a requirement.

They should subsequently become trainers for colleagues in their respective departments.

8.3. International collaboration

Your faculty should establish academic cooperation with:

- Universities conducting AI-based mineral exploration research.
- National geological survey organizations.
- Mineral exploration and mining companies.
- Geospatial and remote sensing research centers.
- Computer science and data science departments.

I particularly recommend joint doctoral supervision between a geoscientist and an AI or computer science specialist.

This arrangement can improve both the geological quality and computational originality of doctoral research.

Part VII. Five-year implementation roadmap

9. Institutional development schedule: 2027–2031

The following schedule is designed for implementation beginning in 2027.

Five-year implementation roadmap

The milestones are proposed institutional targets and should be adjusted after the initial resource and staffing audit.

1. **2027 — Foundation and preparation** Establish capacity Key activities:
   - Establish the AI research center and steering committee.
   - Inventory existing geological, geochemical and geophysical datasets.
   - Select one pilot mineral exploration area.
   - Establish data standards and the common database.
   - Train the initial academic research group.
   - Start one or two pilot research projects.
   - Design the undergraduate AI course.

   Expected outputs:
   - AI research center established.
   - Geological database prototype.
   - Initial AI training program.
   - Two baseline machine-learning models.
   - First research proposals.
2. **2028 — Research model development** Build research capacity Key activities:
   - Develop geochemical anomaly detection.
   - Develop geophysical interpretation models.
   - Develop geological feature classification.
   - Start master's and doctoral research projects.
   - Introduce AI course and practical exercises.
   - Establish international research collaboration.

   Expected outputs:
   - Three or more working AI models.
   - Initial student research projects.
   - Research publications and conference presentations.
   - Initial mineral prospectivity maps.
3. **2029 — Integrated AI system development** Integration phase Key activities:
   - Integrate geological, geochemical and geophysical information.
   - Develop explainable AI models.

- Conduct independent spatial validation.
- Begin field verification of selected targets.
- Introduce AI practical exercises into additional subjects.
- Develop an integrated software prototype.

Expected outputs:

- Integrated mineral prospectivity model.
- AI exploration software prototype.
- Master's and doctoral research results.
- Independent geological validation report.

4. **2030 — Advanced AI and expansion** Advanced development Key activities:
   - Test models in a second geological area.
   - Improve cross-region model transferability.
   - Incorporate remote sensing data where available.
   - Develop 3D geological and subsurface prediction.
   - Expand collaboration with industry.
   - Develop advanced AI research projects.

   Expected outputs:

   - Improved integrated exploration system.
   - Cross-region validation results.
   - Advanced doctoral research.
   - Industry-oriented demonstration projects.
5. **2031 — Institutionalization and application** Operational maturity Key activities:
   - Consolidate the AI research center.
   - Establish a sustainable geological AI database.
   - Release a validated research software platform.
   - Expand graduate and undergraduate AI education.
   - Establish long-term research partnerships.
   - Prepare the next five-year development strategy.

   Expected outputs:

   - Operational faculty AI research center.
   - Integrated mineral exploration decision-support prototype.
   - Established AI curriculum.
   - Sustainable research and postgraduate training program.
   - International research collaboration network.

Part VIII. Equipment, software and financial requirements

10. Required technical infrastructure

Your faculty should begin with an appropriately sized shared AI laboratory.

I would divide the required infrastructure into three stages.

10.1. Initial laboratory



Required facilities:

- 5–10 capable desktop workstations.
- At least one GPU-equipped workstation for initial deep-learning experiments.
- Shared data storage with backup.
- Stable high-speed internet.
- GIS and scientific data-processing software.
- Python and machine-learning environments.

10.2. Intermediate laboratory



Additional facilities:

- Shared GPU server.
- Centralized geological spatial database.
- Network-attached storage (NAS).
- Dedicated GIS and remote-sensing workstations.
- Backup and data security systems.
- Shared software development and model management environment.

10.3. Advanced research laboratory



Possible additional investments:

- Multi-GPU computing server or HPC access.
- Advanced 3D geological modelling workstations.
- Hyperspectral image processing capability.
- Automated mineralogical image analysis equipment.
- Field data collection equipment.
- Cloud computing resources.

My advice: Do not purchase an expensive GPU cluster at the beginning. First determine the computing requirements of your pilot projects. A modest shared system and cloud or university HPC access may be sufficient during the first two years.

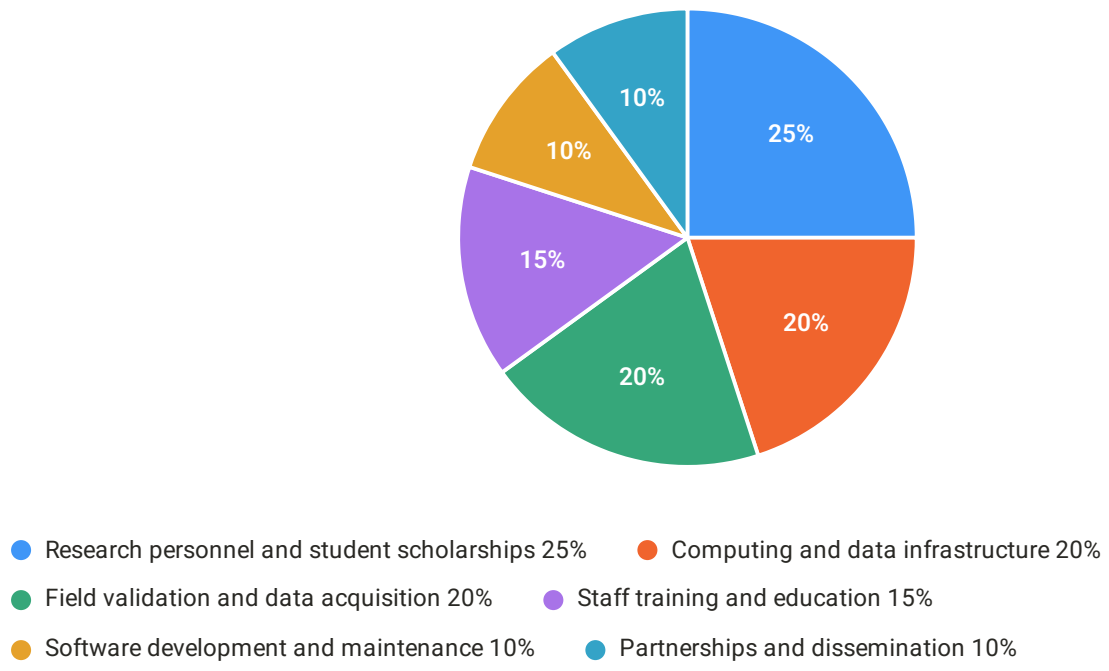
10.4. Indicative budget allocation

Because the country, procurement prices and available university infrastructure are not specified, I would not assign an unreliable monetary figure.

For initial financial planning, consider this illustrative allocation:

Illustrative five-year program budget allocation

Suggested proportions for planning, not quotations or confirmed costs.



The allocation should be adjusted according to existing equipment, staff salaries, access to field areas and external research funding.

Part IX. Management, performance indicators and risk control

11. How should the faculty measure success?

The faculty should establish annual performance indicators.

I recommend the following proposed targets.

Indicator	By end of 2027	By end of 2029	By end of 2031
Organized priority datasets	3	5+	7+
AI research projects	2	5	8+
Academic staff trained	6	12	20
AI-integrated faculty subjects	2	6	10+
AI master's/PhD projects started	2	6	10+
Integrated exploration platforms	Prototype design	Working prototype	Validated platform
External research partnerships	1	3	5+

These are planning targets, not forecasts. They should be revised following a baseline audit and funding assessment.

11.1. Scientific performance indicators

I would also establish a separate evaluation system for the quality of AI research.

The important questions are:

- Does the model perform better than conventional geological and statistical methods?
- Can it identify known mineralized areas?
- Can it predict mineralization in independent geographic areas?
- Can geologists understand why it makes particular predictions?
- Does it provide useful uncertainty information?
- Can its results be reproduced by another researcher?
- Does field investigation support the predicted exploration targets?

A high ROC-AUC value alone should never be considered sufficient evidence of a successful mineral exploration system.

11.2. Major risks and mitigation

Potential problem	Proposed mitigation
Incomplete or inconsistent data	Establish a data quality and harmonization program
Limited known mineral deposits	Use carefully designed positive-unlabelled approaches and geological constraints
Insufficient AI expertise	Organize staff training and international collaboration
Overfitting	Use spatial validation and independent testing
Uninterpretable predictions	Introduce explainable AI
High equipment costs	Use shared computing and cloud resources
Research fragmentation	Coordinate projects through the flagship exploration program
Loss of research data	Establish institutional storage, backup and access policies
Commercially confidential information	Use data-sharing agreements and appropriate access controls

One additional point: mineral exploration datasets may belong to government geological surveys or private exploration companies. Before using them, your faculty should establish clear agreements covering research rights, ownership, confidentiality, publication and reuse.

Part X. The first 90 days: What I recommend doing immediately

I would not wait until the full five-year program has been approved before starting research.

You can make considerable progress during the first three months.

First 90-day action plan

1. **Days 1–30: Institutional organization**
   - Appoint an AI research coordinator.
   - Establish a small interdisciplinary working group.
   - Identify staff with programming and geological data experience.
   - Inventory all existing geological, geochemical and geophysical datasets.
   - Identify two or three possible pilot exploration areas.

   Deliverable: Institutional AI research concept and data inventory.
2. **Days 31–60: Research preparation**
   - Select one pilot exploration area and deposit type.
   - Organize the geological database.
   - Establish a common coordinate reference system and data dictionary.
   - Prepare the first geochemical and geophysical datasets.
   - Select a specific prediction problem.
   - Organize a Python and machine-learning workshop.

   Deliverable: Research-ready pilot dataset and technical research plan.
3. **Days 61–90: First AI experiments**
   - Perform conventional statistical analysis.
   - Develop initial Random Forest and XGBoost models.
   - Compare the AI models with conventional interpretation.
   - Prepare preliminary geological anomaly maps.
   - Identify limitations and additional data requirements.
   - Prepare a master's and doctoral research proposal.

   Deliverable: First AI-based geological exploration research demonstration.

Part XI. Five institutional documents I recommend preparing

To turn this proposal into an actual university program, I suggest preparing five formal documents:

Document	Purpose
1. AI Development Master Plan	Institutional strategy and five-year implementation
2. AI Research Center Establishment Proposal	Organization, staff and responsibilities
3. AI Curriculum Development Proposal	Undergraduate and postgraduate education
4. AI Research Project Portfolio	Research programs and thesis topics

5. AI Laboratory Equipment and Budget Proposal	Infrastructure and financial planning

These documents can be submitted to the dean, faculty council, university management or potential funding organizations.

My overall strategic recommendation

There is one idea I would particularly like to emphasize.

Your faculty should develop one integrated AI-based mineral exploration system as its central research project, while allowing each academic department to contribute its own scientific expertise.

This will create a strong relationship between fundamental geological sciences, applied geological exploration, AI research and university education.

For example:

- The geology department provides geological interpretation.
- The geochemistry department develops geochemical prediction models.
- The geophysics department develops geophysical interpretation models.
- The remote sensing department provides satellite-based geological information.
- The ore deposit geology department provides mineral-system knowledge.
- The mineral exploration department integrates everything into a prospectivity model.
- The computer science and programming specialists develop the software and AI algorithms.

This approach is consistent with the direction of recent mineral exploration research, which increasingly emphasizes the integration of geological, geochemical, geophysical and remote-sensing evidence, together with geological knowledge, explainability and uncertainty assessment.
ScienceDir… +2

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12. One important next step: Develop the formal institutional proposal

I suggest that we take this plan one step further and turn it into a formal document suitable for presentation to your faculty leadership.

I would propose the following title:

MASTER PLAN FOR THE DEVELOPMENT OF ARTIFICIAL INTELLIGENCE IN THE FACULTY OF GEOSCIENCE AND TECHNOLOGY

2027–2031

Proposed document structure

1. Executive Summary
2. Background and Rationale
3. Vision, Mission and Strategic Objectives
4. Establishment of the AI Research Center
5. Research Programs and Flagship Projects
6. AI Education and Curriculum Modernization
7. Master's and Doctoral Research Development
8. Human Resource Development
9. Research Data Infrastructure and AI Laboratory
10. Five-Year Implementation Schedule
11. Budget and Resource Requirements
12. Performance Indicators
13. International Cooperation
14. Risk Management
15. Expected Institutional Outcomes

I can prepare this as a formal, detailed institutional master plan, written in academic and administrative English, with research objectives, implementation tables, budget framework and measurable outcomes.

One detail would help me make the proposal much more practical: Does your faculty already have a designated computer laboratory and GPU-capable computers, or would the university need to establish the AI laboratory from the beginning?

2026-11-06 15:59:09 **You**

OK, please write the detailed institutional master plan, written in academic and administrative English.

ChatGPT gpt-6-thinking

I have prepared the following as a formal institutional master plan, using academic and administrative English. It is structured so that it can serve as a draft for submission to the Dean, Faculty Council, University Management or potential research funding organizations.

The plan assumes a five-year implementation period of **2027–2031** and takes into account your faculty's existing geological, geochemical and geophysical datasets and the programming experience of its academic staff.

MASTER PLAN FOR THE DEVELOPMENT OF ARTIFICIAL INTELLIGENCE IN THE FACULTY OF GEOSCIENCE AND TECHNOLOGY

Five-Year Institutional Development Strategy (2027–2031)

Proposed for: Faculty of Geoscience and Technology

Implementation period: 2027–2031

Document status: Draft for institutional consideration and approval

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EXECUTIVE SUMMARY

Artificial Intelligence (AI) is rapidly transforming scientific research, industrial technologies and higher education. In geological sciences, AI offers significant opportunities to improve geological data interpretation, mineral exploration, geophysical prospecting, geochemical analysis, remote sensing, geological modelling and engineering applications.

The Faculty of Geoscience and Technology recognizes the importance of incorporating AI into its academic, scientific and technological activities. The faculty possesses an important foundation for this development, including academic expertise in geology and related disciplines, existing geological, geochemical and geophysical datasets, and academic staff with programming experience.

This Master Plan proposes the establishment of a comprehensive institutional framework for the development and application of AI in geological sciences and technology over the five-year period 2027–2031.

The central strategic objective is to establish an interdisciplinary research and educational environment in which conventional geological knowledge is integrated with modern AI methods to improve scientific research, mineral exploration, engineering applications and professional education.

The main priority of the proposed program is the development of an **Integrated AI-Based Mineral Exploration System** capable of combining geological, geochemical and geophysical information for mineral prospectivity assessment and exploration target identification.

The Master Plan establishes five principal research programs:

1. AI-Based Geological Mapping and Geological Modelling.
2. AI-Based Geochemical Exploration.
3. AI-Based Geophysical Exploration.
4. AI-Based Remote Sensing and Spatial Analysis.
5. Integrated AI-Based Mineral Prospectivity Mapping and Exploration Decision Support.

In addition to research development, the plan emphasizes the modernization of undergraduate and postgraduate education, development of AI-related master's and doctoral research, systematic training of academic staff, establishment of an AI research laboratory, and development of an integrated geological database.

The proposed implementation will be undertaken in five stages, beginning with institutional organization and data preparation in 2027 and progressing toward the development and validation of integrated AI-based exploration technologies by 2031.

The expected long-term outcome is the establishment of the Faculty of Geoscience and Technology as a recognized center for research, education and technological development in Artificial Intelligence applications in geoscience and mineral exploration.

CHAPTER 1. BACKGROUND AND RATIONALE

1.1. Background

Geological sciences play a fundamental role in the identification, assessment and sustainable utilization of mineral resources, geological resources and the natural environment.

Modern geological investigations increasingly involve the collection and interpretation of large volumes of heterogeneous information obtained through geological mapping, geochemical sampling, geophysical surveys, remote sensing, drilling and laboratory analysis.

Traditional methods of geological interpretation remain essential. However, the increasing volume, complexity and diversity of geological information create opportunities for the application of advanced computational methods.

Artificial Intelligence, particularly machine learning, deep learning, computer vision and data-fusion technologies, can support the interpretation of complex geological relationships, identify previously unrecognized patterns and improve the integration of different exploration datasets.

In mineral exploration, AI can contribute to:

- Identification of geochemical anomalies.
- Interpretation of geophysical anomalies.
- Recognition of geological structures and lithological boundaries.
- Mapping of hydrothermal alteration zones.
- Classification of mineral deposits.
- Prediction of mineralization-favorable geological environments.
- Identification and ranking of exploration targets.
- Three-dimensional geological modelling.
- Optimization of exploration activities.

AI also has considerable potential in mineralogy, petrology, mineral engineering, hydro-geochemistry, remote sensing and civil engineering.

Consequently, the integration of AI into the academic and research activities of the Faculty of Geoscience and Technology represents an important opportunity to modernize its scientific capabilities and strengthen its contribution to geological research and mineral resource development.

1.2. Institutional Background and Existing Capabilities

The faculty has several important conditions for initiating an AI development program:

1. Existing academic expertise in fundamental and applied geological sciences.
2. Availability of geological, geochemical and geophysical datasets.
3. Academic personnel with some programming experience.
4. Existing undergraduate and postgraduate educational programs in relevant disciplines.
5. Opportunities to conduct interdisciplinary research using existing faculty expertise.

These conditions provide a practical foundation for the introduction of AI without requiring the establishment of an entirely new academic discipline or the recruitment of a large computer science department at the initial stage.

However, the effective implementation of AI requires additional institutional capacity, including standardized data management, computing infrastructure, specialized staff training, interdisciplinary research coordination and curriculum development.

1.3. Rationale for the Master Plan

The proposed Master Plan is based on the following considerations:

Scientific rationale: AI can provide additional methods for analyzing complex geological information and developing new approaches to mineral exploration.

Educational rationale: Students should acquire modern data science and AI competencies relevant to contemporary geological and engineering professions.

Institutional rationale: Coordinated AI research can strengthen cooperation among the faculty's academic departments.

Technological rationale: The faculty has the opportunity to develop practical AI-based software and decision-support systems.

Economic and societal rationale: More effective geological assessment and mineral exploration can contribute to responsible mineral resource development, geological risk assessment and sustainable utilization of natural resources.

1.4. Fundamental Principles

The implementation of this Master Plan shall follow these principles:

- Integration of AI with established geological and engineering knowledge.
- Scientific reliability and reproducibility.
- Interdisciplinary cooperation.
- Responsible use of data and computational technologies.
- Transparency and interpretability of AI predictions.
- Independent validation of research results.
- Sustainable development of institutional capacity.
- Continuous improvement of academic curricula.

AI shall be considered a scientific and technological tool that supports, rather than replaces, professional geological interpretation and engineering judgment.

CHAPTER 2. VISION, MISSION AND STRATEGIC OBJECTIVES

2.1. Institutional Vision

To establish the Faculty of Geoscience and Technology as a center of excellence in Artificial Intelligence applications in geological sciences, mineral exploration and earth technology

through the integration of fundamental scientific knowledge, advanced computational methods and practical technological development.

2.2. Institutional Mission

To develop and implement modern AI technologies in geological research, mineral exploration, engineering applications and higher education; to strengthen scientific and technological capabilities; and to prepare highly qualified graduates capable of applying AI in contemporary geoscience and related industries.

2.3. Strategic Objectives

The Master Plan establishes seven strategic objectives.

Objective 1. Development of AI-Based Scientific Research

Develop new research methodologies and applications of AI in geological sciences, mineral exploration and related engineering disciplines.

Objective 2. Development of an Integrated Geological Database

Establish a standardized and accessible institutional database integrating geological, geochemical, geophysical and other relevant scientific information.

Objective 3. Development of AI-Based Mineral Exploration Technologies

Develop integrated machine-learning and deep-learning systems for mineral prospectivity assessment, exploration target identification and geological prediction.

Objective 4. Modernization of Higher Education

Integrate AI, programming, data science and machine learning into undergraduate, master's and doctoral educational programs.

Objective 5. Development of Human Resources

Improve the AI-related scientific and technical competencies of academic staff, researchers and students.

Objective 6. Establishment of Research Infrastructure

Develop an appropriate AI laboratory, computing facilities, software environment and scientific data management system.

Objective 7. International Scientific Cooperation

Strengthen cooperation with universities, geological surveys, research organizations and mineral exploration companies.

CHAPTER 3. ORGANIZATIONAL STRUCTURE AND INSTITUTIONAL MANAGEMENT

3.1. Establishment of the AI for Geoscience Research Center

It is proposed that the faculty establish an interdisciplinary research unit named:

AI FOR GEOSCIENCE AND MINERAL RESOURCES RESEARCH CENTER

The center shall initially operate as a coordinating research and development unit within the faculty.

Its main responsibilities will include:

- Coordination of AI-related research activities.
- Development and maintenance of geological databases.
- Organization of staff training.
- Support for undergraduate and postgraduate research.
- Development of AI-based geological software.
- Coordination of interdisciplinary research projects.
- Preparation of research proposals and funding applications.
- Development of international research cooperation.

3.2. Proposed Organizational Structure

The center shall operate under the supervision of the Dean of the Faculty or an appointed academic authority.

Its organizational structure shall include:

A. Director of the AI Research Center

Responsible for overall management, strategic planning, coordination and reporting.

B. Scientific and Technical Advisory Committee

Responsible for reviewing research priorities, scientific quality, infrastructure requirements and institutional development.

C. Five Research Divisions

1. AI Geological Mapping and Modelling Division.
2. AI Geochemical Exploration Division.
3. AI Geophysical Exploration Division.
4. AI Remote Sensing and GIS Division.
5. Integrated Mineral Exploration AI Division.

D. AI Education and Technical Support Unit

Responsible for staff training, curriculum development, laboratory management, software support and student research activities.

3.3. Academic Staff Participation

Academic staff from the existing departments shall participate in the research divisions according to their expertise.

The center shall promote interdisciplinary research teams involving geologists, geochemists, geophysicists, mineralogists, petrologists, remote sensing specialists, mineral engineers and programming specialists.

The initial research group should preferably include approximately 6–10 active academic staff members, with gradual expansion according to institutional resources.

3.4. Management and Reporting

The center shall prepare:

- Annual research and development plans.
- Annual educational development reports.
- Annual infrastructure and budget proposals.
- Research project progress reports.
- Annual performance assessments.

The Scientific and Technical Advisory Committee shall review the progress of the Master Plan and recommend adjustments when necessary.

CHAPTER 4. PRIORITY RESEARCH PROGRAMS

Five major research programs shall constitute the scientific foundation of the Master Plan.

4.1. Research Program 1: AI-Based Geological Mapping and Modelling

Objective

To develop AI methods for the interpretation, classification and modelling of geological information and the identification of geological environments favorable for mineralization.

Main Research Directions

1. Automatic lithological classification using geological and spatial datasets.
2. AI-based interpretation of geological boundaries.
3. Automatic recognition of faults, folds, fractures and lineaments.
4. Digital geological mapping using computer vision.
5. Three-dimensional geological modelling.
6. Structural geology interpretation.
7. Development of geological knowledge graphs.
8. Integration of geological information with mineral deposit models.

Recommended AI Methods

- Random Forest.
- Support Vector Machine.
- XGBoost.
- Convolutional Neural Networks.
- Computer vision.
- Graph Neural Networks.
- Three-dimensional machine learning.

Expected Results

- AI-based geological classification models.
- Automatic geological interpretation tools.
- Improved digital geological maps.
- Three-dimensional geological models.
- Geological knowledge databases.

Proposed Doctoral Research Topics

- Development of an AI-Based Geological Mapping and Lithological Classification System.
- Application of Deep Learning for Automatic Identification of Geological Structures.
- Development of AI-Based Three-Dimensional Geological Modelling Methods.

4.2. Research Program 2: AI-Based Geochemical Exploration

Objective

To develop AI methodologies for the identification, classification and interpretation of geochemical anomalies associated with mineralization.

Main Research Directions

1. Multielement geochemical data analysis.
2. Identification of geochemical anomalies.
3. Classification of mineralized and background geochemical environments.
4. Identification of pathfinder elements.
5. Analysis of elemental associations.
6. Geochemical dispersion modelling.
7. Prediction of concealed mineralization.
8. Integration of geochemical and geological information.

Recommended AI Methods

- Random Forest.
- XGBoost.
- Support Vector Machine.
- K-means clustering.
- DBSCAN.
- Autoencoders.
- Anomaly detection algorithms.
- Explainable AI.

Expected Results

- AI-based geochemical anomaly detection algorithms.
- Multielement geochemical classification models.
- Geochemical prospectivity maps.
- Improved interpretation of geochemical exploration data.

Proposed Doctoral Research Topics

- Development of an AI-Based System for Multielement Geochemical Anomaly Detection.
- Application of Explainable Machine Learning for Identification of Geochemical Signatures of Mineral Deposits.
- Development of Machine Learning Models for Prediction of Concealed Mineralization Using Surface Geochemical Data.

4.3. Research Program 3: AI-Based Geophysical Exploration

Objective

To improve the interpretation of geophysical survey data and develop AI-based methods for predicting subsurface geological structures and mineralization.

Main Research Directions

1. Interpretation of magnetic anomalies.
2. Analysis of gravity data.
3. Interpretation of electrical resistivity and induced polarization data.
4. Electromagnetic data analysis.
5. Automatic recognition of geophysical anomalies.
6. AI-assisted geophysical inversion.
7. Integration of multiple geophysical methods.
8. Three-dimensional subsurface geological modelling.

Recommended AI Methods

- Artificial Neural Networks.
- Convolutional Neural Networks.
- Random Forest.
- XGBoost.
- Deep Learning.
- Physics-Informed Neural Networks.
- Bayesian modelling.

Expected Results

- AI-based geophysical interpretation algorithms.
- Improved geological structure identification.

- Three-dimensional subsurface prediction models.
- Integrated geophysical interpretation software.

Proposed Doctoral Research Topics

- Development of Deep Learning Models for Integrated Geophysical Data Interpretation.
- AI-Based Interpretation of Magnetic and Electrical Geophysical Anomalies for Mineral Exploration.
- Development of Physics-Informed Machine Learning Methods for Subsurface Geological Prediction.

4.4. Research Program 4: AI-Based Remote Sensing and Spatial Analysis

Objective

To develop AI applications for geological mapping, lithological classification, alteration identification and regional mineral exploration.

Main Research Directions

1. Satellite image classification.
2. Lithological mapping using multispectral data.
3. Identification of hydrothermal alteration zones.
4. Hyperspectral mineral identification.
5. Automatic extraction of geological lineaments.
6. Identification of iron oxide and clay mineral indicators.
7. Integration of remote sensing and geochemical data.
8. GIS-based mineral exploration analysis.

Recommended AI Methods

- Convolutional Neural Networks.
- Transformer models.
- Spectral classification algorithms.
- Random Forest.
- Support Vector Machine.
- Deep Learning.
- Spatial machine learning.

Expected Results

- AI-based geological remote sensing maps.
- Hydrothermal alteration maps.
- Lithological classification systems.
- GIS-based exploration applications.

Proposed Doctoral Research Topics

- Development of a Deep Learning-Based System for Hydrothermal Alteration Mapping.
- AI-Based Lithological Classification Using Multispectral and Hyperspectral Satellite Data.
- Integration of Remote Sensing and Geochemical Information for Mineral Prospectivity Assessment.

4.5. Research Program 5: Integrated AI-Based Mineral Prospectivity Mapping

Objective

To develop an integrated AI system that combines geological, geochemical, geophysical and, where available, remote sensing information to identify and evaluate mineralization-favorable areas.

This program shall be the central flagship research program of the faculty.

Main Research Directions

1. Development of an integrated geological exploration database.
2. Preparation and standardization of multisource datasets.
3. Development of individual AI models for different geological datasets.
4. Integration of geological, geochemical and geophysical evidence.
5. Development of multisource data-fusion algorithms.
6. Application of explainable AI for geological interpretation.
7. Assessment of prediction uncertainty.
8. Production of mineral prospectivity maps.
9. Independent spatial validation.
10. Field verification of selected exploration targets.

Expected Results

- Integrated mineral exploration database.

- Multisource machine-learning models.
- AI-based mineral prospectivity maps.
- Exploration target ranking system.
- Uncertainty and interpretation maps.
- Prototype exploration decision-support software.

Proposed Flagship Doctoral Research Topic

Development of an Explainable Multisource Artificial Intelligence System for Integrated Mineral Prospectivity Mapping and Exploration Targeting.

Expected Institutional Contribution

This research program will provide a common scientific framework for the participation of different academic departments and will serve as the principal technological demonstration of the faculty's AI development activities.

CHAPTER 5. DEVELOPMENT OF THE INTEGRATED AI-BASED MINERAL EXPLORATION SYSTEM

5.1. General Objective

To develop a practical AI-based mineral exploration platform capable of processing, integrating and interpreting geological information to support mineral prospectivity assessment and exploration decision-making.

5.2. System Architecture

The proposed system shall consist of six functional components.

Component 1. Data Acquisition

- Geological maps and field observations.
- Geochemical sampling and analytical results.
- Magnetic and gravity survey data.
- Electrical and electromagnetic survey data.
- Remote sensing imagery, when available.
- Mineral occurrence and drilling information.

Component 2. Data Preparation

- Data cleaning and quality assessment.
- Coordinate system standardization.
- Spatial data integration.
- Missing-value assessment.
- Feature extraction.
- GIS-based data processing.

Component 3. AI Model Development

- Random Forest.
- XGBoost.
- Support Vector Machine.
- Artificial Neural Networks.
- Deep Learning.
- Anomaly detection.
- Bayesian modelling.

Component 4. Data Integration and Interpretation

- Multisource data fusion.
- Geological knowledge integration.
- Explainable AI.
- Feature importance assessment.
- Uncertainty estimation.

Component 5. Prediction and Visualization

- Mineral prospectivity maps.
- Predicted exploration targets.
- Geological interpretation maps.
- Probability and uncertainty maps.
- Three-dimensional geological visualization.

Component 6. Field Verification and Model Updating

- Geological field investigations.
- Additional geochemical sampling.
- Geophysical verification.

- Drilling and laboratory results, when available.
- Updating and retraining of AI models.

5.3. Initial Pilot Project

The first institutional project shall focus on a selected mineralized district with sufficient geological, geochemical and geophysical information.

The pilot project shall proceed through the following stages:

1. Selection of the study area and target deposit type.
2. Compilation of existing geological datasets.
3. Data quality assessment and harmonization.
4. Conventional geological and statistical interpretation.
5. Development of individual machine-learning models.
6. Comparison of different AI algorithms.
7. Development of an integrated prediction model.
8. Application of explainable AI.
9. Spatial validation and uncertainty assessment.
10. Field verification and preparation of a technical report.

The first pilot should prioritize the available datasets. Remote sensing, drilling and additional data sources may be incorporated as they become available.

5.4. Scientific Validation Requirements

All research projects shall apply appropriate scientific validation procedures.

These shall include:

- Comparison with conventional geological methods.
- Independent testing of model performance.
- Spatially separated validation where appropriate.
- Assessment of class imbalance and sampling bias.
- Analysis of prediction uncertainty.
- Geological interpretation of important model variables.
- Documentation of data provenance and model versions.

Randomly splitting nearby geological samples into training and testing datasets shall not be accepted as sufficient evidence of generalization to unexplored regions.

AI predictions shall be interpreted as exploration hypotheses and shall not be treated as confirmed mineral discoveries or substitutes for independent geological investigation.

CHAPTER 6. INTEGRATION OF AI INTO EDUCATIONAL PROGRAMS

6.1. General Objective

To modernize the faculty's educational programs through the systematic integration of AI, programming, machine learning and scientific data analysis into fundamental and specialized subjects.

6.2. Introduction of New Courses

It is proposed to introduce the following courses, subject to university academic approval.

Course 1. Introduction to Artificial Intelligence and Data Science for Geoscientists

Recommended level: Undergraduate.

Suggested credit value: 3 credits.

Main contents:

1. Introduction to Artificial Intelligence.
2. Machine learning and deep learning fundamentals.
3. Python programming.
4. Scientific data management.
5. Statistical analysis.
6. GIS and spatial data processing.
7. Classification and regression.
8. Clustering and anomaly detection.
9. Introduction to neural networks.
10. Geological applications of AI.

Course 2. Machine Learning Applications in Mineral Exploration

Recommended level: Master's.

Suggested credit value: 3 credits.

Main contents:

1. Geological exploration databases.
2. Geochemical anomaly detection.
3. Geophysical data interpretation.
4. Geological feature extraction.
5. Random Forest and XGBoost.
6. Support Vector Machines.
7. Deep learning applications.
8. Mineral prospectivity mapping.
9. Explainable AI.
10. Scientific validation of exploration models.

Course 3. Advanced AI and Multisource Data Integration in Geoscience

Recommended level: Doctoral.

Suggested credit value: 3 credits.

Main contents:

1. Advanced machine learning.
2. Deep neural networks.
3. Multisource data fusion.
4. Spatial machine learning.
5. Geological knowledge graphs.
6. Explainable and interpretable AI.
7. Physics-informed machine learning.
8. Bayesian uncertainty modelling.
9. Advanced mineral exploration applications.
10. Research software development.

6.3. Integration into Existing Faculty Subjects

AI practical exercises shall be introduced progressively into existing subjects.

Existing subject	Proposed AI educational application
Geology	Automatic geological classification and digital mapping
Mineralogy	Image-based mineral identification
Petrology	Automated petrographic image analysis
Geological Structure	Fault, fracture and lineament recognition
Geology of Ore Deposits	AI-based mineral deposit classification
Mineral Exploration	Mineral prospectivity mapping
Mineral Engineering	Prediction and optimization of mineral processing
Applied Geochemistry	Multielement anomaly detection
Geophysical Prospecting	Magnetic, gravity and electrical data interpretation
Remote Sensing	Satellite image classification and alteration mapping
Hydro-geochemistry	Groundwater chemistry prediction
Civil Engineering	Geotechnical prediction and geological hazard assessment

6.4. Educational Development Principles

The faculty shall ensure that:

- AI education is integrated with fundamental geological knowledge.
- Students acquire practical programming competencies.
- Students use actual geological and engineering datasets.
- Research projects encourage interdisciplinary cooperation.
- AI predictions are evaluated scientifically.
- Students understand the limitations and uncertainties of computational methods.

AI shall not replace fundamental geological education. Rather, it shall strengthen students' ability to apply geological knowledge using modern computational technologies.

CHAPTER 7. MASTER'S AND DOCTORAL RESEARCH DEVELOPMENT

7.1. General Objective

To establish a coordinated graduate research program in AI applications in geoscience and mineral exploration.

7.2. Proposed Doctoral Research Topics

The following topics are recommended for initial doctoral research development:

1. Development of Machine Learning Algorithms for Multielement Geochemical Anomaly Detection.
2. Deep Learning-Based Interpretation of Geophysical Data for Concealed Mineral Exploration.
3. AI-Based Geological and Structural Modelling of Mineralization-Favorable Environments.
4. Explainable AI for Geological Interpretation and Mineral Prospectivity Assessment.
5. Development of Multisource AI Data-Fusion Models for Mineral Prospectivity Mapping.
6. Development of an Intelligent Geological Exploration Decision-Support System.
7. Physics-Informed Artificial Intelligence for Subsurface Geological Prediction.
8. Geological Knowledge Graphs and Large Language Models for Mineral Exploration.

7.3. Proposed Master's Research Topics

1. Application of Random Forest for Copper Geochemical Anomaly Detection.
2. Application of XGBoost for Mineral Prospectivity Mapping.
3. Machine Learning-Based Classification of Magnetic Anomalies.
4. Automatic Extraction of Geological Lineaments from Satellite Images.
5. Application of Support Vector Machines for Lithological Classification.
6. GIS-Based Integration of Geological and Geochemical Information.
7. Explainable AI for Interpretation of Geochemical Data.
8. Prediction of Mineralized Zones Using Geological and Geophysical Information.
9. AI-Based Identification of Minerals from Microscopic Images.
10. Machine Learning-Based Prediction of Mineral Processing Performance.

7.4. Integrated Graduate Research Approach

Where appropriate, several master's and doctoral students shall participate in a common research program using shared datasets and complementary research objectives.

For example, a group of doctoral students may investigate geochemical, geophysical, geological and data-fusion components of the same mineral exploration district.

This arrangement will strengthen interdisciplinary research and increase the practical value of individual academic projects.

7.5. Research Quality Requirements

Graduate research involving AI shall demonstrate:

- A clearly defined geological or engineering problem.
- Scientific justification for the selected AI methods.
- Appropriate data quality and documentation.
- Comparison with conventional methods.
- Independent model evaluation.
- Geological interpretation.
- Assessment of limitations and uncertainty.
- Original scientific contributions.

A doctoral dissertation shall demonstrate an original scientific contribution and shall not consist solely of applying an existing algorithm to a new dataset.

CHAPTER 8. HUMAN RESOURCE DEVELOPMENT

8.1. General Objective

To strengthen the scientific and technical competencies of academic staff and students in AI and data science.

8.2. Academic Staff Training Program

A structured staff training program shall be organized in four stages.

Level 1. Programming and Scientific Data Analysis

- Python programming.
- Pandas and NumPy.

- Scientific data management.
- Statistical analysis.
- GIS and spatial data processing.

Level 2. Machine Learning

- Scikit-learn.
- Random Forest.
- XGBoost.
- Support Vector Machines.
- Clustering.
- Anomaly detection.

Level 3. Deep Learning and Advanced AI

- Neural networks.
- Convolutional Neural Networks.
- PyTorch.
- Image processing.
- Multisource data integration.

Level 4. Advanced Geological AI Research

- Explainable AI.
- Physics-informed machine learning.
- Geological knowledge graphs.
- Bayesian uncertainty modelling.
- Research software engineering.

8.3. Training Implementation

The faculty shall organize:

- Regular internal training workshops.
- Practical machine-learning courses.
- Interdepartmental research seminars.
- Joint training with computer science specialists.
- International scientific workshops.

- Short-term academic exchanges.
- Collaborative doctoral supervision.

8.4. Development of Young Researchers

Undergraduate and postgraduate students shall be encouraged to participate in AI-related research through:

- Student research projects.
- Scientific seminars.
- Research competitions.
- Master's and doctoral programs.
- Participation in field investigations.
- Collaborative software development.

CHAPTER 9. RESEARCH INFRASTRUCTURE AND AI LABORATORY DEVELOPMENT

9.1. General Objective

To establish an appropriate technical environment for AI research, education and software development.

9.2. Establishment of the AI Laboratory

The proposed laboratory shall provide facilities for:

1. Scientific programming.
2. Geological database management.
3. GIS and spatial analysis.
4. Machine learning.
5. Deep learning.
6. Remote sensing data processing.
7. Geological visualization.
8. Software development.

9.3. Required Computing Infrastructure

Initial Stage

- Five to ten capable computer workstations.
- At least one GPU-equipped workstation.
- Shared data storage.
- Reliable internet access.
- Backup facilities.
- Open-source scientific software.

Intermediate Stage

- Shared GPU server.
- Centralized geological database.
- Network-attached storage.
- Dedicated GIS workstations.
- Model development and tracking environment.
- Data backup and security infrastructure.

Advanced Stage

- Multi-GPU server or access to high-performance computing.
- Advanced three-dimensional geological visualization.
- Hyperspectral data processing capability.
- Additional specialized computing resources as justified by research requirements.

9.4. Recommended Software

Application	Recommended software
Programming	Python
Data analysis	Pandas, NumPy
Machine learning	Scikit-learn, XGBoost
Deep learning	PyTorch
GIS	QGIS

Application	Recommended software
Spatial database	PostgreSQL and PostGIS
Raster processing	GDAL, Rasterio
Remote sensing	Google Earth Engine and appropriate open-source tools
Model management	MLflow
Version control	Git
Scientific visualization	Python visualization libraries and QGIS

The use of open-source software shall be prioritized where technically and institutionally appropriate.

9.5. Infrastructure Development Policy

Infrastructure investments shall be made progressively.

The faculty shall first assess existing computers, servers, network capacity and software resources before purchasing additional equipment.

Expensive computing infrastructure shall be acquired only after the computing requirements of the initial research projects have been established.

CHAPTER 10. INTEGRATED GEOLOGICAL DATA MANAGEMENT

10.1. General Objective

To establish a standardized institutional geological database supporting research, education and AI system development.

10.2. Main Database Components

Geological Database

- Geological maps.

- Lithological information.
- Stratigraphic information.
- Faults and structural data.
- Geological field observations.

Geochemical Database

- Sample identification.
- Sampling coordinates.
- Element concentrations.
- Analytical methods.
- Laboratory information.
- Detection limits and quality control information.

Geophysical Database

- Magnetic measurements.
- Gravity data.
- Electrical resistivity.
- Induced polarization.
- Electromagnetic data.
- Survey parameters and metadata.

Mineral Occurrence Database

- Known mineral deposits.
- Mineral prospects.
- Mineral occurrences.
- Deposit classifications.
- Ore and gangue mineral information.

Drilling Database

Where available:

- Borehole locations.
- Borehole trajectories.
- Lithological logs.
- Assay results.

- Mineralized intervals.

AI Research Database

- Training datasets.
- Model versions.
- Model parameters.
- Validation results.
- Prediction maps.
- Research documentation.

10.3. Data Management Standards

The faculty shall establish institutional requirements for:

- Standardized coordinate reference systems.
- Common data formats.
- Consistent sample identification.
- Metadata documentation.
- Data quality control.
- Data access and permissions.
- Version control.
- Long-term data preservation.

The database should follow the principles of findability, accessibility, interoperability and reusability wherever possible.

10.4. Data Ownership and Security

All datasets shall be managed in accordance with university regulations, applicable national laws and contractual obligations.

The faculty shall establish procedures for:

- Data ownership.
- Confidential exploration information.
- Research access permissions.
- Publication rights.
- Data sharing with external organizations.

- Backup and recovery.

CHAPTER 11. FIVE-YEAR IMPLEMENTATION PLAN

The implementation period shall cover 2027–2031.

11.1. Year 1: 2027 — Institutional Foundation

Main Objectives

- Establish the AI Research Center.
- Organize the initial interdisciplinary research group.
- Assess existing scientific datasets.
- Select the pilot exploration area.
- Establish basic computing infrastructure.
- Initiate staff training.

Main Activities

1. Appoint the AI research coordinator.
2. Establish the Scientific and Technical Advisory Committee.
3. Inventory existing geological, geochemical and geophysical datasets.
4. Select priority mineral exploration problems.
5. Prepare the initial institutional database.
6. Organize Python and machine-learning training.
7. Develop initial statistical and machine-learning models.
8. Prepare proposals for AI-related educational courses.

Expected Outputs

- AI Research Center established.
- Initial geological database framework.
- Staff training program initiated.
- Pilot research area selected.
- At least two initial AI research projects.
- Initial research proposals prepared.

11.2. Year 2: 2028 — Research Capacity Development

Main Objectives

Develop individual AI applications for geological, geochemical and geophysical datasets.

Main Activities

1. Develop geochemical anomaly detection models.
2. Develop geophysical interpretation models.
3. Develop geological classification models.
4. Start master's and doctoral research projects.
5. Introduce AI-related courses.
6. Organize interdisciplinary research seminars.
7. Establish initial external research partnerships.

Expected Outputs

- Several functioning AI models.
- Initial mineral prospectivity maps.
- Master's and doctoral research projects underway.
- AI educational courses initiated.
- Initial scientific publications and conference presentations.

11.3. Year 3: 2029 — Integrated AI System Development

Main Objectives

Integrate individual research outputs into a common mineral exploration platform.

Main Activities

1. Integrate geological, geochemical and geophysical datasets.
2. Develop multisource machine-learning models.
3. Apply explainable AI.
4. Conduct spatial validation.
5. Assess prediction uncertainty.
6. Begin field verification of selected targets.
7. Develop the first software prototype.

Expected Outputs

- Integrated mineral prospectivity model.
- Initial exploration decision-support prototype.
- Independent validation results.
- Graduate research outputs.
- Expanded AI curriculum.

11.4. Year 4: 2030 — Advanced Research and Expansion

Main Objectives

Improve the integrated system and expand research applications.

Main Activities

1. Evaluate model performance in an additional geological area.
2. Develop advanced deep-learning methods.
3. Integrate remote sensing datasets where available.
4. Develop three-dimensional geological prediction methods.
5. Expand industry and institutional collaboration.
6. Improve the exploration software prototype.
7. Develop advanced doctoral research projects.

Expected Outputs

- Improved integrated exploration platform.
- Cross-region model evaluation.
- Advanced AI research publications.
- Additional external research projects.
- Stronger international cooperation.

11.5. Year 5: 2031 — Institutional Consolidation

Main Objectives

Establish sustainable institutional AI research and educational capacity.

Main Activities

1. Consolidate the AI Research Center.
2. Complete the integrated geological database.

3. Validate and document the exploration software prototype.
4. Expand AI-related education across faculty departments.
5. Establish long-term research partnerships.
6. Evaluate the five-year development program.
7. Prepare the next institutional development strategy.

Expected Outputs

- Operational AI research center.
- Integrated mineral exploration research platform.
- Established AI-related curriculum.
- Sustainable graduate research program.
- Expanded research cooperation network.
- Five-year institutional evaluation report.

CHAPTER 12. FINANCIAL AND RESOURCE REQUIREMENTS

12.1. General Financial Policy

The financial requirements of the Master Plan shall be determined through annual budgeting and institutional resource assessment.

Because equipment costs, salaries, software requirements and national procurement conditions vary, the financial plan shall be developed in accordance with actual local prices and university financial regulations.

12.2. Indicative Budget Distribution

The following distribution is proposed as an initial planning framework.

Budget category	Proposed allocation
Research personnel and student scholarships	25%
Computing and data infrastructure	20%
Field validation and data acquisition	20%

Budget category	Proposed allocation
Staff training and educational development	15%
Software development and maintenance	10%
International cooperation and research dissemination	10%
Total	**100%**

12.3. Funding Sources

Potential funding sources shall include:

1. University and faculty development budgets.
2. National scientific research grants.
3. Government geological and mineral resource development programs.
4. International scientific cooperation programs.
5. Joint research projects with universities.
6. Research agreements with mineral exploration and mining companies.
7. Industrial research and technical service contracts.
8. Competitive research funding from relevant international organizations.

12.4. Financial Management

Annual financial plans shall be prepared based on approved research priorities, equipment requirements, staffing needs and expected project outputs.

All expenditure shall follow the applicable university procurement and financial management procedures.

CHAPTER 13. INTERNATIONAL COOPERATION AND INDUSTRIAL PARTNERSHIPS

13.1. General Objective

To establish international scientific and technological cooperation in AI applications in geological sciences and mineral exploration.

13.2. Priority Cooperation Areas

The faculty shall seek cooperation with:

- Universities specializing in economic geology and mineral exploration.
- Universities with established AI and data science research groups.
- National geological surveys.
- Mineral exploration companies.
- Mining and mineral processing companies.
- Remote sensing and geospatial research organizations.
- International scientific research networks.

13.3. Proposed Cooperation Activities

1. Joint research projects.
2. Collaborative doctoral supervision.
3. Academic staff exchange.
4. International workshops and conferences.
5. Joint scientific publications.
6. Shared geological datasets, subject to appropriate agreements.
7. Development of joint AI software.
8. Joint applications for international research funding.

13.4. Industrial Cooperation

Cooperation with exploration and mining companies should focus on practical problems, including:

- Geochemical anomaly detection.
- Geophysical data interpretation.
- Mineral prospectivity mapping.
- Drill target evaluation.
- Geological data management.
- Mineral processing optimization.

Such cooperation shall be established through formal agreements that specify data ownership, confidentiality, research responsibilities and intellectual property rights.

CHAPTER 14. PERFORMANCE INDICATORS AND MONITORING

14.1. General Monitoring Framework

The implementation of the Master Plan shall be evaluated annually.

The evaluation shall cover research development, education, infrastructure, human resources, scientific outputs and external cooperation.

14.2. Proposed Institutional Performance Indicators

The following targets are proposed for institutional planning.

Performance indicator	2027	2029	2031
Organized priority datasets	3	5 or more	7 or more
Active AI research projects	2	5	8 or more
Academic staff trained	6	12	20
Faculty subjects integrating AI	2	6	10 or more
Master's and doctoral AI projects started	2	6	10 or more
Integrated exploration software	Design	Working prototype	Validated prototype
External research partnerships	1	3	5 or more

These indicators shall be reviewed after the first institutional resource assessment and adjusted to reflect the actual size and capacity of the faculty.

14.3. Scientific Quality Indicators

Research quality shall be evaluated using the following criteria:

- Scientific originality.
- Quality of geological datasets.
- Appropriate application of AI methods.

- Comparison with conventional approaches.
- Independent validation.
- Geological interpretability.
- Assessment of uncertainty.
- Reproducibility of results.
- Scientific publications.
- Practical usefulness of research outputs.

The number of AI models or publications alone shall not be considered sufficient evidence of research quality.

14.4. Annual Review

The AI Research Center shall prepare an annual report containing:

1. Research activities completed.
2. Educational activities completed.
3. Progress in database development.
4. Infrastructure development.
5. Staff and student training.
6. Publications and research outputs.
7. Budget utilization.
8. External partnerships.
9. Problems encountered.
10. Recommendations for the following year.

CHAPTER 15. RISK MANAGEMENT AND QUALITY ASSURANCE

15.1. Potential Risks

The main risks associated with implementation include:

- Incomplete or inconsistent geological datasets.
- Insufficient AI expertise.
- Limited financial resources.

- Inadequate computing infrastructure.
- Data access and confidentiality restrictions.
- Overfitting and unreliable model predictions.
- Limited interdisciplinary cooperation.
- Difficulties in maintaining software and databases.

15.2. Risk Mitigation Measures

Risk	Mitigation measure
Inadequate data quality	Establish data standards and quality control
Limited AI expertise	Implement structured staff training
Financial limitations	Develop infrastructure progressively
Insufficient computing capacity	Use shared computing and cloud resources
Unreliable AI predictions	Require independent scientific validation
Limited interdisciplinary cooperation	Establish common research projects
Data confidentiality	Develop institutional data governance policies
Software maintenance problems	Establish version control and documentation

15.3. Quality Assurance

All AI-related research projects shall follow institutional scientific quality requirements.

The faculty shall promote reproducible research, documented datasets, appropriate validation methods and transparent reporting of AI model limitations.

Research involving geological exploration predictions shall retain professional geological interpretation and field verification as essential components of the research process.

CHAPTER 16. EXPECTED INSTITUTIONAL OUTCOMES

By the end of the five-year implementation period, the faculty is expected to achieve the following outcomes.

16.1. Scientific Outcomes

- Establishment of a coordinated AI research program.
- Development of new AI methods for mineral exploration.
- Improved interpretation of geological, geochemical and geophysical information.
- Development of integrated mineral prospectivity models.
- Increased scientific publications and postgraduate research.
- Expansion of interdisciplinary scientific cooperation.

16.2. Educational Outcomes

- Introduction of AI-related courses.
- Integration of AI practical exercises into existing subjects.
- Development of AI-related master's and doctoral research.
- Improved programming and data science competencies among students.
- Modernization of geological and engineering education.

16.3. Technological Outcomes

- Establishment of an AI research laboratory.
- Development of an integrated geological database.
- Development of machine-learning and deep-learning models.
- Development of an AI-based mineral exploration prototype.
- Improved institutional capacity for scientific software development.

16.4. Institutional Outcomes

- Establishment of the AI for Geoscience Research Center.
- Stronger interdisciplinary cooperation.
- Increased international research collaboration.
- Improved opportunities for competitive research funding.
- Stronger cooperation with geological surveys and mineral exploration companies.
- Development of a sustainable institutional AI research environment.

CHAPTER 17. SUSTAINABILITY AND FUTURE DEVELOPMENT

The sustainability of the proposed program shall be ensured through long-term institutional commitment, continuous academic training, systematic data management and integration of AI research into the faculty's regular academic activities.

The faculty shall prioritize the maintenance and continuous improvement of its geological database, computing infrastructure, research software and academic competencies.

Following the completion of the 2027–2031 Master Plan, the faculty should consider further development in the following areas:

1. AI-based three-dimensional and four-dimensional geological modelling.
2. Physics-informed AI for geophysical inversion and geological prediction.
3. Geological knowledge graphs and large language models.
4. AI-based mineralogical and petrographic image analysis.
5. Advanced mineral resource and mineral processing applications.
6. AI-based geological hazard assessment.
7. Integrated digital geological information systems.
8. Development of operational exploration decision-support technologies.

The faculty should also assess opportunities for establishing a more advanced Center of Excellence in AI for Geoscience and Mineral Resources, depending on the scientific achievements and institutional capacity developed during the first five years.

CHAPTER 18. CONCLUSION

The development of Artificial Intelligence in the Faculty of Geoscience and Technology represents an important strategic opportunity to modernize scientific research, higher education and technological applications.

The faculty already possesses several essential foundations for this development, including expertise in geological sciences, access to geological, geochemical and geophysical information, and academic staff with programming experience.

The proposed Master Plan provides a coordinated institutional framework for utilizing these capabilities and progressively developing advanced AI research and educational activities.

The central priority shall be the development of an integrated AI-based mineral exploration system that combines geological, geochemical and geophysical information to improve mineral

prospectivity assessment and exploration target identification.

At the same time, AI shall be incorporated into the faculty's fundamental and specialized subjects, ensuring that students and academic staff acquire the scientific and computational competencies required for contemporary geoscience research.

Successful implementation will require institutional leadership, interdisciplinary cooperation, appropriate financial support, systematic human resource development and sustained cooperation with national and international research organizations.

By implementing this Master Plan, the Faculty of Geoscience and Technology will establish a foundation for becoming a modern scientific and educational institution in which conventional geological knowledge and advanced AI technologies are developed and applied together.

The ultimate strategic goal is to establish a sustainable institutional center that contributes to the advancement of geological science, intelligent mineral exploration and responsible utilization of mineral resources through Artificial Intelligence.

ANNEX 1. PRIORITY PROJECT PORTFOLIO

Project	Priority	Proposed implementation
AI-Based Geochemical Anomaly Detection	High	2027–2028
AI-Based Geophysical Data Interpretation	High	2027–2029
AI-Based Geological Classification	High	2027–2028
Integrated Mineral Prospectivity Mapping	Very high	2028–2031
Explainable AI for Mineral Exploration	High	2028–2030
AI-Based Remote Sensing and Alteration Mapping	Medium–high	2028–2030
Three-Dimensional Geological Modelling	Medium–high	2029–2031

Project	Priority	Proposed implementation
Geological Knowledge Graph Development	Medium	2029–2031
AI-Based Mineral Processing Optimization	Medium	2029–2031
AI-Based Geological and Geotechnical Assessment	Medium	2029–2031

ANNEX 2. PROPOSED FIRST-YEAR ACTION PLAN

Period	Main activity	Responsible unit	Expected deliverable
Months 1–2	Establish AI Research Center	Faculty leadership	Approved organizational framework
Months 1–2	Identify research personnel	Departments	Initial research team
Months 1–3	Inventory geological datasets	Research Center	Data inventory
Months 2–3	Select pilot exploration area	Research team	Pilot project proposal
Months 3–4	Establish data standards	Data management unit	Data management framework
Months 3–6	Staff training	Education and Technical Support Unit	Initial training program
Months 4–7	Prepare pilot datasets	Research divisions	Research-ready datasets
Months 6–9	Develop initial AI models	Research divisions	Baseline AI models

Period	Main activity	Responsible unit	Expected deliverable
Months 8–10	Prepare new educational courses	Academic departments	Curriculum proposals
Months 10–12	Evaluate initial research results	Advisory Committee	Annual progress report

ANNEX 3. PROPOSED INSTITUTIONAL APPROVAL AND RESPONSIBILITY

The implementation of this Master Plan shall be subject to approval by the relevant university authorities.

Proposed approval authorities

- Rector / President of the University.
- Dean of the Faculty of Geoscience and Technology.
- Faculty Council.
- Relevant academic and research committees.

Proposed implementation responsibility

- Overall responsibility: Dean of the Faculty.
- Scientific coordination: Director of the AI for Geoscience Research Center.
- Research implementation: Academic departments and research divisions.
- Educational implementation: Relevant academic departments and curriculum committees.
- Financial and infrastructure support: Relevant university administrative units.

Proposed review schedule: Annual institutional review, with a comprehensive mid-term assessment in 2029 and final assessment in 2031.

END OF MASTER PLAN

Document designation: Institutional Strategic Development Plan
Planning period: 2027–2031
Proposed implementing institution: Faculty of Geoscience and Technology

Recommended next steps before formal submission

I recommend making three adjustments before submitting this document for approval:

1. **Insert the university and country information.** This will allow the plan to refer to national mineral resource priorities, geological survey organizations and relevant research funding opportunities.
2. **Conduct an infrastructure and staffing audit.** Establish exactly what computers, GPU resources, software, datasets and technical personnel are already available.
3. **Prepare a detailed financial annex.** Convert the proposed budget percentages into actual annual costs using local equipment prices, salaries, scholarships and research expenses.

My strongest recommendation is to obtain institutional approval for the AI Research Center and the integrated mineral exploration pilot project first. These two initiatives can provide the organizational and scientific foundation for implementing the entire five-year master plan.

2026-11-06 16:20:16 **You**

Based on the current situation, we can obtain the following data from the mineral exploration from the target area.

- Geological maps with ground survey data
- Geochemical sampling data such as ionic conductivity of soil
- Geophysical data such as magnetic, gamma-ray spectrometric data, MT survey data, resistivity CT data, gravity data
- Remote sensing data
- Some of drill data with collar, survey, lithology and assay tables.

From those data, we have to complete the 3d mineral prospectivy mapping with the 3d shape of ore body and the grade estimation by using the integrated AI technology you suggested.

Please suggest the detailed progress to develop such AI system with python and QGIS or ArcEngine.

My initial plan is:

- The program interface for 2d map display will be developed by using QGIS or ArcGIS Engine
- AI will be developed with python's machine learning packages
- I am not sure how to display 3d geological body including ore body and display the grade estimation results as we can see in surpac or

3d mine or gocad.

Please suggest the progress of the implementation of AI-support offline version of mineral prospectivity mapping system.

- The method of preparing and constructing the geological database from obtained such geological survey data as described above for AI training and estimating by AI.
- The brief python code for AI-support 3d mineral prospectivity mapping.
- how to develop and integrate AI program and interface to display 2d geological maps and 3d orebodies and 3d grade estimation results.

ChatGPT gpt-6-thinking

Technical Development Plan: Offline AI-Based 3D Mineral Prospectivity Mapping and Orebody Modelling System

Your proposed system is technically feasible, and the datasets you have described provide a good foundation for developing a sophisticated mineral exploration platform.

In fact, your project has the potential to go beyond conventional 2D mineral prospectivity mapping. I recommend developing an integrated system that can perform three major functions:

1. **3D mineral prospectivity prediction:** Identify the probability of mineralization throughout the subsurface, including areas between and beyond existing drill holes.
2. **3D orebody modelling:** Construct the geometry, boundaries and volume of predicted mineralized bodies, constrained by geological interpretation and drilling information.
3. **3D grade estimation:** Estimate the distribution of economic elements, such as Cu, Au, Zn, Pb or other target commodities, within the predicted orebody.

These three functions are related, but scientifically they are different modelling problems. They should be developed as separate AI and geological modelling modules and then integrated into one software platform.

I particularly agree with your proposed architecture:

- QGIS or an equivalent GIS environment for 2D geological maps and spatial data management.
- Python for machine learning, deep learning, geostatistics and 3D modelling.
- A dedicated 3D visualization component for displaying geological volumes, orebody surfaces and grade block models.

However, I would make one important modification: **I recommend QGIS rather than ArcGIS Engine for a new system**, and I would use Python with PyVista/VTK for advanced 3D visualization. This combination allows you to develop a largely open-source, locally installed system without depending on a cloud service.

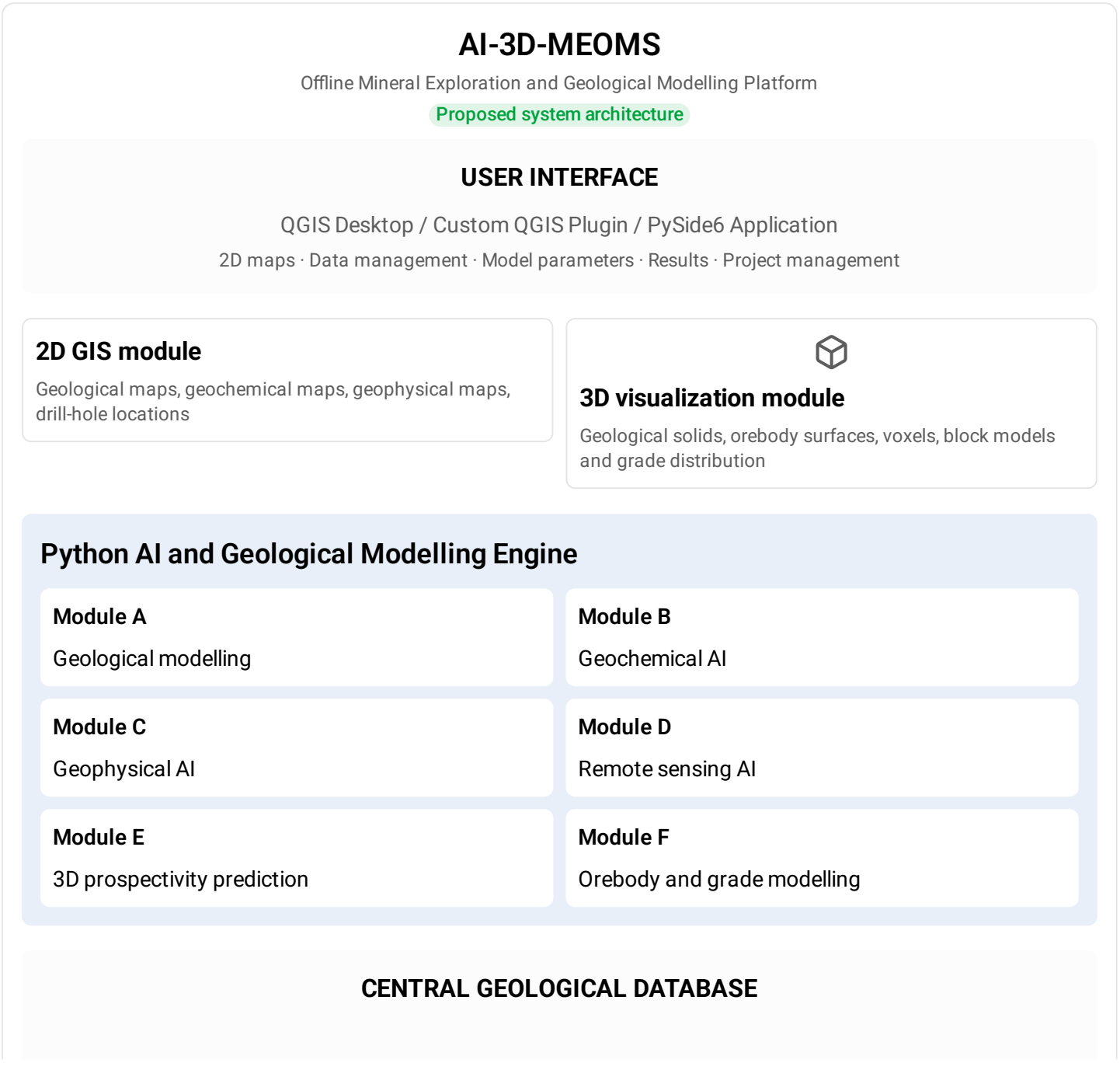
Let me propose the architecture, database design, development stages and initial Python implementation.

PART I. Overall architecture of the proposed system

1. Proposed software concept

I suggest the following name:

Integrated AI-Based 3D Mineral Exploration and Orebody Modelling System (AI-3D-MEOMS)



Drilling · Assays · Geology · Geochemistry · Geophysics · Remote sensing · Model results

FINAL EXPLORATION PRODUCTS

3D mineral prospectivity · 3D orebody geometry · 3D grade block model · Target ranking · Uncertainty maps

All data processing, model training, prediction and visualization can be performed locally on the institution's computers or internal network.

2. Recommended technology stack

I recommend separating the software into four technological layers.

Layer	Recommended technology	Function
2D GIS interface	QGIS / PyQGIS	Geological maps and spatial data
AI engine	Python, Scikit-learn, XGBoost, PyTorch	Machine learning and deep learning
3D modelling	GemPy, LoopStructural, PyVista, VTK	Geological solids and 3D volumes
Database	PostgreSQL/PostGIS or GeoPackage	Geological information management

Additional packages:

Package	Application
NumPy / Pandas	Numerical and tabular data
GeoPandas	Vector geological data
Rasterio / GDAL	Raster and satellite data
SciPy	Scientific computation and interpolation
GSTools / PyKrige	Geostatistics and kriging
PyVista	3D visualization and block models
GemPy	Implicit geological modelling
LoopStructural	Structural geological modelling
PyTorch	Advanced neural networks
SHAP	AI interpretation
Lasio	LAS-format borehole logs, if used

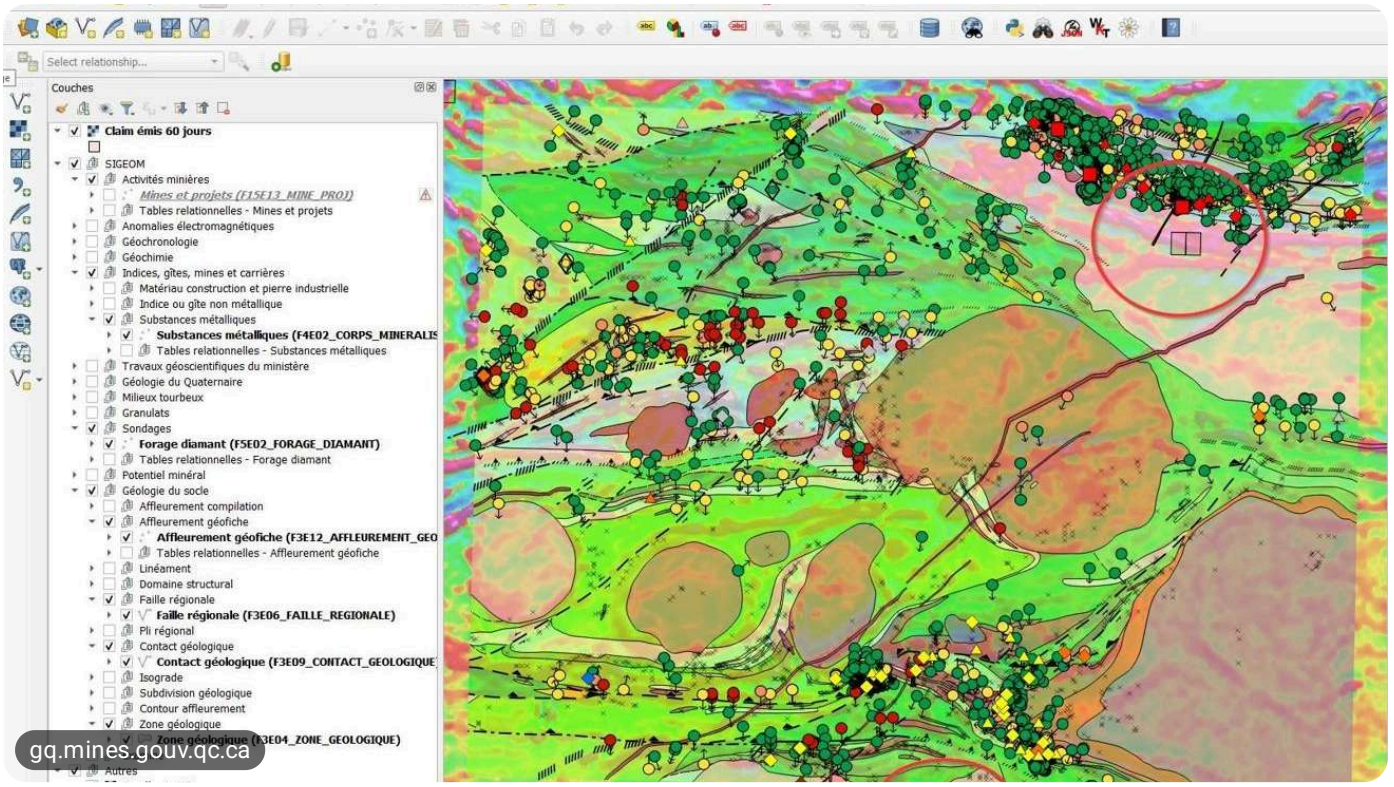
A particularly important architectural decision is that the AI engine should be independent of the GIS interface. This means that the same Python algorithms can be executed through QGIS, a standalone application or batch processing.

3. Important decision: QGIS, ArcGIS Engine or a custom application?

Your initial idea is reasonable, but there is an important software consideration.

QGIS + Python

Recommended



Advantages:

- Open-source and suitable for offline deployment.
- Supports geological maps, raster images, vector data and spatial analysis.
- Python-based plugin development.
- Can be extended with custom processing algorithms.
- Supports 3D terrain and vector visualization.
- No need to develop a complete GIS application from scratch.

QGIS supports custom Processing plugins, which can be added to the Processing Toolbox and used in batch processing and graphical workflows. QGIS Docume...

Official documentation:

ArcGIS Engine

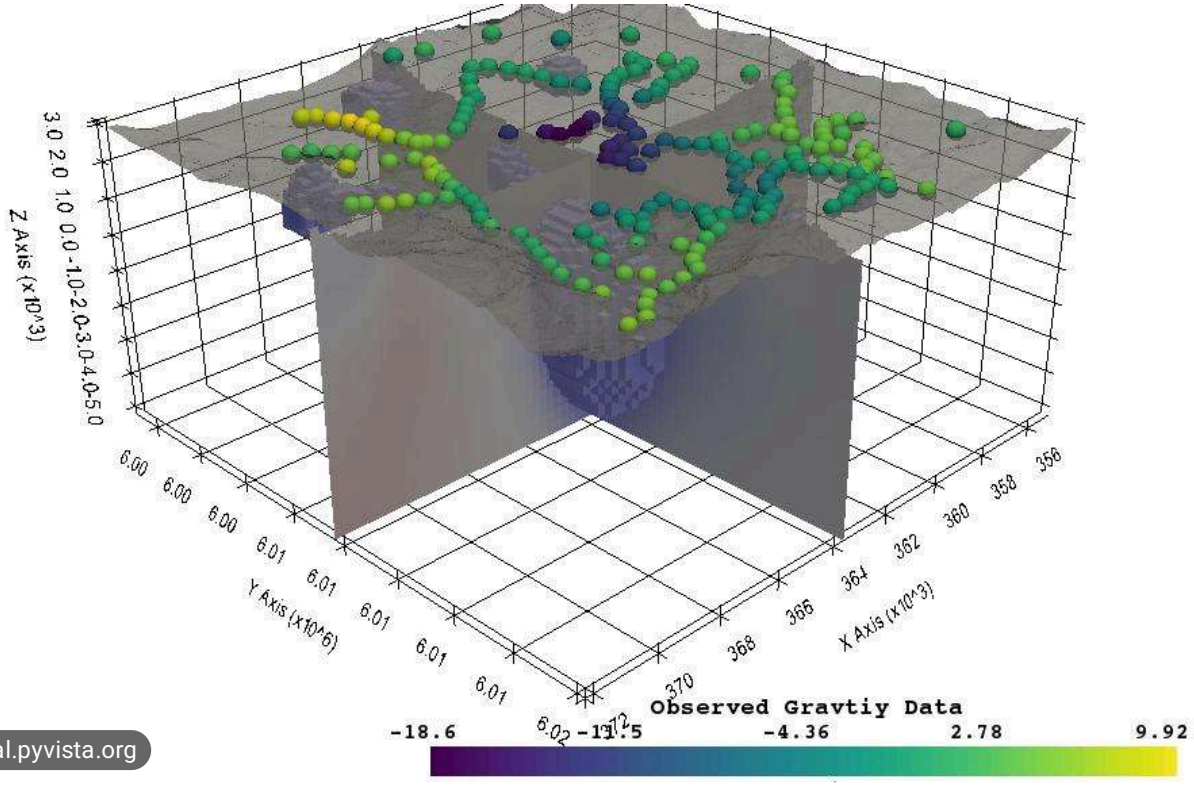
Not recommended for a new project

ArcGIS Engine was officially retired on March 1, 2026. Therefore, I would not recommend beginning a new five-year software development project using ArcGIS Engine. Esri Support

If your university already has an ESRI environment, investigate ArcGIS Pro SDK or the modern ArcGIS Maps SDKs instead.

PyVista + VTK

Recommended for 3D modelling



tutorial.pyvista.org

Advantages:

- Three-dimensional volume rendering.
- Voxel and block model visualization.
- Orebody surface rendering.
- Drill-hole trajectory visualization.
- 3D cross-sections and clipping.
- Interactive rotation, zooming and identification of model values.

PyVista supports volumetric visualization of structured and unstructured 3D datasets, including `ImageData` and `UnstructuredGrid`. PyVista 0.49.2...

Official documentation:

My proposed combination

Function	Software
2D geological map	QGIS
Geological data import	Python, GDAL, GeoPandas
Geochemical data analysis	Python, Pandas, Scikit-learn
Geophysical data analysis	Python, NumPy, SciPy
Geological 3D modelling	GemPy
3D structural modelling	GemPy or LoopStructural
3D mineral prospectivity	XGBoost, Random Forest, PyTorch

Grade estimation	GSTools, PyKrige, custom geostatistical algorithms
3D visualization	PyVista / VTK
3D interface integration	PyVistaQt
Database	PostgreSQL/PostGIS or GeoPackage

There is no need to develop an entirely new 3D visualization engine. PyVista already provides the functionality required to create interactive scientific 3D models.

PART II. Construction of the geological database

4. The most important step: organize the exploration data correctly

I consider database development to be the first major technical task.

The quality of the final AI prediction will depend heavily on the quality of the input database.

You have several different types of data, and they should not simply be combined into one large spreadsheet.

I recommend developing a structured geological database with individual tables for each data category.

4.1. Proposed database structure

Central geological exploration database

PROJECT MASTER TABLE

Project ID · Study area · CRS · Vertical datum · Survey dates

Geological data
Maps · Lithology · Faults · Attitudes

Geochemical data
Samples · Ionic conductivity · Elements

Geophysical data
Magnetic · Gamma · MT · ERT · Gravity

Remote sensing
Bands · Indices · Alteration maps

Drilling data
Collars · Surveys · Lithology · Assays

AI data
Training · Features · Labels · Predictions

↓

3D INTEGRATED MODEL DATABASE

Voxel coordinates · Geological domain · Geophysical attributes · Mineralization probability · Estimated grade · Uncertainty

I recommend two levels of database storage:

- **PostgreSQL/PostGIS:** For a centralized multi-user institutional database.
- **GeoPackage, Parquet and GeoTIFF:** For individual offline exploration projects and exchange of data.

Large three-dimensional arrays should be stored in dedicated scientific formats such as VTK XML, NetCDF or Zarr, rather than as thousands of independent records in an ordinary spreadsheet.

4.2. Geological mapping database

Your geological maps and ground survey data should be separated into the following layers.

Table/layer	Required information
Geological units	Unit ID, unit name, lithology, geological age
Geological boundaries	Boundary geometry, boundary type, confidence
Faults	Fault ID, geometry, type, movement, confidence
Structural measurements	X, Y, Z, strike, dip, dip direction
Field observations	Location, lithology, alteration, mineralization
Geological samples	Sample ID, location, rock type, observations

An example of a structural measurement table:

Point ID	Easting	Northing	Elevation	Strike	Dip	Dip direction
ST001	500100	7500200	325	045	65	135
ST002	500250	7500350	310	090	75	180

The actual values are illustrative.

These measurements are particularly important for developing 3D geological models.

You should preserve the original field observations, rather than storing only interpreted geological boundaries.

4.3. Geochemical database

For your soil ionic conductivity data, I recommend a dedicated table.

Field	Description

Sample_ID	Unique sample identification
Easting	X coordinate
Northing	Y coordinate
Elevation	Ground elevation
Sample_Depth	Sampling depth
Soil_Type	Soil classification
Conductivity	Measured conductivity
Conductivity_Unit	Measurement unit
Measurement_Method	Instrument and method
Temperature	Measurement temperature
Moisture	Soil moisture, if available
Survey_Date	Measurement date
Quality_Flag	Data quality indicator

An important scientific consideration:

Soil ionic conductivity is not equivalent to conventional multielement geochemical analysis.

It can be affected by:

- Soil moisture.
- Salinity.
- Clay content.
- Soil composition.
- Temperature.
- Measurement frequency and instrument characteristics.

Therefore, I recommend treating conductivity as an independent geochemical or near-surface physical indicator.

If you also have ICP-MS, ICP-OES, XRF or other elemental analysis results, include them in the same geochemical data framework but retain their individual analytical metadata.

4.4. Geophysical database

I recommend maintaining separate datasets for each geophysical method.

Method	Main parameters to store	AI application
Magnetic	Total field, residual, RTP, derivatives	Structure and magnetic body identification

Gamma-ray spectrometry	K, eU, eTh, total count, ratios	Lithology and alteration classification
MT	Frequency, apparent resistivity, phase, tensor components	Subsurface conductivity interpretation
Resistivity CT / ERT	Electrode geometry, apparent resistivity, inversion resistivity	Geological and mineralized zone interpretation
Gravity	Observed gravity, Bouguer anomaly, terrain correction	Density contrast and geological structure interpretation

Some additional considerations are very important.

Magnetic data: Keep the original survey data and derived products separately. For example, reduction to pole, analytical signal, vertical derivative and tilt derivative should be stored as separate derived layers.

Gamma-ray spectrometry: Retain the K, eU and eTh channels and their correction information. These measurements are predominantly near-surface indicators and should not be treated as direct deep subsurface measurements.

MT data: Raw frequency-dependent station measurements should be stored separately from the 2D or 3D inversion results. For 3D AI, the inverted resistivity volume is particularly valuable. Store its resolution, sensitivity and depth-of-investigation information if available.

Resistivity CT: If the existing data consist of 2D survey lines, they should not automatically be considered a complete 3D subsurface volume. A 3D interpretation requires appropriate multi-line coverage or 3D inversion.

Gravity data: Preserve terrain corrections, Bouguer reduction density assumptions and processing history. Gravity anomalies are integrated surface-field observations, not direct measurements of density at each depth.

These distinctions are necessary to avoid creating misleading 3D features.

4.5. Remote sensing database

Your remote sensing data should be organized as georeferenced raster datasets.

Recommended inputs:

- Original satellite bands.
- Surface reflectance data.
- Digital elevation model.
- Lithological classification.
- Iron oxide indices.

- Clay mineral indices.
- Alteration maps.
- Structural lineament maps.

For example:

Feature	Description
Band_2	Blue reflectance
Band_3	Green reflectance
Band_4	Red reflectance
Band_8	Near infrared
SWIR_1	Short-wave infrared
SWIR_2	Short-wave infrared
Iron_Index	Derived iron oxide indicator
Clay_Index	Derived clay mineral indicator
Alteration_Index	Derived alteration indicator

The actual bands and indices depend on the satellite sensor.

Remote sensing information is primarily a surface constraint. It should not be copied unchanged into every subsurface voxel.

4.6. Drilling database: particularly important

Your drill data will provide the most direct information for constructing and validating the 3D mineralization and grade models.

I recommend using the standard four-table structure.

Drill-hole database structure

1. COLLAR TABLE

Hole_ID · Easting · Northing · RL · Total_Depth

2. SURVEY TABLE

Hole_ID · Depth · Azimuth · Dip

3. LITHOLOGY TABLE

Hole_ID · From · To · Lithology · Alteration · Mineralization

4. ASSAY TABLE

Hole_ID · From · To · Sample_ID · Cu · Au · Zn · Pb · Other Elements

The tables should be linked using the common `Hole_ID`.

Example collar table

Hole_ID	Easting	Northing	RL	Total_Depth
DH001	500100	7500200	325	250
DH002	500250	7500350	310	300

Example survey table

Hole_ID	Depth	Azimuth	Dip
DH001	0	180	-60
DH001	50	182	-62
DH001	100	185	-65
DH002	0	090	-70
DH002	50	092	-72

The example uses a dip convention in which downward drilling is negative. Your system should convert all survey measurements into one consistent convention.

Example lithology table

Hole_ID	From	To	Lithology	Alteration
DH001	0	30	Andesite	Weak
DH001	30	75	Diorite	Moderate
DH001	75	120	Altered Diorite	Strong
DH001	120	180	Mineralized Zone	Strong

Example assay table

Hole_ID	From	To	Cu (%)	Au (ppm)
DH001	30	32	0.15	0.10
DH001	32	34	0.28	0.25

DH001	34	36	0.75	0.42
DH001	36	38	1.10	0.65

These examples are illustrative and do not represent actual exploration results.

Important: drill-hole desurveying

The collar table provides the starting coordinates of the borehole, but the survey table describes how the borehole deviates underground.

Therefore, you must calculate the 3D coordinates of the drilling trajectory before integrating lithology and assay intervals with the other 3D datasets.

For each sample interval, calculate:

- Starting 3D coordinates.
- Ending 3D coordinates.
- Midpoint X, Y and Z coordinates.
- Interval length.
- Geological domain.
- Assay values.

The resulting midpoint coordinates are particularly important for AI training and grade estimation.

Use a recognized minimum-curvature desurvey algorithm rather than assuming every drill hole is a straight line.

Also retain laboratory method, detection limits, sample recovery, QA/QC results and density measurements where available.

PART III. CONSTRUCTION OF THE 3D GEOLOGICAL DATABASE

5. Establish a common 3D coordinate system

Before applying AI, all datasets must be brought into a common spatial reference framework.

I recommend the following:

Horizontal coordinates: One appropriate projected coordinate reference system, preferably in meters.

Vertical coordinates: Consistent elevation or reduced level (RL), with a documented vertical datum.

Coordinate convention:

- X = Easting.
- Y = Northing.

- Z = Elevation, positive upward.
- Depth below ground = Ground elevation minus elevation.

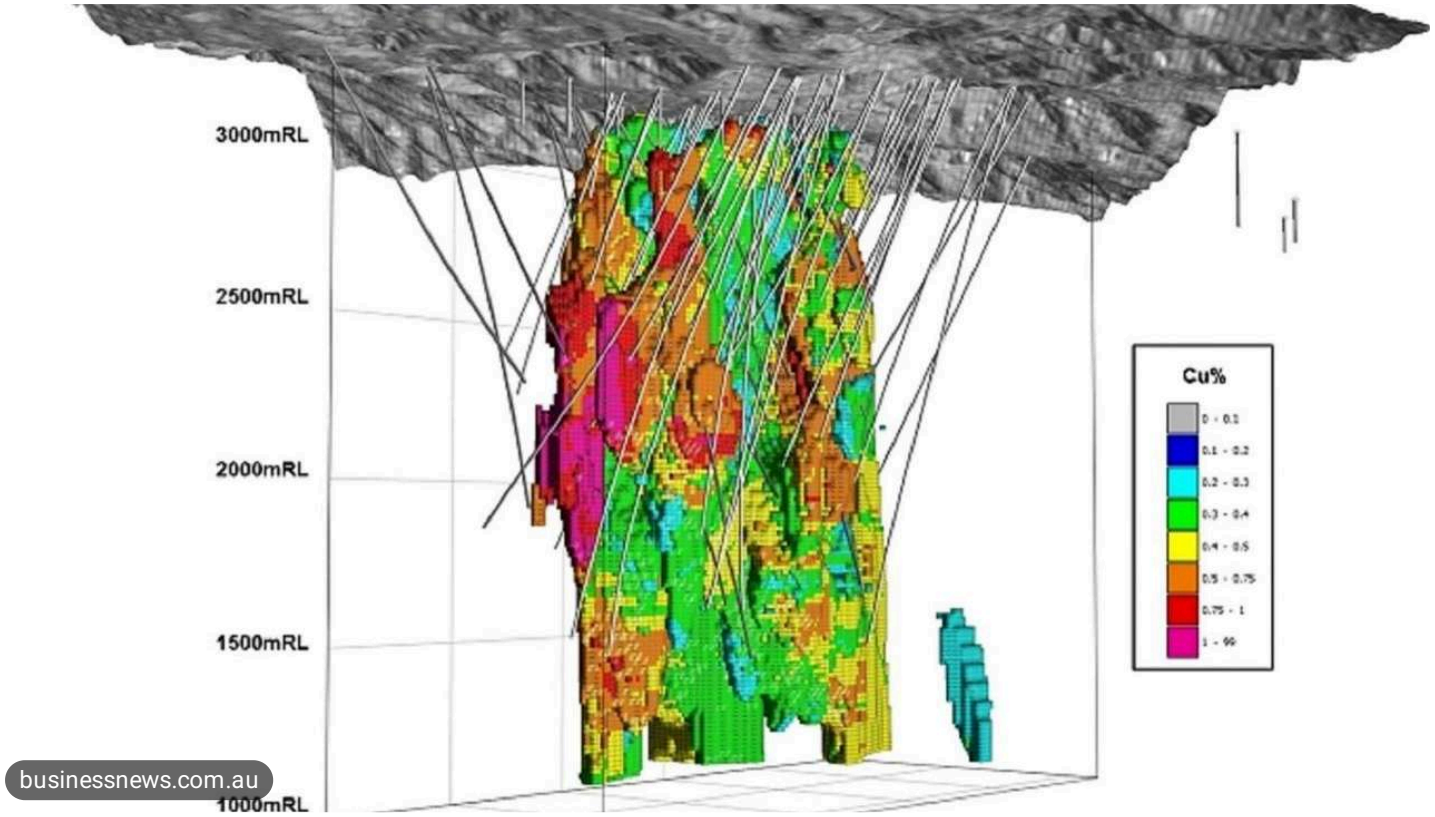
This is important because some geological and geophysical software uses depth-positive-down coordinates, while other software uses elevation-positive-up coordinates.

Do not mix these conventions.

5.1. Establish the 3D voxel model

The next step is to divide the entire exploration area into small 3D cells.

These are commonly called voxels or blocks.



For example, consider a hypothetical exploration area:

Parameter	Example
X extent	2,000 m
Y extent	2,000 m
Z extent	1,000 m
X cell size	25 m
Y cell size	25 m
Z cell size	10 m

The number of voxels would be:

$$N = N_x N_y N_z$$

$$N = 80 \times 80 \times 100 = 640,000$$

Each voxel has a center coordinate:

$$X_i = X_0 + (i + 0.5)\Delta X$$

$$Y_j = Y_0 + (j + 0.5)\Delta Y$$

$$Z_k = Z_0 + (k + 0.5)\Delta Z$$

Each voxel becomes a potential unit of prediction.

The dimensions above are examples only. The actual voxel size must be selected based on survey spacing, drill-hole density, geological continuity and the resolution of the geophysical inversions. A finer grid does not create additional geological information.

5.2. Construct the feature database for each voxel

This is the most important step for developing 3D AI prediction.

For each voxel, you need to calculate or assign a set of geological attributes.

I recommend the following feature categories.

Category	Example AI features
Geological	Lithology code, geological domain
Structural	Distance to fault, distance to contact, fracture density
Geochemical	Soil conductivity, geochemical anomaly indices
Magnetic	Magnetic intensity, analytical signal, vertical derivative
Gamma-ray	K, eU, eTh, K/eTh, eU/eTh
MT	Log resistivity, conductivity and inversion confidence
Resistivity CT	Inverted resistivity, depth and sensitivity
Gravity	Bouguer anomaly, gravity derivatives, inversion density
Remote sensing	Alteration indices, lithological classification
Drilling	Lithology, mineralization indicators, distance to drill observations

An example of the resulting voxel database:

X	Y	Z	Lithology	Distance to fault	Magnetic	Resistivity	Conductivity	Mineralization
500100	7500200	250	Diorite	35	125	180	45	1
500125	7500200	250	Andesite	110	95	450	30	0
500150	7500200	250	Diorite	25	150	120	55	1

Illustrative values only. The final column represents a training label, not an automatically known property of every voxel.

There is one critical distinction in this database:

Surface data and subsurface data must be treated differently.

For example:

- Satellite imagery is primarily a surface dataset.
- Soil conductivity is generally a near-surface measurement.
- Gamma-ray spectrometry is a near-surface measurement.
- Magnetic and gravity measurements are surface fields that may be transformed or inverted to provide subsurface models.
- MT and resistivity inversion results can provide depth-dependent information.
- Drill-hole data provide direct subsurface constraints.

Do not assign the same surface measurement to every voxel below it as though it were a direct measurement at depth.

Instead, use surface information as a surface predictor, a geological boundary constraint or a carefully modelled covariate with appropriate uncertainty.

For example, if a 3D inversion provides resistivity values at different depths, those values can be used directly in the corresponding voxels, subject to the inversion's resolution and sensitivity.

---

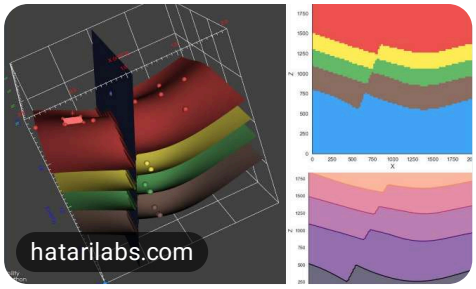
6. Construct a 3D geological model before AI prediction

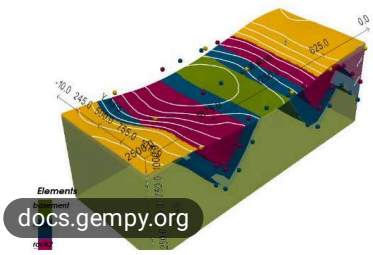
I strongly recommend that you construct a geological framework model before developing the final mineral prospectivity model.

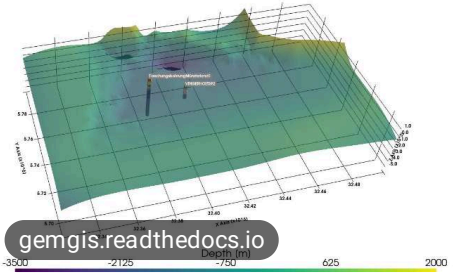
The reason is that geological boundaries, faults, folds and intrusive bodies strongly control the distribution of mineralization.

An AI model without a proper geological framework may identify statistical relationships but fail to respect the actual geological structure.

Recommended software: GemPy







GemPy is an open-source Python package for implicit 3D geological modelling.

It can construct geological models using:

- Geological contact points.
- Structural orientation measurements.
- Stratigraphic relationships.
- Fault systems.
- Fold structures.
- Geological unconformities.

It also supports 3D visualization through PyVista. GemPy +2

Official documentation: .

An alternative is LoopStructural, which provides implicit structural geological modelling using geological observations and orientation constraints. LoopStruct... +1

My recommendation

For your initial system:

- Use GemPy to construct the stratigraphic and structural geological framework.
- Use the drill-hole lithology information to constrain the geological boundaries.
- Incorporate ground survey contacts and structural measurements.
- Construct fault surfaces and intrusive bodies where sufficient data exist.
- Export the resulting geological model into a common 3D voxel framework.

Do not initially attempt to model every geological feature. Start with the major geological units, faults and mineralization-related intrusive or host-rock units.

PART IV. DEVELOPMENT OF THE 3D AI MINERAL PROSPECTIVITY MODEL

7. How to develop the AI model

Once the geological database and 3D voxel model have been established, you can begin developing the AI algorithms.

I recommend using two levels of AI modelling.

Level 1: Individual AI models

Develop separate models using geological, geochemical, magnetic, gamma-ray, MT, resistivity and gravity information.

Level 2: Integrated AI model

Combine the information from the individual datasets to predict mineralization probability throughout the 3D exploration volume.

This is important because you need to understand which datasets contribute useful information to the final prediction.

7.1. Define the AI training data

For supervised machine learning, you need training examples with known outcomes.

Your drill-hole information can provide these labels.

For example:

Geological condition	Training label
Verified mineralized interval	1
Verified barren interval	0
Undrilled location	Unknown

The labels must be defined according to your target commodity and deposit model.

For example, if you are exploring a copper deposit, you could define a mineralized interval using a geological and assay-based rule.

However, do not treat all undrilled locations as barren.

This is one of the most important issues in mineral exploration AI. An unexplored area is not necessarily a non-mineralized area.

I recommend:

- Positive samples: verified mineralized intervals.
- Negative samples: sufficiently investigated, appropriately logged and verified barren intervals.
- Unlabelled samples: locations without sufficient evidence.

Where negative samples are limited, consider positive-unlabelled learning or carefully constructed geological background samples.

7.2. Develop the first AI models

For the initial stage, I recommend comparing these algorithms:

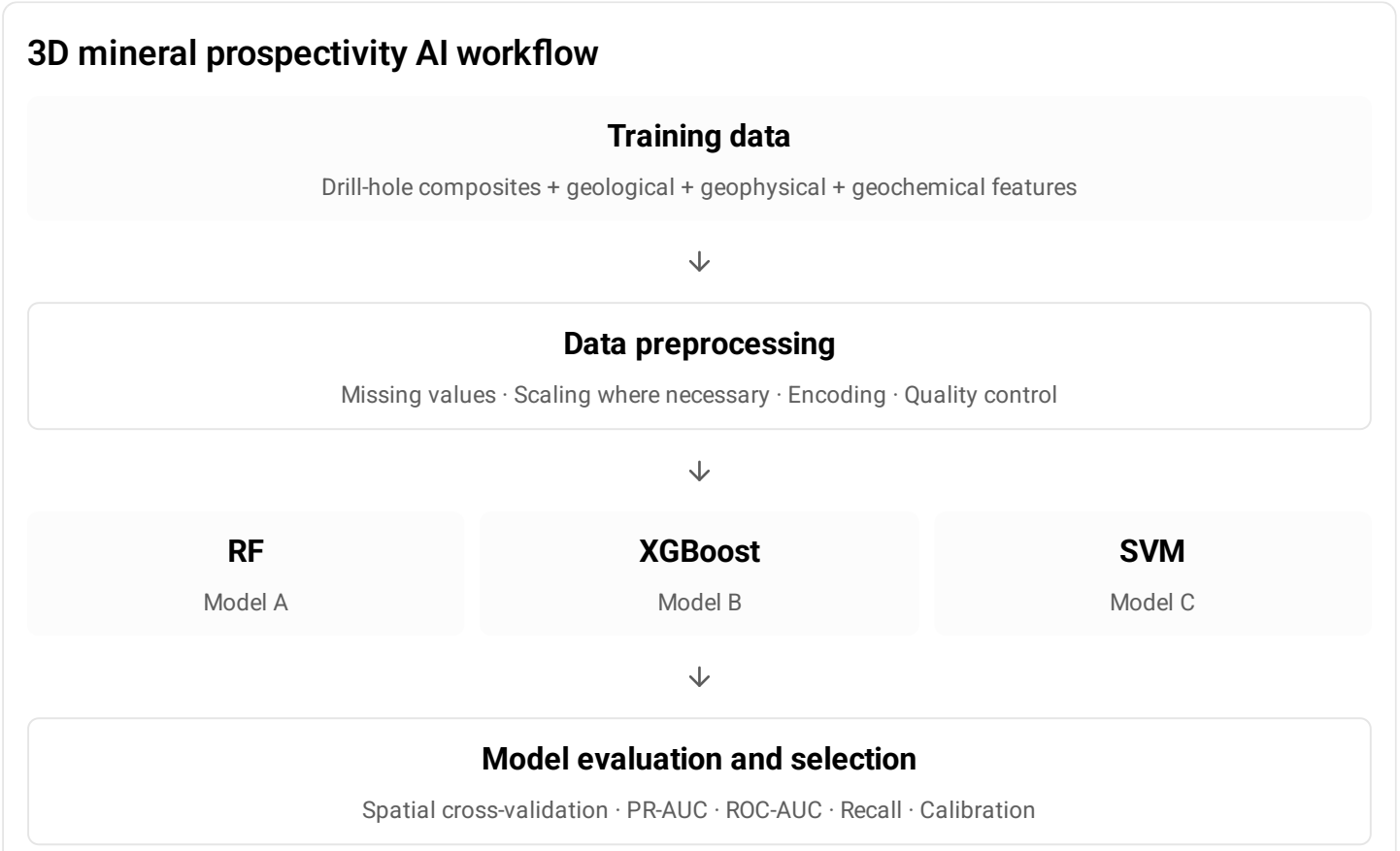
Algorithm	Main application	Priority
Random Forest	3D mineralization classification	Very high
XGBoost	3D prospectivity prediction	Very high
SVM	Classification of mineralized environments	High
ANN	Nonlinear geological relationships	Medium
CNN	Spatial patterns in 3D geological data	Medium
3D CNN	Volumetric feature recognition	Advanced
Autoencoder	Anomaly detection	Advanced

I would begin with Random Forest and XGBoost.

The reason is that your initial dataset is likely to consist primarily of structured numerical and categorical geological attributes rather than millions of labeled images.

Tree-based models are well suited to this type of dataset and can be interpreted using feature importance and SHAP.

7.3. Recommended model development procedure



↓

Prediction of all 3D voxels

Probability of mineralization for each voxel

↓

3D mineral prospectivity volume

Probability distribution · High-potential zones · Uncertainty

7.4. An important point about model validation

Do not randomly divide individual assay intervals from the same drill holes into training and testing datasets.

That can cause data leakage because nearby intervals are spatially correlated.

Instead, I recommend:

- Grouping training and testing data by drill hole.
- Using spatial block cross-validation.
- Holding out entire geographical areas when possible.
- Performing an independent validation in another part of the exploration district.

The model should be evaluated on its ability to predict mineralization in locations that it has not previously seen. Spatially grouped validation is important for avoiding overly optimistic performance estimates. scikit-learn Bl…

Also evaluate precision-recall performance because mineralized intervals may be much less frequent than barren intervals.

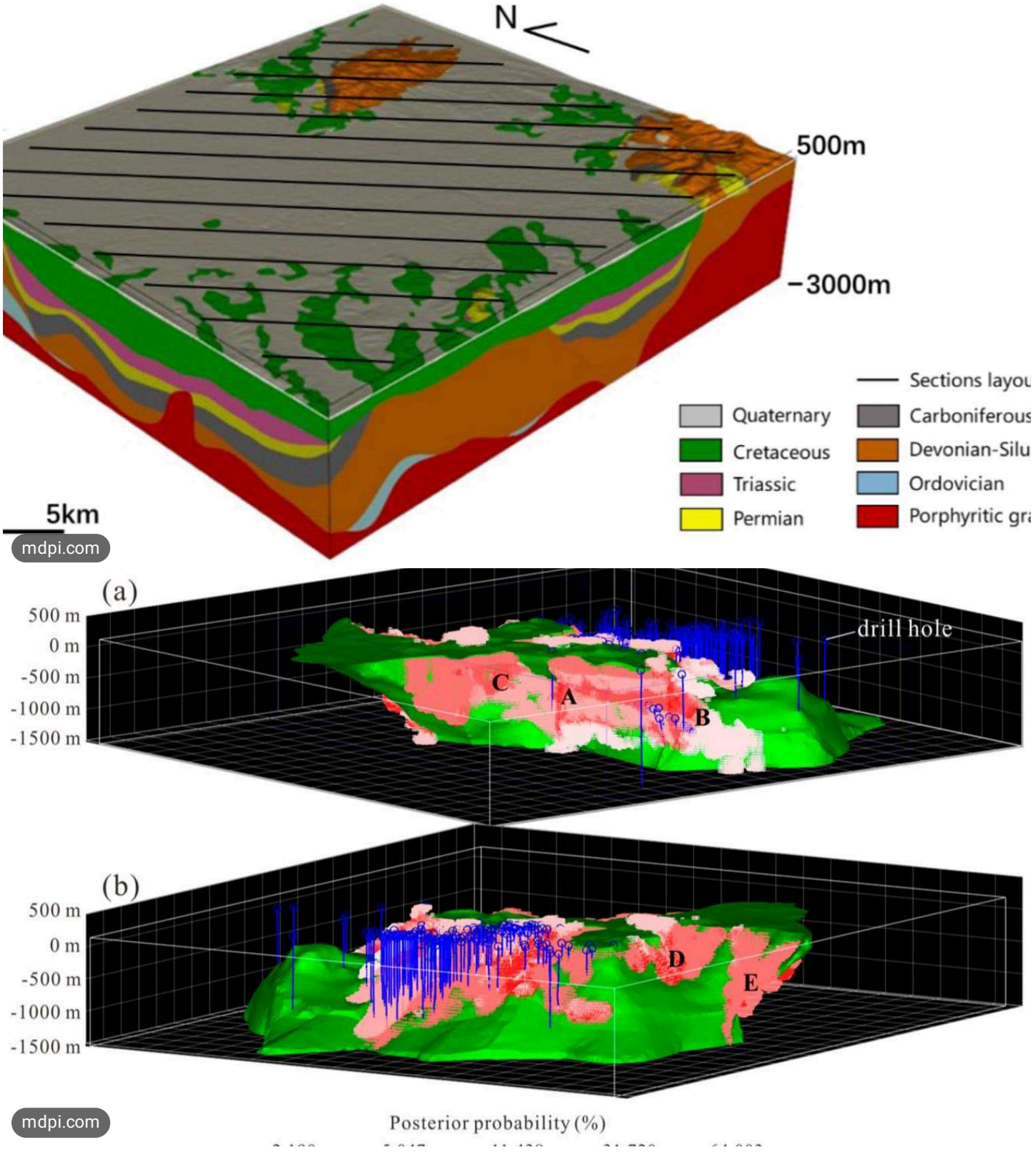
PART V. 3D OREBODY MODELLING

8. How to obtain the 3D shape of the orebody

This is an important question because you want the final visualization to resemble the 3D geological models displayed in Surpac, 3D Mine or GOCAD.

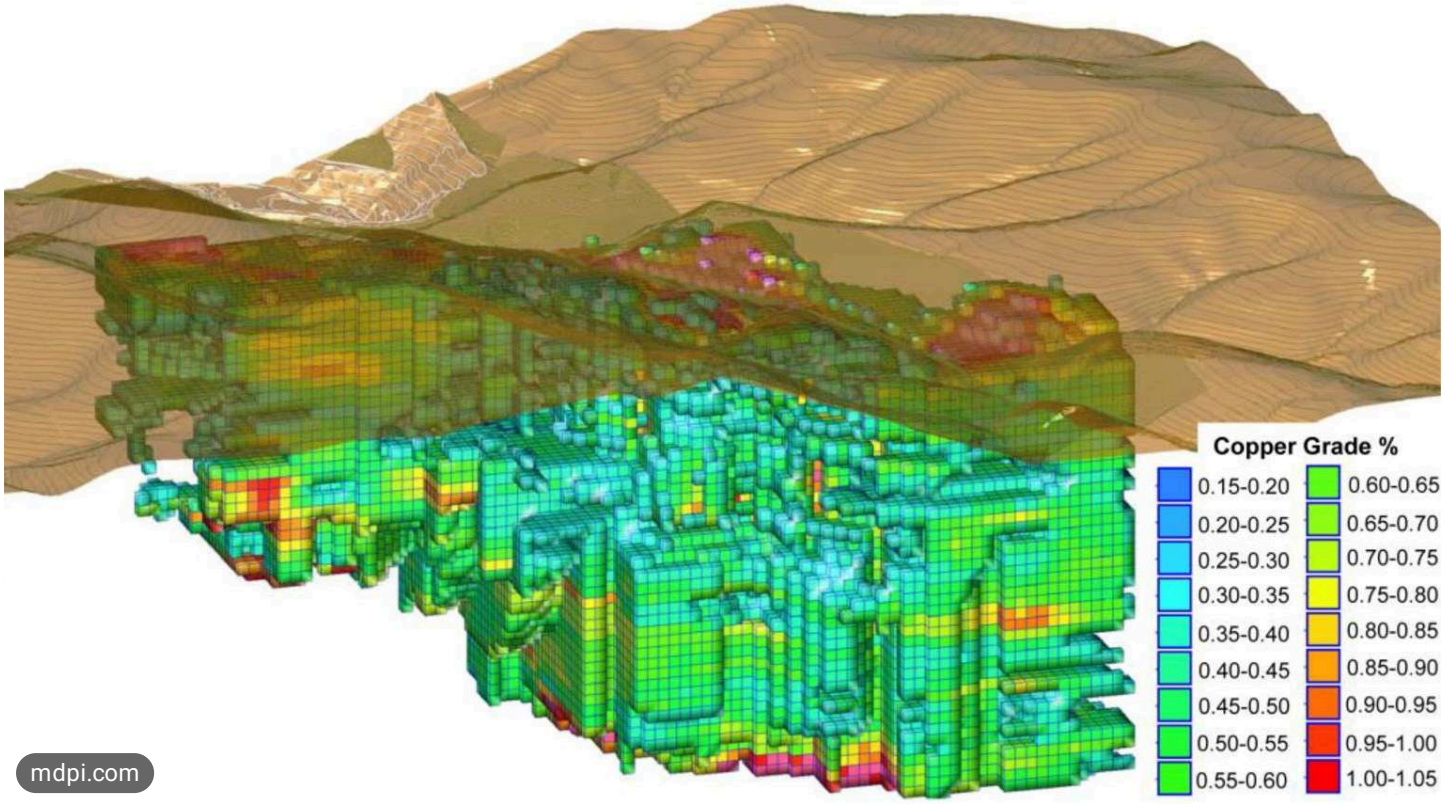
I recommend distinguishing three types of 3D objects.

A. Geological framework model



Represents the spatial probability or favorability of mineralization in each voxel.

C. Orebody and grade block model



Represents the interpreted mineralized volume and the estimated grade distribution within it.

8.1. Method 1: Construct an orebody surface from the 3D AI prediction

Suppose your AI model predicts the following probabilities:

Voxel	Mineralization probability
A	0.15
B	0.32
C	0.58
D	0.81
E	0.94

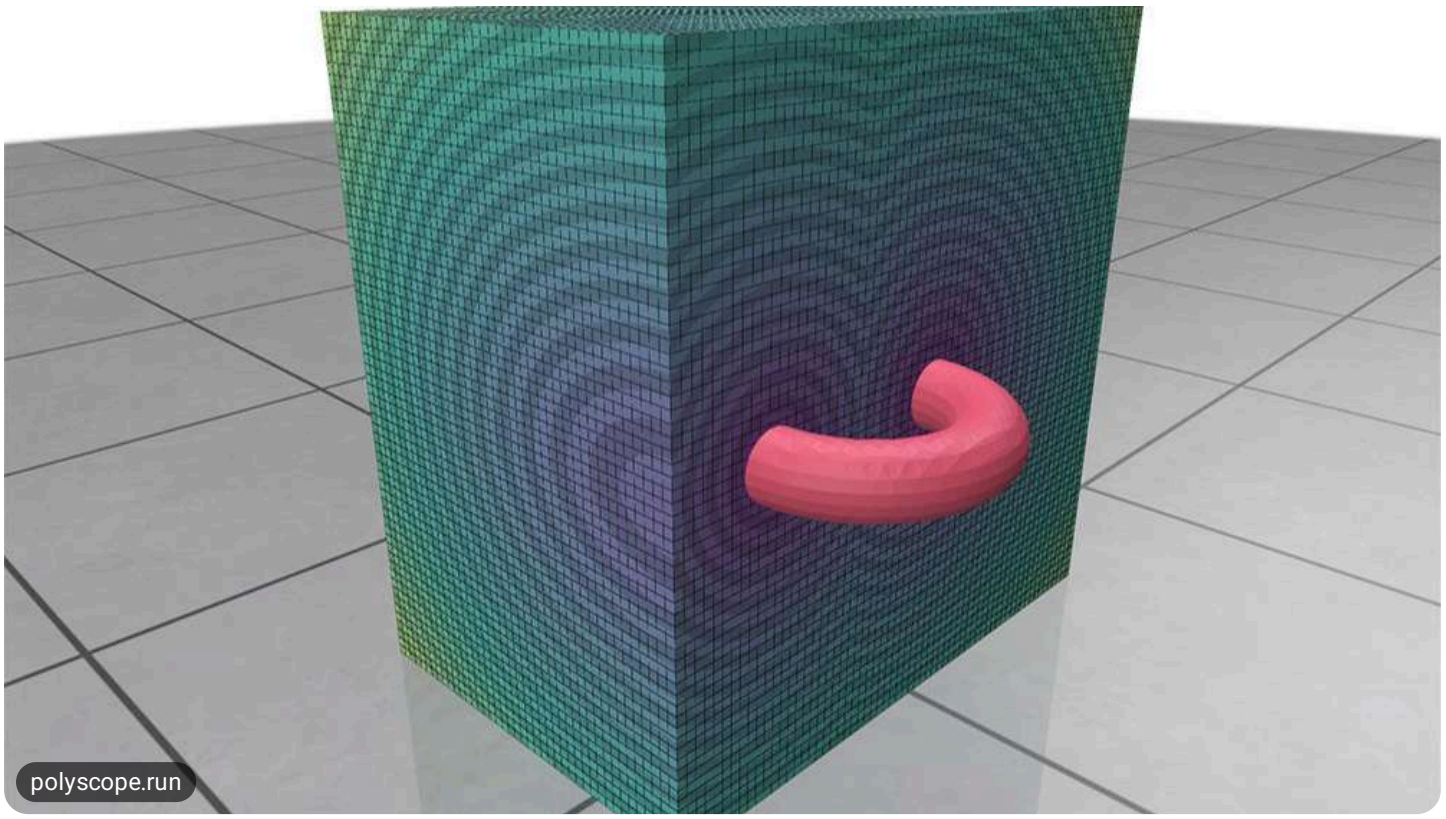
You can select a probability threshold, for example:

$$P(\text{mineralization}) \geq 0.70$$

All voxels satisfying this condition become part of the high-prospectivity volume.

Then:

1. Select the high-probability voxels.
2. Identify connected volumes.
3. Remove isolated noise according to geological criteria.
4. Extract the external surface using the Marching Cubes algorithm.
5. Construct a triangular surface mesh.
6. Display the surface in PyVista.



This is a practical method for visualizing predicted mineralization zones.

However, the resulting surface should initially be called a **3D high-prospectivity envelope**, not a confirmed orebody.

The probability threshold should be selected through geological validation and the intended exploration purpose, rather than using 0.70 universally.

8.2. Method 2: Construct the orebody using geological and drilling constraints

This is the more scientifically rigorous approach for obtaining an interpreted orebody geometry.

Use the following information together:

- Geological framework.
- Lithological contacts.
- Fault and structural surfaces.
- Mineralized drill-hole intervals.
- Assay results.
- AI mineralization probability.

For example, consider a mineralized zone occurring within a particular intrusive rock and controlled by a fault system.

The modelling procedure would be:

1. Construct the intrusive body using geological observations and drill-hole intersections.
2. Construct the fault surface.

3. Identify mineralized intervals within the intrusive body.
4. Predict mineralization probability in the surrounding 3D voxels.
5. Apply geological domain constraints.
6. Integrate grade estimates.
7. Construct an interpreted orebody solid.

This approach is more reliable than using an AI probability threshold alone.

8.3. Method 3: Define orebody boundaries using grade estimation

Where sufficient drilling and assay information is available, you can use grade-based modelling.

For example, if the target commodity is copper, you may define a conceptual orebody boundary based on:

$$G_{\mathrm{Cu}} \geq G_{\mathrm{cutoff}}$$

where:

- G_{Cu} is estimated copper grade.
- G_{cutoff} is a project-defined cut-off grade.

In practice, the cut-off grade depends on economic, metallurgical, mining and environmental considerations. It should not be chosen merely to make the AI model produce an attractive shape.

The final orebody interpretation should consider:

- Geological continuity.
- Mineralization continuity.
- Structural controls.
- Assay distribution.
- Geological domains.
- Grade estimation uncertainty.

My recommendation: Use the AI probability model to identify prospective volumes, but use geological constraints and grade estimation to develop the final interpreted orebody geometry.

PART VI. 3D GRADE ESTIMATION

9. How to estimate the grade distribution in 3D

Grade estimation should be developed as a separate module from mineral prospectivity prediction.

This is particularly important because the two models require different training information.

Model	Main training information	Result
Mineral prospectivity model	Geological, geochemical, geophysical and mineralization labels	Probability of mineralization
Grade estimation model	Desurveyed and composited drill-hole assays	Estimated grade of each block

For example, the prospectivity model might predict that a particular volume has a high probability of copper mineralization.

The grade estimation model then predicts the copper concentration within that volume using drill-hole assay information and appropriate geological constraints.

9.1. Prepare assay data for grade estimation

Do not directly use the original assay intervals without checking sample lengths and geological domains.

The preparation should include:

1. Drill-hole desurveying.
2. Assay interval validation.
3. Geological domaining.
4. Exploratory statistical analysis.
5. Sample compositing.
6. Outlier analysis and top-cut assessment.
7. Variogram analysis.
8. Estimation.

For example, suppose your assay samples have lengths ranging from 0.5 m to 3 m.

You may decide to composite them to a common support length, such as 2 m, where geologically and statistically appropriate.

Do not composite across major geological boundaries or incompatible mineralization domains.

9.2. Develop a geological domain model

Before grade estimation, divide the mineralized area into appropriate estimation domains.

Examples:

- Primary copper mineralization.
- Secondary or supergene copper mineralization.
- Oxide mineralization.
- Sulphide mineralization.

- Different lithological or structural domains.

Different domains may have different grade distributions and spatial continuity.

A single grade estimation model for the entire deposit may be inappropriate.

The CIM Mineral Resource and Mineral Reserve Best Practice Guidelines emphasize geological and statistical domaining, sample support, estimation and validation as important components of mineral resource estimation. mrmr.cim.o... +1

9.3. Which grade estimation method should be used?

I recommend comparing conventional geostatistical methods with AI methods.

Method	Advantages	Recommended role
Inverse Distance Weighting	Simple and relatively fast	Baseline comparison
Ordinary Kriging	Uses spatial continuity	Main conventional estimator
Indicator Kriging	Models grade thresholds and probability	Grade distribution analysis
Random Forest Regression	Captures nonlinear relationships	AI grade prediction
XGBoost Regression	Strong nonlinear predictive modelling	AI grade prediction
Regression Kriging	Combines ML trend and spatial residuals	Advanced hybrid estimator
Conditional Simulation	Models spatial uncertainty	Advanced resource modelling

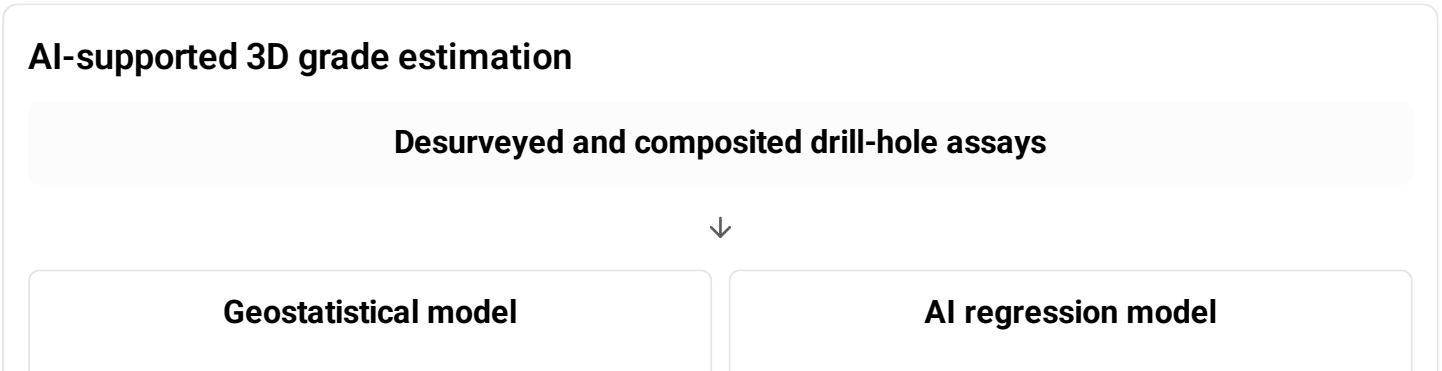
My preferred development strategy would be:

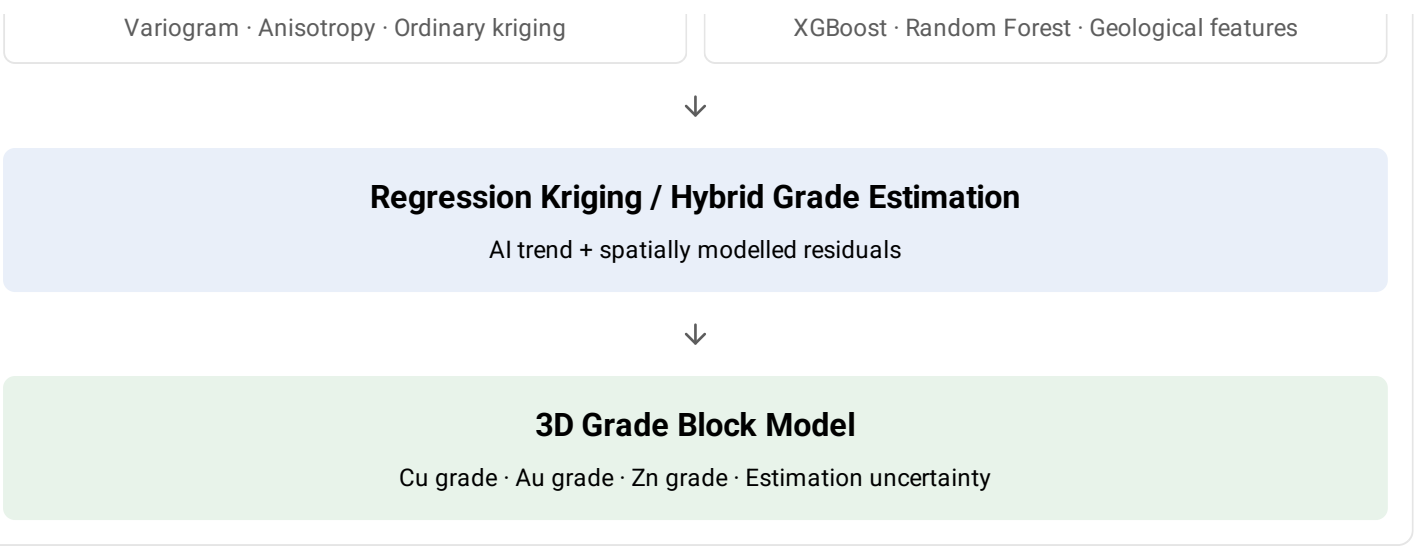
Ordinary Kriging + XGBoost + Regression Kriging.

The reason is that conventional geostatistics provides an important scientific reference, while AI can identify complex relationships between grade, lithology, geophysical properties and other geological variables.

A hybrid model can combine these strengths.

9.4. Proposed hybrid grade estimation architecture





This is a recommended research architecture, not a claim that the hybrid will always outperform ordinary kriging. The methods should be compared using independent validation.

9.5. Important limitation of grade estimation

If your drill-hole coverage is limited, you can still develop an experimental 3D grade prediction model.

However, the uncertainty may be high, particularly away from drilled locations.

The model should therefore display:

- Estimated grade.
- Distance to the nearest drill hole.
- Number of informing samples.
- Estimation uncertainty or kriging variance.
- Geological domain.
- Estimation method.

The AI system must not present speculative grades as reliable resource estimates.

Formal Mineral Resource or Mineral Reserve reporting requires compliance with the applicable national reporting code and the involvement of appropriately qualified professionals. The CIM guidelines provide a useful reference for this work. mrmr.cim.org

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PART VII. PYTHON IMPLEMENTATION: INITIAL CODE FOR 3D AI MINERAL PROSPECTIVITY

10. Proposed Python program structure

I recommend developing the AI engine as a collection of independent Python modules.

Do not put all algorithms into one large Python script.

A suggested project structure is:

```
</> Plain text

AI_3D_MEOMS/
 │
 ├──── main.py
 │
 ├──── database/
 │     ├──── database_manager.py
 │     ├──── import_geology.py
 │     ├──── import_geochemistry.py
 │     ├──── import_geophysics.py
 │     ├──── import_remote_sensing.py
 │     └──── import_drilling.py
 │
 ├──── preprocessing/
 │     ├──── coordinate_transform.py
 │     ├──── drillhole_desurvey.py
 │     ├──── assay_compositing.py
 │     ├──── data_quality.py
 │     └──── feature_engineering.py
 │
 ├──── geological_model/
 │     ├──── geological_3d.py
 │     ├──── structural_model.py
 │     └──── voxel_model.py
 │
 ├──── ai_models/
 │     ├──── prospectivity_rf.py
 │     ├──── prospectivity_xgboost.py
 │     ├──── prospectivity_cnn.py
 │     ├──── model_validation.py
 │     └──── explainable_ai.py
 │
 ├──── grade_estimation/
 │     ├──── variogram.py
 │     ├──── kriging_3d.py
 │     ├──── ai_grade.py
 │     └──── regression_kriging.py
 │
 ├──── orebody_modelling/
 │     ├──── orebody_extraction.py
 │     ├──── surface_generation.py
 │     └──── block_model.py
 │
 ├──── visualization/
 │     ├──── view_2d.py
```

```
│     ├──── view_3d.py
│     ├──── drillhole_3d.py
│     └──── grade_visualization.py
│
├──── qgis_plugin/
│     ├──── plugin.py
│     ├──── dock_widget.py
│     └──── processing_provider.py
│
├──── models/
│     ├──── prospectivity_model.joblib
│     └──── model_metadata.json
│
├──── outputs/
│
└──── configuration/
      ├──── project_settings.json
      └──── database_settings.json
```

Each module can be developed and tested independently.

10.1. Python environment

For initial development, I recommend the following packages:

```bash
pip install numpy pandas scipy
pip install scikit-learn xgboost
pip install torch
pip install geopandas rasterio
pip install pyvista pyvistaqt
pip install pykrige gstools
pip install shap shapely
pip install gempy gempy-viewer
pip install sqlalchemy psycopg2-binary
```

This is an indicative package list, not a guaranteed compatible single installation command. GemPy, PyVista, Qt and QGIS versions have interdependencies, so pin and test a complete environment before deployment.

For QGIS integration, do not blindly install a separate PySide6 or PyQt version into the QGIS Python environment. QGIS supplies its own Qt bindings.

I recommend keeping the GIS application environment and the AI computation environment separate during initial development.

10.2. Python code example 1: Preparing a 3D voxel grid

The following code generates a regular 3D grid.

```python
import numpy as np
import pandas as pd

# Study area limits
x_min, x_max = 500000, 502000
y_min, y_max = 7500000, 7502000
z_min, z_max = -500, 500

# Voxel dimensions (meters)
dx, dy, dz = 25, 25, 10

# Grid cell centers
xc = np.arange(x_min + dx/2, x_max, dx)
yc = np.arange(y_min + dy/2, y_max, dy)
zc = np.arange(z_min + dz/2, z_max, dz)

# Generate 3D coordinates
zz, yy, xx = np.meshgrid(
    zc, yc, xc, indexing="ij"
)

# Create voxel table
voxels = pd.DataFrame({
    "x": xx.ravel(),
    "y": yy.ravel(),
    "z": zz.ravel()
})

voxels["i"] = np.tile(
    np.arange(len(xc)),
    len(yc) * len(zc)
)

voxels["j"] = np.tile(
    np.repeat(np.arange(len(yc)), len(xc)),
    len(zc)
)

voxels["k"] = np.repeat(
```

```
    np.arange(len(zc)),
    len(yc) * len(xc)
)

voxels.to_parquet(
    "voxel_coordinates.parquet",
    index=False
)

print(voxels.head())
print("Total voxels:", len(voxels))
```

This generates the voxel coordinates and cell indices.

The next step is to populate each voxel with the geological and geophysical attributes.

For example:

```
</> Python

FEATURES = [
    "lithology",
    "distance_fault",
    "magnetic_rtp",
    "magnetic_analytic_signal",
    "gravity_bouguer",
    "gamma_k",
    "gamma_uth",
    "gamma_ue",
    "mt_log_resistivity",
    "ert_log_resistivity",
    "soil_conductivity"
]
```

These are example feature names.

Your actual features should be selected according to data availability and geological relevance.

The feature-generation process should use the appropriate spatial interpolation, geological modelling or geophysical inversion results.

Importantly, the surface measurements must not simply be extrapolated unchanged through the entire 3D volume.

10.3. Python code example 2: Training the 3D mineral prospectivity model

Suppose you have two prepared datasets:

`training_composites.parquet`

Contains:

- X, Y, Z coordinates.
- Geological and geophysical attributes.
- Mineralization labels.
- Spatial block identification.

`voxel_features.parquet`

Contains:

- X, Y, Z coordinates.
- Voxel indices i, j, k.
- The same predictive features for each voxel.

The following code develops a Random Forest model.

```python
import numpy as np
import pandas as pd

from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GroupShuffleSplit
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

from sklearn.metrics import (
    roc_auc_score,
    average_precision_score,
    classification_report
)

# Load training data
train = pd.read_parquet(
    "training_composites.parquet"
)

# Load 3D voxel features
voxels = pd.read_parquet(
    "voxel_features.parquet"
)
```

```
# Predictive features
numeric_features = [
    "distance_fault",
    "magnetic_rtp",
    "magnetic_analytic_signal",
    "gravity_bouguer",
    "gamma_k",
    "gamma_uth",
    "gamma_ue",
    "mt_log_resistivity",
    "ert_log_resistivity",
    "soil_conductivity"
]

categorical_features = [
    "lithology"
]

features = numeric_features + categorical_features

# Target
target = "ore_flag"

# Spatially grouped validation
splitter = GroupShuffleSplit(
    n_splits=1,
    test_size=0.25,
    random_state=42
)

train_idx, test_idx = next(
    splitter.split(
        train[features],
        train[target],
        groups=train["spatial_block"]
    )
)

X_train = train.iloc[train_idx][features]
X_test = train.iloc[test_idx][features]

y_train = train.iloc[train_idx][target]
y_test = train.iloc[test_idx][target]

# Preprocessing
preprocessor = ColumnTransformer(
    transformers=[
        (
```

```
            "numeric",
            SimpleImputer(
                strategy="median",
                add_indicator=True
            ),
            numeric_features
        ),
        (
            "categorical",
            Pipeline([
                (
                    "imputer",
                    SimpleImputer(
                        strategy="most_frequent"
                    )
                ),
                (
                    "encoder",
                    OneHotEncoder(
                        handle_unknown="ignore"
                    )
                )
            ]),
            categorical_features
        )
    ]
)

# AI model
model = Pipeline([
    ("preprocessing", preprocessor),

    ("classifier", RandomForestClassifier(
        n_estimators=500,
        max_depth=15,
        min_samples_leaf=3,
        class_weight="balanced_subsample",
        random_state=42,
        n_jobs=-1
    ))
])

# Training
model.fit(X_train, y_train)

# Evaluation
prediction = model.predict(X_test)
probability = model.predict_proba(X_test)[:, 1]
```

```
print(classification_report(
    y_test,
    prediction
))

print(
    "ROC-AUC:",
    roc_auc_score(y_test, probability)
)

print(
    "PR-AUC:",
    average_precision_score(y_test, probability)
)

# Predict all 3D voxels
voxels["P_mineralization"] = (
    model.predict_proba(
        voxels[features]
    )[:, 1]
)

# Save model predictions
voxels.to_parquet(
    "3d_prospectivity_results.parquet",
    index=False
)
```

This is a research prototype, not a validated production model.

A few important requirements:

- The training data must be constructed from desurveyed and appropriately composited drill intervals.
- The spatial blocks must be defined before model validation.
- The spatial split should include appropriate geographical separation or buffering where possible.
- The example does not perform probability calibration.
- The feature preparation must ensure that surface-only data are not falsely interpreted as depth-resolved measurements.
- The model should be compared with XGBoost and conventional geological prospectivity methods.

Also, a probability output from Random Forest should not automatically be interpreted as a calibrated geological probability.

For your actual research, I recommend probability calibration and independent validation before using the prediction values for exploration decisions.

10.4. Python code example 3: Exporting the 3D prospectivity model

This is where PyVista becomes particularly important.

It can convert the AI predictions into a 3D volume.

```python
import numpy as np
import pandas as pd
import pyvista as pv

# Load AI prediction results
df = pd.read_parquet(
    "3d_prospectivity_results.parquet"
)

# Sort voxels: X varies fastest
df = df.sort_values(
    ["k", "j", "i"]
)

# Grid dimensions
nx = int(df["i"].max()) + 1
ny = int(df["j"].max()) + 1
nz = int(df["k"].max()) + 1

# Create 3D volume
volume = pv.ImageData(
    dimensions=(nx + 1, ny + 1, nz + 1),
    spacing=(25, 25, 10),
    origin=(500000, 7500000, -500)
)

# Add AI probability as cell data
volume.cell_data["P_mineralization"] = (
    df["P_mineralization"].to_numpy(
        dtype=np.float32
    )
)

# Save complete 3D volume
volume.save(
    "3D_mineral_prospectivity.vti"
)

# Convert cell values to point values
```

```python
point_volume = volume.cell_data_to_point_data()

# Extract high-prospectivity isosurface
surface = point_volume.contour(
    isosurfaces=[0.70],
    scalars="P_mineralization"
)

if surface.n_points > 0:
    surface.save(
        "high_prospectivity_surface.vtp"
    )

print("3D prospectivity model exported.")
```

The code assumes that the input rows have a complete, regular voxel grid, correct `i`, `j`, `k` indices and the stated origin and cell dimensions.

You must change the grid dimensions, origin and spacing to match your actual project.

The output files are:

File	Purpose
`3D_mineral_prospectivity.vti`	Complete 3D probability volume
`high_prospectivity_surface.vtp`	Extracted high-prospectivity surface

These files can be loaded into a PyVista viewer.

10.5. Python code example 4: Displaying the 3D prospectivity volume

```python
</> Python

import pyvista as pv

# Read 3D prediction
volume = pv.read(
    "3D_mineral_prospectivity.vti"
)

# Convert cell values to point values
volume = volume.cell_data_to_point_data()

# Create interactive 3D viewer
plotter = pv.Plotter()
```

```
# Display 3D volume
plotter.add_volume(
    volume,
    scalars="P_mineralization",
    cmap="viridis",
    opacity="sigmoid",
    show_scalar_bar=True
)

# Display high-potential surface
surface = pv.read(
    "high_prospectivity_surface.vtp"
)

plotter.add_mesh(
    surface,
    color="orange",
    opacity=0.4
)

plotter.add_axes()

plotter.show_grid()

plotter.show()
```

The result is an interactive 3D scientific visualization.

You can rotate, zoom and inspect the volume.

The same visualization can later be integrated into your QGIS application.

---

11. Python code example 5: Initial 3D grade estimation

For grade estimation, I recommend initially comparing Ordinary Kriging with AI regression models.

The following example demonstrates 3D Ordinary Kriging using the PyKrige library.

PyKrige provides a dedicated `OrdinaryKriging3D` class for estimation from 3D sample coordinates and measured values. PyKrige 1.7.3....

Suppose you have the following prepared assay composite table:

X	Y	Z	Cu
500100	7500200	200	0.45
500120	7500210	190	0.72

500150	7500230	180	1.15

The coordinates are assumed to have already been desurveyed.

```python
import numpy as np
import pandas as pd

from pykrige.ok3d import OrdinaryKriging3D

# Load composited drill-hole assays
df = pd.read_parquet(
    "copper_composites.parquet"
)

# Use one geological estimation domain
df = df[
    df["domain"] == "PRIMARY_CU"
].dropna(
    subset=["x", "y", "z", "Cu_pct"]
)

# Sample coordinates and copper grade
x = df["x"].to_numpy()
y = df["y"].to_numpy()
z = df["z"].to_numpy()
grade = df["Cu_pct"].to_numpy()

# Voxel center coordinates
xc = np.arange(500012.5, 502000, 25)
yc = np.arange(7500012.5, 7502000, 25)
zc = np.arange(-495, 500, 10)

# 3D Ordinary Kriging
OK = OrdinaryKriging3D(
    x,
    y,
    z,
    grade,
    variogram_model="spherical"
)

# Estimate grade
estimated_grade, kriging_variance = OK.execute(
    "grid",
    xc,
    yc,
```

```
    zc
)

# Convert results to arrays
estimated_grade = np.ma.filled(
    estimated_grade,
    np.nan
)

kriging_variance = np.ma.filled(
    kriging_variance,
    np.nan
)

# Save results
np.save(
    "3d_copper_grade.npy",
    estimated_grade
)

np.save(
    "3d_copper_kriging_variance.npy",
    kriging_variance
)

print("3D copper grade estimation completed.")
```

Important technical qualification: This code is a basic demonstration of 3D kriging. It does not implement the full workflow required for a defensible mineral resource estimate.

For an actual project, you must determine the directional variogram, geological anisotropy, estimation neighborhood, sample support, search ellipsoid, top cuts and domain boundaries. The default variogram fitting is not a substitute for geological and geostatistical analysis.

For large voxel grids, use local search neighborhoods and an appropriately optimized implementation rather than attempting to krige every block against every assay simultaneously.

11.1. Integrating AI regression with kriging

After the conventional kriging model is working, develop an XGBoost regression model.

The model can use:

- Lithology.
- Distance to faults.
- Distance to intrusive contacts.
- Magnetic attributes.

- Resistivity.
- Gravity-derived attributes.
- Geochemical indicators.
- Other geologically justified predictors.

A possible hybrid formulation is:

$$\widehat{G}(x) = \widehat{G}_{AI}(x) + \widehat{R}_{Kriging}(x)$$

Where:

- $\widehat{G}_{AI}$ is the grade trend estimated by AI.
- $\widehat{R}_{Kriging}$ is the kriged residual correction.

This is regression kriging.

You should compare the hybrid results against ordinary kriging and XGBoost regression using spatially separated drill-hole validation.

11.2. Generate a grade-colored 3D block model

Once the grade estimates have been assigned to the voxel grid, PyVista can display them.

```python
import numpy as np
import pyvista as pv

# Load block model
grade = np.load(
    "3d_copper_grade.npy"
)

# Dimensions returned by PyKrige are (nx, ny, nz)
grade_zyx = np.transpose(
    grade,
    (2, 1, 0)
)

nz, ny, nx = grade_zyx.shape

# Create 3D block model
block_model = pv.ImageData(
    dimensions=(nx + 1, ny + 1, nz + 1),
    spacing=(25, 25, 10),
    origin=(500000, 7500000, -500)
)
```

```
# Assign copper grade
block_model.cell_data["Cu_grade"] = (
    grade_zyx.ravel(order="C")
)

# Save block model
block_model.save(
    "3D_copper_grade_block_model.vti"
)

# Display model
viewer = pv.Plotter()

viewer.add_mesh(
    block_model,
    scalars="Cu_grade",
    cmap="turbo",
    show_edges=False
)

viewer.add_axes()
viewer.show_grid()
viewer.show()
```

This creates a 3D grade-colored block model.

For a final application, I recommend adding:

- Grade legends and units.
- Geological domain filters.
- Drill-hole trajectories.
- Section slicing.
- Clipping planes.
- Minimum and maximum grade filters.
- Grade statistics.
- Block volume and tonnage calculations, using validated density information.

The example assumes that the kriging result uses the same cell-center coordinates, grid dimensions and ordering as the block model.

PART VIII. DEVELOPING THE USER INTERFACE AND INTEGRATING 2D AND 3D DISPLAY

12. Proposed main program interface

I recommend that the final system resemble a professional mineral exploration application, rather than simply a Python program with separate plotting windows.

The main interface should contain:

AI-3D-MEOMS — Main interface concept

AI-3D-MEOMS

Project: Exploration_001

Project | Database | GIS | 3D Model | AI | Grade | Reports

Project Layers

- ☑ Geological map
- ☑ Geological structures
- ☑ Geochemical samples
- ☑ Magnetic map
- ☑ Gravity map
- ☑ MT / resistivity
- ☑ Drill holes

AI Tools

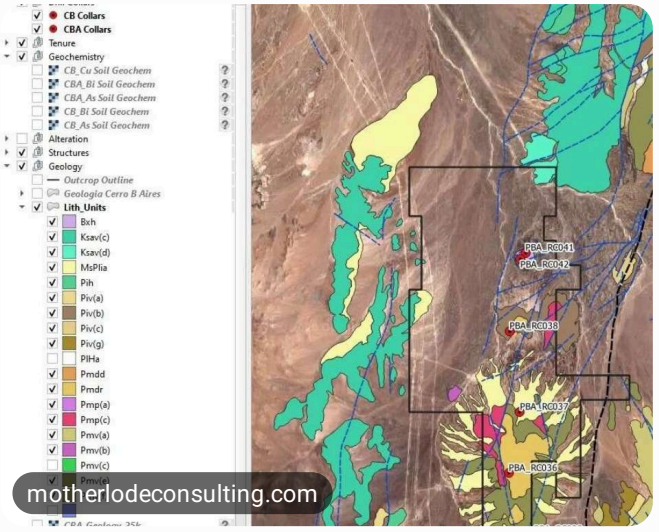
Data preparation

Model training

3D prediction

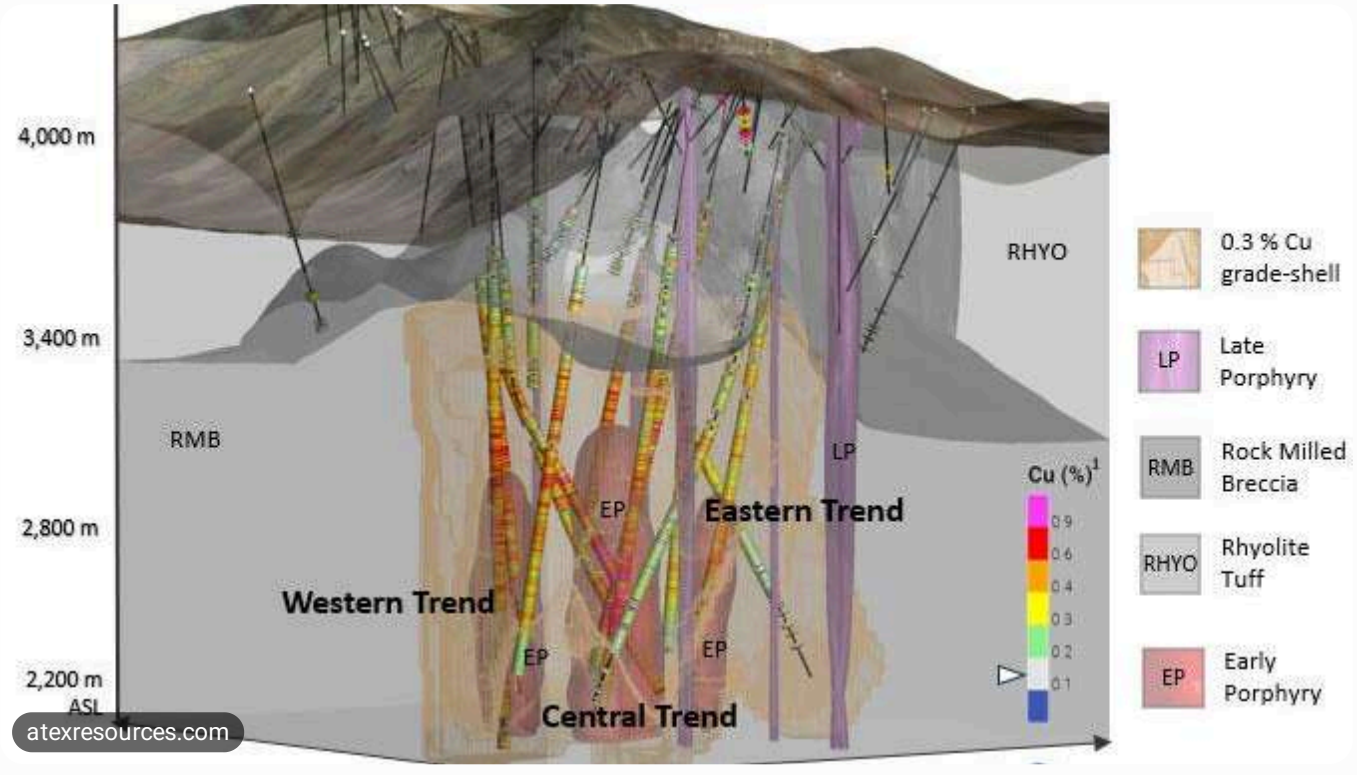
Grade estimation

2D Geological Map



Layer control · Map navigation · Geological interpretation

3D Geological and Mineral Prospectivity Viewer



3D geological solids · Orebody · Grade distribution · Probability volume · Drill-hole trajectories

Model Status and Results

Model	Validation	Output
Random Forest	**Spatial CV**	**3D Volume**

Conceptual interface design, not an existing or functioning application.

12.1. Main interface modules

I suggest the following modules.

Module	Main functions
Project Management	Create, save and load exploration projects
Database Management	Import, organize and validate geological data
Geological Mapping	2D map visualization and editing
Geochemical Analysis	Anomaly maps and statistical analysis
Geophysical Analysis	Geophysical maps and inversion data
Drill-hole Management	Collar, survey, lithology and assay visualization
Geological 3D Modelling	Geological units, faults and contacts
AI Model Training	Select algorithms and training parameters
3D Prospectivity Mapping	Calculate and display probability volumes
Orebody Modelling	Extract and interpret mineralized solids
Grade Estimation	Kriging and AI-based grade prediction
3D Visualization	Display and investigate 3D models
Reporting	Export maps, model statistics and technical reports

12.2. How to integrate QGIS and PyVista

This is the central software engineering question.

I recommend two development stages.

Stage A. Separate applications communicating through files

Recommended for the first prototype

Develop:

- QGIS as the 2D GIS application.
- Python as the AI processing engine.

- PyVista as a separate 3D visualization application.

The interaction works as follows:



For example, a QGIS plugin can execute the AI engine using Python's `subprocess` module.

A simplified example:

```python
import subprocess

ai_program = r"C:\AI_MEOMS\python.exe"

ai_script = r"C:\AI_MEOMS\run_prospectivity.py"

project = r"D:\Projects\Exploration_001"

subprocess.Popen([
    ai_program,
    ai_script,
    "--project",
    project
])
```

The AI engine reads the project data and writes its results to the project's output directory.

QGIS then loads the resulting 2D maps and the 3D viewer opens the VTK files.

This approach is particularly suitable for offline deployment because the two programs communicate using local files.

Stage B. Integrate the 3D viewer into the main application

Second development stage

After the separate applications are working correctly, you can embed PyVista into a Qt-based user interface.

The relevant package is `pyvistaqt`.

It provides a `QtInteractor` component that allows PyVista's 3D rendering functionality to be placed inside a Qt application. PyVistaQt … +1

For example:

```python
from PySide6.QtWidgets import (
    QApplication,
    QMainWindow,
    QVBoxLayout,
    QWidget
)

from pyvistaqt import QtInteractor

class GeologicalViewer(QMainWindow):

    def __init__(self):
        super().__init__()

        self.setWindowTitle(
            "AI 3D Mineral Exploration System"
        )

        self.widget = QWidget()
        self.layout = QVBoxLayout()

        self.plotter = QtInteractor(self.widget)

        self.layout.addWidget(
            self.plotter.interactor
        )
```

```
        self.widget.setLayout(
            self.layout
        )

        self.setCentralWidget(
            self.widget
        )

        self.show_geology()

    def show_geology(self):

        import pyvista as pv

        model = pv.Sphere()

        self.plotter.add_mesh(
            model,
            color="lightblue"
        )

        self.plotter.add_axes()

if __name__ == "__main__":

    app = QApplication([])

    window = GeologicalViewer()

    app.exec()
```

This example demonstrates the 3D visualization component. It is not yet connected to QGIS.

For your final application, you have two choices:

Option 1: Keep QGIS as the main application and embed PyVista using the same Qt/PyQt runtime that the installed QGIS version uses.

Option 2: Develop a custom PySide6 application with a QGIS map component and a PyVista 3D component.

I recommend Option 1 initially because it reduces the amount of GIS development required.

However, there is an important compatibility issue: QGIS has its own Qt/PyQt runtime, while PyVistaQt can use different Qt bindings. Do not mix incompatible Qt libraries in the same process. During early development, separate processes are safer; later, test a single compatible Qt environment before embedding.

12.3. Can QGIS itself display 3D geological models?

Yes, to a certain extent.

QGIS includes a 3D Map View that supports terrain, 3D vector features and mesh-based visualization. It also supports exporting 3D scenes in OBJ format. QGIS Docu… +1

However, for your particular requirements, I would use:

3D object	Preferred visualization
Ground surface / DEM	QGIS 3D
Geological map features	QGIS 3D
Fault surfaces	PyVista or QGIS 3D
Geological solids	PyVista
Drill-hole trajectories	PyVista
Mineral prospectivity voxels	PyVista
Orebody solid	PyVista
Grade block model	PyVista
Cross-sectional slices	PyVista

In other words, QGIS can handle much of the 3D map context, but PyVista should be the main 3D modelling and block-model visualization engine.

12.4. Suggested 3D output formats

File format	Purpose
`.vti`	Regular voxel and scalar volumes
`.vtu`	Unstructured 3D block models
`.vtp`	Orebody surfaces and geological meshes
`.vtk`	General scientific 3D data
`.obj`	Surface model exchange
`.stl`	Triangular surface geometry
`.csv` / `.parquet`	Block coordinates and attributes
`.gpkg`	2D and 3D GIS vector information
`.tif`	2D geological and geophysical raster results

For interoperability with Surpac, 3D Mine or GOCAD, I recommend first developing reliable exports in neutral formats and then adding version-specific import/export converters.

Do not assume that all software will directly interpret the same block-model format or geological surface representation.

PART IX. OFFLINE SYSTEM DEVELOPMENT AND DEPLOYMENT

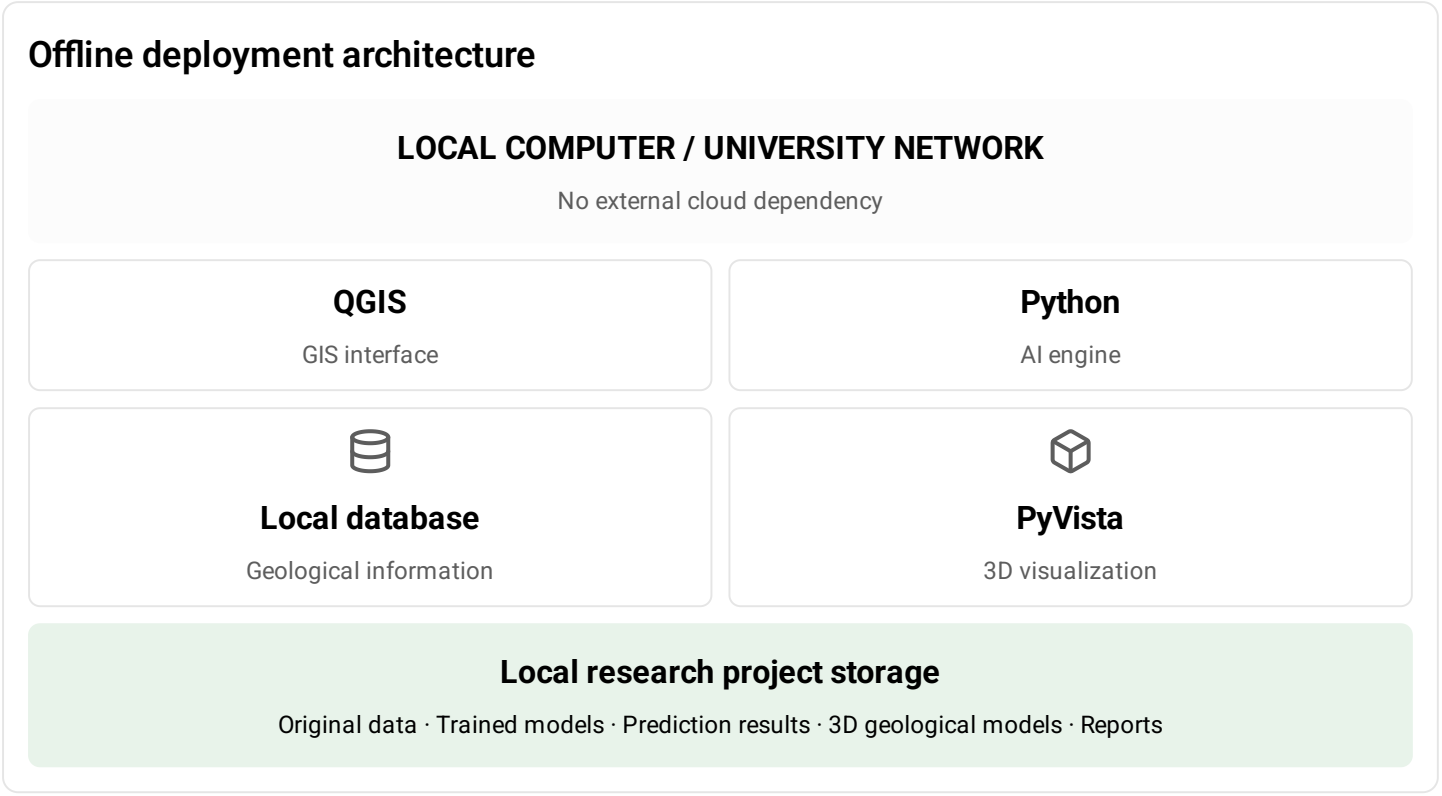
13. How to develop a completely offline system

I understand that offline operation is an important requirement.

I recommend that the system be designed so that no internet connection is required for:

- Data import.
- Data preprocessing.
- AI training.
- AI prediction.
- 3D geological modelling.
- Grade estimation.
- Visualization.
- Report generation.

13.1. Offline architecture



13.2. Recommended installation procedure

I recommend the following deployment procedure:

1. Install the QGIS standalone offline installer.
2. Install the required Python environment.
3. Install the AI packages from a locally prepared package repository.
4. Install the QGIS plugin.
5. Install the PyVista 3D viewer.
6. Configure the local database.
7. Copy the trained AI models and model metadata.
8. Test all functions without an internet connection.

QGIS provides standalone installers that include the software dependencies and do not require an internet connection during installation. QGIS

For the Python packages, prepare a local installation directory on a connected development computer:

```bash
pip download -r requirements.txt -d wheelhouse
```

Then transfer the wheelhouse to the offline computer and install:

```bash
pip install \
    --no-index \
    --find-links=wheelhouse \
    -r requirements.txt
```

For complex environments, a tested Conda environment packaged using `conda-pack` is another option.

All package versions should be pinned and tested together before deployment.

13.3. Model and project management

Every AI model should be saved with its associated information.

For example:

```plain text
```

```
models/
 |
 ├──── model_001/
 |     ├──── rf_model.joblib
 |     ├──── xgb_model.json
 |     ├──── feature_list.json
 |     ├──── training_info.json
 |     ├──── validation_report.pdf
 |     └──── model_version.txt
```

The metadata should record:

- Training dataset version.
- Geological study area.
- Target mineral commodity.
- Training labels.
- Feature definitions.
- Model parameters.
- Validation methodology.
- Model performance.
- Date of training.
- Software versions.

This will allow the faculty to reproduce and compare different AI prediction results.

PART X. RECOMMENDED IMPLEMENTATION SCHEDULE

14. Development stages for the complete software system

I recommend an initial development period of approximately 15–18 months.

Do not try to complete the entire system simultaneously. Build a working system progressively, validating each component before integrating the next.

Period	Development task	Expected deliverable
Months 1–2	Data inventory and database design	Geological database structure
Months 2–4	Data import and quality control	Organized geological datasets
Months 3–6	Drill-hole desurveying and 3D geological modelling	Initial 3D geological model
Months 5–7	Voxel construction and feature engineering	3D AI training database
Months 7–10	RF, XGBoost and SVM development	First 3D prospectivity models
Months 9–12	Orebody geometry extraction	Initial 3D mineralized volumes

Months 10–13	Grade estimation and geostatistics	3D grade block model
Months 11–15	QGIS and PyVista integration	Integrated software prototype
Months 15–18	Independent validation and offline packaging	First operational offline version

Some activities should proceed in parallel, particularly database development, staff training and interface design.

14.1. Development milestones

Five development milestones

1. **Milestone 1: Geological database** First 3 months Import all available datasets, validate coordinate systems, organize drill-hole information and establish the database.
2. **Milestone 2: 3D geological model** Months 3–6 Construct geological units, faults, drill-hole trajectories and initial 3D geological surfaces.
3. **Milestone 3: AI prospectivity model** Months 6–10 Train and validate Random Forest and XGBoost models, then generate the 3D prospectivity volume.
4. **Milestone 4: Orebody and grade modelling** Months 9–13 Develop geological constraints, extract mineralized solids and construct the grade block model.
5. **Milestone 5: Integrated offline software** Months 13–18 Integrate the 2D GIS, AI processing and 3D visualization components and validate the complete system.

14.2. Suggested research and development team

For the first version, I would recommend the following team.

Position	Main responsibility
Project leader / economic geologist	Overall scientific direction
Exploration geologist	Geological interpretation and orebody modelling
Geochemist	Geochemical data processing
Geophysicist	Geophysical data preparation and interpretation
Python AI developer 1	Machine learning and data processing
Python AI developer 2	3D modelling and visualization
GIS specialist	QGIS and spatial database
Geostatistician	Variography and grade estimation

Some roles can initially be combined, depending on your available academic staff.

It is particularly important to have both an experienced exploration geologist and a geostatistics specialist involved in the project.

PART XI. IMPORTANT TECHNICAL RECOMMENDATIONS

15. Seven issues that I would pay particular attention to

Before concluding, I want to emphasize several technical matters that can determine whether your system becomes a genuinely useful exploration tool or merely a sophisticated visualization program.

15.1. Do not mix mineral prospectivity and grade estimation

These are separate scientific problems.

A prospectivity model predicts where mineralization is likely to occur.

A grade model predicts the concentration of an economic element within an established geological and mineralization domain.

The final software should therefore contain separate algorithms for these two functions.

15.2. Do not use surface geophysical data as direct depth measurements

This is particularly important for your magnetic, gravity, gamma-ray and soil conductivity information.

If you have 3D geophysical inversion results, use the corresponding depth-dependent model values.

If you only have surface measurements, use them as surface evidence or geological constraints, with appropriate treatment of their depth sensitivity.

15.3. Do not begin with 3D CNN

Although 3D CNNs are attractive, they require sufficient labeled training examples and carefully constructed volumetric data.

I recommend:

1. Random Forest.
2. XGBoost.
3. Spatial validation.
4. Explainable AI.
5. 3D CNN after obtaining sufficient training data.

This will reduce development complexity and make the initial research easier to evaluate scientifically.

15.4. Geological modelling should constrain AI

AI predictions should be consistent with:

- Geological contacts.
- Fault geometry.
- Host-rock distribution.
- Structural controls.
- Mineral deposit models.

For example, if a particular mineral deposit type is known to occur within a specific intrusive complex, that geological relationship should be considered during modelling.

15.5. Use actual drilling information to constrain the orebody

The drill holes are essential for constructing the orebody geometry.

The AI model can predict the likely continuation of mineralization between drill holes, but geological interpretation and drilling verification are necessary.

15.6. Use uncertainty as an important output

Your final 3D system should not display only one predicted orebody.

It should also show where the prediction is reliable and where additional drilling is required.

I recommend displaying:

- Prospectivity probability.
- Grade estimate.
- Kriging variance or other estimation uncertainty.
- Distance to the nearest drill hole.
- Data coverage.
- Geological model uncertainty.

15.7. Keep the entire workflow reproducible

Each prediction should be associated with:

- Input data versions.
- Model parameters.
- Geological model version.
- Training and validation results.
- Prediction date.
- Output grid definition.

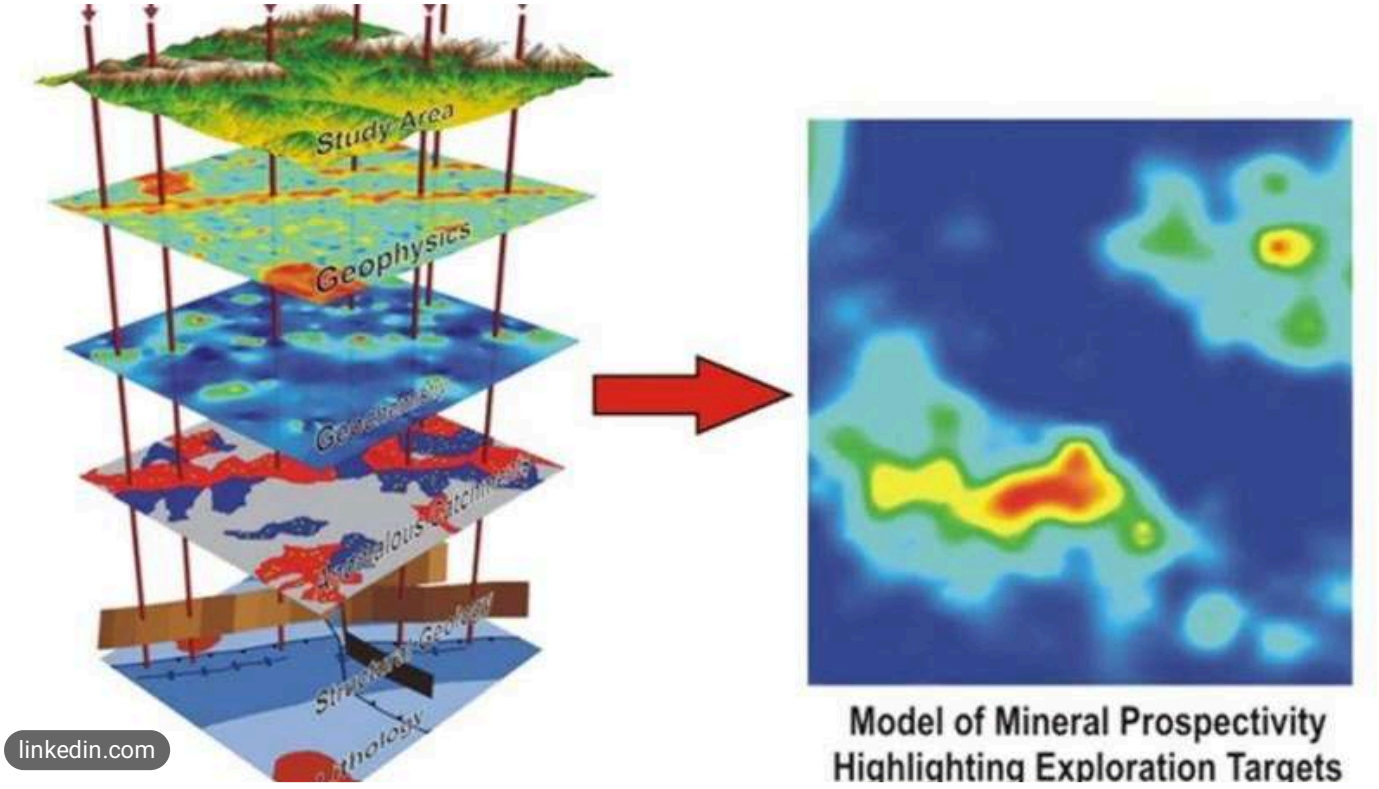
This is particularly important when new drill holes are added and the AI model is retrained.

PART XII. MY FINAL RECOMMENDED DEVELOPMENT STRATEGY

Considering all the information you have provided, I would organize the development into three major products.

Product 1

2D Geological Exploration and Data Management System



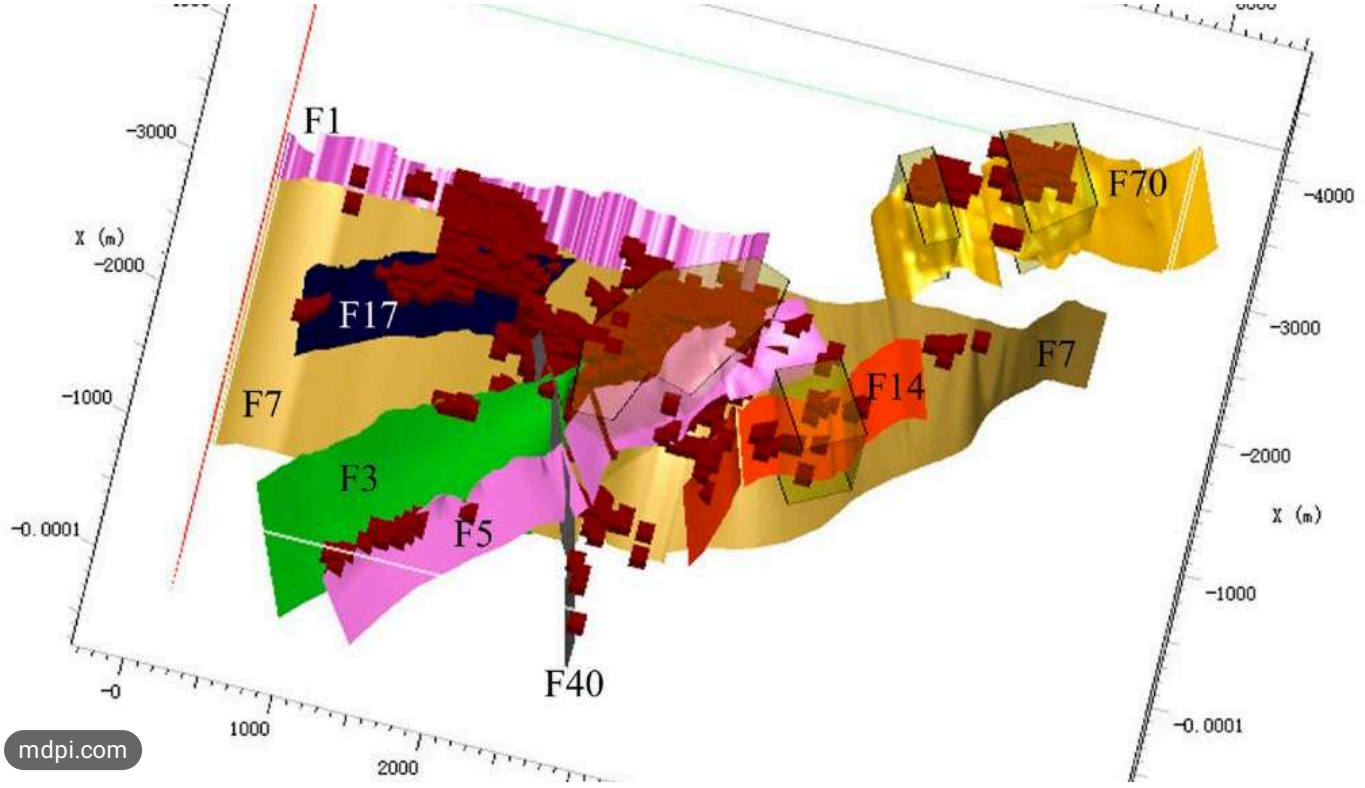
Main software: QGIS + Python

Main functions:

- Geological map display.
- Geochemical anomaly maps.
- Magnetic and gravity maps.
- MT and resistivity data display.
- Remote sensing imagery.
- Drill-hole collar and trace display.
- Database management.

Product 2

AI-Based 3D Mineral Prospectivity and Orebody Modelling System



mdpi.com

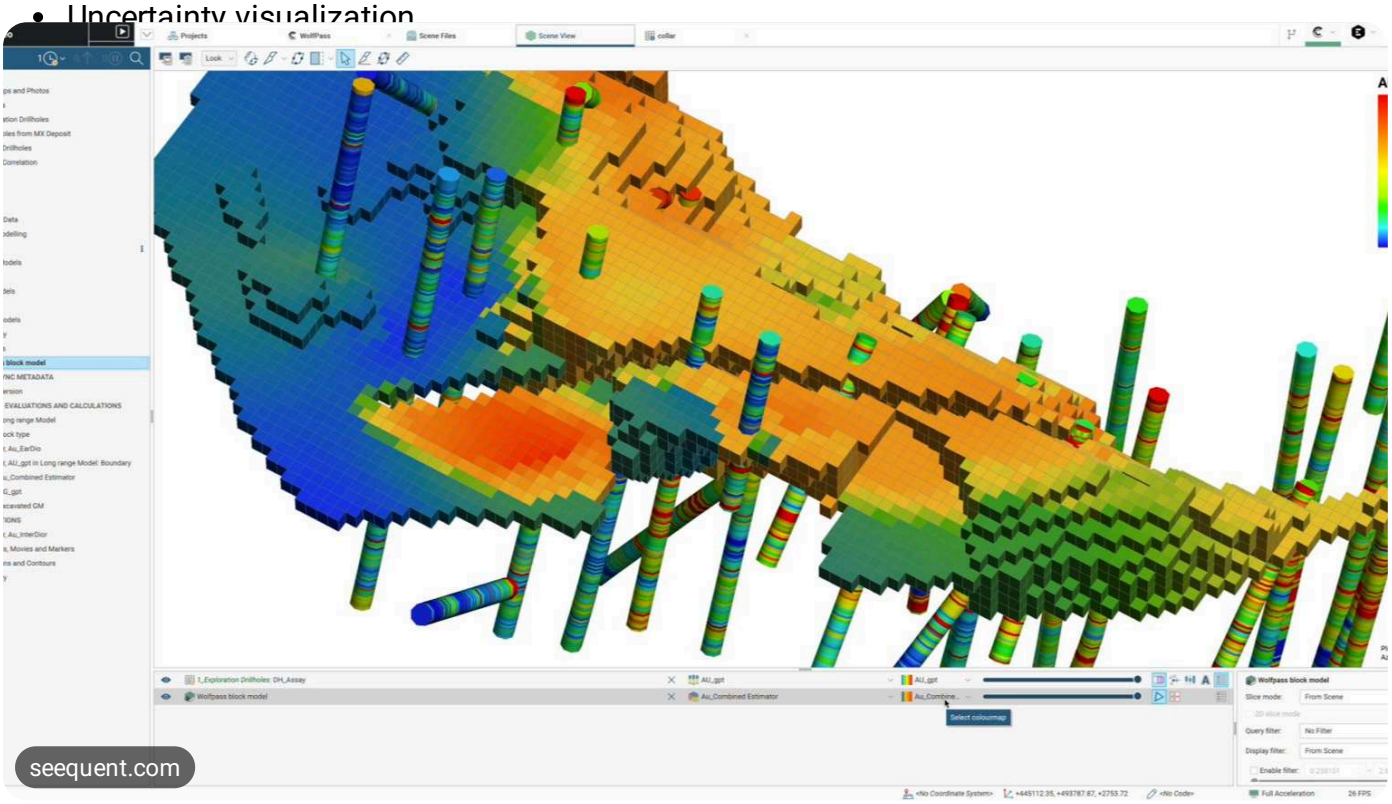
Main software: Python + Scikit-learn + XGBoost + GemPy + PyVista

Main functions:

- 3D geological modelling.
- AI-based mineralization probability.
- 3D prospectivity volume.
- Orebody surface extraction.
- Geological constraints.
- Uncertainty visualization.

Products

3D Grade Estimation and Exploration Decision-Support System



seequent.com

Main software: Python + GSTools / PyKrige + XGBoost + PyVista

Main functions:

- 3D grade estimation.
- Grade block model.
- Geological domain modelling.
- Mineralized volume calculation.
- Drill-hole evaluation.
- Exploration target ranking.
- Prediction uncertainty.

The three products should eventually be integrated into one offline software package.

16. My recommendation for your first actual development project

I would start with the following project rather than attempting all the functions at once:

Recommended pilot project

Development of an AI-Based 3D Mineral Prospectivity Mapping System Using Integrated Geological, Geochemical, Geophysical and Drilling Data

Initial scope

1. Import geological maps, ground survey information and existing geochemical and geophysical datasets.
2. Import drill-hole collar, survey, lithology and assay tables.
3. Construct a standardized 3D geological database.
4. Desurvey drill holes and generate assay composites.
5. Construct a 3D geological framework.
6. Develop Random Forest and XGBoost models.
7. Generate 3D mineral prospectivity maps.
8. Display the prediction volumes and drill holes in PyVista.
9. Validate the prediction against independent drilling and geological evidence.

Initial software: QGIS + Python + GemPy + PyVista.

Initial output: A functioning offline prototype of the 3D mineral prospectivity system.

Once this prototype works, develop the orebody extraction and grade estimation modules.

This sequence will minimize technical risks and allow your academic staff to publish research results at each development stage.

17. Five technical questions before designing the detailed software specification

I can go one step further and prepare the actual software development specification, including the complete database schema, Python source-code structure, QGIS plugin design and 3D viewer

integration.

However, five details will significantly affect the technical design.

1. Which operating system will be used?

- (●) Windows 11
- ( ) Windows 10
- ( ) Linux
- ( ) Both Windows and Linux

2. What is the main target mineral deposit type?

- (●) Porphyry copper
- ( ) Epithermal gold and silver
- ( ) Orogenic gold
- ( ) Volcanogenic massive sulfide (VMS)
- ( ) Skarn deposits
- ( ) Other or multiple deposit types

3. Approximately how many drill holes are currently available?

- ( ) Fewer than 10
- (●) 10–30
- ( ) 30–100
- ( ) 100–500
- ( ) More than 500

4. What is the condition of the geophysical data?

- ( ) Mainly 2D survey data
- ( ) Mainly 3D inversion models
- (●) Both 2D survey data and 3D inversion models
- ( ) Not yet confirmed

5. What is your software licensing preference?

- (•) Completely open-source software
- ( ) Open-source plus existing commercial software
- ( ) Commercial software is acceptable

Submit technical requirements

With these answers, I can develop the next level of the project: a **detailed technical software development blueprint with database table definitions, data-processing algorithms, Python source-code modules and interface specifications**.

My strongest advice is to keep the first version focused on the complete workflow from geological database to 3D prospectivity volume. Once that is functioning and independently validated, the orebody geometry and grade block-model functions can be added without having to redesign the entire software architecture.

2026-11-06 16:32:57 **You**

Please prepare the detailed technical software development specification for AI-3D-MEOMS based on these requirements:
Operating system: Windows 10
Deposit type: Porphyry copper
Available drill holes: 10–30 drill holes
Geophysical data: Both 2D survey data and 3D inversion models
Software preference: Completely open-source software
Include complete database schemas, drillhole desurveying and compositing, 3D voxel feature engineering, AI model training, orebody extraction, grade estimation, Python project structure, QGIS plugin implementation, PyVista integration, and offline deployment.

ChatGPT gpt-6-thinking

AI-3D-MEOMS

Detailed Technical Software Development Specification

Integrated AI-Based 3D Mineral Exploration, Prospectivity Mapping and Orebody Modelling System

Technical development specification Proposed baseline

Specification	Selected requirement
Operating system	Windows 10
Target deposit	Porphyry copper
Drilling database	10–30 drill holes
Geophysical data	2D surveys and 3D inversion models
Software policy	Completely open-source
GIS interface	QGIS
AI development	Python
Geological modelling	GemPy / LoopStructural
3D visualization	PyVista / VTK
Database	PostgreSQL/PostGIS or GeoPackage
Deployment	Completely offline
Development approach	Modular, expandable architecture

1. Executive technical recommendation

Based on your requirements, I recommend developing AI-3D-MEOMS as a locally installed, modular geological exploration application specifically designed for porphyry copper exploration.

The system should integrate five scientific functions:

1. Geological interpretation and 3D geological framework modelling.
2. Integrated geological, geochemical, geophysical and remote sensing data analysis.
3. Three-dimensional mineral prospectivity prediction.
4. Three-dimensional mineralized body interpretation and geometry extraction.
5. Three-dimensional copper grade estimation and block-model visualization.

The most important consideration is your available drilling information: **10–30 drill holes**.

This is sufficient to develop an initial 3D geological interpretation and a research-oriented AI prospectivity prototype, provided the holes contain useful mineralized and barren intersections and the supporting datasets are of good quality.

However, it is a relatively small dataset for training a general-purpose 3D deep-learning model.

Therefore, I recommend the following scientific strategy:

- Use geological knowledge and geological framework modelling as the foundation.

- Use Random Forest and XGBoost as the initial AI models.
- Use carefully selected geochemical and geophysical attributes as predictors.
- Apply spatially separated validation and leave-one-drill-hole-out experiments.
- Use conventional geostatistics as the baseline for grade estimation.
- Develop hybrid AI-geostatistical estimation only after the baseline has been established.
- Treat the resulting 3D mineralized geometry as an exploration interpretation, not automatically as a mineral resource model.

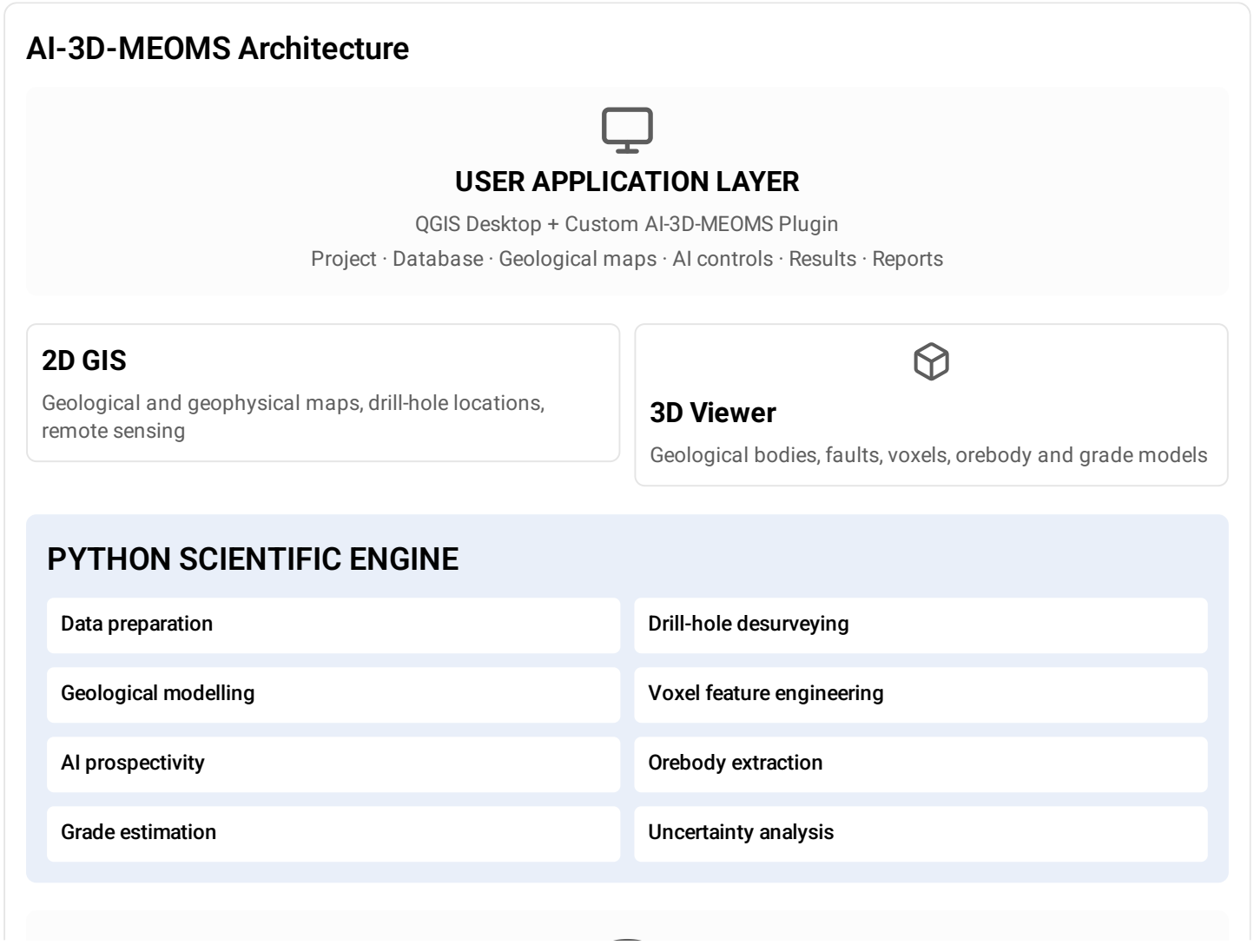
There is also one infrastructure issue that needs attention: Windows 10 reached the end of general Microsoft support on 14 October 2025. In 2026, a new institutional system should ideally be designed to migrate to Windows 11 or a supported Linux environment. If Windows 10 is an unavoidable requirement, the deployment should use a carefully frozen, tested and security-managed software environment.

---

2. Overall system architecture

2.1. Proposed architecture

I recommend a three-layer architecture with a common project database and a separate scientific processing engine.





CENTRAL GEOLOGICAL DATABASE

PostgreSQL/PostGIS · GeoPackage · Parquet · GeoTIFF · VTK

FINAL SCIENTIFIC OUTPUTS

3D geological framework · Mineral prospectivity · Interpreted orebody · Copper grade blocks · Uncertainty · Exploration targets

All processing, database access, model training, predictions and visualization are performed locally.

2.2. Software selection

The following is the proposed open-source software stack.

Component	Selected software	Purpose
Main GIS	QGIS	2D geological interface
GIS programming	PyQGIS	QGIS plugin development
Main programming language	Python	Scientific engine
Numerical processing	NumPy, SciPy	Numerical calculations
Tabular processing	Pandas, PyArrow	Geological tables
Spatial processing	GeoPandas, Shapely	Vector processing
Raster processing	GDAL, Rasterio	Raster and remote sensing
Database	PostgreSQL/PostGIS	Institutional database
Local project database	GeoPackage	Standalone offline projects
Machine learning	Scikit-learn	Initial AI models
Gradient boosting	XGBoost	Mineral prospectivity and grade regression
Deep learning	PyTorch	Future advanced AI
Geological modelling	GemPy	3D geological framework
Structural modelling	LoopStructural	Advanced structural interpretation
Geostatistics	GSTools, PyKrige	Variograms and kriging
3D visualization	PyVista, VTK	Orebody and block-model display
3D Qt integration	PyVistaQt	Embedded 3D viewer
GUI development	QGIS Qt bindings	Interface components
Model interpretation	SHAP	Explainable AI

Model management	MLflow	Local experiment tracking
Software version control	Git	Source-code management

The software shall not require commercial licenses or an internet connection during normal operation.

For offline reproducibility, every software dependency shall be pinned to a tested version. The final version numbers must be selected after testing the complete stack on the target computer.

The technical specification below is designed as an implementation document for your faculty's software development team. It goes beyond a general research proposal and defines the database, data-processing rules, AI workflow, software interfaces, source-code organization, testing requirements and deployment procedures.

A few design decisions are particularly important. The USGS porphyry copper deposit model provides an appropriate geological foundation for selecting alteration, structural, geochemical and geophysical features. GemPy supports implicit 3D geological modelling, QGIS supports Python Processing plugins, and PyVista provides scientific 3D visualization. These are suitable foundations for the proposed open-source system. U.S. Geolog... +3

One operating-system qualification: Windows 10 reached end of support on 14 October 2025. The system can still be developed for Windows 10, but I recommend making the Python engine portable to Windows 11 and Linux as well. Microsoft ... +1

Document

TECHNICAL SOFTWARE DEVELOPMENT SPECIFICATION

AI-3D-MEOMS: Integrated AI-Based 3D Mineral Exploration, Prospectivity Mapping and Orebody Modelling System

Project type: Scientific software research and development
Target deposit: Porphyry copper
Operating environment: Windows 10 (64-bit), with portability to Windows 11 and Linux
Software licensing policy: Open-source software exclusively
Development language: Python
GIS platform: QGIS
3D visualization platform: PyVista / VTK
Database platform: PostgreSQL/PostGIS and GeoPackage
Proposed development period: 18 months
Document status: Technical development specification, Version 1.0

SECTION 1. PROJECT DEFINITION AND DEVELOPMENT OBJECTIVES

1.1. Project definition

AI-3D-MEOMS shall be developed as an integrated, offline-capable software system for processing geological exploration information, predicting mineralization potential, constructing three-dimensional geological and mineralized bodies, and estimating the spatial distribution of copper grade.

The software shall integrate:

- Geological mapping and ground survey information.
- Geochemical sampling and analytical data.
- Magnetic survey data.
- Gamma-ray spectrometric data.
- Magnetotelluric (MT) survey data.
- Electrical resistivity tomography (ERT) and resistivity cross-section data.
- Gravity survey data.
- Satellite and other remote sensing information.
- Drill-hole collar, survey, lithology and assay information.

The system shall utilize machine learning, geostatistical analysis, three-dimensional geological modelling and spatial data-fusion techniques.

1.2. Main development objectives

The software shall provide the following functions.

ID	Function	Main output
F01	Geological database management	Integrated exploration database
F02	2D geological mapping	GIS geological and geophysical maps
F03	Drill-hole management	Desurveyed drill-hole database

ID	Function	Main output
F04	3D geological modelling	Geological framework model
F05	Geochemical analysis	Geochemical anomaly maps
F06	Geophysical interpretation	Geophysical attribute and inversion models
F07	3D feature engineering	Integrated voxel feature database
F08	AI model training	Trained mineral prospectivity models
F09	3D prospectivity prediction	3D mineral prospectivity volume
F10	Orebody interpretation	3D interpreted mineralized solid
F11	Grade estimation	3D copper grade block model
F12	3D visualization	Interactive geological and mineralization viewer
F13	Exploration assessment	Ranked exploration targets and uncertainty maps
F14	Reporting	Maps, tables and technical reports

1.3. Scientific scope

The system shall initially focus on porphyry copper mineralization.

The geological interpretation and AI feature selection shall therefore consider the following geological characteristics:

- Porphyritic intrusive rocks.
- Intrusive contacts and associated host rocks.
- Faults and fracture systems.
- Stockwork quartz veins.
- Hydrothermal breccias.
- Potassic alteration.
- Phyllic or sericitic alteration.
- Propylitic alteration.
- Argillic alteration.
- Sodic-calcic alteration, where present.
- Disseminated and vein-controlled copper sulfides.
- Primary and supergene copper mineralization.

These characteristics shall constitute geological knowledge constraints for AI development.

The system shall not assume that every porphyry copper deposit exhibits identical alteration, geochemical or geophysical signatures. Geological features shall be selected according to the actual deposit model and field evidence.

1.4. Main scientific products

Three independent but interconnected 3D models shall be generated.

Model A: 3D Geological Framework Model

Contains:

- Geological units.
- Intrusive bodies.
- Fault surfaces.
- Geological contacts.
- Structural information.
- Alteration domains.

Model B: 3D Mineral Prospectivity Model

Contains:

- Mineralization prospectivity score.
- Calibrated probability, if sufficient validation data are available.
- Favorable geological volumes.
- Spatial uncertainty.
- Exploration target ranking.

Model C: 3D Orebody and Grade Model

Contains:

- Interpreted mineralized volumes.
- Estimated copper grade.
- Other commodity grades where available.
- Geological domains.
- Estimation uncertainty.
- Drill-hole constraints.

These models shall be stored and visualized independently.

SECTION 2. SOFTWARE ARCHITECTURE

2.1. General architecture

The system shall consist of three principal layers.

Layer 1: User Interface

Main software: QGIS.

Responsibilities:

- Project management.
- 2D geological map display.
- Data import.
- Database management.
- AI parameter configuration.
- Processing control.
- Result visualization.

- Report generation.

Layer 2: Scientific Processing Engine

Main software: Python.

Responsibilities:

- Data preprocessing.
- Drill-hole desurveying.
- Geochemical analysis.
- Geophysical attribute extraction.
- 3D geological modelling.
- Voxel construction.
- AI training.
- 3D prediction.
- Orebody extraction.
- Grade estimation.

Layer 3: 3D Visualization Engine

Main software: PyVista and VTK.

Responsibilities:

- Geological surface visualization.
- Drill-hole trajectory display.
- 3D voxel rendering.
- Prospectivity volume visualization.
- Orebody surface rendering.
- Grade block model visualization.
- Interactive geological sections.

2.2. Software communication

The first version shall use separate software processes.

QGIS shall communicate with the Python AI engine through local configuration files and process execution.

The AI engine shall generate standardized outputs.

The PyVista viewer shall read these outputs and display the corresponding 3D models.

The system shall not require direct network communication or cloud services.

2.3. Data interchange formats

Data type	Preferred format
Geological vector data	GeoPackage
Geological raster data	GeoTIFF
Geochemical tables	CSV / Parquet
Drill-hole tables	CSV / Parquet
3D regular volumes	VTK ImageData (.vti)
3D unstructured blocks	VTK UnstructuredGrid (.vtu)
Orebody surfaces	VTK PolyData (.vtp)
General surface models	OBJ / STL
Project configuration	JSON
Scientific arrays	NumPy / Zarr
Central database	PostgreSQL/PostGIS

Large raster and 3D datasets shall normally be stored as files with references in the database rather than inserted as large binary objects into ordinary relational tables.

SECTION 3. DATABASE DEVELOPMENT AND COMPLETE DATA SCHEMA

3.1. Database design principles

The database shall follow a normalized relational structure.

Each exploration project shall possess a unique Project ID.

All spatial data shall use a common horizontal coordinate reference system and a documented vertical datum.

The database shall preserve original observations separately from processed and interpreted datasets.

The following principles are mandatory:

1. Original survey data shall never be overwritten by processed data.
2. Every sample and drill hole shall have a unique identification code.
3. Geological and geophysical observations shall retain their original measurement units.
4. Analytical detection limits shall be preserved.
5. All derived data shall retain information about their source.
6. Model outputs shall be associated with a particular model version.
7. Missing measurements shall not be automatically interpreted as zero.
8. Uncertain or interpolated information shall be identified separately from direct observations.

3.2. Proposed database organization

The PostgreSQL/PostGIS database shall contain the following schemas:

Schema	Contents
core	Project and data asset management
geology	Geological maps and structural observations
geochemistry	Soil samples and chemical analyses
geophysics	Geophysical survey and inversion data
remote_sensing	Satellite and raster information
drilling	Drill-hole and assay information
derived	Desurveyed holes, composites and processed features
modelling	Geological models, AI training and predictions

For individual offline projects, the same logical structure may be implemented using GeoPackage and Parquet files.

3.3. Core project database

Table: `core.project`

Field	Data type	Description
project_id	UUID	Primary key
project_name	VARCHAR	Project name
study_area	TEXT	Study area description
target_deposit	VARCHAR	Porphyry copper
target_commodity	VARCHAR	Cu
horizontal_srid	INTEGER	Project horizontal CRS
vertical_datum	VARCHAR	Elevation reference
coordinate_units	VARCHAR	Normally meters
z_convention	VARCHAR	Elevation positive upward
created_date	TIMESTAMP	Project creation
project_status	VARCHAR	Active / archived

Table: `core.data_asset`

Field	Data type	Description
asset_id	UUID	Primary key
project_id	UUID	Project reference
asset_name	VARCHAR	Dataset name

Field	Data type	Description
asset_type	VARCHAR	Geological / geochemical / geophysical etc.
file_path	TEXT	Relative project file path
file_format	VARCHAR	CSV / SHP / TIFF / VTI etc.
source_description	TEXT	Original source
acquisition_date	DATE	Survey or data date
coordinate_system	VARCHAR	Original CRS
checksum_sha256	VARCHAR	File integrity
processing_status	VARCHAR	Raw / processed / validated

The file path shall be relative to the project directory wherever practical, so that complete projects can be moved between computers.

3.4. Geological database

Table: `geology.lithology_unit`

Field	Description
unit_id	Unique unit identifier
unit_code	Geological code
unit_name	Geological unit name
lithology_type	Lithological classification
geological_age	Geological age
intrusive_status	Intrusive / extrusive / sedimentary etc.
porphyry_relevance	Geological interpretation

Field	Description
description	Geological description

Table: `geology.geological_feature`

Field	Description
feature_id	Unique identifier
project_id	Project ID
feature_type	Polygon / line / point
unit_id	Geological unit reference
contact_type	Intrusive / fault / stratigraphic etc.
geometry	Spatial geometry
confidence	Interpretation confidence
source_asset_id	Original dataset reference

Table: `geology.structure`

Field	Description
structure_id	Structure identifier
structure_type	Fault / fracture / fold / lineament
strike	Strike in degrees
dip	Dip in degrees
dip_direction	Dip direction
movement_type	Geological movement

Field	Description
confidence	Interpretation confidence
geometry	Spatial geometry

Table: `geology.structural_measurement`

Field	Description
measurement_id	Unique identifier
observation_id	Field observation
easting	X coordinate
northing	Y coordinate
elevation	Z coordinate
strike	Strike
dip	Dip
measurement_type	Bedding / foliation / fracture / fault
measurement_method	Field measurement method

Table: `geology.field_observation`

Field	Description
observation_id	Unique identifier
observation_date	Date
easting	X coordinate
northing	Y coordinate

Field	Description
elevation	Z coordinate
lithology	Observed rock type
alteration	Observed alteration
mineralization	Observed mineralization
vein_density	Vein observation, if measured
description	Field notes
observer	Observer identification

3.5. Geochemical database

Table: `geochemistry.sample`

Field	Description
sample_id	Unique sample identifier
project_id	Project ID
easting	X coordinate
northing	Y coordinate
elevation	Ground elevation
sample_depth	Sampling depth
sample_type	Soil / rock / stream sediment
soil_type	Soil classification
sampling_date	Date

Field	Description
laboratory	Laboratory
sample_method	Sampling method
quality_flag	Quality status

Table: geochemistry.assay_result

This table shall store elemental analyses in a normalized long format.

Field	Description
result_id	Unique result identifier
sample_id	Sample reference
analyte	Cu / Mo / Au / Ag / As / Sb etc.
result_value	Analytical result
unit	ppm / ppb / percent
detection_limit	Analytical detection limit
qualifier	Below detection / above range / normal
analytical_method	ICP-MS / ICP-OES / XRF etc.
laboratory_batch	Batch identifier

Table: geochemistry.soil_physics

Field	Description
measurement_id	Unique identifier
sample_id	Sample reference

Field	Description
ionic_conductivity	Measured conductivity
conductivity_unit	mS/m or other actual unit
measurement_temperature	Temperature
moisture_content	Moisture
soil_ph	pH, if measured
measurement_frequency	Instrument frequency, if applicable
instrument_id	Instrument
measurement_method	Measurement method

Important: Ionic conductivity shall be treated as a soil physical-chemical indicator. It shall not be considered equivalent to copper concentration or a direct subsurface conductivity measurement.

The original measurement unit and instrument calibration information must be retained.

3.6. Geophysical database

Table: `geophysics.survey`

Field	Description
survey_id	Survey identifier
project_id	Project reference
method	Magnetic / gamma / MT / ERT / gravity
survey_name	Survey name
acquisition_date	Survey date
survey_operator	Organization

Field	Description
coordinate_system	Original CRS
processing_method	Processing description
quality_status	Quality assessment

Table: geophysics.station

Field	Description
station_id	Station identifier
survey_id	Survey reference
easting	X coordinate
northing	Y coordinate
elevation	Z coordinate
line_id	Survey line
station_number	Station number

Table: geophysics.observation

This table shall store point measurements from magnetic, gamma-ray and gravity surveys.

Field	Description
observation_id	Unique identifier
station_id	Station reference
parameter_code	TMI / RTP / K / eU / eTh / Bouguer etc.
value	Measured or processed value

Field	Description
unit	Original unit
processing_level	Raw / corrected / derived
uncertainty	Measurement uncertainty

Table: `geophysics.mt_spectrum`

Field	Description
mt_id	Unique identifier
station_id	Station reference
frequency_hz	Frequency
apparent_resistivity	Apparent resistivity
phase	Phase angle
impedance_component	Tensor component
error_estimate	Measurement error
processing_level	Raw / processed

Where the full impedance tensor is available, its individual components shall be retained rather than reducing all MT information to one resistivity value.

Table: `geophysics.ert_measurement`

Field	Description
ert_id	Unique identifier
survey_id	Survey reference
line_id	Survey line

Field	Description
electrode_a	Electrode A
electrode_b	Electrode B
electrode_m	Potential electrode M
electrode_n	Potential electrode N
apparent_resistivity	Measured apparent resistivity
geometric_factor	Geometric factor
quality_flag	Measurement quality

Table: `geophysics.inversion_run`

Field	Description
inversion_id	Unique identifier
survey_id	Original survey
inversion_method	Inversion algorithm
inversion_dimension	2D or 3D
software	Software used
model_date	Processing date
regularization	Inversion parameters
rms_error	RMS misfit
model_status	Accepted / preliminary

Table: `geophysics.inversion_cell`

Field	Description
cell_id	Unique identifier
inversion_id	Inversion reference
x_center	Cell center X
y_center	Cell center Y
z_center	Cell center Z
dx	Cell width
dy	Cell length
dz	Cell height
property_code	Resistivity / susceptibility / density
property_value	Inversion result
uncertainty	Model uncertainty
sensitivity	Sensitivity or resolution indicator

The inversion-cell table shall be used to represent depth-dependent geophysical properties.

For large inversion models, the actual cell arrays should be stored in scientific files, while the database stores their metadata and file references.

3.7. Remote sensing database

Table: `remote_sensing.raster_asset`

Field	Description
raster_id	Unique identifier
project_id	Project reference

Field	Description
sensor	Satellite or airborne sensor
acquisition_date	Image date
band_name	Band name
raster_path	File path
spatial_resolution	Pixel size
coordinate_system	CRS
processing_level	Original / corrected / derived
cloud_percentage	Cloud coverage
quality_flag	Quality status

Derived raster features should include:

- Iron oxide indices.
- Clay alteration indices.
- Hydrothermal alteration indicators.
- Lithological classification.
- Lineament density.
- Digital elevation model.
- Slope and terrain derivatives.

3.8. Drill-hole database

The drill-hole database shall follow the standard separation of collar, survey, lithology and assay information.

Table: `drilling.collar`

Field	Description
hole_id	Primary drill-hole ID
project_id	Project reference
easting	Collar X
northing	Collar Y
elevation	Collar RL
total_depth	Final hole depth
azimuth_initial	Initial azimuth
dip_initial	Initial dip
drilling_method	Drilling method
hole_status	Completed / abandoned etc.

Table: `drilling.survey_station`

Field	Description
survey_id	Unique identifier
hole_id	Drill-hole reference
measured_depth	Measured depth
azimuth	Azimuth
dip	Dip
survey_method	Survey instrument
quality_flag	Survey quality

Table: `drilling.lithology_interval`

Field	Description
interval_id	Unique identifier
hole_id	Drill-hole reference
from_m	Starting measured depth
to_m	Ending measured depth
lithology_code	Lithology
geological_domain	Geological domain
confidence	Logging confidence
description	Geological description

Table: `drilling.alteration_interval`

Field	Description
alteration_id	Unique identifier
hole_id	Drill-hole reference
from_m	Starting depth
to_m	Ending depth
alteration_type	Potassic / phyllic / propylitic etc.
alteration_intensity	Weak / moderate / strong
mineral_assemblage	Observed minerals
confidence	Confidence

Table: `drilling.assay_interval`

Field	Description
assay_interval_id	Unique identifier
hole_id	Drill-hole reference
sample_id	Laboratory sample
from_m	Starting depth
to_m	Ending depth
qa_status	QA/QC status

Table: `drilling.assay_value`

Field	Description
assay_value_id	Unique identifier
assay_interval_id	Assay interval reference
analyte	Cu / Mo / Au etc.
result_value	Analytical value
unit	ppm / percent
detection_limit	Detection limit
qualifier	Analytical qualifier

Table: `drilling.density_interval`

Field	Description
density_id	Unique identifier

Field	Description
hole_id	Drill-hole reference
from_m	Starting depth
to_m	Ending depth
density_value	Measured density
density_unit	g/cm³ or equivalent
measurement_method	Laboratory method

Table: `drilling.qaqc`

Field	Description
qaqc_id	Unique identifier
sample_id	Sample reference
control_type	Blank / standard / duplicate
expected_value	Reference value
measured_value	Result
acceptance_status	Pass / fail
comments	QA/QC remarks

3.9. Derived geological database

Table: `derived.desurveyed_point`

Field	Description
hole_id	Drill-hole ID

Field	Description
measured_depth	Measured depth
x	Desurveyed X
y	Desurveyed Y
z	Desurveyed elevation
interpolation_method	Minimum curvature
calculation_version	Algorithm version

Table: `derived.desurveyed_interval`

Field	Description
hole_id	Drill-hole ID
from_m	Starting depth
to_m	Ending depth
x_start	Starting X
y_start	Starting Y
z_start	Starting elevation
x_end	Ending X
y_end	Ending Y
z_end	Ending elevation
x_mid	Midpoint X
y_mid	Midpoint Y

Field	Description
z_mid	Midpoint elevation

Table: `derived.composite`

Field	Description
composite_id	Unique identifier
hole_id	Drill-hole ID
from_m	Composite start
to_m	Composite end
x_mid	Midpoint X
y_mid	Midpoint Y
z_mid	Midpoint elevation
geological_domain	Domain
alteration_code	Alteration
Cu_pct	Copper grade
Mo_ppm	Molybdenum grade
Au_ppm	Gold grade
sample_coverage	Valid composite coverage

The composite table shall also retain the original sample identifiers or a linked provenance table.

3.10. AI and modelling database

Table: `modelling.model_run`

Field	Description
model_id	Unique identifier
project_id	Project
model_type	Classification / regression / geological model
algorithm	RF / XGBoost / kriging etc.
target_variable	Mineralization / Cu grade
training_dataset_version	Dataset version
parameter_json	Model parameters
validation_method	Spatial CV / grouped CV
performance_json	Performance statistics
model_file	Saved model location
creation_date	Date
model_status	Experimental / validated

Table: `modelling.feature_definition`

Field	Description
feature_id	Unique feature identifier
feature_name	Feature name
data_source	Original data source
property_type	Continuous / categorical
spatial_support	Surface / 2D / 3D / point

Field	Description
unit	Measurement unit
generation_method	Interpolation / derivative / distance
uncertainty_method	Uncertainty definition

Table: `modelling.training_observation`

Field	Description
observation_id	Unique identifier
hole_id	Drill-hole group
x	X coordinate
y	Y coordinate
z	Z coordinate
target_label	1 / 0 / unknown
target_grade	Cu grade
label_source	Assay / geology / interpretation
dataset_version	Training dataset version

Table: `modelling.prediction_asset`

Field	Description
prediction_id	Unique identifier
model_id	Model reference
output_type	Prospectivity / grade / uncertainty

Field	Description
output_path	VTK or other file
grid_definition	Grid reference
prediction_date	Date
interpretation_status	Preliminary / reviewed

3.11. Database implementation SQL

The following SQL establishes the principal database schemas and core tables. The complete implementation shall use the field definitions in Sections 3.3–3.10.

```sql
CREATE EXTENSION IF NOT EXISTS postgis;

CREATE SCHEMA IF NOT EXISTS core;
CREATE SCHEMA IF NOT EXISTS geology;
CREATE SCHEMA IF NOT EXISTS geochemistry;
CREATE SCHEMA IF NOT EXISTS geophysics;
CREATE SCHEMA IF NOT EXISTS remote_sensing;
CREATE SCHEMA IF NOT EXISTS drilling;
CREATE SCHEMA IF NOT EXISTS derived;
CREATE SCHEMA IF NOT EXISTS modelling;

CREATE TABLE core.project (
    project_id UUID PRIMARY KEY,
    project_name VARCHAR(200) NOT NULL,
    target_deposit VARCHAR(100),
    target_commodity VARCHAR(50),
    horizontal_srid INTEGER NOT NULL,
    vertical_datum VARCHAR(100),
    z_convention VARCHAR(30)
        DEFAULT 'elevation_positive_up',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE core.data_asset (
    asset_id UUID PRIMARY KEY,
    project_id UUID NOT NULL
        REFERENCES core.project(project_id),
    asset_name VARCHAR(200) NOT NULL,
    asset_type VARCHAR(100),
```

```
    file_path TEXT NOT NULL,
    file_format VARCHAR(30),
    checksum_sha256 VARCHAR(64),
    processing_status VARCHAR(30),
    acquisition_date DATE
);

CREATE TABLE drilling.collar (
    hole_id VARCHAR(50) PRIMARY KEY,
    project_id UUID NOT NULL
        REFERENCES core.project(project_id),
    easting DOUBLE PRECISION NOT NULL,
    northing DOUBLE PRECISION NOT NULL,
    elevation DOUBLE PRECISION NOT NULL,
    total_depth DOUBLE PRECISION NOT NULL,
    CHECK (total_depth > 0)
);

CREATE TABLE drilling.survey_station (
    survey_id BIGSERIAL PRIMARY KEY,
    hole_id VARCHAR(50) NOT NULL
        REFERENCES drilling.collar(hole_id),
    measured_depth DOUBLE PRECISION NOT NULL,
    azimuth DOUBLE PRECISION NOT NULL,
    dip DOUBLE PRECISION NOT NULL,
    UNIQUE (hole_id, measured_depth),
    CHECK (measured_depth >= 0)
);

CREATE TABLE drilling.lithology_interval (
    interval_id BIGSERIAL PRIMARY KEY,
    hole_id VARCHAR(50) NOT NULL
        REFERENCES drilling.collar(hole_id),
    from_m DOUBLE PRECISION NOT NULL,
    to_m DOUBLE PRECISION NOT NULL,
    lithology_code VARCHAR(50),
    geological_domain VARCHAR(100),
    CHECK (to_m > from_m)
);

CREATE TABLE drilling.assay_interval (
    assay_interval_id BIGSERIAL PRIMARY KEY,
    hole_id VARCHAR(50) NOT NULL
        REFERENCES drilling.collar(hole_id),
    sample_id VARCHAR(100),
    from_m DOUBLE PRECISION NOT NULL,
    to_m DOUBLE PRECISION NOT NULL,
    qa_status VARCHAR(30),
    CHECK (to_m > from_m)
```

```
);

CREATE TABLE drilling.assay_value (
    assay_value_id BIGSERIAL PRIMARY KEY,
    assay_interval_id BIGINT NOT NULL
        REFERENCES drilling.assay_interval(assay_interval_id),
    analyte VARCHAR(30) NOT NULL,
    result_value DOUBLE PRECISION,
    unit VARCHAR(30),
    detection_limit DOUBLE PRECISION,
    qualifier VARCHAR(30)
);

CREATE TABLE derived.desurveyed_interval (
    interval_id BIGSERIAL PRIMARY KEY,
    hole_id VARCHAR(50) NOT NULL,
    from_m DOUBLE PRECISION,
    to_m DOUBLE PRECISION,
    x_mid DOUBLE PRECISION,
    y_mid DOUBLE PRECISION,
    z_mid DOUBLE PRECISION,
    calculation_version VARCHAR(30)
);

CREATE TABLE modelling.model_run (
    model_id UUID PRIMARY KEY,
    project_id UUID NOT NULL
        REFERENCES core.project(project_id),
    model_type VARCHAR(50),
    algorithm VARCHAR(100),
    target_variable VARCHAR(100),
    training_dataset_version VARCHAR(100),
    parameter_json JSONB,
    performance_json JSONB,
    model_file TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE modelling.prediction_asset (
    prediction_id UUID PRIMARY KEY,
    model_id UUID NOT NULL
        REFERENCES modelling.model_run(model_id),
    output_type VARCHAR(100),
    output_path TEXT NOT NULL,
    grid_definition JSONB,
    prediction_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Additional tables shall be implemented according to the preceding database specifications.

Database validation shall include foreign-key checks, interval overlap detection, duplicate sample detection, missing coordinates and invalid measurement units.

SECTION 4. DRILL-HOLE DESURVEYING AND ASSAY COMPOSITING

4.1. General requirements

The system shall transform the drill-hole collar and survey information into three-dimensional borehole trajectories.

The minimum-curvature method shall be the default desurveying method.

It is appropriate for calculating borehole trajectories from successive measured-depth, azimuth and inclination observations.

The system shall support other desurveying methods only when specifically required by the survey data or institutional procedures.

4.2. Coordinate conventions

The system shall use:

- X: Easting.
- Y: Northing.
- Z: Elevation, positive upward.
- Azimuth: Clockwise from geographic or grid north, as specified.
- Measured depth: Positive downhole.
- Dip: Negative downward from horizontal, if that is the original survey convention.

If the survey uses inclination measured from vertical, the conversion shall be performed before processing.

The original survey convention shall always be recorded.

4.3. Minimum-curvature algorithm

For two consecutive survey stations, calculate the dogleg angle:

[\cos D = \cos I_1\cos I_2+ \sin I_1\sin I_2\cos(A_2-A_1)]

where:

- (I_1,I_2) are inclinations measured from vertical.
- (A_1,A_2) are azimuths.
- (D) is the dogleg angle.

The ratio factor is:

[RF=\frac{2}{D}\tan\left(\frac{D}{2}\right)]

For very small dogleg angles:

[RF\approx 1]

For measured-depth difference (\Delta MD):

[\Delta N = \frac{\Delta MD}{2}RF (\sin I_1\cos A_1+\sin I_2\cos A_2)]

[\Delta E = \frac{\Delta MD}{2}RF (\sin I_1\sin A_1+\sin I_2\sin A_2)]

[\Delta TVD = \frac{\Delta MD}{2}RF (\cos I_1+\cos I_2)]

The coordinates are updated as:

[X_2=X_1+\Delta E]

[Y_2=Y_1+\Delta N]

[Z_2=Z_1-\Delta TVD]

The final equation assumes elevation-positive-up coordinates.

4.4. Python implementation

```python
import numpy as np
import pandas as pd


def minimum_curvature_desurvey(
    collar,
    survey,
    step_m=1.0
):
    """
    Minimum-curvature borehole desurveying.
```

```
    collar:
        x, y, z

    survey:
        md, azimuth_deg, dip_deg

    Convention:
        Dip is negative downward from horizontal.
        Azimuth is clockwise from north.
        Z is elevation positive upward.
    """

    s = survey.sort_values("md").copy()

    if s["md"].duplicated().any():
        raise ValueError("Duplicate survey depths.")

    if s["md"].iloc[0] != 0:
        raise ValueError(
            "A survey station at MD=0 is required."
        )

    md_s = s["md"].to_numpy(dtype=float)

    if np.any(np.diff(md_s) <= 0):
        raise ValueError("Invalid survey depths.")

    # Convert dip to inclination from vertical.
    inc_s = np.deg2rad(
        90.0 + s["dip_deg"].to_numpy(dtype=float)
    )

    # Handle azimuth crossing 360/0 degrees.
    azi_s = np.unwrap(
        np.deg2rad(
            s["azimuth_deg"].to_numpy(dtype=float)
        )
    )

    max_md = md_s[-1]

    # Densify the trajectory.
    md = np.unique(
        np.concatenate([
            np.arange(0, max_md, step_m),
            md_s,
            [max_md]
        ])
    )
```

```
    inc = np.interp(md, md_s, inc_s)
    azi = np.interp(md, md_s, azi_s)

    x = np.zeros(len(md))
    y = np.zeros(len(md))
    z = np.zeros(len(md))

    x[0] = collar["x"]
    y[0] = collar["y"]
    z[0] = collar["z"]

    for k in range(1, len(md)):

        dmd = md[k] - md[k-1]

        i1, i2 = inc[k-1], inc[k]
        a1, a2 = azi[k-1], azi[k]

        cos_dl = (
            np.cos(i1) * np.cos(i2)
            +
            np.sin(i1) * np.sin(i2)
            * np.cos(a2 - a1)
        )

        dogleg = np.arccos(
            np.clip(cos_dl, -1.0, 1.0)
        )

        if dogleg < 1e-8:
            rf = 1.0
        else:
            rf = (
                2.0 * np.tan(dogleg / 2.0)
                / dogleg
            )

        dE = (
            dmd / 2.0
            * rf
            * (
                np.sin(i1) * np.sin(a1)
                +
                np.sin(i2) * np.sin(a2)
            )
        )

        dN = (
```

```
            dmd / 2.0
            * rf
            * (
                np.sin(i1) * np.cos(a1)
                +
                np.sin(i2) * np.cos(a2)
            )
        )

        dTVD = (
            dmd / 2.0
            * rf
            * (
                np.cos(i1) + np.cos(i2)
            )
        )

        x[k] = x[k-1] + dE
        y[k] = y[k-1] + dN
        z[k] = z[k-1] - dTVD

    return pd.DataFrame({
        "md": md,
        "x": x,
        "y": y,
        "z": z
    })
```

The implementation shall also check that the survey trajectory does not exceed the recorded total depth and that all input values fall within valid angular ranges.

The original survey records shall remain unchanged.

4.5. Desurveyed assay intervals

For each assay interval:

[MD_{mid}=\frac{MD_{from}+MD_{to}}{2}]

The coordinates shall be obtained from the desurveyed trajectory.

The system shall retain:

- Starting coordinates.
- Ending coordinates.
- Midpoint coordinates.
- Measured depth.

- Interval length.
- Geological domain.
- Assay values.

A sufficiently dense trajectory shall be generated so that interpolation of interval coordinates does not introduce unacceptable positional errors.

4.6. Assay compositing

The initial recommended composite lengths are 1 m and 2 m for comparison.

The final composite length shall be selected according to:

- Original assay interval lengths.
- Geological continuity.
- Expected mining or estimation support.
- Drill-hole density.
- Mineralization domain characteristics.

Composites shall not cross major geological or mineralization domain boundaries.

For a particular element:

[G_c= \frac{\sum_{i=1}^{n}G_iL_i} {\sum_{i=1}^{n}L_i}]

where:

- (G_i) is the assay grade.
- (L_i) is the contributing interval length.
- (G_c) is the composite grade.

The system shall calculate composite coverage:

[C= \frac{\text{Valid assayed length}} {\text{Target composite length}}]

A recommended initial acceptance threshold is 80%, subject to adjustment according to the original sampling protocol.

Intervals with inadequate assay coverage shall be flagged rather than automatically assigned zero grade.

4.7. Assay quality control

Before compositing, the system shall perform:

- Duplicate interval detection.
- Overlapping interval detection.
- Missing interval identification.
- Below-detection-limit assessment.
- Laboratory QA/QC review.
- Unit harmonization.
- Outlier identification.

Below-detection-limit values shall be treated as censored analytical observations, not automatically as zero.

SECTION 5. 3D GEOLOGICAL FRAMEWORK MODELLING

5.1. Objective

To develop a three-dimensional geological model representing the distribution of geological units, intrusive bodies, faults, structural boundaries and hydrothermal alteration zones.

5.2. Software

Primary: GemPy.

Alternative or complementary: LoopStructural.

Visualization: PyVista.

GemPy shall be used to construct implicit geological surfaces from geological contact points and structural orientation information.

5.3. Input data

- Geological map boundaries.
- Ground survey observations.
- Lithological contacts.
- Structural measurements.
- Fault traces.
- Drill-hole lithology intervals.
- Intrusive contacts.

- Alteration observations.

5.4. Proposed geological model

The initial porphyry copper model shall include:

1. Regional host-rock units.
2. Porphyritic intrusive complex.
3. Major faults.
4. Secondary structural features.
5. Intrusive contacts.
6. Mineralization-related alteration domains.
7. Known mineralized drill-hole intervals.

The model shall distinguish observed geological boundaries from inferred boundaries.

5.5. Geological model validation

The model shall be evaluated against:

- Surface geological mapping.
- Drill-hole lithological logs.
- Structural measurements.
- Geological cross-sections.
- Known intrusive boundaries.
- Independent geological interpretation.

The model shall not be treated as accurate merely because the interpolation algorithm converges.

Geological plausibility and data support must be reviewed by the exploration geologist.

5.6. Geological model outputs

- Geological unit volumes.
- Fault surfaces.
- Intrusive body surfaces.
- Lithological block model.
- Alteration domain model.

- Geological uncertainty model, where feasible.

Recommended output formats: VTI, VTP and VTU.

SECTION 6. 3D VOXEL CONSTRUCTION AND FEATURE ENGINEERING

6.1. General principle

The exploration area shall be represented by a regular three-dimensional voxel grid.

Each voxel shall contain its spatial coordinates and a set of geological, geochemical and geophysical attributes.

The voxel database shall be the principal input for 3D mineral prospectivity prediction.

6.2. Voxel grid definition

Each grid shall be defined by:

- X origin.
- Y origin.
- Z origin.
- Number of cells in X, Y and Z.
- Cell dimensions.
- Horizontal CRS.
- Vertical datum.
- Grid orientation.
- Geological model version.

The voxel dimensions shall be selected based on the actual data resolution and geological scale.

For the first pilot, the following may be tested:

Parameter	Initial test range
Horizontal X cell size	25–50 m
Horizontal Y cell size	25–50 m

Parameter	Initial test range
Vertical Z cell size	10–25 m

These are initial experimental values, not universal recommended block sizes.

The grid shall not be substantially finer than the resolution of the supporting geophysical inversion and drilling information.

6.3. Feature categories

A. Geological features

Feature	Definition
Lithology_Code	Geological unit classification
Distance_Intrusion	Distance to intrusive contact
Distance_Fault	Distance to major fault
Fault_Density	Local fault density
Fracture_Density	Structural fracture density
Contact_Distance	Distance to favorable lithological contact
Alteration_Code	Alteration classification
Intrusive_Proximity	Distance to interpreted porphyry center

B. Geochemical features

Where available:

- Cu concentration.
- Mo concentration.
- Au concentration.
- Ag concentration.
- As concentration.

- Sb concentration.
- Bi concentration.
- Te concentration.
- W concentration.
- Sn concentration.
- Pb concentration.
- Zn concentration.
- Selected elemental ratios.

Geochemical values shall be subjected to appropriate statistical treatment.

Log transformations and compositional data analysis shall be considered according to the analytical characteristics of the dataset.

C. Magnetic features

- Total magnetic intensity.
- Reduced-to-pole magnetic field.
- Residual magnetic field.
- Analytical signal.
- First vertical derivative.
- Tilt derivative.
- 3D magnetic susceptibility inversion.

D. Gamma-ray spectrometric features

- K concentration.
- Equivalent uranium.
- Equivalent thorium.
- Total count.
- K/eTh ratio.
- eU/eTh ratio.

Ratios shall be calculated only where the denominator is sufficiently reliable.

E. MT features

- Log apparent resistivity.
- MT inversion resistivity.

- Phase.
- Inversion sensitivity.
- Model resolution.
- Depth of investigation.

F. Resistivity CT / ERT features

- Inverted resistivity.
- Log resistivity.
- Depth.
- Inversion sensitivity.
- Distance to survey line.
- Inter-line interpolation confidence.

G. Gravity features

- Bouguer anomaly.
- Regional-residual gravity anomaly.
- Horizontal derivatives.
- Vertical derivatives.
- 3D density inversion.
- Inversion confidence.

H. Remote sensing features

- Iron oxide indices.
- Clay alteration indices.
- Lithological classification.
- Alteration classification.
- Lineament density.
- Digital elevation model derivatives.

6.4. Important treatment of surface and subsurface information

The system shall distinguish between:

Surface measurements:

- Soil ionic conductivity.

- Soil geochemistry.
- Satellite imagery.
- Surface gamma-ray spectrometry.
- Surface magnetic measurements.
- Surface gravity measurements.

Depth-dependent information:

- 3D MT inversion.
- 3D ERT inversion.
- 3D magnetic susceptibility inversion.
- 3D density inversion.
- Drill-hole geological observations.

Surface measurements shall not be copied unchanged into every subsurface voxel.

The system shall maintain separate surface feature layers and depth-dependent geological or geophysical features.

Where surface observations are incorporated into a 3D prospectivity model, the spatial relationship must be explicitly justified and documented.

For example, a surface alteration anomaly may be used to constrain a geological interpretation, but it must not automatically be treated as evidence of identical alteration at depth.

6.5. Geophysical data integration

For 3D inversion models, the system shall:

1. Import the original inversion cells.
2. Validate their coordinates.
3. Validate the vertical datum.
4. Identify valid model regions.
5. Import sensitivity and uncertainty information.
6. Resample the model to the common voxel grid.
7. Preserve areas outside the valid inversion domain as missing values.

For 2D geophysical surveys, the system shall preserve the survey-line geometry.

Interpolation between lines shall only be undertaken where the line spacing and geological setting justify it.

The system shall not interpret a 2D resistivity section as a complete 3D volume without adequate supporting information.

6.6. Feature interpolation

The following interpolation methods may be used:

Data type	Proposed processing
Geological contacts	Geological framework modelling
Faults	Structural surface modelling
Lithology	Geological domain assignment
Point geochemical data	2D spatial interpolation
Magnetic surface data	2D raster processing
MT 3D inversion	3D resampling
ERT 2D lines	Line-based section modelling
Gravity 3D inversion	3D resampling
Drill-hole observations	Desurveyed point and interval sampling
Remote sensing	Raster resampling

Every derived feature shall contain a source identifier and an uncertainty or quality indicator.

6.7. Python feature extraction example

The following example samples a 3D inversion volume onto voxel coordinates.

```python
import numpy as np
import pandas as pd
```

```
from scipy.interpolate import RegularGridInterpolator

# Voxel coordinate table
voxels = pd.read_parquet(
    "voxel_coordinates.parquet"
)

# Original inversion volume
# Array order: Z, Y, X
resistivity_3d = np.load(
    "mt_resistivity_3d.npy"
)

# Coordinates of the inversion grid
x_axis = np.load("mt_x.npy")
y_axis = np.load("mt_y.npy")
z_axis = np.load("mt_z.npy")

# Log-transform positive resistivity
log_resistivity = np.log10(
    np.where(
        resistivity_3d > 0,
        resistivity_3d,
        np.nan
    )
)

# Build interpolation function
interpolator = RegularGridInterpolator(
    (z_axis, y_axis, x_axis),
    log_resistivity,
    method="linear",
    bounds_error=False,
    fill_value=np.nan
)

# Sample all voxel centers
query = voxels[
    ["z", "y", "x"]
].to_numpy()

voxels["mt_log_resistivity"] = (
    interpolator(query)
)

voxels.to_parquet(
    "voxel_features.parquet",
    index=False
```

```
)
```

This example assumes a regular inversion grid. The coordinate axes must be strictly increasing and must match the actual array ordering. For irregular inversion cells, spatial cell lookup or interpolation methods appropriate to the inversion mesh shall be used instead.

SECTION 7. AI TRAINING DATASET CONSTRUCTION

7.1. Main training principle

The AI model shall be trained using geological observations and drill-hole evidence.

The drill-hole composites shall provide the principal subsurface training locations.

Each composite shall be associated with the corresponding geological and geophysical features at its desurveyed midpoint.

The system shall not automatically classify every undrilled voxel as barren.

7.2. Training labels

Three data categories shall be maintained.

Category	Label	Meaning
Verified mineralized interval	1	Positive training observation
Verified barren interval	0	Negative training observation
Undrilled or uncertain location	Unknown	Not used as a negative label

The definition of mineralization shall be established by the project geologist.

It may incorporate:

- Copper assay results.
- Geological logging.
- Sulfide mineralogy.
- Alteration intensity.
- Geological domain.

- Target deposit model.

The copper threshold used to define a mineralized interval shall be project-specific.

The threshold shall not automatically be treated as an economic cut-off grade.

7.3. Small-data strategy

Because the project has only 10–30 drill holes, the initial system shall prioritize models that can work with relatively small structured datasets.

The proposed order is:

1. Geological knowledge-based prospectivity model.
2. Random Forest classification.
3. XGBoost classification.
4. Logistic regression as a transparent baseline.
5. Positive-unlabelled learning, where negative observations are insufficient.
6. Advanced deep learning only after substantially more independent training information becomes available.

A large number of assay composites from a small number of drill holes does not constitute an equally large number of independent geological observations.

7.4. Training dataset format

Each training observation shall contain:

Feature	Description
Hole_ID	Drill-hole grouping identifier
X	Midpoint Easting
Y	Midpoint Northing
Z	Midpoint elevation
Lithology	Geological unit
Alteration	Alteration domain

Feature	Description
Distance_Fault	Fault proximity
Distance_Intrusion	Intrusive contact proximity
Magnetic_3D	Magnetic inversion attribute
Gravity_3D	Density inversion attribute
MT_3D	MT inversion resistivity
ERT_3D	ERT inversion attribute
Gamma_Features	Relevant radiometric attributes
Geochemical_Features	Available geochemical attributes
Ore_Flag	Training label
Cu_Grade	Copper assay composite

Features shall be included only where their measurement support and quality are appropriate.

SECTION 8. AI MODEL DEVELOPMENT AND VALIDATION

8.1. Initial algorithms

Model 1: Random Forest

Purpose:

- Mineralization classification.
- Nonlinear feature relationships.
- Feature importance assessment.

Model 2: XGBoost

Purpose:

- Nonlinear geological prediction.
- Multisource feature integration.
- Mineral prospectivity scoring.

Model 3: Logistic Regression

Purpose:

- Baseline statistical comparison.
- Transparent geological relationships.

Model 4: Explainable AI

Purpose:

- SHAP-based interpretation.
- Identification of important predictors.
- Assessment of individual prediction explanations.

Model 5: Advanced models

The following may be developed in later stages:

- 3D CNN.
- Autoencoders.
- Graph Neural Networks.
- Physics-informed neural networks.

These shall not be required for the first operational version.

8.2. Initial Random Forest implementation

```python
import pandas as pd
import numpy as np
import joblib

from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import StratifiedGroupKFold
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
```

```
from sklearn.metrics import (
    roc_auc_score,
    average_precision_score
)

# Load composite training dataset
df = pd.read_parquet(
    "training_composites.parquet"
)

# Verified training observations only
df = df[
    df["ore_flag"].isin([0, 1])
].copy()

numeric_features = [
    "distance_fault",
    "distance_intrusion",
    "magnetic_3d",
    "gravity_3d",
    "mt_log_resistivity",
    "ert_log_resistivity",
    "gamma_k",
    "gamma_uth"
]

categorical_features = [
    "lithology",
    "alteration"
]

features = (
    numeric_features +
    categorical_features
)

X = df[features]
y = df["ore_flag"]
groups = df["hole_id"]

# Data preprocessing
preprocessor = ColumnTransformer([
    (
        "numeric",
        SimpleImputer(
            strategy="median",
            add_indicator=True
        ),
        numeric_features
```

```
    ),
    (
        "categorical",
        Pipeline([
            (
                "imputer",
                SimpleImputer(
                    strategy="most_frequent"
                )
            ),
            (
                "encoder",
                OneHotEncoder(
                    handle_unknown="ignore"
                )
            )
        ]),
        categorical_features
    )
])

# Model
pipeline = Pipeline([
    ("preprocessor", preprocessor),

    ("classifier", RandomForestClassifier(
        n_estimators=500,
        max_depth=8,
        min_samples_leaf=3,
        class_weight="balanced_subsample",
        random_state=42,
        n_jobs=-1
    ))
])

# Grouped cross-validation
n_groups = groups.nunique()

if n_groups < 5:
    raise ValueError(
        "Insufficient drill-hole groups for the "
        "specified five-fold validation design."
    )

cv = StratifiedGroupKFold(
    n_splits=5,
    shuffle=True,
    random_state=42
)
```

```
oof_probability = np.full(
    len(df),
    np.nan
)

for train_idx, test_idx in cv.split(
    X,
    y,
    groups=groups
):

    X_train = X.iloc[train_idx]
    X_test = X.iloc[test_idx]

    y_train = y.iloc[train_idx]

    pipeline.fit(
        X_train,
        y_train
    )

    oof_probability[test_idx] = (
        pipeline.predict_proba(X_test)[:, 1]
    )

valid = np.isfinite(oof_probability)

if y[valid].nunique() == 2:

    print(
        "Grouped ROC-AUC:",
        roc_auc_score(
            y[valid],
            oof_probability[valid]
        )
    )

    print(
        "Grouped PR-AUC:",
        average_precision_score(
            y[valid],
            oof_probability[valid]
        )
    )

# Fit final model after evaluation
pipeline.fit(X, y)
```

```
joblib.dump(
    pipeline,
    "random_forest_prospectivity.joblib"
)
```

This is a baseline implementation.

The number of folds shall be reduced if the number of independent holes or class distributions do not support five-fold stratified grouping. If a training fold loses one of the classes, the validation design must be revised and documented.

For final research validation, spatially separated test areas and leave-one-hole-out experiments shall also be considered.

The final model shall not be accepted merely because it produces a high random-split accuracy.

8.3. XGBoost implementation

A second model shall be developed using XGBoost.

The model shall use the same training observations and validation partitions as Random Forest to ensure a fair comparison.

Suggested initial parameters:

Parameter	Initial value
n_estimators	300
max_depth	3
learning_rate	0.03
subsample	0.8
colsample_bytree	0.8
reg_lambda	5
random_state	42

These parameters are starting values and shall be tuned only within the training folds.

8.4. Model evaluation

The following indicators shall be calculated:

- ROC-AUC.
- Precision-recall AUC.
- Precision.
- Recall.
- F1-score.
- Confusion matrix.
- Top-ranked target capture rate.
- Probability calibration, where justified.

The model shall also be compared against a geological knowledge-based baseline.

8.5. Model uncertainty

The system shall generate additional uncertainty indicators.

These shall include:

1. Model disagreement.
2. Bootstrap prediction variability.
3. Distance from known drill-hole observations.
4. Missing feature coverage.
5. Geophysical inversion sensitivity.
6. Geological model uncertainty.

A prediction with a high prospectivity score but little supporting data shall be flagged as requiring further investigation.

8.6. Explainable AI

The software shall include SHAP or equivalent feature interpretation.

For example, the system should be able to indicate whether a predicted prospective voxel is associated with:

- Proximity to an intrusive contact.
- Favorable host lithology.
- A fault intersection.

- A particular magnetic signature.
- A resistivity anomaly.
- Geochemical evidence.

Feature importance shall be interpreted as model association, not proof of geological causation.

SECTION 9. 3D MINERAL PROSPECTIVITY PREDICTION

9.1. Prediction process

The trained AI model shall be applied to the complete voxel feature database.

For each voxel, the model shall generate a prospectivity score.

Where probability calibration has been independently validated, the output may be displayed as a calibrated probability.

Otherwise, the system shall identify the result as an uncalibrated model score.

9.2. Prediction outputs

Output	Description
Prospectivity score	AI model output
Calibrated probability	Probability, if validated
Model uncertainty	Prediction variability
Geological domain	Host-rock classification
Data coverage	Available predictor information
Nearest drill distance	Distance to nearest drill-hole data
Target rank	Relative exploration ranking

9.3. Prediction grid

The system shall produce a regular 3D model containing:

- X, Y and Z coordinates.
- Voxel dimensions.
- Geological domain.
- Prospectivity score.
- Model uncertainty.
- Data coverage.

The primary output shall be a VTK ImageData file (.vti).

SECTION 10. 3D OREBODY EXTRACTION AND MODELLING

10.1. Scientific distinction

The system shall distinguish between:

High-prospectivity volume: A volume predicted by AI to have favorable mineralization characteristics.

Interpreted mineralized body: A geological solid supported by geological interpretation, drilling and assay evidence.

These terms shall not be used interchangeably.

10.2. Initial orebody extraction procedure

1. Load the 3D prospectivity volume.
2. Apply a selected prospectivity threshold.
3. Apply geological domain constraints.
4. Identify connected high-prospectivity volumes.
5. Remove isolated computational noise where justified.
6. Extract the external surface.
7. Compare the result with known drill-hole intersections.
8. Review the resulting interpretation.
9. Export the surface model.

The prospectivity threshold shall be selected through model validation and geological interpretation.

A fixed threshold of 0.70 shall not be used automatically.

10.3. Connected-volume extraction

</> Python

```python
import numpy as np
import pandas as pd

from scipy import ndimage
import pyvista as pv

# Load predicted voxel table
df = pd.read_parquet(
    "3d_prospectivity_results.parquet"
)

nx = int(df["i"].max()) + 1
ny = int(df["j"].max()) + 1
nz = int(df["k"].max()) + 1

# Construct 3D prospectivity array
p = np.full(
    (nz, ny, nx),
    np.nan,
    dtype=np.float32
)

p[
    df["k"].to_numpy(),
    df["j"].to_numpy(),
    df["i"].to_numpy()
] = df[
    "prospectivity_score"
].to_numpy()

# Example threshold
threshold = 0.70

# Geological domain constraint
# Replace with the actual geological mask.
geological_mask = np.isfinite(p)

mask = (
    (p >= threshold)
    &
    geological_mask
)
```

```
# Identify connected 3D components
structure = np.ones(
    (3, 3, 3),
    dtype=np.uint8
)

labels, number = ndimage.label(
    mask,
    structure=structure
)

# Remove small isolated components
counts = np.bincount(
    labels.ravel()
)

minimum_voxels = 20

keep = counts >= minimum_voxels
keep[0] = False

filtered = keep[labels]

# Export as a 3D volume
grid = pv.ImageData(
    dimensions=(nx + 1, ny + 1, nz + 1),
    spacing=(25, 25, 10),
    origin=(500000, 7500000, -500)
)

grid.cell_data["prospectivity_score"] = (
    np.nan_to_num(p, nan=-1.0).ravel(order="C")
)

grid.cell_data["high_prospectivity"] = (
    filtered.astype(np.uint8).ravel(order="C")
)

grid.save(
    "prospectivity_volume.vti"
)

# Create a high-prospectivity surface
point_grid = grid.cell_data_to_point_data()

surface = point_grid.contour(
    isosurfaces=[threshold],
    scalars="prospectivity_score"
```

```
)

if surface.n_points > 0:

    surface.save(
        "prospectivity_surface.vtp"
    )
```

This code is a prototype and assumes a complete regular grid with the stated origin and spacing, and consistent index ordering. The geological mask must be constructed from the actual geological framework. The thresholded volume is an exploration prospectivity envelope, not a confirmed orebody.

10.4. Geological constraints

The interpreted orebody shall be constrained by:

- Mineralized drill-hole intervals.
- Geological units.
- Intrusive contacts.
- Fault systems.
- Alteration zones.
- Copper assay distribution.
- Geological continuity.

The system shall permit manual geological editing and interpretation of the final mineralized solid.

The software shall preserve both the original AI prediction and the geologist-reviewed interpretation.

10.5. Final orebody outputs

- Orebody surface mesh.
- Orebody volume.
- Geological domain identification.
- Drill-hole intersection information.
- Prospectivity score.
- Interpretation confidence.
- Model version.

SECTION 11. 3D COPPER GRADE ESTIMATION

11.1. General principle

Copper grade estimation shall be performed independently of mineral prospectivity classification.

The principal estimation data shall be desurveyed and composited drill-hole assays.

The initial software shall compare conventional geostatistics and AI regression.

11.2. Grade estimation procedures

The workflow shall be:

1. Validate assay information.
2. Complete desurveying.
3. Complete compositing.
4. Define geological domains.
5. Perform exploratory data analysis.
6. Evaluate grade distributions.
7. Assess outliers and top-cut requirements.
8. Analyze spatial continuity.
9. Construct directional variograms where justified.
10. Select an estimation method.
11. Estimate grades within appropriate domains.
12. Validate against independent drill holes.
13. Export the grade block model.

11.3. Proposed estimation methods

Method	Purpose
Nearest Neighbor	Baseline estimate
Inverse Distance Weighting	Baseline interpolation
Ordinary Kriging	Geostatistical estimate

Method	Purpose
Indicator Kriging	Threshold probability
Random Forest Regression	AI grade prediction
XGBoost Regression	AI grade prediction
Regression Kriging	Hybrid estimation
Conditional Simulation	Advanced uncertainty analysis

11.4. Special consideration for 10–30 drill holes

The limited drilling information is a significant constraint.

A large number of composite samples does not automatically provide sufficient independent spatial information to estimate a reliable three-dimensional variogram.

The faculty shall therefore:

- Examine the spatial distribution of the drill holes.
- Assess whether sufficient directional sample pairs exist.
- Compare several variogram models.
- Avoid fitting complex anisotropy models without evidence.
- Use simple baseline estimates where variogram modelling is unstable.
- Restrict extrapolation beyond geological and drilling support.
- Display uncertainty and distance to supporting drill holes.

Ordinary kriging shall only be accepted as a defensible estimation method where the geological and statistical assumptions are adequately supported.

11.5. Ordinary Kriging implementation

PyKrige provides a three-dimensional ordinary kriging implementation.

The following is an example of how the estimation module shall be structured.

```
</> Python
```

```
import numpy as np
import pandas as pd

from pykrige.ok3d import OrdinaryKriging3D

# Load composited drill-hole assays
samples = pd.read_parquet(
    "copper_composites.parquet"
)

# Select one geological domain
samples = samples[
    samples["domain"] == "PRIMARY_CU"
].dropna(
    subset=["x", "y", "z", "Cu_pct"]
)

# Drill-hole sample coordinates
x = samples["x"].to_numpy()
y = samples["y"].to_numpy()
z = samples["z"].to_numpy()

grade = samples["Cu_pct"].to_numpy()

# Load block centers
blocks = pd.read_parquet(
    "estimation_blocks.parquet"
)

# Parameters must come from an approved variogram fit.
validated_variogram = {
    "sill": VALIDATED_SILL,
    "range": VALIDATED_RANGE,
    "nugget": VALIDATED_NUGGET
}

OK = OrdinaryKriging3D(
    x,
    y,
    z,
    grade,
    variogram_model="spherical",
    variogram_parameters=validated_variogram,
    pseudo_inv=True
)

# Estimate only the permitted estimation blocks
estimated, variance = OK.execute(
    "points",
```

```
    blocks["x"].to_numpy(),
    blocks["y"].to_numpy(),
    blocks["z"].to_numpy(),
    backend="loop",
    n_closest_points=16
)

blocks["Cu_estimated"] = np.asarray(
    estimated
)

blocks["Kriging_variance"] = np.asarray(
    variance
)

blocks.to_parquet(
    "copper_grade_estimation.parquet",
    index=False
)
```

`VALIDATED_SILL`, `VALIDATED_RANGE` and `VALIDATED_NUGGET` are placeholders and must be replaced with parameters obtained from an approved variogram analysis. They must never be assigned arbitrary values merely to execute the code.

The number of neighboring samples shall be determined by the estimation search strategy.

11.6. Hybrid AI-geostatistical estimation

Following successful baseline estimation, develop regression kriging.

The general formulation is:

[\hat{G}(x)= \hat{G}*{AI}(x)+ \hat{R}*{K}(x)]

where:

- (\hat{G}_{AI}(x)) is the AI-estimated grade trend.
- (\hat{R}_{K}(x)) is the kriged residual correction.

The AI regression model may incorporate geological, geophysical and structural information.

However, the model shall be trained using only data that would be available at the prediction location without using its own assay result as a predictor.

11.7. Grade block model structure

Each block shall contain:

Field	Description
Block_ID	Unique block identifier
X	Block center X
Y	Block center Y
Z	Block center Z
DX	Block width
DY	Block length
DZ	Block height
Domain	Geological domain
Cu_Estimate	Estimated Cu grade
Mo_Estimate	Estimated Mo grade
Density	Measured or approved density
Estimation_Method	Kriging / IDW / AI etc.
Number_Samples	Number of informing samples
Nearest_Hole_Distance	Distance to nearest drill hole
Estimation_Variance	Estimation variance
Model_Confidence	Model confidence indicator

The block dimensions shall be based on drill spacing, geological continuity, estimation support and intended application.

The system shall support regular blocks initially and sub-blocking in a later version.

11.8. Mineralized volume and tonnage

Where measured density information is available:

[V=\sum_{i=1}^{n}V_i]

[T=\sum_{i=1}^{n}V_i\rho_i]

where:

- (V_i) is block volume.
- (\rho_i) is bulk density.
- (T) is total estimated tonnage.

The system may calculate preliminary mineralized volume and illustrative tonnage statistics.

However, these calculations shall not be presented as Mineral Resource or Mineral Reserve estimates without the required geological, geostatistical, QA/QC, density, economic and professional reporting procedures.

SECTION 12. PYVISTA 3D VISUALIZATION ENGINE

12.1. General requirements

PyVista shall be the principal scientific visualization library.

The 3D viewer shall support:

- Geological surfaces.
- Intrusive bodies.
- Fault surfaces.
- Drill-hole trajectories.
- Prospectivity voxel volumes.
- Mineralized surfaces.
- Copper grade block models.
- Geological cross-sections.
- Clipping and slicing.
- Attribute inspection.
- Export of 3D models.

12.2. Main viewer components

The interface shall contain:

Scene control

- Display and hide geological objects.
- Change object transparency.
- Change color schemes.
- Set vertical exaggeration.
- Set camera direction.

Data control

- Select prospectivity score.
- Select copper grade.
- Select geological domain.
- Select uncertainty.
- Select individual geophysical properties.

Analysis control

- Create horizontal sections.
- Create vertical sections.
- Clip geological volumes.
- Extract selected blocks.
- Inspect individual voxel values.
- Display drill-hole intersections.

12.3. Basic 3D visualization code

```python
import pyvista as pv

# Read prospectivity volume
volume = pv.read(
    "prospectivity_volume.vti"
)

# Read interpreted mineralized surface
surface = pv.read(
    "prospectivity_surface.vtp"
)
```

```
plotter = pv.Plotter()

plotter.add_volume(
    volume,
    scalars="prospectivity_score",
    cmap="viridis",
    opacity="sigmoid"
)

plotter.add_mesh(
    surface,
    color="orange",
    opacity=0.5
)

plotter.add_axes()
plotter.show_grid()

plotter.show()
```

For large models, the viewer shall provide surface extraction and block filtering rather than rendering every voxel at maximum detail.

12.4. Drill-hole visualization

Drill holes shall be displayed as 3D polylines using desurveyed coordinates.

The display shall support:

- Hole ID labels.
- Assay interval coloring.
- Lithology interval coloring.
- Alteration interval coloring.
- Selected-hole highlighting.
- Intersection with interpreted mineralized volumes.

The system shall allow users to select a drill hole in the 3D viewer and identify its corresponding record in the 2D GIS and database.

12.5. Grade block visualization

The grade block model shall support selectable color attributes.

Examples:

- Cu grade.
- Mo grade.
- Geological domain.
- Estimation variance.
- Nearest drill-hole distance.

The color legend shall include the correct grade units.

The viewer shall support clipping planes and section views to investigate internal grade distribution.

SECTION 13. QGIS PLUGIN DEVELOPMENT

13.1. General concept

The main software interface shall be developed as a QGIS Python plugin named:

AI-3D-MEOMS Plugin

The plugin shall add a dedicated toolbar, dockable control panel and Processing Toolbox algorithms.

13.2. Main interface layout

The proposed interface shall include the following sections.

A. Project Management

- Create project.
- Open project.
- Save project.
- Set coordinate system.
- Define exploration area.
- Configure project directory.

B. Database Management

- Import geological maps.
- Import geochemical samples.
- Import geophysical data.

- Import remote sensing data.
- Import drill-hole data.
- Validate database.

C. Geological Mapping

- Display geological units.
- Display faults and structures.
- Display geochemical maps.
- Display magnetic maps.
- Display gravity maps.
- Display MT and ERT sections.
- Display drill-hole locations.

D. AI Processing

- Prepare training data.
- Select target commodity.
- Select predictive features.
- Select AI algorithm.
- Configure model parameters.
- Train AI model.
- Validate AI model.
- Generate 3D prediction.

E. 3D Modelling

- Open 3D geological model.
- Display drill holes.
- Display prospectivity volume.
- Generate mineralized envelope.
- Open grade block model.
- Display uncertainty.

F. Reporting

- Export 2D maps.
- Export 3D models.

- Export statistical tables.
- Generate technical reports.

13.3. Plugin directory structure

```
</> Plain text

AI3D_MEOMS_PLUGIN/
 |
 ├──── __init__.py
 ├──── metadata.txt
 ├──── plugin.py
 ├──── resources.py
 |
 ├──── icons/
 |
 ├──── ui/
 |     ├──── main_dock.py
 |     ├──── project_dialog.py
 |     ├──── database_dialog.py
 |     ├──── ai_dialog.py
 |     └──── model_dialog.py
 |
 ├──── processing/
 |     ├──── provider.py
 |     ├──── import_data.py
 |     ├──── validate_database.py
 |     ├──── prepare_voxels.py
 |     ├──── train_ai.py
 |     ├──── predict_3d.py
 |     └──── grade_estimation.py
 |
 ├──── bridge/
 |     ├──── engine_process.py
 |     ├──── project_config.py
 |     └──── output_loader.py
 |
 └──── styles/
```

13.4. Plugin metadata

```
</> INI

[general]

name=AI-3D-MEOMS
```

```
qgisMinimumVersion=3.34
description=AI-Based 3D Mineral Exploration
version=1.0.0
author=Faculty of Geoscience and Technology
email=
category=Analysis
hasProcessingProvider=yes
```

The minimum QGIS version shall be adjusted after the final compatibility tests.

13.5. Plugin initialization

```python
def classFactory(iface):

    from .plugin import AI3DMEOMS

    return AI3DMEOMS(iface)
```

13.6. Main plugin class

```python
from qgis.PyQt.QtWidgets import QAction


class AI3DMEOMS:

    def __init__(self, iface):

        self.iface = iface
        self.action = None

    def initGui(self):

        self.action = QAction(
            "AI-3D-MEOMS",
            self.iface.mainWindow()
        )

        self.action.triggered.connect(
            self.open_main_window
        )

        self.iface.addToolBarIcon(
            self.action
```

```
        )

        self.iface.addPluginToMenu(
            "&AI-3D-MEOMS",
            self.action
        )

    def open_main_window(self):

        print(
            "AI-3D-MEOMS interface activated"
        )

    def unload(self):

        if self.action:

            self.iface.removeToolBarIcon(
                self.action
            )

            self.iface.removePluginMenu(
                "&AI-3D-MEOMS",
                self.action
            )
```

The main window shall subsequently be replaced by a dockable interface containing the project and AI controls.

13.7. Integration with the Processing Toolbox

The following processing algorithms shall be registered:

1. Import Geological Data.
2. Import Geochemical Data.
3. Import Geophysical Data.
4. Import Drill-Hole Data.
5. Validate Geological Database.
6. Desurvey Drill Holes.
7. Composite Assay Data.
8. Construct 3D Voxel Grid.
9. Generate 3D Features.
10. Train Mineral Prospectivity Model.

11. Predict 3D Prospectivity.
12. Extract High-Prospectivity Volumes.
13. Estimate Copper Grade.
14. Export 3D Model.

The Processing Toolbox shall allow individual algorithms to be executed separately and combined into repeatable processing workflows.

13.8. Communication with the Python engine

The QGIS plugin shall not directly import the complete AI environment.

Instead, it shall communicate with the independent Python engine using QProcess or another non-blocking process interface.

Example:

```python
from qgis.PyQt.QtCore import QProcess


class AIEngineBridge:

    def __init__(self, parent=None):

        self.process = QProcess(parent)

        self.process.readyReadStandardOutput.connect(
            self.read_output
        )

        self.process.finished.connect(
            self.process_finished
        )

    def run(self, python_exe, config_file):

        self.process.setProgram(
            python_exe
        )

        self.process.setArguments([
            "-m",
            "ai3d_meoms.cli",
            "run",
```

```
            "--config",
            config_file
        ])

        self.process.start()

    def read_output(self):

        output = bytes(
            self.process.readAllStandardOutput()
        ).decode(
            "utf-8",
            errors="replace"
        )

        print(output)

    def process_finished(self, code, status):

        print(
            "AI processing completed:",
            code,
            status
        )
```

The actual executable path shall point to the dedicated AI Python environment.

The plugin shall monitor progress and display processing errors.

The main QGIS interface shall remain responsive while the AI engine is operating.

13.9. Processing configuration

The plugin shall generate a JSON configuration file for every processing task.

Example:

</> JSON

```
{
  "project_id": "PORPHYRY_001",
  "operation": "prospectivity_prediction",
  "training_file": "training_composites.parquet",
  "voxel_file": "voxel_features.parquet",
  "model_file": "random_forest_prospectivity.joblib",
  "output_file": "prospectivity_volume.vti",
  "target": "copper_mineralization",
  "coordinate_system": "PROJECT_DEFINED_CRS"
```

```
}
```

The AI engine shall return a processing status and a list of output files.

13.10. Loading outputs into QGIS

The plugin shall automatically load appropriate outputs:

- GeoTIFF prospectivity maps.
- GeoPackage geological features.
- Drill-hole location layers.
- Geological boundaries.
- Target locations.
- Cross-section traces.

Three-dimensional VTK outputs shall be opened in the PyVista viewer.

SECTION 14. PYVISTA INTEGRATION AND 3D VIEWER DEVELOPMENT

14.1. Initial implementation

For Version 1.0, the 3D viewer shall be a separate application.

This is recommended because QGIS and the AI engine may require different Python and Qt environments.

The 3D viewer shall be launched from the QGIS plugin.

14.2. Viewer project structure

```
</> Plain text

viewer/
 │
 ├──── main.py
 ├──── viewer_window.py
 ├──── geological_scene.py
 ├──── drillhole_display.py
 ├──── prospectivity_display.py
 ├──── orebody_display.py
 ├──── grade_display.py
```

```
└──── section_tools.py
```

14.3. 3D scene management

The viewer shall manage the following scene objects:

Object	Data source
Ground surface	DEM
Geological units	GemPy output
Fault surfaces	Geological model
Intrusive bodies	Geological model
Drill holes	Desurveyed trajectories
Prospectivity volume	AI prediction
Prospectivity surface	Extracted isosurface
Orebody solid	Geological interpretation
Grade blocks	Grade estimation
Uncertainty volume	Model uncertainty

14.4. Later integration into QGIS

After Version 1.0 is stable, the faculty may investigate embedding the PyVista viewer using PyVistaQt.

However, the Qt binding and VTK dependencies must be compatible with the exact QGIS environment.

The separate-process approach shall remain the reference implementation until successful integration tests have been completed.

SECTION 15. PYTHON SCIENTIFIC ENGINE DIRECTORY STRUCTURE

The following structure is the recommended development standard.

```
</> Plain text

AI_3D_MEOMS/
 |
 ├──── pyproject.toml
 ├──── README.md
 ├──── requirements.lock
 |
 ├──── configuration/
 |     ├──── project.json
 |     ├──── grid.json
 |     ├──── features.json
 |     └──── model_parameters.json
 |
 ├──── data/
 |     ├──── raw/
 |     ├──── validated/
 |     ├──── processed/
 |     ├──── training/
 |     └──── prediction/
 |
 ├──── database/
 |     ├──── connection.py
 |     ├──── schema.sql
 |     ├──── validation.py
 |     └──── asset_manager.py
 |
 ├──── imports/
 |     ├──── geological_import.py
 |     ├──── geochemical_import.py
 |     ├──── magnetic_import.py
 |     ├──── gamma_import.py
 |     ├──── mt_import.py
 |     ├──── ert_import.py
 |     ├──── gravity_import.py
 |     ├──── remote_sensing_import.py
 |     └──── drillhole_import.py
 |
 ├──── preprocessing/
 |     ├──── coordinate_transform.py
 |     ├──── drillhole_desurvey.py
 |     ├──── assay_compositing.py
 |     ├──── geochemical_processing.py
```

```
│   ├── geophysical_processing.py
│   └── data_quality.py
│
├── geological_model/
│   ├── gempy_model.py
│   ├── structural_model.py
│   ├── geological_domains.py
│   └── model_validation.py
│
├── voxel_model/
│   ├── grid_builder.py
│   ├── geology_features.py
│   ├── geochemistry_features.py
│   ├── geophysics_features.py
│   ├── remote_sensing_features.py
│   └── integrated_features.py
│
├── ai/
│   ├── training_data.py
│   ├── random_forest.py
│   ├── xgboost_model.py
│   ├── validation.py
│   ├── explainable_ai.py
│   ├── calibration.py
│   └── uncertainty.py
│
├── orebody/
│   ├── prospectivity_volume.py
│   ├── connected_components.py
│   ├── surface_extraction.py
│   ├── geological_constraints.py
│   └── interpretation.py
│
├── grade/
│   ├── exploratory_analysis.py
│   ├── variogram_analysis.py
│   ├── kriging.py
│   ├── idw.py
│   ├── xgboost_regression.py
│   ├── regression_kriging.py
│   └── grade_validation.py
│
├── visualization/
│   ├── pyvista_viewer.py
│   ├── geological_display.py
│   ├── drillhole_display.py
│   ├── prospectivity_display.py
│   └── grade_display.py
│
```

```
├──── reporting/
│     ├──── map_export.py
│     ├──── model_report.py
│     └──── statistics.py
│
├──── cli/
│     ├──── __main__.py
│     └──── commands.py
│
├──── tests/
│     ├──── test_database.py
│     ├──── test_desurvey.py
│     ├──── test_compositing.py
│     ├──── test_voxel.py
│     ├──── test_ai.py
│     └──── test_visualization.py
│
├──── models/
│
└──── outputs/
```

Each module shall have a clearly defined input, output and error-handling procedure.

SECTION 16. OFFLINE DEPLOYMENT FOR WINDOWS 10

16.1. Deployment requirements

The system shall be capable of operation without internet connectivity.

The following functions shall work offline:

- Geological database access.
- GIS display.
- AI model training.
- AI prediction.
- 3D geological modelling.
- Orebody extraction.
- Grade estimation.
- 3D visualization.
- Report generation.

No cloud-based AI service shall be required.

16.2. Recommended deployment architecture

The QGIS installation and the Python AI environment shall be maintained separately.

Environment A: QGIS

Contains:

- QGIS Desktop.
- PyQGIS.
- GIS plugins.
- AI-3D-MEOMS QGIS plugin.

Environment B: AI Python environment

Contains:

- Python.
- NumPy.
- Pandas.
- Scikit-learn.
- XGBoost.
- SciPy.
- GemPy.
- GSTools.
- PyKrige.
- PyVista.
- PyVistaQt.

Environment C: Database

Contains:

- PostgreSQL/PostGIS, if centralized database operation is required.
- GeoPackage for individual offline projects.

Environment D: 3D viewer

Contains:

- PyVista.
- VTK.

- Qt binding compatible with the viewer application.

16.3. Python package preparation

On an internet-connected development computer:

```bash
pip download -r requirements.txt -d wheelhouse
```

Transfer the wheelhouse to the offline computer.

Install using:

```bash
pip install \
    --no-index \
    --find-links=wheelhouse \
    -r requirements.txt
```

For complicated scientific environments, a locked Conda environment or a packaged environment using conda-pack may be preferable.

16.4. Software version management

The faculty shall maintain:

- Tested Python version.
- Tested QGIS version.
- Tested GDAL version.
- Tested PyVista version.
- Tested VTK version.
- Tested GemPy version.
- Tested Scikit-learn version.
- Tested XGBoost version.
- Tested PyTorch version, if used.

The exact versions shall be selected during the initial compatibility test.

No untested automatic software updates shall be performed on the operational offline system.

16.5. Security and maintenance

Since Windows 10 general support ended in October 2025, the faculty should assess migration to Windows 11 or a supported Linux distribution.

If Windows 10 must be retained:

- Use a controlled offline network environment.
- Restrict external data transfer.
- Apply available security maintenance.
- Scan installation media.
- Maintain offline backups.
- Avoid exposing the system to untrusted networks.

The software architecture shall not depend on Windows-specific components beyond the user-interface and deployment layer.

SECTION 17. HARDWARE REQUIREMENTS

17.1. Initial development workstation

The following is a practical starting specification.

Component	Recommended configuration
CPU	8-core modern processor
RAM	32 GB
Storage	1 TB SSD
GPU	Optional dedicated GPU
GPU memory	8 GB or more if deep learning is planned
Display	Full HD or higher
Operating system	Windows 10 64-bit for initial compatibility testing

17.2. Research workstation

For larger 3D models:

Component	Recommended configuration
CPU	12–16 cores
RAM	64 GB
Storage	2 TB NVMe SSD
GPU	Dedicated GPU
GPU memory	12–16 GB or more
Backup	External or network storage

The initial Random Forest and XGBoost models do not require a powerful GPU.

The GPU should be purchased primarily for future deep-learning development and intensive 3D processing, not simply because the system is described as AI-based.

17.3. Data-volume management

The system shall avoid unnecessary duplication of large voxel tables.

For large arrays, use NumPy, Zarr or other appropriate scientific storage.

For tabular data, use Parquet rather than repeatedly exporting large CSV files.

SECTION 18. SOFTWARE TESTING AND SCIENTIFIC QUALITY ASSURANCE

18.1. Unit testing

The development team shall test individual functions.

Examples:

- Coordinate transformations.

- Drill-hole desurveying.
- Assay compositing.
- Geological domain assignment.
- Geophysical interpolation.
- Voxel indexing.
- AI feature extraction.
- Prospectivity prediction.
- Surface extraction.
- Grade estimation.

18.2. Integration testing

The following complete workflows shall be tested:

Test A: Geological database

Import geological maps and verify coordinates and attribute integrity.

Test B: Drill-hole database

Import collar, survey, lithology and assay information and verify desurveyed trajectories.

Test C: 3D voxel database

Generate voxels and verify grid coordinates, cell dimensions and feature coverage.

Test D: AI prospectivity

Train the model and evaluate independent drill-hole prediction.

Test E: Orebody modelling

Generate a prospectivity envelope and compare it with geological and drilling evidence.

Test F: Grade estimation

Generate a block model and validate estimated grades against held-out drill holes.

Test G: Offline deployment

Disconnect the computer from the internet and execute the complete workflow.

18.3. Scientific validation

The software shall maintain a comparison between:

- Geological expert interpretation.
- Conventional statistical analysis.
- Conventional geophysical interpretation.
- AI prospectivity predictions.
- Geostatistical grade estimation.
- AI grade estimation.

The AI system shall not be accepted as scientifically reliable based solely on visual agreement with known mineralization.

Independent validation shall be performed wherever sufficient data exist.

18.4. Acceptance criteria

The initial system shall be considered technically operational when it can:

1. Import all specified input data types.
2. Store the original data without loss.
3. Desurvey all valid drill holes.
4. Construct and display a 3D geological framework.
5. Construct a 3D voxel database.
6. Generate reproducible AI training datasets.
7. Train and evaluate Random Forest and XGBoost models.
8. Predict mineral prospectivity throughout the defined valid volume.
9. Extract and display high-prospectivity surfaces.
10. Estimate copper grade using at least one approved baseline method.
11. Display 3D geological bodies and drill-hole trajectories.
12. Export grade and prospectivity models.
13. Operate without internet connectivity.

Scientific acceptance additionally requires documented model limitations, validation results and geological review.

SECTION 19. IMPLEMENTATION SCHEDULE

19.1. Proposed 18-month schedule

Period	Development activity	Main deliverable
Months 1–2	Requirements and database design	Approved technical architecture
Months 2–4	Database and data import modules	Functional geological database
Months 3–5	Drill-hole desurveying and compositing	3D drill-hole database
Months 4–7	Geological framework modelling	Initial 3D geological model
Months 5–8	Voxel and feature engineering	3D AI training dataset
Months 7–10	Random Forest and XGBoost	Initial prospectivity models
Months 9–12	3D prediction and extraction	Prospectivity volume and surfaces
Months 10–13	Grade estimation	Initial grade block model
Months 11–15	QGIS and PyVista integration	Integrated application
Months 14–16	Scientific validation	Validation report
Months 16–18	Offline packaging and acceptance	Operational prototype

19.2. Development priorities

The development shall be divided into three releases.

Version 0.1 — Geological Data Management

Expected functions:

- Data import.
- Database validation.

- 2D geological mapping.
- Drill-hole desurveying.
- Assay compositing.
- Basic 3D geological visualization.

Version 0.5 — AI Prospectivity Prototype

Expected functions:

- 3D voxel construction.
- Geological feature extraction.
- Geophysical data integration.
- Random Forest and XGBoost training.
- Spatial validation.
- 3D prospectivity mapping.
- Prospectivity surface extraction.

Version 1.0 — Integrated Exploration System

Expected functions:

- Full QGIS interface.
- Independent AI processing engine.
- Integrated 3D geological viewer.
- Interpreted orebody modelling.
- Copper grade block modelling.
- Uncertainty visualization.
- Offline deployment.
- Technical reporting.

SECTION 20. PROJECT TEAM AND RESPONSIBILITIES

Position	Main responsibility
Project Director	Overall management and scientific supervision
Economic Geologist	Porphyry copper deposit model and interpretation

Position	Main responsibility
Structural Geologist	Structural and geological framework modelling
Geochemist	Geochemical processing and interpretation
Geophysicist	Magnetic, gamma, MT, ERT and gravity modelling
AI Developer	Machine learning and model validation
Python Developer	Scientific software architecture
GIS Developer	QGIS plugin and spatial database
Geostatistician	Variogram analysis and grade estimation
3D Visualization Developer	PyVista and VTK integration

The same individual may undertake more than one role initially.

However, geological interpretation, AI development and grade estimation should each receive appropriate specialist review.

SECTION 21. DEVELOPMENT RISKS AND MITIGATION

Potential problem	Mitigation
Limited number of drill holes	Use geological knowledge, grouped validation and cautious AI models
Insufficient barren samples	Use positive-unlabelled approaches and verified negative evidence
Inconsistent survey coordinates	Standardize CRS and vertical datum
Poor drill-hole survey quality	Apply trajectory validation and retain uncertainty

Potential problem	Mitigation
Incomplete geophysical coverage	Apply coverage masks and uncertainty indicators
Inconsistent geological logging	Establish common geological codes
Unreliable variograms	Compare baseline estimators and restrict kriging
AI overfitting	Use spatial validation and regularization
Inaccurate 3D geological geometry	Incorporate geological expert review
Poor visualization performance	Use data filtering and level-of-detail rendering
Qt compatibility problems	Keep QGIS and 3D viewer in separate processes initially
Offline package dependency problems	Freeze and test software environments
Uncertain grade estimates	Display estimation variance and drilling support

SECTION 22. EXPECTED DELIVERABLES

At the end of the initial development project, the faculty shall possess:

D01. Geological Exploration Database

A structured database containing geological, geochemical, geophysical, remote sensing and drill-hole information.

D02. Drill-Hole Processing Module

A Python module for desurveying, interval calculation and assay compositing.

D03. 3D Geological Framework

A three-dimensional model of the main geological units, intrusive bodies and faults.

D04. 3D Voxel Feature Database

An integrated database of geological and geophysical attributes.

D05. AI Mineral Prospectivity Models

Random Forest and XGBoost models with documented validation results.

D06. 3D Prospectivity Mapping Module

A program for predicting and visualizing mineralization prospectivity.

D07. Orebody Interpretation Module

A program for generating and reviewing three-dimensional mineralized volumes.

D08. 3D Grade Estimation Module

A program for estimating and displaying copper grade distribution.

D09. QGIS Application Plugin

A GIS interface for managing and controlling the scientific engine.

D10. PyVista 3D Viewer

A three-dimensional geological and mineralization visualization application.

D11. Offline Installation Package

A complete local installation package and software dependency archive.

D12. Technical Documentation

User manuals, developer documentation, database documentation and scientific validation reports.

SECTION 23. FUTURE DEVELOPMENT

Following successful completion of Version 1.0, the system may be expanded to include:

1. Three-dimensional CNN-based geological feature recognition.
2. Advanced multisource data-fusion neural networks.
3. Physics-informed geophysical inversion.

4. Bayesian geological uncertainty modelling.
5. Conditional grade simulation.
6. Geological knowledge graphs.
7. Automated geological report preparation.
8. AI-assisted drilling target optimization.
9. Multi-deposit exploration assessment.
10. Mineral processing and metallurgical prediction.

The system architecture shall be sufficiently modular to support these future developments without requiring a complete redesign.

SECTION 24. FINAL TECHNICAL RECOMMENDATION

The faculty should develop AI-3D-MEOMS according to the following implementation sequence:

First: Establish the geological database and ensure the quality and compatibility of all input datasets.

Second: Develop the drill-hole desurveying and compositing modules.

Third: Construct the 3D geological framework and voxel database.

Fourth: Develop the initial Random Forest and XGBoost mineral prospectivity models.

Fifth: Generate the 3D prospectivity volume and extract geological exploration targets.

Sixth: Develop the orebody interpretation and copper grade estimation modules.

Seventh: Integrate all functions into the QGIS and PyVista offline application.

The central scientific principle shall be that AI predictions must remain consistent with geological evidence and must be validated against independent observations.

The software shall support professional exploration geologists rather than replace geological interpretation.

The initial 10–30 drill holes shall be treated as a valuable but limited subsurface dataset. Consequently, the first version shall emphasize geological constraints, interpretable machine learning and rigorous spatial validation rather than unnecessarily complex deep-learning architectures.

The successful completion of this project will provide the Faculty of Geoscience and Technology with an institutional foundation for advanced AI-based mineral exploration research and education.

TECHNICAL REFERENCES AND SOFTWARE RESOURCES

1. Ayuso, R. A., et al. (2010). *Porphyry Copper Deposit Model*. U.S. Geological Survey, Scientific Investigations Report 2010–5070–B. https://doi.org/10.3133/sir20105070B
2. GemPy Project. *Open-source 3D Geological Modelling Software*. https://github.com/gempy-project/gempy
3. QGIS Development Team. *PyQGIS Developer Cookbook*. https://docs.qgis.org/latest/en/docs/pyqgis_developer_cookbook/
4. PyVista Project. *3D Visualization and Mesh Analysis*. https://docs.pyvista.org/
5. PyVistaQt Project. *PyVista Integration with Qt*. https://qt.pyvista.org/
6. PyKrige Project. *Three-Dimensional Ordinary Kriging*. https://geostat-framework.readthedocs.io/projects/pykrige/en/stable/
7. GSTools Project. *Geostatistical Modelling and Spatial Interpolation*. https://geostat-framework.readthedocs.io/projects/gstools/en/stable/
8. Microsoft. *Windows 10 End of Support*. https://support.microsoft.com/en-us/windows/deployment/updates-lifecycle/windows-10-support-has-ended-on-october-14-2025

DOCUMENT END

Document designation: Technical Software Development Specification
System: AI-3D-MEOMS
Version: 1.0
Proposed implementing institution: Faculty of Geoscience and Technology

Final engineering considerations

Before the development team begins implementation, I would establish four mandatory technical decisions:

1. **Freeze the software environment.** GemPy's current Version 3 architecture has evolved, so the compatible GemPy, engine and viewer versions must be tested and pinned together before developing the geological modelling module. GitHub +1

2. **Keep the AI engine independent of QGIS.** This will reduce Python and Qt dependency conflicts and make future migration to Windows 11 or Linux considerably easier.

3. **Make drilling validation the scientific priority.** With 10–30 holes, spatially independent validation and geological constraints are more important than increasing AI model complexity. PyKrige can provide 3D ordinary kriging, but variogram and anisotropy assumptions must be established from the actual data. PyKrige 1.7… +1

4. **Use PyVista as the principal 3D scientific viewer.** Its Qt integration provides a future route to an embedded interface, while the separate viewer is the safer initial implementation. PyVistaQt … +1

My recommended first development target

I would make the first demonstrable software milestone:

Version 0.1 target

A working geological exploration data and 3D visualization system

- Import the geological maps.
- Import magnetic, gamma-ray, MT, ERT and gravity datasets.
- Import remote sensing information.
- Import 10–30 drill holes.
- Desurvey drill-hole trajectories.
- Display lithology and assay intervals.
- Construct the initial 3D geological framework.
- Display the geological model and drill holes in PyVista.

This milestone establishes the data foundation required for all subsequent AI development.

I recommend treating the complete system as a faculty research and software engineering project, with an exploration geologist, geophysicist, geochemist, AI developer and geostatistician jointly responsible for scientific validation.

The next practical development step should be the actual Python source-code implementation of Version 0.1, beginning with the geological database, drill-hole desurveying module and QGIS interface.

2026-11-06 16:48:23 You

For the last time, please explain about the explainable AI for the mineral exploration project, which you suggested.

I think It is the final goal of our project, but I am not sure what it is.

So, please provide the detailed information with the following contents:

- The concept and functionality of explainable AI for mineral exploration
- Data preparation for the development of explainable AI
- It will utilze LLM? if true, how to construct Chinese LLM?
- explainable AI can generate the report of mineral exploration from the database and the result of mineral prospectivy mapping?
- The progress of the development of explainable AI for mineral exploration

ChatGPT gpt-6-thinking

Explainable AI for Mineral Exploration

Scientific concept, data preparation, Chinese LLM development, automatic reporting and implementation roadmap for AI-3D-MEOMS

I am very pleased that you have raised this question. In fact, I consider Explainable Artificial Intelligence (XAI) one of the most important components of your proposed mineral exploration project.

Your concern is entirely reasonable. You have already planned to integrate geological, geochemical, geophysical, remote sensing and drilling data, and to use AI to produce 3D mineral prospectivity maps, interpreted orebody models and copper grade distributions.

However, there is another important question:

When the AI system identifies a particular underground area as a promising copper exploration target, can it explain to the exploration geologist WHY that area is promising, WHICH geological

evidence supports the prediction, HOW reliable the prediction is, and WHAT should be investigated next?

That is the central purpose of Explainable AI.

I would actually recommend expanding the final objective of AI-3D-MEOMS from simply producing a 3D mineral prospectivity map to developing an:

Proposed ultimate research objective

Explainable and Knowledge-Assisted AI for Intelligent Mineral Exploration

An integrated system capable of:

1. Integrating geological exploration information.
2. Predicting concealed mineralization using AI.
3. Explaining the geological evidence behind each prediction.
4. Quantifying prediction uncertainty.
5. Comparing AI results with established mineral-system knowledge.
6. Identifying additional exploration requirements.
7. Generating scientifically traceable mineral exploration reports.
8. Supporting exploration decisions through interaction between geologists and AI.

There is an important distinction, however: **XAI is not itself the final mineral exploration objective. It is the scientific transparency and interpretation layer that makes the complete AI exploration system understandable, testable and useful.**

I will explain the subject from the perspective of your Faculty of Geoscience and Technology, particularly for the porphyry copper exploration project and the limited availability of 10–30 drill holes.

PART 1. CONCEPT AND FUNCTIONALITY OF EXPLAINABLE AI FOR MINERAL EXPLORATION

1. What is Explainable AI?

Explainable AI (XAI) is a collection of methods that enables human users to understand and examine how an AI model produces its predictions.

Ordinary machine learning can identify complex relationships between input data and predicted mineralization.

For example, your AI model might receive:

- Geological lithology.
- Distance to intrusive contacts.

- Fault density.
- Magnetic susceptibility.
- Gravity anomalies.
- MT resistivity.
- Soil ionic conductivity.
- Copper and molybdenum geochemistry.
- Remote sensing alteration indices.

After processing these variables, the model might produce the following prediction:

Example: AI prediction for an underground exploration target

Voxel location **X = 500,250; Y = 7,500,350; Z = 450 m**

Model prospectivity score **0.87**

Illustrative model output, not a calibrated probability.

High-priority exploration target

But why did the AI produce this result?

- Favorable porphyritic intrusive host rock.
- Close spatial relationship to a major fault.
- Favorable magnetic inversion response.
- A resistivity anomaly at depth.
- Supporting Cu–Mo geochemical evidence.

An ordinary AI system may only show the score of 0.87.

An explainable AI system should also show the contribution of the different input variables to that prediction, the uncertainty of the prediction and the geological interpretation of those contributions.

The geologist can then investigate whether the prediction is consistent with the known characteristics of porphyry copper mineral systems.

2. Difference between ordinary AI and Explainable AI

Function	Ordinary AI	Explainable AI
Integrate exploration data	Yes	Yes
Predict mineral prospectivity	Yes	Yes
Generate 3D prospectivity maps	Yes	Yes
Identify important geological factors	Limited	Yes
Explain individual exploration targets	Limited	Yes

Show positive and negative feature contributions	Usually not	Yes
Compare predictions with geological knowledge	Requires additional work	Integrated functionality
Examine prediction stability	Not necessarily	Can be integrated
Explain model uncertainty	Not necessarily	Can be integrated
Support geological decision-making	Limited transparency	Stronger transparency

The distinction is important.

XAI does not necessarily improve the predictive accuracy of a model. Instead, it allows the exploration team to examine whether the model's behavior is scientifically reasonable.

Recent research on explainable mineral prospectivity mapping emphasizes that predictive performance, geological plausibility, explanation stability, spatial validation and exploration usefulness must be assessed separately. MDPI +2

3. The five principal functions of XAI in mineral exploration

I recommend implementing five major XAI functions in AI-3D-MEOMS.

Function 1. Global explanation: What controls mineral prospectivity in the entire exploration area?

This function investigates the overall behavior of the AI model.

For example, suppose the model has been trained using geological, geochemical, geophysical and remote sensing data.

XAI might identify the following illustrative ranking:

Rank	Predictive feature	Relative importance
1	Distance to porphyritic intrusion	22%
2	Fault density	19%
3	MT resistivity	17%
4	Magnetic susceptibility	15%
5	Cu geochemical anomaly	12%
6	Mo geochemical anomaly	8%
7	Gamma-ray alteration indicator	5%
8	Remote sensing alteration index	2%

Illustrative figures only. Actual importance values must be calculated from the trained model and must not be interpreted automatically as percentages of geological causation.

This result could help your geoscientists answer questions such as:

- Is structural control important in this exploration area?
- Is the intrusive complex an important predictor?
- Does resistivity contribute more strongly than magnetic information?
- Are geochemical anomalies useful for identifying concealed mineralization?
- Is the AI relying excessively on one particular dataset?

Scientific application: Evaluating the geological relationships learned by the AI model.

Function 2. Local explanation: Why is this particular underground target promising?

This is probably the most useful XAI function for practical mineral exploration.

Suppose the system identifies three high-priority targets: A, B and C.

Although all three may have high prospectivity scores, their geological evidence may be different.

Example of individual target explanations

Target A High priority

0.91

Main supporting evidence:

- Strong intrusive-contact relationship.
- High fault intersection density.
- Favorable magnetic response.
- Supporting Cu–Mo geochemistry.Illustrative uncalibrated model score.

Target B High priority

0.86

Main supporting evidence:

- Strong MT resistivity contrast.
- Favorable host lithology.
- Close proximity to a major fault.
- Limited direct geochemical evidence.Illustrative uncalibrated model score.

Target C Requires verification

0.82

Main supporting evidence:

- Strong magnetic anomaly.
- Favorable regional structure.

- Limited drill-hole control.
- Insufficient geochemical coverage.Illustrative uncalibrated model score.

The system can therefore explain why different targets receive high prospectivity scores.

This helps geologists understand the geological basis of each prediction rather than simply selecting the highest numerical score.

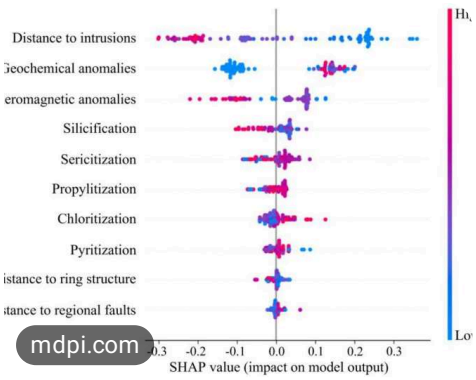
Function 3. Spatial explanation: Where do the important geological factors occur in 3D?

This function is particularly important for your project.

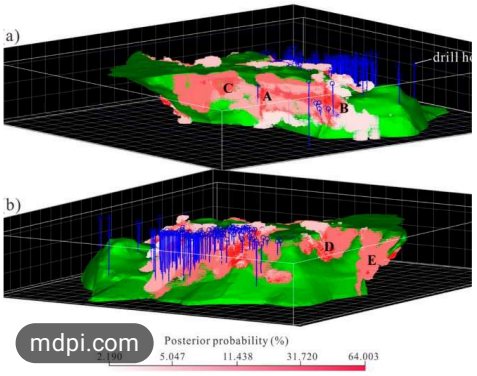
Your AI system will predict mineral prospectivity throughout a 3D voxel model.

XAI should also produce 3D spatial explanations showing where each predictor contributes positively or negatively to the prediction.

For example:







The 3D viewer should allow users to display:

- 3D prospectivity distribution.
- Positive contribution of fault density.
- Positive or negative contribution of magnetic susceptibility.
- MT resistivity contribution.
- Geochemical contribution.
- Intrusive-contact contribution.
- Combined explanation of the selected target.

For example, a geologist could select a voxel in the 3D viewer and ask:

"Why does the AI consider this location favorable for porphyry copper mineralization?"

The system could display its individual feature contributions and their supporting data.

This is one of the most valuable research functions of the proposed AI-3D-MEOMS.

Function 4. Geological explanation: Is the AI prediction geologically reasonable?

This function compares AI results with established porphyry copper deposit models.

The system should contain a geological knowledge base describing:

Geological concept	Expected evidence
Porphyry intrusion	Intrusive rocks and intrusive contacts
Structural pathways	Faults, fractures and structural intersections
Potassic alteration	K-feldspar, biotite and associated alteration indicators
Phyllic alteration	Quartz-sericite-pyrite assemblages
Propylitic alteration	Chlorite, epidote and carbonate assemblages
Mineralization	Chalcopyrite, bornite and associated sulfides
Geochemical signature	Cu, Mo and appropriate pathfinder elements
Geophysical signature	Deposit-dependent magnetic, density and electrical properties

The AI explanation can then be compared with the geological knowledge base.

For example:

- Does the predicted target occur within a favorable intrusive complex?
- Is it structurally controlled?
- Is there supporting alteration evidence?
- Are the geophysical anomalies consistent with the interpreted geology?
- Is the predicted depth compatible with the drilling information?

There is an important scientific qualification: porphyry systems have variable geophysical and alteration signatures. For example, magnetite destruction can produce magnetic lows in some systems, whereas magnetite-rich alteration can produce magnetic highs in others.

Therefore, the geological knowledge base must contain context-dependent rules rather than rigid universal rules.

Function 5. Uncertainty explanation: How reliable is the prediction?

A high prospectivity score does not necessarily mean that the prediction is reliable.

For example:

- Target A may have high prospectivity and extensive supporting data.
- Target B may have high prospectivity but very limited drilling information.
- Target C may have high prospectivity because of an uncertain geophysical inversion.

The system should therefore show:

1. AI prediction score.
2. Prediction uncertainty.
3. Drill-hole coverage.
4. Geophysical data quality.
5. Geological interpretation confidence.
6. Explanation stability.

This is especially important because you currently have only 10–30 drill holes.

I recommend displaying prospectivity and uncertainty side by side rather than combining them into one misleading confidence number.

PART 2. TECHNICAL METHODS FOR DEVELOPING EXPLAINABLE AI

4. Which XAI algorithms should be used?

For your project, I recommend combining several XAI algorithms rather than relying on only one.

The following methods are particularly suitable for Random Forest and XGBoost, which we previously selected as the principal AI algorithms.

XAI method	Main purpose	Application in AI-3D-MEOMS
SHAP	Explain feature contributions	Main explanation algorithm
Permutation Importance	Evaluate predictor importance	Global model assessment
Partial Dependence Plot	Examine predictor-response relationships	Geological interpretation
Accumulated Local Effects (ALE)	Examine nonlinear effects with correlated features	Geochemical and geophysical interpretation
LIME	Explain individual predictions	Additional target-level interpretation
Counterfactual Analysis	Examine alternative conditions affecting predictions	Exploration scenario analysis
Ensemble/Bootstrap Analysis	Evaluate prediction stability	Uncertainty assessment

I recommend making **SHAP the principal XAI algorithm**, supported by permutation importance, ALE and uncertainty analysis.

4.1. SHAP: SHapley Additive exPlanations

SHAP is a mathematical method for explaining the contribution of individual input variables to a model prediction.

It is based on Shapley values from cooperative game theory.

The basic formulation is:

$$f(x) = \phi_0 + \sum_{i=1}^{n} \phi_i$$

where:

- $f(x)$ is the model prediction.
- ϕ_0 is the reference or baseline prediction.
- ϕ_i is the contribution of feature i.
- n is the number of input features.

The interpretation is straightforward:

- Positive SHAP value: the feature pushes the prediction upward relative to the baseline.
- Negative SHAP value: the feature pushes the prediction downward.
- Large absolute SHAP value: the feature has a large contribution to that particular prediction.

For example:

Feature	SHAP contribution
Favorable intrusive lithology	+0.18
Fault density	+0.15
MT resistivity	+0.12
Magnetic susceptibility	+0.08
Cu geochemical anomaly	+0.07
Unfavorable alteration	-0.05
Baseline model output	0.32
Final model output	0.87

Illustrative contributions expressed in probability units for a probability-explained model. Real values depend on the selected model, reference data and SHAP configuration.

The contributions sum to the final model output in this example.

This allows a geologist to understand how the AI arrived at its prediction.

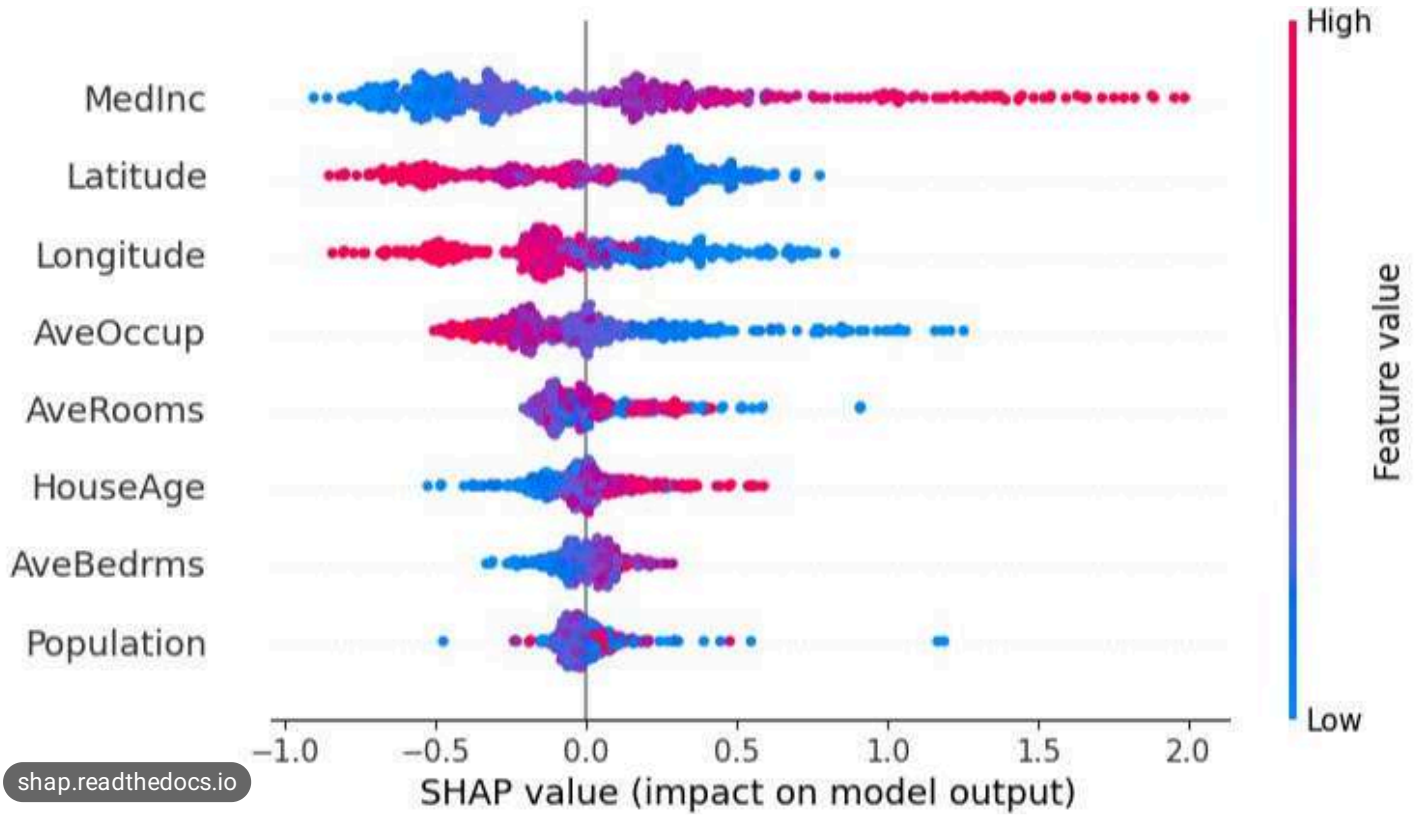
An important technical issue: geological variables are often correlated. For example, total magnetic intensity, reduced-to-pole magnetic intensity and magnetic derivatives are related measurements. SHAP attribution can be sensitive to how such dependencies are handled. Therefore, the system

should group related features and examine alternative explanation settings. The SHAP documentation explicitly discusses these feature-dependence assumptions. SHAP lates... +1

4.2. Global SHAP analysis

Global SHAP analysis examines the importance and contribution patterns of all features across the study area or validation dataset.

The main output should be a SHAP summary plot.



The plot normally shows:

- Vertical axis: input variables.
- Horizontal axis: SHAP contribution.
- Each point: one observation.
- Point color: actual feature value.
- Horizontal position: positive or negative contribution.

For mineral exploration, it can reveal, for example, whether high fault density, a particular resistivity range or a favorable geochemical signature consistently increases the predicted prospectivity.

I recommend generating separate global explanations for:

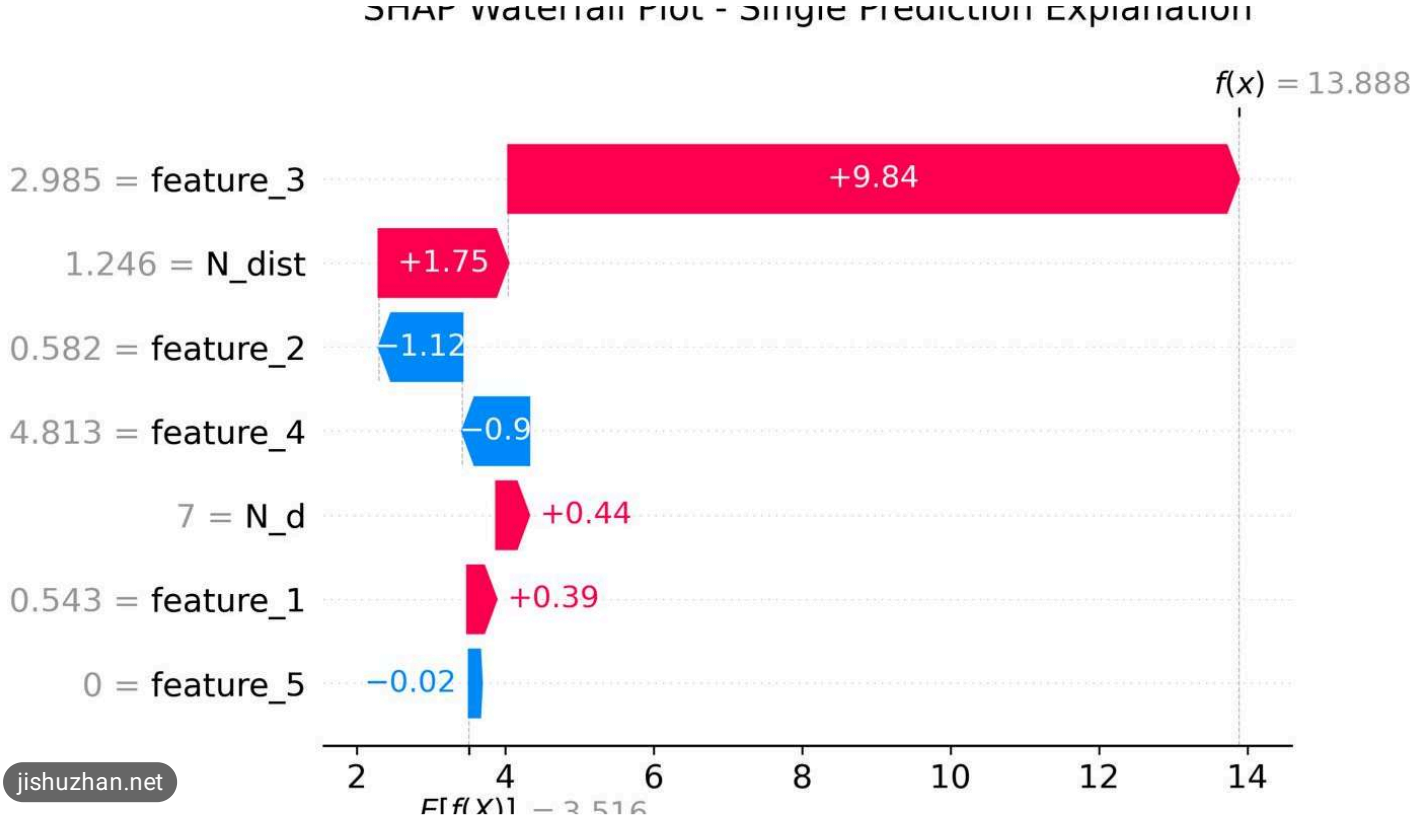
1. The complete exploration area.
2. Individual geological domains.
3. Individual depth intervals.
4. Independently validated drill-hole groups.

This is more informative than producing only one feature-importance ranking.

4.3. Local SHAP analysis

Local SHAP analysis explains an individual prediction.

The system should generate a SHAP waterfall plot for each selected exploration target.



For example, a user selects Target A in the PyVista 3D viewer.

The system retrieves its feature vector and generates:

- Baseline prediction.
- Positive feature contributions.
- Negative feature contributions.
- Final model prediction.
- Missing-data indicators.
- Distance to known drill holes.
- Prediction uncertainty.

The explanation should also show the original units and actual feature values.

For example:

Geological evidence	Actual value	Contribution
Distance to intrusive contact	45 m	Positive
Distance to major fault	70 m	Positive

MT resistivity	180 Ωm	Positive
Magnetic susceptibility	0.012 SI	Negative
Cu soil anomaly	350 ppm	Positive

These values are illustrative and must be replaced with actual project measurements.

4.4. Spatial SHAP analysis

This is where XAI becomes especially valuable for your 3D mineral exploration system.

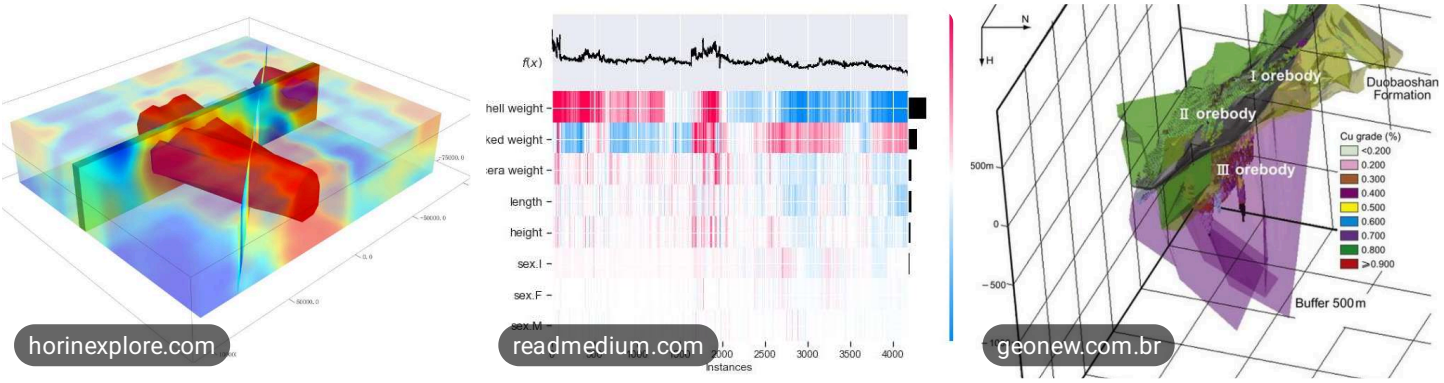
Your existing AI model will predict prospectivity for thousands or millions of voxels.

We can calculate SHAP values for selected voxels and construct separate spatial contribution volumes.

For example:

3D model	Description
SHAP_Intrusion.vti	Intrusive-contact contribution
SHAP_Fault.vti	Fault-density contribution
SHAP_Magnetic.vti	Magnetic contribution
SHAP_MT.vti	MT contribution
SHAP_Geochemistry.vti	Geochemical contribution
SHAP_Alteration.vti	Alteration contribution

These files can be opened in PyVista.



A geologist could turn individual contribution volumes on and off to examine which evidence supports a predicted mineralized volume.

For computational efficiency, the first version should calculate spatial SHAP explanations for:

- High-priority exploration targets.
- Representative voxels at different depths.

- Selected geological domains.
- Spatially distributed validation samples.

It should not necessarily calculate SHAP for every voxel in a very large grid.

4.5. Permutation importance

Permutation importance provides another way of examining the importance of individual features.

The procedure is:

1. Train the AI model.
2. Calculate its validation performance.
3. Randomly shuffle one input feature.
4. Calculate the new validation performance.
5. Measure the performance reduction.
6. Repeat for all features.

If shuffling the MT resistivity feature substantially reduces model performance, this suggests that MT resistivity contains useful predictive information for that particular model and validation dataset.

However, correlated features can mask one another's importance. I recommend conducting both individual-feature and geological feature-group permutation tests.

4.6. Partial dependence and ALE analysis

These methods answer another important geological question:

How does the AI model's prediction change when a particular geological or geophysical variable changes?

For example:

- How does mineral prospectivity change with increasing distance from an intrusive contact?
- At what range of MT resistivity does the model predict the highest prospectivity?
- Does increasing fault density consistently increase prospectivity?
- Does the Cu–Mo geochemical relationship contribute to the prediction?

ALE is particularly useful to consider where geological variables are correlated.

These curves describe the model's learned response, not necessarily the true physical or geological relationship.

4.7. Counterfactual analysis

Counterfactual analysis investigates how a prediction would change under an alternative input scenario.

For example:

"Would this target still be ranked highly if the magnetic anomaly were absent?"

Or:

"Would the prediction remain high if the intrusive-contact distance were greater?"

This can help the geologist identify which evidence is essential to a target's ranking.

However, the system must distinguish hypothetical changes from actual geological changes. It must not suggest that changing an input measurement would physically change the underlying mineral deposit.

PART 3. DATA PREPARATION FOR EXPLAINABLE AI

5. Does XAI require a separate training dataset?

This is an important point.

XAI generally does not require training an entirely new machine learning model.

It normally explains an existing trained model.

Therefore, your existing AI training database can also be used for XAI.

However, additional information must be prepared so that the explanation is meaningful and traceable.

I recommend creating three related datasets.

Dataset A: AI training dataset

Contains the geological, geochemical and geophysical features and the mineralization labels.

Dataset B: XAI interpretation dataset

Contains:

- Feature names.
- Feature descriptions.
- Units.
- Data sources.

- Feature-generation methods.
- Geological meanings.
- Expected relationships with porphyry copper mineral systems.
- Spatial uncertainty.
- Data quality.

Dataset C: Geological knowledge database

Contains:

- Porphyry copper deposit models.
- Geological interpretations.
- Alteration assemblages.
- Structural controls.
- Geochemical signatures.
- Geophysical interpretation principles.
- Institutional exploration reports.
- Published scientific references.
- Faculty geological terminology.

These three datasets will work together.

5.1. Preparing the geological data

The following table provides my recommendations for the geological features.

Original data	XAI feature	Geological meaning
Geological map	Lithology	Host-rock characteristics
Intrusive contact	Distance to intrusion	Relationship to porphyry intrusion
Fault mapping	Distance to fault	Structural control
Structural measurements	Fault orientation	Structural geometry
Fault intersections	Structural intersection density	Potential fluid pathways
Field observations	Alteration intensity	Hydrothermal alteration
Drill-hole lithology	3D lithological domain	Subsurface host rocks
Drill-hole alteration	3D alteration domain	Mineralization-related alteration

The geological knowledge database should explain why each feature is relevant to porphyry copper exploration.

For example:

Feature: Distance to intrusive contact.

Scientific meaning: Many porphyry copper systems are spatially associated with porphyritic intrusions and related hydrothermal alteration.

Potential interpretation: A high model contribution may indicate that the model has learned an association between intrusive proximity and known mineralization.

Limitation: Proximity to an intrusion alone does not establish mineralization.

5.2. Preparing geochemical data

The geochemical database should include:

- Cu.
- Mo.
- Au.
- Ag.
- As.
- Sb.
- Bi.
- Te.
- Pb.
- Zn.
- Other elements appropriate to the deposit model.

For XAI, it is useful to prepare:

- Individual elemental concentrations.
- Log-transformed concentrations.
- Selected elemental ratios.
- Geochemical anomaly indicators.
- Multielement association indicators.

The system should retain the original laboratory results, detection limits and units.

Below-detection-limit observations must be treated appropriately.

Geochemical data are frequently strongly skewed and may be compositional. Transformations and anomaly thresholds should be justified rather than applied automatically.

A particular caution for your project:

Soil geochemistry and soil ionic conductivity are surface observations. They should not be treated as directly measured underground voxel properties.

They can support surface prospectivity interpretation or provide additional evidence through a clearly defined spatial relationship, but their use at depth must be justified.

5.3. Preparing geophysical data

The following attributes should be considered.

Geophysical method	Features for XAI
Magnetic	TMI, RTP, residual, analytical signal, derivatives, susceptibility
Gamma-ray	K, eU, eTh, selected ratios
MT	Apparent resistivity, phase, 3D inverted resistivity, sensitivity
ERT	Apparent and inverted resistivity, depth and sensitivity
Gravity	Bouguer anomaly, residual anomaly, density inversion
Remote sensing	Iron oxide, clay alteration, lineament density and alteration indices

For every feature, the system must preserve:

- Data source.
- Survey date.
- Measurement unit.
- Processing method.
- Spatial resolution.
- Inversion method, if applicable.
- Sensitivity or uncertainty.

This is necessary because an AI model can otherwise give high importance to an unreliable or poorly processed geophysical layer.

For example, a high SHAP contribution from MT resistivity does not automatically mean that the identified resistivity anomaly represents a hydrothermal alteration zone.

It must be examined alongside geological information, inversion quality and other evidence.

5.4. Preparing drilling data

The drilling database is particularly important because it provides the principal subsurface observations.

The system should use:

- Desurveyed collar and trajectory information.
- Lithological intervals.
- Alteration intervals.

- Mineralization intervals.
- Composite Cu grades.
- Other elemental assays.
- Density measurements, if available.

The XAI training database should link each training observation to:

- Drill-hole ID.
- Composite ID.
- X, Y and Z coordinates.
- Geological domain.
- Assay information.
- Original data source.

Training labels

I recommend three categories:

Category	Description	AI treatment
Mineralized	Verified mineralized intervals	Positive
Confirmed barren	Adequately sampled and verified barren intervals	Negative
Unknown	Undrilled or insufficiently characterized areas	Unlabeled

Do not classify all undrilled voxels as barren.

This would introduce substantial bias into your AI model and consequently into the explanations.

Also, avoid dividing adjacent composites from the same drill hole randomly between training and testing. This can cause data leakage and produce misleadingly optimistic performance and unstable explanations.

5.5. Constructing the XAI feature dictionary

I recommend developing a separate CSV or database table called:

`xai_feature_dictionary.csv`

Example:

Feature code	Feature name	Data source	Geological interpretation
GEO01	Lithology	Geological map	Host-rock type
GEO02	Distance_Intrusion	Geological model	Intrusive proximity
GEO03	Fault_Density	Structural map	Structural pathways

GEO04	Alteration	Geological logging	Hydrothermal alteration
GEO05	Vein_Density	Drill logging	Stockwork development
GCH01	Cu_Anomaly	Geochemistry	Copper enrichment
GCH02	Mo_Anomaly	Geochemistry	Molybdenum enrichment
GPH01	Magnetic_Susceptibility	Magnetic inversion	Magnetic property distribution
GPH02	MT_Resistivity	MT inversion	Electrical property distribution
GPH03	Density_Inversion	Gravity inversion	Density distribution
RS01	Clay_Index	Remote sensing	Surface alteration indicator

Additional fields should include:

- Unit.
- Source file.
- Source record or raster identifier.
- Processing algorithm.
- Spatial resolution.
- Missing-data percentage.
- Quality indicator.
- Expected geological relationship.
- Known limitations.

This dictionary will be very important when the LLM eventually generates geological explanations in natural language.

5.6. Important issue: stability of explanations with limited drilling

Because your project has only 10–30 drill holes, the system must test whether the explanations remain similar when different drill holes are excluded.

For example:

- Train using 25 drill holes.
- Calculate SHAP values.
- Exclude several drill holes.
- Retrain the model.
- Calculate SHAP values again.
- Compare the feature rankings and contribution patterns.

If the importance of intrusive proximity changes drastically between experiments, the system should report that the explanation is unstable.

I recommend using:

- Leave-one-drill-hole-out validation where feasible.
- Grouped spatial cross-validation.
- Bootstrap experiments.
- Alternative mineralization-label definitions.
- Feature-group sensitivity analysis.

The independent unit is the drill hole or spatial group, not every assay composite.

With your current dataset, the initial XAI results should be described as exploratory and hypothesis-generating until independently supported by additional drilling.

PART 4. WILL AI-3D-MEOMS UTILIZE AN LLM?

6. What is the relationship between XAI and LLM?

This is probably the most important conceptual distinction in your question.

XAI and LLM are different technologies, but they can be integrated into one system.

XAI explains the numerical behavior of the machine learning model.

LLM (Large Language Model) interprets structured information and communicates it through natural language.

For example:

Technology	Responsibility
Random Forest	Predict mineralization prospectivity
XGBoost	Predict mineralization prospectivity
SHAP	Explain the contribution of input features
Geostatistics	Estimate grade and spatial uncertainty
Geological knowledge base	Provide established geological knowledge
LLM	Explain results in natural language and draft reports

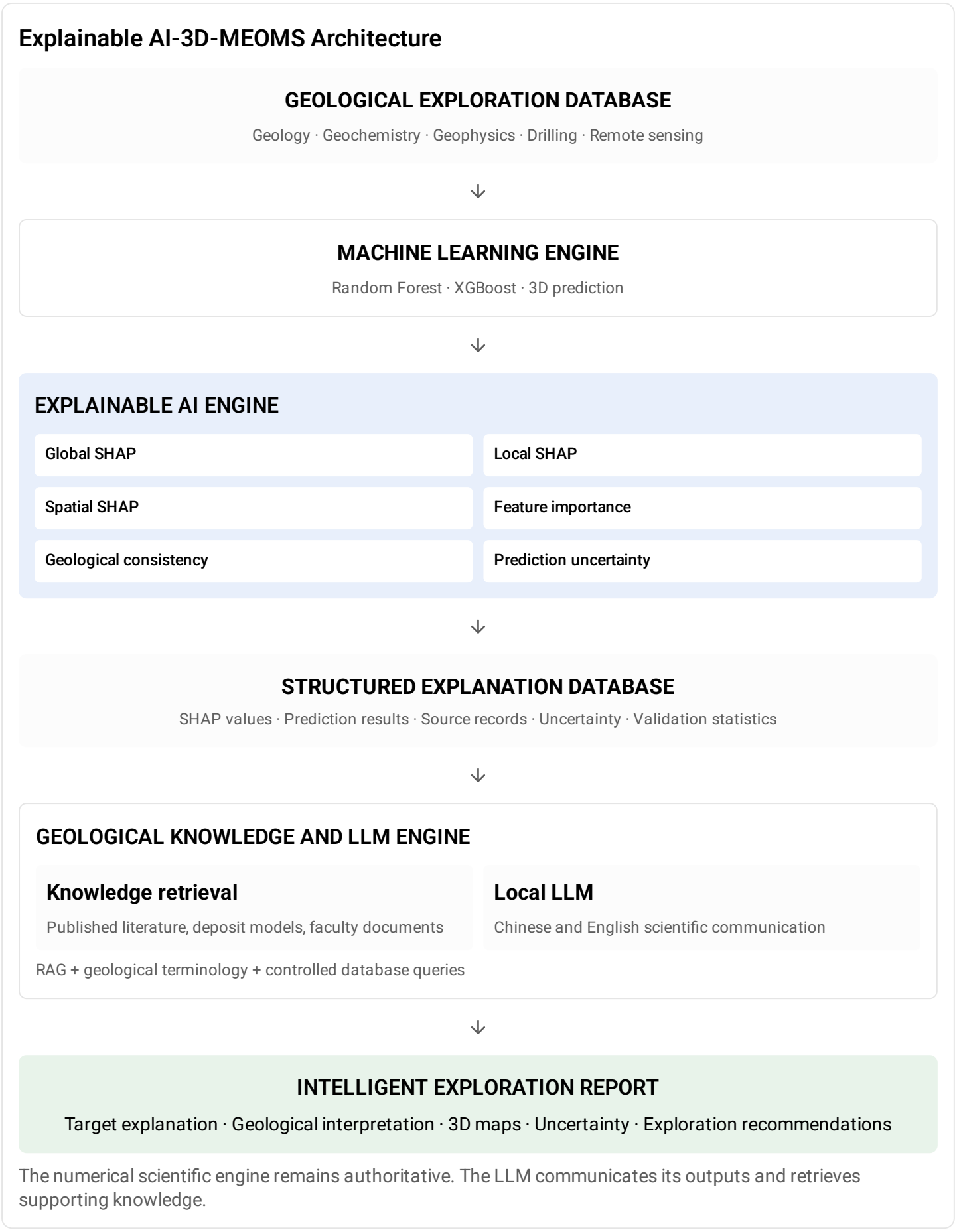
The LLM should not replace SHAP or the actual geological and geostatistical calculations.

Instead, it should use the outputs of those methods.

6.1. Proposed integrated architecture

I recommend adding a new module to AI-3D-MEOMS called:

Geoscience Intelligence and Reporting Module (GIRM).



The numerical scientific engine remains authoritative. The LLM communicates its outputs and retrieves supporting knowledge.

6.2. Should you develop your own Chinese LLM?

My recommendation is yes, if your faculty needs a Chinese-language geological assistant, but **do not develop a completely new LLM from the beginning**.

That would require enormous training datasets, GPU resources, computational infrastructure and substantial financial investment.

Instead, develop a specialized Chinese-English geoscience LLM using an existing open-weight model.

I would call this:

AI-3D-MEOMS Geoscience Assistant (Chinese-English Edition).

Its principal functions would be:

- Understand Chinese geological questions.
- Understand English geological questions.
- Explain AI prospectivity results.
- Interpret SHAP outputs.
- Explain porphyry copper geological concepts.
- Search institutional exploration reports.
- Generate Chinese and English mineral exploration reports.
- Assist students and researchers in geological data interpretation.

6.3. Recommended LLM: Qwen

I recommend evaluating the Qwen3 model family.

Its official repository documents multilingual capabilities, Chinese-language support, several model sizes, local inference options and fine-tuning frameworks. Its open-weight models are distributed under Apache 2.0 according to the project repository. GitHub +1



unsloth.ai

For your offline institutional system, I would initially evaluate:

Model size	Recommended application
4B	Lightweight local geological assistant

8B	Main initial Chinese-English assistant
14B	More advanced scientific interpretation, if hardware permits
30B or larger	Future research workstation or server

These are evaluation candidates, not guaranteed performance rankings. The final choice should be based on Chinese and English geological question-answering tests, available memory and inference speed.

6.4. How to construct the specialized Chinese LLM

There are three possible approaches.

Approach A: Use an existing Chinese-capable LLM directly

Download an appropriate open-weight model and run it locally.

This is the simplest approach.

Approach B: Develop a geological knowledge-enhanced LLM using RAG

This is my strongest recommendation.

Approach C: Fine-tune an existing model using geological training examples

This should be considered after developing the knowledge base and testing the basic system.

I would not recommend pretraining a new Chinese LLM from scratch.

6.5. RAG: Retrieval-Augmented Generation

RAG is a technique that allows an LLM to answer questions using a collection of external documents and knowledge sources.

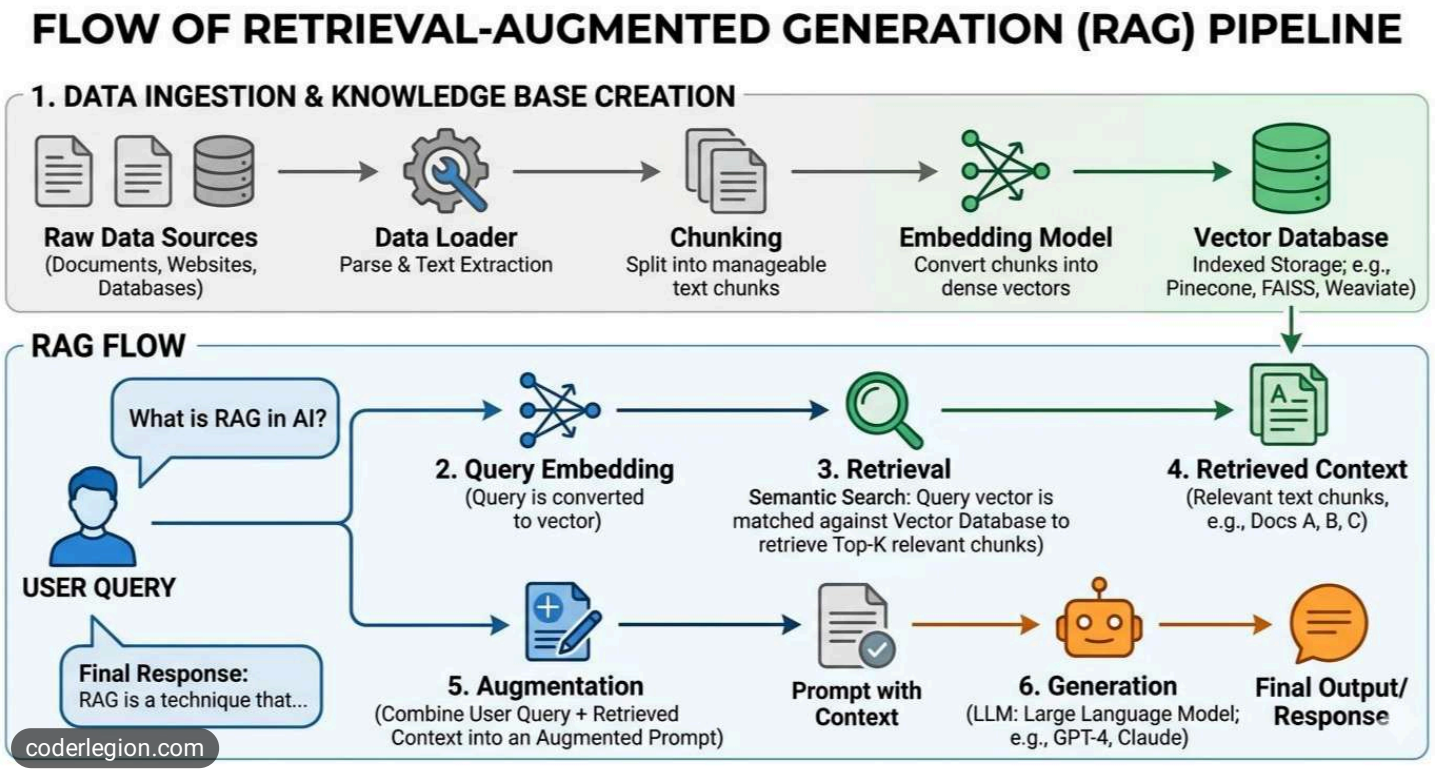
For your project, RAG would be particularly useful because your faculty may have:

- Geological textbooks.
- Mineralogy and petrology lecture notes.
- Porphyry copper deposit models.
- Exploration reports.
- Geological survey reports.
- Geochemical interpretation manuals.
- Geophysical interpretation manuals.
- Published research papers.
- Institutional geological databases.

Instead of asking the LLM to remember all this information through model training, we provide access to the documents through a local knowledge retrieval system.

The process is:

1. Import approved geological documents.
2. Extract text, tables and metadata.
3. Divide documents into manageable sections.
4. Convert sections into numerical embeddings.
5. Store the embeddings in a local searchable index.
6. Retrieve relevant passages when the user asks a question.
7. Provide the retrieved passages to the LLM.
8. Generate an answer with source references.



Recommended RAG software

Component	Recommended software
Language model	Qwen3
Embedding model	BGE-M3
Vector search	FAISS
Document extraction	PyMuPDF, python-docx, openpyxl
Database	PostgreSQL / SQLite
RAG development	Python

User interface	QGIS plugin

The BGE-M3 model is particularly interesting because it supports more than 100 languages, including Chinese and English, and offers dense, sparse and multi-vector retrieval capabilities. It is available under the MIT license. BAAI/bge-... +1

FAISS can provide local vector similarity search without depending on an online service. Faiss docume...

6.6. How to prepare the Chinese geological knowledge base

I recommend preparing five knowledge collections.

Collection	Contents
KB01: Geological fundamentals	Geology, mineralogy and petrology
KB02: Ore deposit geology	Porphyry copper, epithermal, skarn and other deposits
KB03: Exploration techniques	Geochemical and geophysical prospecting
KB04: AI and geostatistics	Machine learning, SHAP, variograms and 3D modelling
KB05: Institutional knowledge	Your faculty's exploration reports and field observations

A particularly important component is a bilingual geological terminology dictionary.

For example:

English	Chinese
Mineral prospectivity mapping	矿产远景评价
Porphyry copper deposit	斑岩铜矿床
Hydrothermal alteration	热液蚀变
Potassic alteration	钾化蚀变
Phyllic alteration	绢英岩化蚀变
Propylitic alteration	青磐岩化蚀变
Mineralized body	矿化体
Drill-hole desurveying	钻孔轨迹测斜计算
Grade estimation	品位估算
Explainable AI	可解释人工智能

The terminology database should be reviewed by bilingual geologists rather than relying exclusively on automatic translation.

6.7. Fine-tuning the LLM

After RAG has been established, you can fine-tune the model using geological question-and-answer examples.

I recommend using LoRA or QLoRA rather than retraining all model parameters.

An open-source framework called LLaMA-Factory supports Qwen models and parameter-efficient fine-tuning methods. LlamaFact... +1

Your training examples could look like this:

```JSON
{
  "instruction": "Explain the geological significance of a magnetic anomaly in porphyry copper exploration.",
  "input": "The target area has a magnetic high associated with an intrusive complex and nearby faults.",
  "output": "The magnetic anomaly may reflect magnetite-bearing intrusive rocks or hydrothermal alteration. Its significance should be evaluated together with geological mapping, alteration observations, magnetic inversion results, and drilling evidence. A magnetic high alone does not confirm copper mineralization."
}
```

I recommend developing supervised examples covering:

- Geological interpretation.
- Mineralogy and petrology.
- Porphyry copper alteration.
- Geochemical anomaly interpretation.
- Geophysical interpretation.
- SHAP explanation.
- 3D prospectivity interpretation.
- Exploration report writing.
- Technical Chinese-English translation.

Start with a small, expert-reviewed training set and expand it based on measured performance.

Important distinction: RAG updates the knowledge available to the LLM without retraining its weights. Fine-tuning changes the model's behavior, terminology and response style. For your project, RAG should be developed first.

6.8. Offline deployment of the Chinese LLM

Because your system is intended to operate completely offline, the LLM must be installed on a local workstation or institutional server.

I recommend evaluating llama.cpp as the inference engine.

It supports local CPU/GPU inference, quantized models and an OpenAI-compatible HTTP interface. This allows the QGIS plugin and Python scientific engine to communicate with the local LLM.
GitHub +1

The architecture will be:

1. Download the selected Qwen model and all required files on an authorized internet-connected computer.
2. Convert or obtain a compatible GGUF model.
3. Install llama.cpp and dependencies.
4. Transfer all files to the offline workstation.
5. Run the model locally.
6. Connect AI-3D-MEOMS to the local model interface.

Example Windows command:

```powershell
</> Powershell

llama-server.exe `
  -m models\qwen3-8b-instruct-q4.gguf `
  -c 8192 `
  --host 127.0.0.1 `
  --port 8080
```

The model filename above is illustrative; obtain a compatible GGUF checkpoint and use its actual filename and recommended chat template.

The Python application can then communicate with the local service:

```python
</> Python

import requests

url = "http://127.0.0.1:8080/v1/chat/completions"

payload = {
    "model": "local-geoscience-model",
    "messages": [
        {
            "role": "system",
```

```
            "content": (
                "You are a geological exploration "
                "research assistant. Use only the "
                "provided scientific evidence."
            )
        },
        {
            "role": "user",
            "content": (
                "Explain the geological significance "
                "of this mineral prospectivity result."
            )
        }
    ],
    "temperature": 0.1
}

response = requests.post(
    url,
    json=payload,
    timeout=180
)

response.raise_for_status()

answer = response.json()[
    "choices"
][0]["message"]["content"]

print(answer)
```

This example illustrates the communication mechanism. The actual system will send structured model outputs, SHAP contributions and retrieved geological references rather than only a general question.

Offline operation

Once the model weights, embedding models, software packages and knowledge database have been installed, the entire reporting and question-answering process can run without internet connectivity.

PART 5. AUTOMATIC MINERAL EXPLORATION REPORT GENERATION USING XAI AND LLM

7. Can Explainable AI generate mineral exploration reports?

Yes. This is one of the most promising applications of integrating XAI and LLM in your project.

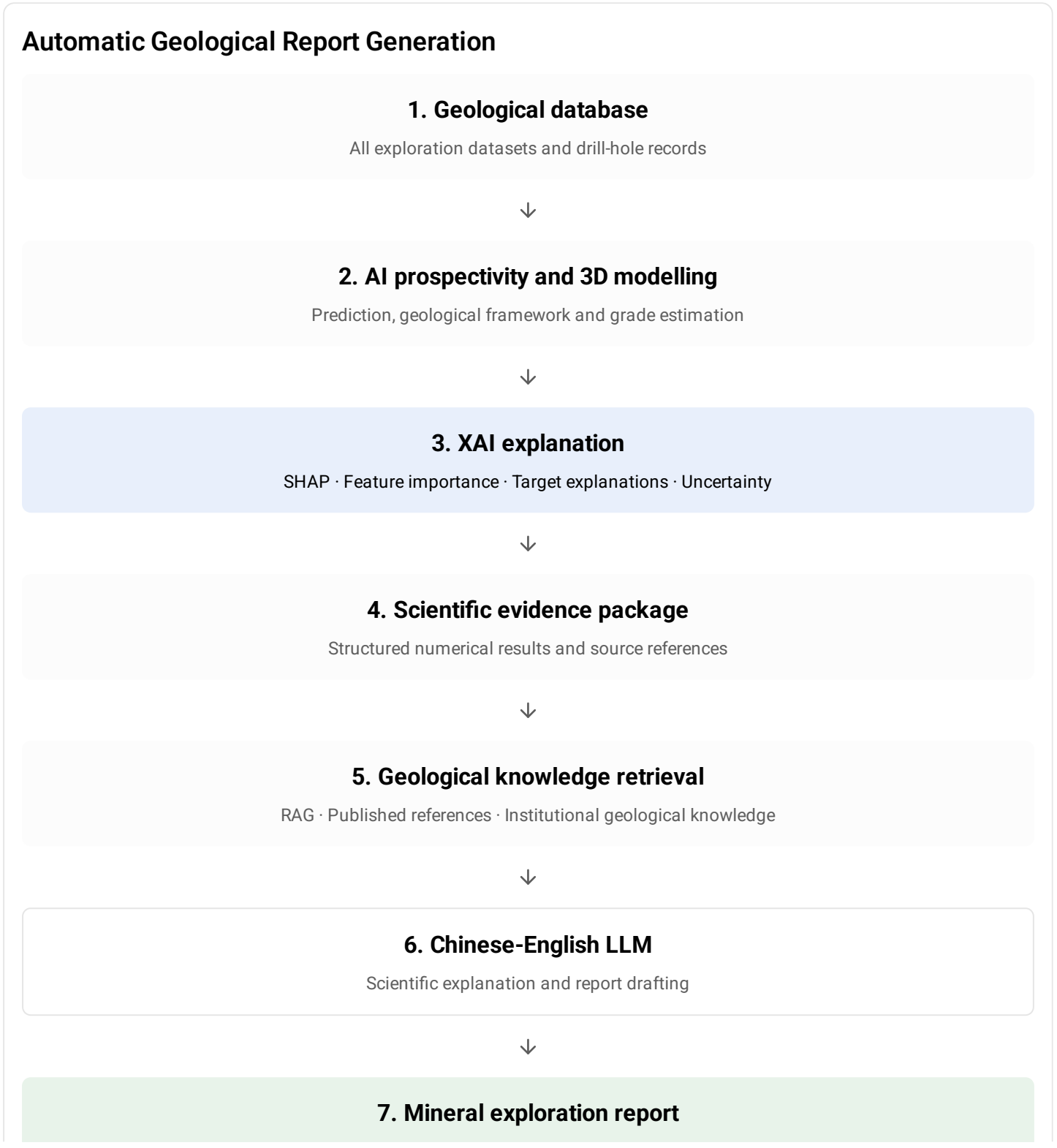
However, it is useful to distinguish two processes:

- XAI generates quantitative explanations of the AI model.
- LLM converts those explanations and the geological information into readable scientific and administrative reports.

The LLM does not independently calculate the mineral prospectivity or determine the actual geological significance of a deposit.

It prepares a scientifically grounded narrative using the verified outputs of your existing software.

7.1. Proposed automatic report generation workflow



DOCX · PDF · Tables · 2D maps · 3D figures

↓

8. Geological expert review and approval

The report generation process should be repeatable and linked to the exact model version and database version.

7.2. Example: Automatic explanation of a mineral exploration target

Suppose your AI-3D-MEOMS system identifies Target A.

The scientific engine generates the following structured information:

</> JSON

```
{
  "target_id": "TARGET_A",
  "model_id": "RF_CU_001",
  "model_score": 0.87,
  "calibrated_probability": false,

  "location": {
    "x": 500250,
    "y": 7500350,
    "z": 450
  },

  "global_validation": {
    "spatial_cv_roc_auc": 0.78,
    "validation_method": "grouped_spatial_cv"
  },

  "explanation": {
    "intrusive_contact": 0.18,
    "fault_density": 0.15,
    "mt_resistivity": 0.12,
    "magnetic_susceptibility": 0.08,
    "cu_geochemistry": 0.07
  },

  "data_quality": {
    "geology": "good",
    "geochemistry": "moderate",
    "geophysics": "moderate",
    "drilling": "limited"
```

```
  },

  "uncertainty": {
    "model_uncertainty": "moderate",
    "drilling_support": "limited"
  }
}
```

All values are illustrative.

The LLM receives this information together with relevant geological knowledge and produces a report.

Example of the resulting automatic report

Illustrative generated report

TARGET A: GEOLOGICAL AND AI-BASED EXPLORATION ASSESSMENT

1. General assessment

The AI-based mineral prospectivity model identifies Target A as an area of relatively high prospectivity within the investigated exploration area. The model produces an uncalibrated prospectivity score of 0.87.

2. Principal geological evidence

SHAP analysis indicates that proximity to the porphyritic intrusive contact and local fault density make substantial positive contributions to the model prediction.

The geological relationships are consistent with important characteristics of porphyry copper systems, in which intrusive complexes and structural pathways may influence hydrothermal fluid circulation.

3. Geophysical evidence

The MT resistivity attribute contributes positively to the model prediction. This relationship should be evaluated together with the inversion sensitivity, geological framework and magnetic data.

4. Geochemical evidence

The copper geochemical feature provides additional positive predictive evidence. Further investigation of the spatial distribution of Cu and Mo is recommended.

5. Reliability and limitations

Although Target A receives a relatively high model score, drilling support is limited. The result should therefore be considered an exploration hypothesis rather than confirmation of an economic mineralized body.

6. Recommended follow-up

- Detailed geological mapping.

- Structural verification.
- Ground geochemical sampling.
- Geophysical interpretation and infill surveys where justified.
- Drill targeting after geological review.

This example demonstrates the intended report format, not an actual exploration result.

Notice that the LLM does more than summarize the AI score.

It organizes the evidence, explains its geological significance, identifies limitations and proposes follow-up work.

However, every geological interpretation and recommendation must be supported by the actual data and reviewed by a qualified exploration geologist.

7.3. What types of reports should the system generate?

I recommend developing five automatic reporting functions.

Report	Contents	Main user
Report 1	Geological and exploration database assessment	Exploration team
Report 2	AI model training and validation	AI researchers
Report 3	XAI interpretation of mineral prospectivity	Exploration geologists
Report 4	Individual 3D exploration target assessment	Exploration managers
Report 5	Integrated mineral exploration assessment	Institutional management

Report 1. Geological database assessment

Automatically summarize:

- Available datasets.
- Data quality.
- Spatial coverage.
- Missing information.
- Drill-hole distribution.
- Geophysical survey coverage.
- Geochemical anomaly distribution.

Report 2. AI model performance

Automatically summarize:

- Training dataset size.

- Positive and negative observations.
- Selected predictors.
- Random Forest results.
- XGBoost results.
- Spatial validation results.
- Model limitations.

Report 3. Explainable AI interpretation

Automatically summarize:

- Global SHAP results.
- Feature importance.
- Positive and negative contributions.
- Feature interactions, where calculated.
- Explanation stability.
- Geological interpretation.

Report 4. Individual target assessment

For each selected target:

- Target location.
- Depth range.
- Prospectivity score.
- Local SHAP explanation.
- Geological setting.
- Geochemical support.
- Geophysical support.
- Drill-hole proximity.
- Uncertainty.
- Recommended additional investigation.

Report 5. Integrated exploration assessment

This should be the most comprehensive report.

It should integrate:

- Geological framework.
- Geochemical interpretation.
- Geophysical interpretation.
- Remote sensing interpretation.
- AI prospectivity.
- XAI explanation.

- 3D geological model.
- Interpreted mineralized geometry.
- Copper grade estimation.
- Uncertainty assessment.
- Exploration recommendations.

The report may include automatic 2D maps and selected screenshots from the PyVista 3D viewer.

7.4. How should the LLM obtain information from the database?

I recommend two separate information pathways.

Pathway A: Structured database queries

The Python engine retrieves numerical information from PostgreSQL/PostGIS.

Examples:

- Number of drill holes.
- Number of samples.
- Mean copper grade.
- Maximum copper grade.
- Prospectivity score.
- SHAP contributions.
- Estimated volume.
- Grade estimation variance.

These values must be calculated using Python or SQL, not by asking the LLM to calculate them from unstructured text.

Pathway B: Geological knowledge retrieval

The RAG system retrieves explanatory information from geological reports, textbooks, research publications and institutional documents.

The LLM combines the two information sources to generate the report.

For example:

- Database: Target A is 45 m from an intrusive contact.
- SHAP: Intrusive-contact proximity makes a positive contribution.
- Geological knowledge: Intrusive contacts are relevant in many porphyry copper systems.
- LLM: Generates an explanation of the target's geological significance, with appropriate qualifications.

This is the correct division of responsibilities.

7.5. Preventing incorrect AI-generated reports

This issue is extremely important for scientific applications.

The LLM can generate incorrect statements even when it produces fluent academic English or Chinese.

Therefore, I recommend implementing the following controls:

1. All numerical values must originate from verified database queries or scientific model outputs.
2. All scientific claims derived from literature must have retrievable source references.
3. Every target explanation must be linked to its original SHAP results.
4. The system must distinguish measured observations from interpolated values and model predictions.
5. The system must distinguish prospectivity scores from calibrated probabilities.
6. The system must distinguish interpreted orebody volumes from confirmed mineral resources.
7. The LLM must identify missing information rather than inventing it.
8. All formal exploration reports must be reviewed by a qualified geologist before approval.

The LLM should generate a *draft scientific interpretation*, not independently certify an exploration target or mineral resource.

PART 6. DEVELOPMENT PROGRESS OF EXPLAINABLE AI FOR MINERAL EXPLORATION

8. Recommended development roadmap

I recommend developing the XAI and LLM components after the initial mineral prospectivity AI model has been established.

There is no need to develop everything simultaneously.

The following is a proposed 15-month development program that can overlap with the existing 18-month AI-3D-MEOMS implementation schedule.

8.1. Development phases

15-month XAI development roadmap

1. **Phase 1 — Data preparation and geological knowledge organization** Months 1–2 Activities:
    - Prepare the XAI feature dictionary.
    - Establish porphyry copper geological knowledge.
    - Organize geological and geophysical data provenance.

- Define mineralization labels.
- Establish data quality indicators.

Deliverable: XAI-ready geological database and knowledge dictionary.

2. **Phase 2 — Baseline AI model and validation** Months 3–4 Activities:
   - Train Random Forest and XGBoost.
   - Establish spatial validation.
   - Evaluate prediction performance.
   - Investigate data leakage and class imbalance.
   - Identify correlated predictors.

   Deliverable: Validated baseline AI models.
3. **Phase 3 — Global and local XAI development** Months 5–6 Activities:
   - Implement SHAP.
   - Develop global feature-importance analysis.
   - Develop local target explanations.
   - Implement permutation importance.
   - Develop ALE plots.
   - Investigate explanation stability.

   Deliverable: Functional XAI interpretation engine.
4. **Phase 4 — 3D spatial XAI development** Months 7–8 Activities:
   - Generate voxel-level SHAP explanations.
   - Develop 3D contribution volumes.
   - Integrate SHAP outputs with PyVista.
   - Develop target-selection and inspection tools.
   - Display prediction uncertainty.

   Deliverable: Interactive 3D Explainable Mineral Prospectivity System.
5. **Phase 5 — Chinese-English geological LLM development** Months 9–10 Activities:
   - Install a local Qwen model.
   - Prepare the geological knowledge base.
   - Implement BGE-M3 embeddings.
   - Implement FAISS retrieval.
   - Develop RAG.
   - Prepare bilingual geological terminology.
   - Evaluate optional LoRA fine-tuning.

   Deliverable: Local Chinese-English Geoscience Assistant.
6. **Phase 6 — Automatic report generation** Months 11–12 Activities:
   - Connect the database to the LLM.
   - Connect SHAP and prediction outputs.
   - Generate target-specific reports.
   - Generate integrated exploration reports.
   - Export DOCX and PDF.
   - Develop report verification procedures.

Deliverable: Automatic Mineral Exploration Reporting System.

7. **Phase 7 — Integration and scientific verification** Months 13–15 Activities:
   - Integrate the XAI engine with QGIS and PyVista.
   - Implement interactive geological questions.
   - Conduct geological expert evaluation.
   - Compare predicted targets with field evidence.
   - Complete offline deployment.
   - Prepare technical documentation and publications.

   Deliverable: Integrated Explainable AI-3D-MEOMS prototype.

8.2. Required development modules

I recommend adding the following directory to the existing Python project.

```
</> Plain text

AI_3D_MEOMS/
 │
 ├──── xai/
 │     ├──── feature_dictionary.py
 │     ├──── global_shap.py
 │     ├──── local_shap.py
 │     ├──── spatial_shap.py
 │     ├──── permutation_importance.py
 │     ├──── ale_analysis.py
 │     ├──── counterfactual.py
 │     ├──── explanation_stability.py
 │     ├──── geological_consistency.py
 │     └──── xai_report_data.py
 │
 ├──── llm/
 │     ├──── local_llm.py
 │     ├──── document_import.py
 │     ├──── document_processing.py
 │     ├──── embedding.py
 │     ├──── vector_database.py
 │     ├──── rag_retrieval.py
 │     ├──── geological_assistant.py
 │     ├──── report_generation.py
 │     └──── report_validation.py
 │
 ├──── knowledge_base/
 │     ├──── porphyry_copper/
 │     ├──── geological_fundamentals/
 │     ├──── geochemistry/
 │     ├──── geophysics/
```

```
│   ├── mineral_exploration/
│   ├── xai_methods/
│   └── terminology/
│
└── outputs/
    ├── shap/
    ├── xai_maps/
    ├── xai_3d/
    ├── target_reports/
    └── exploration_reports/
```

8.3. Initial Python implementation of SHAP

The following example demonstrates how to introduce XAI into your existing Random Forest pipeline.

```python
import shap
import numpy as np

from scipy import sparse

# Existing trained pipeline
preprocessor = pipeline.named_steps[
    "preprocessor"
]

rf_model = pipeline.named_steps[
    "classifier"
]

# Transform the original training data
X_background = preprocessor.transform(
    X_train
)

# Transform target locations
X_explain = preprocessor.transform(
    X_targets
)

# Convert sparse matrices if necessary
if sparse.issparse(X_background):
    X_background = X_background.toarray()

if sparse.issparse(X_explain):
    X_explain = X_explain.toarray()
```

```
feature_names = (
    preprocessor.get_feature_names_out()
)

# SHAP explanation
explainer = shap.TreeExplainer(
    rf_model,
    data=X_background[
        :min(100, len(X_background))
    ],
    feature_perturbation="interventional",
    model_output="probability",
    feature_names=feature_names
)

shap_values = explainer(
    X_explain
)

# Binary classification:
# select the positive class if the installed
# SHAP version returns a separate class dimension.
if shap_values.values.ndim == 3:
    shap_values = shap_values[:, :, 1]

# Global explanation
shap.plots.beeswarm(
    shap_values
)

# Global feature ranking
shap.plots.bar(
    shap_values
)

# Local explanation of the first target
shap.plots.waterfall(
    shap_values[0]
)
```

This example assumes that `pipeline` has already been fitted and that `X_train` and `X_targets` use exactly the same input features.

For a spatial validation experiment, the background data must come from the corresponding training fold, not the held-out test holes.

For correlated geological and geophysical features, the explanation should also be repeated using appropriate background and feature-grouping strategies.

The official SHAP implementation provides TreeExplainer specifically for tree-based models, including Random Forest and XGBoost. SHAP lates... +1

8.4. Evaluation of the XAI system

The faculty should evaluate XAI using five independent criteria.

Evaluation criterion	Scientific question
Model fidelity	Does the explanation correctly represent the trained model?
Explanation stability	Does the explanation remain similar across training experiments?
Geological plausibility	Is the explanation consistent with geological evidence?
Data traceability	Can the original source of every important feature be identified?
Exploration usefulness	Does the explanation improve target evaluation or follow-up decisions?

I particularly recommend examining explanation stability through the following experiment:

- Train RF using one spatial training configuration.
- Calculate global SHAP values.
- Change the training drill-hole groups.
- Retrain RF.
- Calculate global SHAP again.
- Compare the rankings and signs of the important features.
- Repeat for XGBoost.

A feature that is consistently important across multiple independent experiments is generally more useful for geological interpretation than one that appears important in only one experiment.

The acceptance criteria should be established before the final pilot evaluation, rather than chosen after seeing favorable results.

PART 7. INTEGRATION OF EXPLAINABLE AI INTO QGIS AND PYVISTA

9. Proposed new software interface

I recommend adding a new major section to the existing AI-3D-MEOMS QGIS interface:

Explainable AI and Geological Intelligence.

The interface should contain six functions.

Interface function	Operation
Global XAI	Display overall SHAP and feature importance
Local XAI	Explain selected exploration targets
3D XAI	Display spatial feature contributions
Geological Interpretation	Compare AI results with geological knowledge
Geological AI Assistant	Ask questions in Chinese or English
Automatic Reporting	Generate scientific exploration reports

9.1. Example of an interactive geological AI assistant

Imagine that your exploration geologist is viewing a 3D prospectivity model in PyVista.

The geologist selects a voxel or exploration target and asks:

AI-3D-MEOMS Geological Assistant Offline

Geologist's question

Why does the AI predict high mineral prospectivity at this location? Explain the geological and geophysical evidence.

AI assistant's response

The selected location has a high prospectivity score relative to the model's reference predictions.

SHAP analysis identifies three principal positive contributions:

1. Proximity to the porphyritic intrusive complex.
2. High structural density associated with interpreted faults.
3. The MT resistivity attribute.

These features are compatible with some geological characteristics of porphyry copper systems.

However, the prediction is not independently confirmed by drilling. The available data support an exploration hypothesis rather than a confirmed mineralized body.

Recommended investigation:

- Verify intrusive contacts and structural intersections.
- Examine MT inversion sensitivity.
- Conduct detailed alteration mapping.
- Consider additional drilling after geological review.

Illustrative interface and response. Actual responses will be generated from the selected voxel's measured features, SHAP values and retrieved geological references.

This is a very attractive educational and research function for your faculty.

Students and researchers could interact with the 3D geological model using ordinary geological questions rather than having to understand all the underlying Python algorithms.

9.2. The role of the geologist

I recommend establishing a human-in-the-loop approach.

The software should allow the geologist to:

- Accept an AI interpretation.
- Reject an AI interpretation.
- Modify a geological interpretation.
- Add field observations.
- Record alternative geological hypotheses.
- Recalculate the AI prediction after updating the database.
- Compare old and new prediction results.

This is important because the geological interpretation should remain scientifically controlled by geoscientists.

PART 8. FINAL RECOMMENDATIONS FOR YOUR FACULTY

10. What should be developed first?

Although the complete system may eventually include XAI, Chinese LLM, RAG and automatic report generation, I strongly recommend the following order.

Recommended institutional development priorities

Priority 1

Explainable machine learning

Implement SHAP, global and local explanations, feature importance and spatial validation.

Priority 2

3D spatial explainability

Display individual feature contributions in the 3D geological model.

Priority 3

Geological knowledge base

Establish porphyry copper knowledge, bilingual terminology and institutional scientific documents.

Priority 4

Chinese-English LLM and RAG

Install a local Qwen model and connect it to the geological knowledge base.

Priority 5

Automatic scientific reporting

Generate target-specific and integrated mineral exploration reports.

I would not start by training a Chinese LLM. I would first establish the numerical XAI system and then develop the LLM as an intelligent communication and knowledge-retrieval component.

11. Recommended research projects for academic staff and postgraduate students

This development program can also become an important research and doctoral education platform.

I recommend dividing the work into the following research topics.

Research topic	Possible doctoral thesis title
1	Development of Explainable Machine Learning for Integrated Mineral Prospectivity Mapping
2	Three-Dimensional SHAP-Based Interpretation of Integrated Geophysical and Geological Data for Porphyry Copper Exploration
3	Geological Knowledge-Guided Explainable AI for Mineral Exploration
4	Uncertainty-Aware Explainable Machine Learning for 3D Mineral Prospectivity Prediction
5	Development of a Chinese-English Geological Large Language Model Using Retrieval-Augmented Generation
6	Intelligent Automatic Generation of Mineral Exploration Reports Using Explainable AI and Large Language Models
7	Integration of Geological Knowledge Graphs and Explainable AI for Porphyry Copper Exploration
8	Human-in-the-Loop Explainable AI for 3D Mineral Exploration Target Optimization

These are potentially independent doctoral research topics, provided each has a sufficiently original research question, appropriate data and rigorous validation.

12. How to measure the final success of the project

I recommend establishing the following final acceptance criteria.

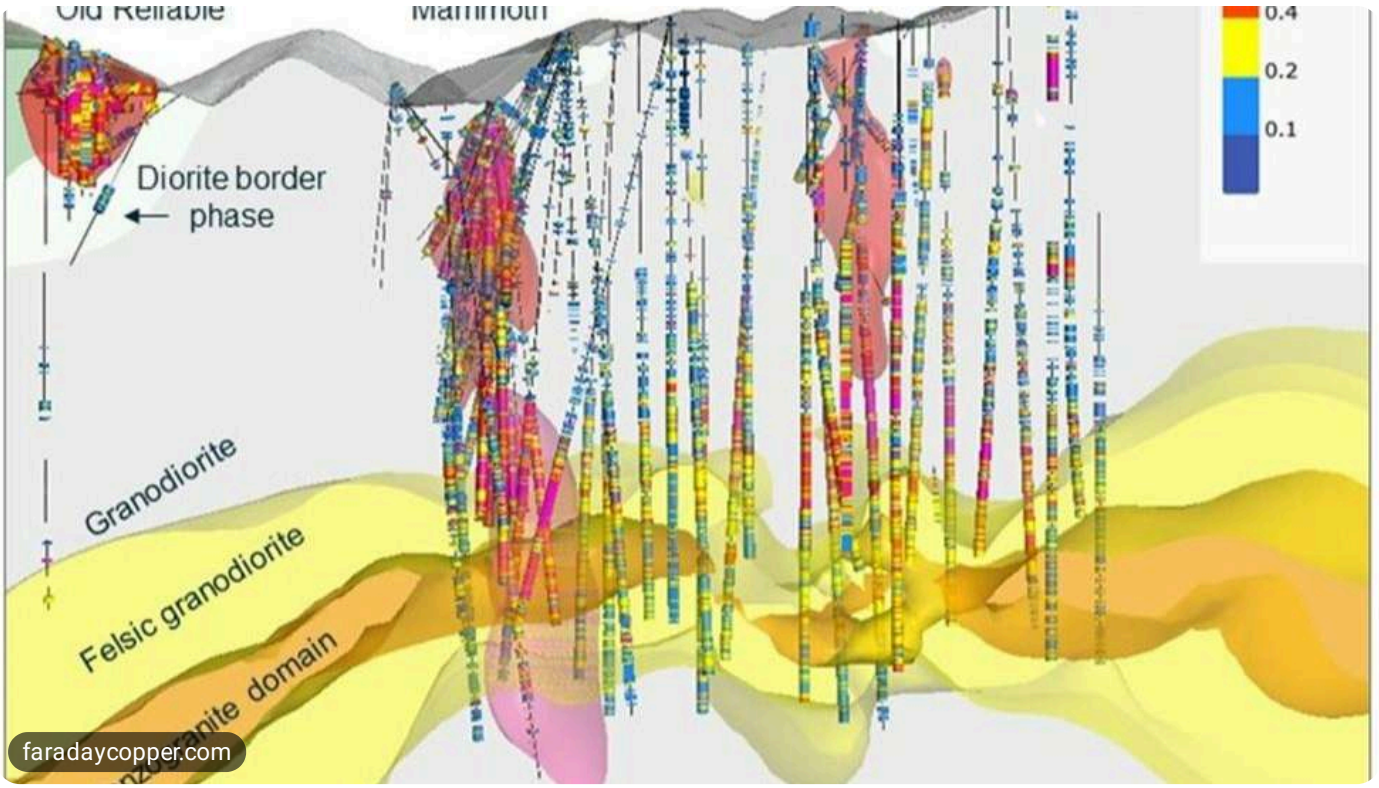
Area	Required result
AI prediction	Reproducible, independently evaluated prospectivity models
Global explanation	Quantitative identification of important predictors
Local explanation	Individual explanation of selected targets
3D explanation	Spatial visualization of predictor contributions
Geological consistency	Comparison with porphyry copper geological knowledge
Uncertainty	Separate prediction and explanation reliability indicators
Chinese LLM	Reliable Chinese and English geological communication
Knowledge retrieval	Traceable references to institutional and scientific documents
Automatic reporting	Generation of evidence-supported exploration reports
Offline operation	All functions operate without internet connectivity
Expert evaluation	Geological review and independent field verification

PART 9. MY FINAL CONCEPT FOR AI-3D-MEOMS

After considering your entire project, I would propose that its ultimate development objective be formulated as follows.

Proposed final system

AI-3D-MEOMS: An Explainable, Knowledge-Assisted and Intelligent Mineral Exploration System



My most important scientific recommendation

The system will have six interconnected capabilities.

I want to emphasize one point.

The final achievement of your project should not simply be a system that says:

"Target A has 87% prospectivity."

It should be a system that can answer the following questions:

1. Why does the AI identify Target A?
2. Which geological, geochemical and geophysical observations support the prediction?
3. Which factors contribute positively or negatively?
4. How stable is the explanation?
5. Is the prediction consistent with the porphyry copper geological model?
6. What information is missing?
7. What additional exploration should be conducted?
8. What evidence would confirm or contradict the prediction?

If your system can answer these questions using traceable data, explainable algorithms and expert-reviewed geological knowledge, it will represent a substantial advance over a conventional mineral prospectivity mapping program.

In my view, the best strategy for your faculty is to develop XAI as the scientific core, a Chinese-English LLM as the communication and knowledge interface, and QGIS/PyVista as the operational platform.

This would provide not only a practical mineral exploration software package but also a long-term research platform for doctoral students, academic staff and collaborative geological exploration

1. **Geological data intelligence:** Organize and integrate all geological exploration information.
2. **Predictive intelligence:** Predict 3D mineral prospectivity and estimate copper grade.
3. **Explanatory intelligence:** Explain why particular underground locations are identified as prospective.
4. **Geological knowledge intelligence:** Compare predictions with established geological knowledge.
5. **Language intelligence:** Communicate with geologists in Chinese and English using a local LLM.
6. **Reporting intelligence:** Generate scientifically traceable exploration reports and recommendations.

projects.

Selected technical references

- Lu et al. (2026), *Explainable Machine Learning in Mineral Prospectivity Mapping: A Critical Review of Methods, Geological Knowledge Embedding, Validation, and Future Directions*. .
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